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A Framework for Research
and Curriculum Development
in Mathematics Education

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Development in Mathematics Education

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Preface

The interest in producing this book arose out of a concern for the education of graduate students and young researchers in mathematics education. A few years ago, as a result of her experience in giving seminars and workshops about APOS Theory, María Trigueros, a member of the writing team for this book, raised the issue in a conversation that took place among some of the authors. The conversation centered around the question of why novice researchers find it so difficult to learn and then to apply APOS Theory. In her view, published articles were not sufficiently didactical to teach the theory in an effective manner. She believed a solution could be offered by writing a book on APOS. Since then there has been considerable reflection about this project among the members of the writing team. This led to a detailed plan that was followed by a long process of writing and revision and finally to the book itself through a contract with Springer.

The fact that Ed Dubinsky, the founder of APOS Theory, as well as RUMEC¹ members including two of Dubinsky's former students, who learned the theory from him through working on research projects together, and Solange Roa-Fuentes, a new generation researcher who just finished her doctorate under the supervision of one of the authors, all participated in the writing of this book, makes it unique. Like the theory itself, people who study and apply APOS Theory are in constant interaction, teaching this framework, learning from it and from each other.

The purpose of the book is to present a “portrait” of APOS Theory: to give a detailed explanation of the theory, the basic principles behind it and its various components; to describe the way in which research to develop the theory has been, and continues to be, taking place; to show how it can be, and has been, used in teaching; and to point to studies that report on the effectiveness of APOS-based instruction. In the spirit of one pedagogical strategy most often used in connection with APOS Theory, the authors of this book have worked cooperatively, with each author intimately involved in writing, reviewing and revising every chapter. The authors engaged in a joint process of discussion, writing, revision, discovery, and

¹ The meaning of this acronym will be clarified in Chap. 2.

sometimes even debate, as they interacted with the theory in finding ways to explain and present it in a single volume that encompasses more than 25 years of research on student thinking and teaching. It is hoped that this cooperative work has led to a comprehensive unity in the text that will be helpful to anyone interested in understanding and using APOS Theory: students, researchers, teachers, educators and decision makers.

We thank everybody, especially the graduate students who constantly challenged us through their questions. In particular we thank Yanet González, a graduate student at Cinvestav, for providing some of the information for the annotated bibliography. We would also like to acknowledge Annie Selden who read parts of the draft of this book and made many helpful suggestions.

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Chapter 1

Introduction

The acronym APOS stands for Action, Process, Object, and Schema. APOS Theory is a theory of how mathematical concepts can be learned. Rooted in the work of Jean Piaget, its fundamental ideas were first introduced in the early 1980s (Dubinsky 1984), and since that time, extensive development and application have been carried out by researchers, curriculum developers, and teachers in many countries throughout the world.

APOS Theory focuses on models of what might be going on in the mind of an individual when he or she is trying to learn a mathematical concept and uses these models to design instructional materials and/or to evaluate student successes and failures in dealing with mathematical problem situations. APOS Theory can be used, and in many studies has been used, successfully, as a strictly developmental perspective (e.g., Breidenbach et al. 1992), as a strictly analytical evaluative tool (e.g., Dubinsky et al. 2013), or as both (e.g., Weller et al. 2011). APOS-based research and curriculum development has focused mainly on learning mathematics by students in the secondary and postsecondary grades, but as will be seen in Chap. 9, work has also been done in the context of elementary and middle school mathematics. There is also some preliminary work on applying APOS Theory to areas outside of mathematics, such as computer science.

APOS is a constructivist theory. In Chap. 2 the sense in which that statement is made will be explained. Chapter 2 also contains a description of Piaget's notion of reflective abstraction and the role it has played in the development of APOS Theory. The development of APOS Theory to date is described in terms of three major periods: first thoughts, work done by the Research in Undergraduate Mathematics Education Community (RUMEC), and continuing efforts by small teams that function independently.

Chapter 3 discusses the mental structures that constitute APOS Theory: Action, Process, Object, and Schema and some of the mechanisms by which those mental structures are constructed—interiorization, encapsulation, coordination, reversal, and de-encapsulation. The discussion in this chapter is exemplified by several

specific mathematical concepts that illustrate how APOS Theory can be used to hypothesize the construction of mathematical knowledge by an individual. There is also a discussion of some general issues that should be taken into consideration when working with APOS Theory.

Chapter 4 presents an in-depth description of one of the major tools used in APOS-based research and curriculum development—the genetic decomposition—a hypothetical model of mental constructions that a student may need to make in order to learn a mathematical concept. The chapter includes discussion of what a genetic decomposition is and its role in working with APOS Theory. Several issues regarding genetic decompositions, such as their nonuniqueness, the relation between a preliminary genetic decomposition and its refinement(s), and common misunderstandings about the design of a genetic decomposition, are also considered.

Chapter 5 is concerned with the design and implementation of instruction using APOS Theory. Implementation is usually carried out using the Activities, Class discussions, Exercises (ACE) Teaching Cycle, an instructional approach that supports development of the mental constructions called for by a genetic decomposition.

The ACE cycle includes activities on which students typically work cooperatively using a *mathematical programming language* such as ISETL (Interactive SET Language). All of the components of the ACE Teaching Cycle and some features of ISETL are described and examples are given of APOS-based instruction on groups in abstract algebra for mathematics majors and on infinite repeating decimals in a course for prospective elementary and middle school teachers.

In Chap. 6, the overarching research stance linked to APOS Theory is presented as a paradigm, which differs from most mathematics education research trends in its theoretical approach, methodology, and types of results. Following the ideas of Kuhn (1962), this paradigm contains theoretical, methodological, and pedagogical components that are closely linked together. It is pointed out that not all studies that adopt APOS as a theoretical framework make use of all the elements of the paradigm referred to in this chapter. Rather, it serves as an “ideal” organization of an APOS-based research study.

Schemas and the thematization of a Schema are the central topics of Chap. 7. The chapter begins with a general description of a schema and several examples. Then, there is a description of how Schemas may develop and of the consequences that result from modifications of a Schema, either through the introduction of new information or through the interaction of one Schema with another. The chapter ends by dealing with the issue of thematization, the mechanism involved in constructing an Object conception of a Schema.

Chapter 8 contains three themes: a general discussion of the progression between stages in APOS Theory and related pedagogical strategies, a description of the terms *stages* and *levels* as they appear in the work of Piaget and in APOS-based research, and a summary of the research regarding 0.9 that suggested the need for

levels between stages and for Totality as a new stage between Process and Object.¹ At this point, the status of Totality and the use of levels described here are no more than tentative because evidence for a separate stage and/or the need for levels arose out of just two investigations: the studies of the relation between 0.9 and 1, which is discussed in Chap. 8, and the study of fractions, considered in Chap. 9. Thus, it remains for future research to determine if Totality can be considered as a separate stage, if levels are really needed in these contexts, and to explore what the mental mechanism(s) for constructing them might be. Research is also needed to determine the role of Totality and levels in other contexts, both those involving infinite processes and those involving finite processes.

Chapter 9 discusses a 1990s project that involved the introduction of Piagetian and APOS ideas into the teaching of fractions in grades 4 and 5. The chapter describes the data and conclusions from three studies related to that project (Arnon 1998; Arnon et al. 1999, 2001). In contrast to most APOS-based studies, these investigations describe the possibility of applying APOS Theory to investigate the learning of mathematics in elementary school. In elementary and middle school grades, most students are at what Piaget defined as the stage of concrete operations, which means that the objects acted upon by actions must be concrete (e.g., blocks, cutouts, drawings), that is, they need to be perceivable by one's senses (Piaget 1975, 1974/1976). At the higher grades, however, the objects on which actions are performed are not necessarily concrete objects that belong mainly to the physical world but rather can be abstract objects (e.g., propositions, functions) that exist mainly in the minds of individuals. Thus, from the perspective of APOS Theory, the concreteness of the objects to which actions are applied constitutes the main difference between the elementary or middle grades and the secondary or postsecondary grades. In addition, the chapter describes how imagination plays a substantial role in a young child's interiorization of an Action into a Process. Another study (Arnon et al. 2001), discussed in Chap. 9, describes how using concrete representations following some APOS principles enabled grade 5 students to learn abstract mathematical concepts that are outside the regular syllabi of elementary school curricula.

Chapter 10 gives some questions that have been asked about APOS Theory either in print or in personal communication with the authors of this book and provides suggested responses to these questions. The topics range among specific components of the theory, their relationship to the work of Jean Piaget, and dealing with particular concepts in mathematics courses. Other topics from mathematics education research, such as context, epistemology, intuition, metacognition, metaphors, and representations, are discussed.

Chapter 11 summarizes the entire book by exploring themes and common threads. The notions of the developmental/evaluative dichotomy, the growth of the theory through mechanisms similar to those used in the theory, and the future of

¹ Per convention, the bar over the digits in a decimal expression signifies the digits that repeat in a repeating decimal.

APOS Theory are discussed. The chapter provides “APOS Theory at a glance” and ends with some final thoughts.

Finally, Chap. 12 contains an extensive annotated bibliography of publications related to APOS Theory.

Although the intention has been to make this book as up to date as possible, whatever success has been achieved in that endeavor may not last. APOS Theory is a living, growing body of ideas that attempts to synthesize the thinking of its progenitor, Piaget, with that of current and future workers, along with data resulting from empirical studies of students trying to learn mathematics. The result is that those involved in APOS-based research and curriculum development are continually revising and rethinking various aspects of the theory and making revisions where appropriate. This is in keeping with Piaget who once wrote that “‘Piaget’s Theory’ is not completed at this date and the author of these pages has always considered himself one of the chief revisionists of Piaget” (Piaget 1975, p. 164). This means that this book cannot, and should not, be considered as the “last word” on APOS Theory. Future studies will lead to further revisions. Indeed, the reader can see in this text that this has already occurred. Although almost all of the work done in APOS Theory during the first 30 years of its life is still relevant, the book points out several examples in which thinking about one or another aspect of the theory has changed and, accordingly, various descriptions have been modified. In some cases, material about APOS Theory that appears in published works is used (with attribution), but revisions are made to reflect current thinking and the results of APOS-based research. There is even one example (see the discussion of Totality in Chap. 8) where a major change in the theory has been proposed. It is hoped that this dynamic nature of APOS Theory makes it even more helpful to researchers, teachers, and students.

Chapter 2

From Piaget's Theory to APOS Theory: Reflective Abstraction in Learning Mathematics and the Historical Development of APOS Theory

The aim of this chapter is to explain where APOS Theory came from and when it originated. A discussion of the main components of APOS Theory—the mental stages or structures of Action, Process, Object, and Schema and the mental mechanisms of interiorization, coordination, reversal, encapsulation, and thematization—points to when they first came on the scene and how their meanings developed. The published research of those involved in the development of APOS Theory, which includes some early colleagues and students of Dubinsky as well as those who were members of the Research in Undergraduate Mathematics Education Community (RUMEC), is described. The descriptions in this chapter are very brief and will be expanded in later chapters.

In the seven sections of this chapter, the development of APOS Theory and its application in helping students construct their understanding of various mathematical concepts is traced. This development, which began in the early 1980s, has continued since that time and plays an important role in mathematics education research and curriculum development. The chapter begins with a description of Piaget's notion of reflective abstraction (Sect. 2.1) and how it inspired the development of APOS Theory (Sect. 2.2). Next there are descriptions of the first thoughts about APOS Theory (Sect. 2.3) and its first major period of development (Sect. 2.4). The second major period began with the formation of RUMEC, which produced a large amount of coordinated cooperative research conducted by teams of mathematicians who were moving into education research (Sect. 2.5). During the third major period, from the end of RUMEC as a formal organization to the present, various teams have been functioning independently to conduct APOS-based research and to study its application to the design and implementation of instruction (Sect. 2.6). Finally, there is a very brief mention of two related theoretical perspectives (Sect. 2.7).

2.1 Piaget's Notion of Reflective Abstraction

One of the major ideas of Piaget is what he called *reflective abstraction*, which he saw as both the main mechanism for the mental constructions in the development of thought and the mental mechanism by which all logico-mathematical structures are developed in the mind of an individual. He reiterated this point of view very often in many different contexts. For example, regarding the development of thought, he wrote, "The development of cognitive structures is due to reflective abstraction. . ." (Piaget 1975/1985, p. 143). Regarding mathematics, he considered that reflective abstraction is the mental mechanism by which all logico-mathematical structures are derived (Piaget 1967/1971) and wrote that "...it [reflective abstraction] alone supports and animates the immense edifice of logico-mathematical construction" (Piaget 1974/1980, p. 92). Starting around 1982, a research mathematician, Dubinsky, switched from research in mathematics (functional analysis) to research into the mental activities involved in students' learning of mathematics. It was statements like the above about the edifice of mathematics that first attracted him to Piaget's ideas about reflective abstraction and their application to mathematical thinking.

What is reflective abstraction? Piaget's answer, repeated in many different publications, consists of two parts. The first part involves reflection, in the sense of awareness and contemplative thought, about what Piaget called content and operations on that content, and in the sense of reflecting content and operations from a lower cognitive level or stage to a higher one (i.e., from processes to objects). The second part consists of reconstruction and reorganization of the content and operations on this higher stage that results in the operations themselves becoming content to which new operations can be applied (Piaget 1973). This second step appeared to Dubinsky to be very close to certain mathematical ideas. One of many examples is the case of functions. They are first constructed as operations that transform elements in a set, called the domain, into elements in a set, called the range. Then, at a higher stage, as elements of a function space, functions become content on which new operations are constructed. Integers are another example. At one stage, an integer is an operation or process of forming units (objects that are identical to each other) into a set, counting these objects and ordering them. At a higher stage, integers become objects to which new operations, e.g., those of arithmetic, are applied (Piaget 1965). These types of examples led Dubinsky to believe that reflective abstraction could be a powerful tool in describing the mental development of more advanced mathematical concepts.

2.2 Reflective Abstraction and the Antecedents of APOS Theory

To clarify the above ideas, it might be helpful to look at some of Piaget's examples of reflective abstraction and see how they formed the antecedents of APOS Theory.

Piaget did not believe that the most general and useful abstract ideas come from drawing out common features of a variety of phenomena. Considering an example from advanced mathematics, he wrote that:

[T]he group concept or property is obtained, not by this sort of abstraction [drawing out common features], but by a mode of thought characteristic of modern mathematics and logic—"reflective abstraction"—which does not derive properties from *things*, but from our ways of *acting on things*, the operations we perform on them.... (Piaget 1968/1970, p. 19)

In other words, the development of knowledge about an object, either mental or physical, requires both the object and a subject who acts on the object. In his view, the subject (knower) and the object cannot be dissociated; it is impossible to speak of either of them without the other. Piaget applied these ideas to the full range of topics in mathematics, from the most elementary concepts constructed by the young child to the advanced work of the research mathematician. This general framework—content and operations on content that lead to the operations themselves becoming new content—lays the foundation for more subtle distinctions, such as the distinction between material actions and interiorized operations, that constitute the difference between the mental structures of Action and Process and how mental mechanisms, such as interiorization and encapsulation, lead to the formation of different conceptions that constitute the $A \rightarrow P \rightarrow O \rightarrow S$ progression.

Piaget made many statements such as:

... it follows that when the child discovers by experience the result of an action, for example, that the result of an addition is independent of the order followed (which is a property of the actions of combining and ordering and not a property of the objects as such, which include neither sum nor order independently of the actions carried out on them), reflective abstraction consists of translating a succession of material actions into a system of interiorized operations, the laws of which are simultaneously implied in an act. (Beth and Piaget 1966/1974)

What Piaget seems to be saying here is that properties of objects do not reside in the objects but rather in the actions that are performed on these objects. Thus, properties of objects depend on both the objects and on subjects who know the objects. Dubinsky interpreted "material actions" in the above quote to refer to actions that are performed by a subject but are external to the subject. In the example above, the material actions consisted of transforming physical objects by taking two small sets of objects; counting the number in one set, then the other set, and adding the two results to get the total number of objects; and then repeating the action with the order of the two sets reversed to see that the total is the same. Here, the objects are numbers (integers represented by sets of physical objects), the action applied to these objects is addition, and the property (of the operation, not the numbers) is commutativity. In APOS Theory, Piaget's "interiorized operations" became Processes.¹ The "translation" became the APOS mental mechanism of "interiorization," whereby an

¹ Capitalization is used to differentiate between Piaget's terminology and Dubinsky's use of Action, Process, Object, and Schema.

external, that is, physical, Action ("material" action) is reconstructed in the mind of the subject to become a Process (interiorized operation), that is, an internal, mental construction that does the same thing as the Action, but wholly in the mind of the subject rather than externally. Dubinsky considered Piaget's "system" to refer to a "Schema" which, in this case, is the concept of commutativity, and Piaget's phrase "which are simultaneously implied in an act" was the source of the idea of "coherence," by which a subject decides whether or not a particular Schema is applicable to a particular mathematical problem situation.

Piaget's notion of reflective abstraction also influenced the development, in APOS Theory, of how a Process (interiorized operation or Action) is transformed into an Object (operation to which new higher stage operations can be applied) via the mental mechanism of encapsulation. Piaget applied reflective abstraction to the concept of proportion when he wrote, "A proportion is an equality of relationships, in other words, a specific case of equivalence between relations" (Piaget et al. 1968/1977, p. 186). This is because a proportion begins with a relation between two objects. For example, division of a positive integer a by a positive integer b , written $\frac{a}{b}$, tells us that a certain number of copies of b is contained in a , which is a relation between a and b . Then, there is the same relation between two other positive integers c and d , written $\frac{c}{d}$. Each of these two relations is an action applied to the pairs of positive integers a, b and c, d . Thus, the relations come from a learner's actions on these objects and so are examples of reflective abstraction. A further reflective abstraction transforms these actions to fractions, which are objects to which numerical values can be assigned.

Dubinsky's interpretation of this description is that the relations $\frac{a}{b}$ and $\frac{c}{d}$ are Processes that are encapsulated to become Objects to which another relation, comparison, can be applied. This latter relation is the meaning of a proportion, which is itself an Action, not on physical objects, but on the relations $\frac{a}{b}$ and $\frac{c}{d}$ which become mental Objects as a result of applying the mental mechanism of encapsulation.

Other examples of how APOS Theory arises out of Piaget's studies of reflective abstraction and the development of intelligence concern the notions of schema, thematization of schemas, and coordination of schemas. Consider, again, the concept of positive integer. Piaget (1965) describes a long period of development of this concept, culminating around age 7.² This development involves the construction of a number of schemas, the main two being what Piaget called seriation (ordering) and classification (formation and comparison of sets). The schema of classification is the structure that allows the child to look at some objects (e.g., pieces of fruit) and think of them as indistinguishable units, ignoring all qualities. In

² As Piaget (1972) acknowledged, subsequent research showed that the age at which various cognitive developments occur could vary as a function of the child's culture and other factors such as aptitudes and interests.

applying this schema, the child uses the schema of 1–1 correspondence (previously or simultaneously constructed) to determine that two such sets have the same number of units or uses the set inclusion schema (also previously or simultaneously constructed) to determine that one set has more (or less) elements than another. This latter understanding lays the foundation for development of the schema of seriation by which the child is able to imagine a sequence of sets such as

$$\{1\}, \{1 + 1\}, \{1 + 1 + 1\}, \{1 + 1 + 1 + 1\}, \dots^3$$

With these constructions, the child can name these sets as one, two, three, four, etc., and also name their position in the order as first, second, third, fourth, etc. Finally, the two schemas of classification (set formation) and seriation (ordering) are thematized and then coordinated to form a new schema. The key step in the coordination occurs when the child realizes that, for example, the set with four elements in the above sequence is also the fourth set in the sequence. The resulting schema is the concept of positive integer. Piaget considered all of these constructions to be examples of reflective abstraction.

The coordination referred to by Piaget is an action on two schemas. This is a very general use of the term *coordination* that includes any construction which uses two schemas, such as one schema following another or going back and forth between the two schemas using parts of one and then of the other. In order to do this, the schemas must first be thematized, which means made into objects (as processes are encapsulated into objects) to which the action of coordination can then be applied.

The notion of an individual moving mentally from Action to Process, and from Process to Object, arises clearly in Piaget's discussion of the cognitive development of functions, where he wrote, "Their most general characteristic stems from their passage from qualitative coproperties resulting from elementary 'applications' to operatorily quantifiable covariations, then to variations of variations, etc." (Piaget et al. 1968/1977, p. 186). Here, Piaget's comment can be interpreted to refer to functions as maps (application) which are initially actions and then processes (operatorily quantifiable covariations) and then to objects (making it possible to apply variations to variations.)

Dubinsky interpreted these types of passages as descriptions of cognitive development that begins with Actions (elementary "applications") that are interiorized into Processes (operatorily quantifiable covariations) and then encapsulated into Objects to which new Actions can be applied (variation of variations). This is an example, rooted in Piaget's reflective abstraction, of the development from Action to Process to Object to Schema, the $A \rightarrow P \rightarrow O \rightarrow S$ progression that is the heart of APOS Theory. Note, however, that although this progression is presented, of necessity, as a linear string, the development does not always proceed linearly, one stage after another. Rather, an individual may move back and forth between stages as the situation requires.

³ Piaget is using the symbol "+" here, not as addition, but as "and."

Thus this section has shown examples of how Piaget's theory of reflective abstraction formed the antecedents to APOS Theory—the mental structures of Action, Process, Object, and Schema, and the mental mechanisms of interiorization, coordination, reversal, encapsulation, and thematization as well as their formation into the developmental $A \rightarrow P \rightarrow O \rightarrow S$ progression. This progression, together with the APOS structures and some of the mechanisms, is illustrated in Fig. 2.1 which shows that Actions operate on Objects; Actions are interiorized into Processes; Processes are encapsulated into Objects; and Objects are de-encapsulated back to the Processes whence they came. The entire system is part of a Schema. This diagram and variations of it will be used throughout the book.

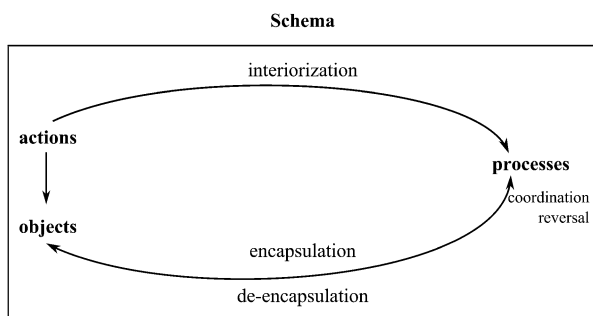


Fig. 2.1 APOS Theory (based on Asiala et al. 1996)

2.3 First Thoughts About APOS Theory, 1983–1984

Dubinsky began, around 1983, to think about applying Piaget's reflective abstraction to postsecondary mathematics and to develop the ideas that later became APOS Theory. The first publication concerning these ideas appeared in 1984 in the proceedings of a conference in Helsinki, Finland, at which he was an invited speaker (Dubinsky 1984). In this talk, he discussed the distinction between thinking about a function as a Process and as an Object and spoke about using the experience of computer programming to help students understand that distinction. He discussed how one applies Actions to mental Objects, gave an example in which a Pascal program represented a proof by mathematical induction, and expressed the view (which he later investigated formally in Dubinsky 1986a, 1989) that if students wrote, debugged, and used such a program, their development of an understanding of induction would be enhanced.

At that time, Dubinsky was particularly interested in the use of computer experiences to help students construct their understanding of mathematical concepts. He wondered, for example, if working with computers in certain ways could help students make reflective abstractions. He observed that the intensive work in writing, debugging, and running computer programs tended to have a profound effect on how the programmer thinks about the content of what he or

she was representing on the computer. He thought that careful choices of computer activities could change students' thinking in ways that would help them learn mathematics. Jack Schwartz, one of the few people elected to the US National Academy of Sciences in both Mathematics and Computer Science and the creator of the programming language SETL (forerunner of ISETL), once observed that the only mathematical background necessary to learn to program in SETL was contained in the content of a standard college first-year course in discrete mathematics. Dubinsky noticed that there seemed to be a lot more people who were successful in learning to program computers than in learning discrete mathematics. So he decided to try to reverse Schwartz' observation and have students learn to program in SETL⁴ in order to learn topics in discrete mathematics. In the Helsinki talk, he reported on a discrete mathematics course that he developed based on that idea.

2.4 First Developments of APOS Theory, 1985–1995

During the period 1985–1988, Dubinsky, with various collaborators, developed pedagogical methods for using programming to induce students to interiorize Actions into Processes, encapsulate Processes into Objects, and apply the mental structures that were constructed as a result of programming to learn various mathematical concepts. This work led to the publication of the first textbook that was based entirely on APOS Theory and the use of computer programming as a pedagogical tool (Baxter et al. 1988). This was a textbook for a college course in discrete mathematics that covered such topics as propositional and predicate calculus, sets and tuples, functions, combinatorics, matrices, determinants, mathematical induction, relations, and graphs. The pedagogical strategy used throughout the text was for the authors to develop, for each concept, a *genetic decomposition*, a description of the mental structures of Actions, Processes, and Objects that students might use in constructing an understanding of the concept and the mental mechanisms (i.e., specific reflective abstractions) such as interiorization and encapsulation by which students might construct these structures. The text included laboratory activities in which students used ISETL to write short computer programs that were designed to help them to make the mental constructions called for by the genetic decomposition. (See Chap. 4 for a more thorough discussion of genetic decomposition.) The two most important programming activities involved the interiorization of an Action to a Process (by writing a computer program that performed the action on any appropriate input) and encapsulation of the Process to an Object (by using the program as input and/or output to other programs). Examples of these programming activities will be discussed in Chap. 5.

⁴Later, SETL was replaced by the interpretive programming language, ISETL, developed by G. Levin.

During the period 1989–1995, Dubinsky continued working with various collaborators to develop the framework that eventually became known as APOS Theory. Following Piaget's dictum that the nature of a concept (epistemology) is inextricably interwoven with how it develops in the mind of an individual (learning), all of the research into the epistemology of mathematics took place in conjunction with one or more applications of the theory to teaching in actual classrooms in various colleges. This preliminary APOS-based research and curriculum development led to publications in refereed journals on such mathematical concepts as mathematical induction, compactness, functions, predicate calculus, and calculus. This research, together with the effectiveness of APOS-based pedagogical strategies, was reported in a number of conference presentations and published papers that appeared during the period 1989–1997. These reports will be described in some detail in Chap. 6. The pedagogical strategies eventually led to the development of the *ACE Teaching Cycle* which is discussed in detail in Chap. 5.

During the two periods, 1983–1984 and 1985–1995, the main components of APOS Theory were introduced, developed, and understood pretty much in the way they are understood today. These components include both the mental structures, Actions, Processes, Objects, and Schemas, and the mental mechanisms for building these structures, interiorization, coordination, reversal, encapsulation, and thematization. At the Helsinki conference, in his first public report on this work, Dubinsky began speaking about Actions, Processes, and Objects and, in particular, about applying an Action to a Process (Dubinsky 1984). The term encapsulation, as the mental mechanism for transforming a Process to a mental Object, was introduced publicly about a year later (Dubinsky 1985). Shortly after, interiorization as the mechanism for converting an Action to a Process was first mentioned in an APOS context (Dubinsky 1986b), although the idea of converting an external Action to an internal Process was present from the beginning, as was the application of an Action to a mental Object (Dubinsky 1984). Also discussed during that time was the idea of transforming a Schema to an Object, which could be acted on by another Schema (Dubinsky 1986b). At the time, this transformation was considered to be an encapsulation, but later in Asiala et al. (1996), the name was changed to thematization, the term presently in use, in order to maintain consistency with Piaget (Piaget and García 1983/1989).

During this period APOS-related research did not pay much attention to the mental structure of a Schema. While Schema was mentioned as a “more or less coherent collection of objects along with actions which the subject can perform on them” (Dubinsky 1986b, p. 2), nothing was said during this period about the meaning of “coherent” Schema nor was any attempt made to distinguish the notion of Schema from the “concept image” of Vinner and his colleagues (see Vinner 1983; Vinner and Dreyfus 1989). In fact, Schema and concept image differ in three important ways. The first difference is illustrated in Vinner and Dreyfus (1989):

In most cases, he or she decides on the basis of a *concept image*, that is, the set of all the mental pictures associated in the student's mind with the concept name, together with all the properties characterizing them. (By mental picture we mean any kind of representation—picture, symbolic form, diagram, graph, etc.) The student's image is a

result of his or her experience with examples and nonexamples of the concept. Hence, the set of mathematical objects considered by the student to be examples of the concept is not necessarily the same as the set of mathematical objects determined by the definition. (p. 356)

As can be seen from this quote, concept image is mainly concerned with the mathematics involved in a concept whereas Schema describes the mental structures involved in the mind of an individual who understands, or is developing an understanding of, that mathematics. The second difference is that a Schema can be thematized to be an Object on which Actions can be performed and which can become parts of other Schemas, whereas no such activities are discussed in the literature on concept image. The third difference is the notion of coherence which concerns the use of a Schema in dealing with mathematical problem situations and which, again, is not considered in the literature on concept image. A more detailed discussion of the nature of Schema, including the distinction between the stage of Schema in APOS Theory and concept image, will be given in Sect. 2.5 and in Chap. 7.

Although Dubinsky spoke of a continuous development from Action to Process to Object to Schema in many places during this period (see, e.g., Dubinsky 1991, for a full discussion that is not very different from the current understanding of this progression), the acronym APOS was not introduced until the next period (Cottrill et al. 1996).

2.5 RUMEC, 1995–2003

During the period 1988–1996, Dubinsky was the recipient of grants from the US National Science Foundation (NSF) to conduct curriculum development projects in undergraduate mathematics courses. The research for the paper on reflective abstraction (Dubinsky 1991) and several reports on calculus (not directly related to APOS Theory) were supported by these grants. However, he had also collected a huge amount of data, mostly from interviews with college students who had been studying mathematical concepts in APOS-based courses in calculus, discrete mathematics, and abstract algebra. It was impossible for one person, or even a team of two or three researchers, to analyze all of this data and report the results. So the data remained unanalyzed at that time.

Then, in 1995, Dubinsky received a 5-year grant from the NSF to conduct summer workshops on professional development for college mathematics faculty to learn more about cooperative learning. The project was titled Cooperative Learning in Undergraduate Mathematics Education (CLUME). In one of the workshop sessions during Summer, 1995, Dubinsky discussed the use of cooperative learning in his curriculum development projects and mentioned in passing the existence of this unanalyzed data. The participants expressed considerable interest in hearing about the research that generated this data and so an “off-line” unofficial evening session was held to talk about research in undergraduate mathematics education.

It turned out that most of the CLUME participants were mathematicians who were interested in changing their research efforts from mathematics to undergraduate mathematics education. So an organization, RUMEC, was formed. Its purpose was to help mathematicians get started in doing education research. The initial method, in the spirit of cooperative learning, was to conduct cooperative research in small teams of three, four, or five researchers, by analyzing Dubinsky's data and reporting the results in papers submitted for publication. This work was funded for 5 years, from 1996 until 2001, by two grants from the Exxon Educational Foundation.

An important feature of the work of RUMEC was something that was unique to education research in mathematics: each team conducted its research and produced a draft of a research paper that was distributed by email to all of the approximately 25 members of the organization. The manuscript was reviewed by all RUMEC members, with suggestions for revisions sent to the authors. It was then discussed by the entire group at its annual meeting. This process was called *internal review*. It was only advisory as the authors made the final decisions on the suggestions, produced a final manuscript, and submitted it for publication. The process was apparently effective because during the period 1995–2003, RUMEC teams submitted 14 manuscripts, and, although some journals requested revisions, every RUMEC submission was eventually accepted for publication!

These publications dealt with the development of APOS Theory (Asiala et al. 1996; Clark et al. 1997; Dubinsky and McDonald 2001; Weller et al. 2003) and with learning specific mathematical concepts including limits (Cottrill et al. 1996); graphing and the derivative (Asiala et al. 1997a; Baker et al. 2000); the chain rule (Clark et al. 1997); cosets, normality, and quotient groups (Asiala et al. 1997b); binary operations, groups, and subgroups (Brown et al. 1997); permutations and symmetries (Asiala et al. 1998); sequences and series (McDonald et al. 2000); fractions (Arnon et al. 2001); and the definite integral (Czarnocha et al. 2001). There was also one study on student attitudes about their experiences with pedagogy based on APOS Theory (Clark et al. 1999). Details of some of these studies will be presented in subsequent chapters, but two publications during this period deserve special mention. The first (Asiala et al. 1996) gives a complete and coherent description of APOS Theory as it stood at that time; a description of the ACE Teaching Cycle, which is the main pedagogical strategy for APOS-based instruction; and the methodology used in APOS-based research. The second (Weller et al. 2003) summarizes the results of APOS-based research and the ACE Teaching Cycle up until that time.

2.6 Beyond RUMEC, 2003–Present

When the Exxon funding ran out in 2001, it was no longer possible to hold meetings of the RUMEC membership. Internal reviews were still conducted for a while, but by 2003, RUMEC no longer existed as an organization. Nevertheless, individual

members of RUMEC continue to form research teams for individual studies. There are two examples in which this “post-RUMEC” research has led to further developments of APOS Theory.

The first development involves the structure of Schema. Although the idea of a Schema becoming an Object that can be acted on by a Process or by another Schema emerged very early in the development of APOS Theory (Dubinsky 1986b), this idea, now referred to as *thematization of a schema*, was not extensively studied until Cooley et al. (2007). This development and the interaction of schemas will be discussed in full detail in Chap. 7.

The second development has to do with a series of studies that use APOS Theory to investigate the development of students’ understanding of the mathematical concept of infinity (Weller et al. 2004; Dubinsky et al. 2005a, b, 2008, 2013; Stenger et al. 2008; Brown et al. 2010; Weller et al. 2009, 2011). These studies led to the introduction of a potential new stage in APOS Theory, namely, Totality, which lies between Process and Object. The terms “totality” and “total entity” had been used since the beginning of APOS Theory (Dubinsky 1984), but it was always considered to be just part of the Object stage. Totality as a separate stage and the reasons for its introduction will be discussed in Chap. 8.

In Mexico a group of researchers are currently conducting an APOS-based project to study mental constructions involved in learning linear algebra concepts (such as vector space, basis, linear transformations, spanning sets, and systems of linear equations) and to make pedagogical suggestions for courses on these topics (see, e.g., Kú et al. 2008; Oktaç and Trigueros 2010; Parraguez and Oktaç 2010; Roa-Fuentes and Oktaç 2010; Trigueros and Oktaç 2005).

2.7 Related Theoretical Perspectives

Shortly after Dubinsky began speaking about Actions, Processes, Objects, and Schemas, Sfard began to speak about operational and structural conceptions (Sfard 1987), which she later changed to process and object (Sfard 1991). Somewhat later, Gray and Tall introduced the notion of a procept, which is an amalgam of three components: a process, an object that is produced by that process, and a symbol which is used to represent either the process or the object (Gray and Tall 1994). Although both of these theoretical perspectives have some commonalities with APOS Theory, there are important differences. For example, neither of them deals with Actions or the construction of Processes (as in interiorizing Actions to Processes). Also, there is no mention of anything like schemas, although Tall and Vinner (1981) discuss concept image, which, as was indicated in Sect. 2.4, is very similar to a Schema, but with some differences.

Finally, these two approaches do not put emphasis on the application of their theoretical perspectives to the design and implementation of instruction.

Chapter 3

Mental Structures and Mechanisms: APOS Theory and the Construction of Mathematical Knowledge

The focus of this chapter is a discussion of the characteristics of the mental structures that constitute APOS Theory, Action, Process, Object, and Schema, and the mechanisms, such as interiorization, encapsulation, coordination, reversal, de-encapsulation, thematization, and generalization, by which those mental structures are constructed.

In Sect. 3.1, general aspects of mental structures, mental mechanisms, and their role in the development of an individual's understanding of mathematical concepts are considered. In Sect. 3.2, these mental constructions and mechanisms are explained and examples are provided to illustrate how APOS Theory describes an individual's construction of mathematical knowledge. These ideas are summarized in Sect. 3.3.

3.1 Preliminary Aspects and Terminology

APOS Theory is principally a model for describing how mathematical concepts can be learned; it is a framework used to explain how individuals mentally construct their understandings of mathematical concepts. From a cognitive perspective, a particular mathematical concept is framed in terms of its *genetic decomposition*, a description of how the concept may be constructed in an individual's mind. This differs from a mathematical formulation of the concept, which deals with how the concept is situated in the mathematical landscape—its role as a mathematical idea.

Individuals make sense of mathematical concepts by building and using certain mental *structures* (or *constructions*) which are considered in APOS Theory to be *stages* in the learning of mathematical concepts (Piaget and García, 1983/1989). These structures arise through instances of reflective abstraction (as discussed in Chap. 2), which, in APOS theory, consists of mental mechanisms such as interiorization, encapsulation, coordination, reversal, de-encapsulation, and thematization. Since a genetic decomposition is hypothesized theoretically and tested empirically,

it can serve as a powerful descriptive and predictive tool to describe an individual's mathematical thinking. By detailing the structures involved in learning a particular concept, a genetic decomposition can help an instructor to uncover sources of difficulty that arise in the learning process. By providing a description of how a concept might develop in the mind of an individual, a genetic decomposition can help to guide the design of instruction. Genetic decompositions are addressed in considerably more detail in Chap. 4, and their role in instruction and research is explained further in Chaps. 5 and 6.

In APOS-based research, the terms *conception* and *concept* appear quite frequently. Although related, they are different ideas. McDonald et al. (2000) describe the distinction as follows:

We distinguish between conception and concept as the first is intrapersonal (i.e., the individual's idea or understanding) and the latter is communal (i.e., a concept as agreed upon by mathematicians). (p. 78)

For a particular piece of mathematical content, a conception develops as a result of reflective activity. The term concept refers to the collective understanding of that content by the community of mathematicians. Thus, a genetic decomposition is a model of the development of those individual conceptions that align with a concept.

3.2 Description of Mental Structures and Mechanisms

General descriptions of the mental structures and mental mechanisms that are used in APOS Theory appear in this section. The construction of different mathematical concepts in algebra, calculus, and statistics is used to illustrate how the structures and the mental mechanisms that give rise to them develop.

Dubinsky (1991) discusses five types of reflective abstraction, or mental mechanisms—interiorization, coordination, reversal, encapsulation, and generalization—that lead to the construction of mental structures: Actions, Processes, Objects, and Schemas. Figure 3.1 illustrates the relationships between these structures and mechanisms (this figure also appears in Chap. 2). In Chap. 8 a

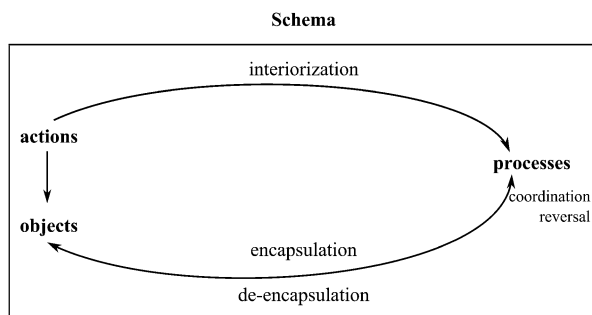


Fig. 3.1 Mental structures and mechanisms for the construction of mathematical knowledge

new version of this figure is presented, which takes into account the possible stage of Totality.

The interaction of the elements in Fig. 3.1 can be described as follows:

... we consider that understanding a mathematical concept begins with manipulating previously constructed mental or physical objects to form actions; actions are then interiorized to form processes which are then encapsulated to form objects. Objects can be de-encapsulated back to the processes from which they were formed. Finally, actions, processes and objects can be organized in schemas. (Asiala et al. 1996, p. 9)

Dubinsky (1991) characterizes the overall relationship among these elements as a “circular feedback system” (p. 106). Although the construction of mathematical knowledge is nonlinear, as will be seen more clearly in Chap. 4, the APOS-based description of the mental construction of a mathematical concept is presented in a hierarchical manner. The depth and complexity of an individual’s understanding of a concept depends on her or his ability to establish connections among the mental structures that constitute it. These connections form the basis of a Schema (described in more detail in Sect. 3.2.5 and in Chap. 7) whose coherence (described briefly in Sect. 3.2.5 and in Chap. 7) is crucial to an individual’s ability to make sense of mathematical situations related to the concept.

Assimilation and accommodation, the mechanisms used by Piaget in his work, and the mechanism of generalization do not appear in Fig. 3.1. Assimilation of knowledge refers to a mechanism by which a subject can apply a cognitive structure, essentially without change, to include a cognitive object the subject has not previously dealt with. Accommodation refers to a mechanism by which a mental structure is reconstructed and modified in order to deal with a new situation. Both mechanisms are related to the APOS idea of generalization.

In the rest of this section, each structure and the way it is constructed is described and illustrated with examples.

3.2.1 *Actions*

According to Piaget and adopted by APOS Theory, a concept is first conceived as an Action, that is, as an externally directed transformation of a previously conceived Object, or Objects. An Action is external in the sense that each step of the transformation needs to be performed explicitly and guided by external instructions; additionally, each step prompts the next, that is, the steps of the Action cannot yet be imagined and none can be skipped. For example, in the case of the function concept, “an individual who requires an explicit expression in order to think about the concept of function and can do little more than substitute for the variable in the expression and manipulate it is considered to have an action understanding of functions” (Dubinsky et al. 2005a, p. 338). Thus, the expression acts as an external cue that indicates how the Action must be performed, step-by-step, by the substitution of specific values.

An individual who is limited to an Action conception relies on external cues. In the case of the composition of two functions, an individual with an Action conception would need to have explicit expressions for each function and could only think about the composition for specific values. As Breidenbach et al. (1992) note: an individual “would probably be unable to compose two functions in more general situations, e.g., when functions had split domains, or if they were not given by expressions at all” (p. 251).

Although the most primitive of structures (and often, the only one stressed in traditional teaching), Actions are fundamental to APOS Theory. An Action conception is necessary for the development of other structures. In particular, Processes are interiorized Actions, and mental Objects arise because of the application of Actions. New Actions lead to the development of higher order structures. For instance, in the case of functions, performing operations on them spurs their encapsulation as Objects.

Actions may be basic or complex depending on the context. Following are some examples:

In Linear Algebra: The construction of the concept of n -tuple may begin by performing the Action that consists in taking a specified quantity of numbers and placing them in a particular order.

In Statistics: In the case of the mean, the Action of calculating the mean for a given set of data is determined by the definition of the data set. Students who are learning how to calculate the mean of a specific set of data make the calculations according to the Actions prescribed by the formula, that is, by adding the values of pieces of data and dividing the sum by the number of data pieces.

In Calculus: Actions are needed to construct an estimate of the definite integral as the area under a curve: for example, in dividing an interval into specific subintervals of a given size, constructing a rectangle under the curve for each subinterval, calculating the area of each rectangle, and calculating the sum of the areas of the rectangles.

3.2.2 *Interiorization and Processes*

Processes are constructed using one of two mental mechanisms: interiorization or coordination. Each of these mechanisms gives rise to new Processes. Interiorization is explained here, and coordination is considered in Sect. 3.2.4.

As Actions are repeated and reflected on, the individual moves from relying on external cues to having internal control over them. This is characterized by an ability to imagine carrying out the steps without necessarily having to perform each one explicitly and by being able to skip steps, as well as reverse them. Interiorization is the mechanism that makes this mental shift possible.

An action must be interiorized. As we have said, this means that some internal construction is made relating to the action. An interiorized action is a process. Interiorization permits one to be conscious of an action, to reflect on it and to combine it with other actions. (Dubinsky 1991, p. 107)

In the same spirit, Dubinsky et al. (2005a) give the following description of a Process and interpret it for the case of functions:

As an individual repeats and reflects on an action, it may be *interiorized* into a mental *process*. A process is a mental structure that performs the same operation as the action being interiorized, but wholly in the mind of the individual, thus enabling her or him to imagine performing the transformation without having to execute each step explicitly. Thus, for example, an individual with a process understanding of function will construct a mental process for a given function and think in terms of inputs, possibly unspecified, and transformations of those inputs to produce outputs. (p. 339)

Although an Action and a Process, when related to a given concept, may involve the same transformation, they differ in the following sense: for an Action, one must actually make the transformation (either physically or mentally); for a Process one can carry out the transformation without the need to go through each step. Following are some examples:

In Linear Algebra: When the Actions involved in the construction of an n -tuple are interiorized into a Process, the subject can construct an n -tuple mentally even when n is not specified; he or she can also consider the construction of n -tuples in any vector space, including infinite dimensional spaces. It is also possible for the individual to think about the elements of the tuple, considering that the elements may repeat, but the order in which they appear cannot be changed.

In Statistics: In the case of the mean, the Action of computing the mean of a set of data points is interiorized into a Process when students can describe in general how to compute the mean with being given a specific data set and grasp the idea that a mean represents a characteristic of a set of numbers as a whole (Mathews and Clark 2003).

In Calculus: For the definite integral, the Action of determining the Riemann sum for a particular partition is interiorized into a Process when an individual can describe how the Riemann sum is determined for an unspecified partition and imagine this process continuing with decreasing *mesh size* (maximum length of a subinterval).

3.2.3 Encapsulation and Objects

Encapsulation occurs when an individual applies an Action to a Process, that is, sees a dynamic structure (Process) as a static structure to which Actions can be applied. Dubinsky et al. (2005a) offer the following explanation:

If one becomes aware of the process as a totality, realizes that transformations can act on that totality and can actually construct such transformations (explicitly or in one's imagination), then we say the individual has *encapsulated* the process into a cognitive *object*. For the function concept, encapsulation allows one to apply transformations of functions such as forming a set of functions, defining arithmetic operations on such a set, equipping it with a topology, etc. (p. 339)

Examples of encapsulation of Processes include the following:

In Linear Algebra: Comparing n -tuples or performing binary operations on n -tuples are Actions on n -tuples. For these Actions to be applied successfully, the Process of forming an n -tuple must be encapsulated into an Object.

In Statistics: To think of the mean as one of several measures of central tendency that gives information about a set of data and to ascertain its properties (which are Actions), the Process of calculating a mean is encapsulated into a mental Object (Mathews and Clark 2003).

In Calculus: The area under the curve for a function on a closed interval is the limit of Riemann sums—an Action applied to the Riemann sum Process. In order to determine the existence of this limit and/or to calculate its value, the student needs to encapsulate the Riemann sum Process into an Object.

As reported in various APOS-based studies, the mechanism of encapsulation is the most difficult. For example, in a study about students' conceptions of the fundamental theorem of statistics, Clark et al. (2007) reported:

However, moving beyond a process conception of mean is much more difficult. Three of the students in this study had not progressed beyond a process conception of the mean. Although they could perform the necessary actions, describe the process of computing the mean of a set of numbers, and in some cases reverse this process, these students appeared unable to conceive of the mean of a data set as an entity itself. They were unable to perform any actions on the output of their processes or to associate any meaningful properties with the means they computed. (p. 5)

In a study about the mental construction of two variable functions, Trigueros and Martínez-Planell (2010) found that only one of their students had constructed an Object conception.

Sfard also wrote about the “inherent difficulty of reification” (similar to encapsulation in APOS Theory), suggesting that:

The ability to see something familiar in a totally new way is never easy to achieve. The difficulties arising when a process is converted into an object are, in a sense, like those experienced during transition from one scientific paradigm to another. . . (Sfard 1991, p. 30)

As will be seen in Chap. 5, APOS-based instruction has had considerable success in dealing with this difficulty.

3.2.4 *De-encapsulation, Coordination, and Reversal of Processes*

Once a Process has been encapsulated into a mental Object, it can be *de-encapsulated*, when the need arises, back to its underlying Process. In other words, by applying the mechanism of de-encapsulation, an individual can go back to the Process that gave rise to the Object.

The mechanism of *coordination* is indispensable in the construction of some Objects. Two Objects can be de-encapsulated, their Processes coordinated, and the coordinated Process encapsulated to form a new Object.

This is what happens mentally with function composition. To compose two functions F and G to obtain $F \circ G$, the two function Objects must be de-encapsulated to the Processes that gave rise to them. These Processes are then coordinated, by applying the Process of F to the elements obtained by applying the Process of G . The resulting Process is then encapsulated into a new Object.

As indicated in Fig. 3.1, a Process can be reversed. For example, Dubinsky (1991) explained how the function Process can be reversed to obtain an inverse function:

It is by reflecting on the totality of a function's process that one makes sense of the notion of a function being onto. Reflection on the function's process and the reversal of that process seem to be involved in the idea of a function being one-to-one. (p. 115)

The idea of a bijective function is constructed mentally and gives rise to an inverse function by applying the mechanism of reversal.

In relation to function composition, the following three types of problems are similar mathematically:

1. Given F and G , find H such that $H = F \circ G$.
2. Given G and H , find F such that $H = F \circ G$.
3. Given F and H , find G such that $H = F \circ G$.

Cognitively speaking, however, they are different. According to Ayers et al. (1988), in the solution of problems of the second and third type, "reversals of the processes seem to be required" (p. 254). Dubinsky provides the following analysis about the difference:

The first kind of problem [1] seems to require only the coordination of two processes that, presumably, have been interiorized by the subject.

The second [2], however may require that the following be done for each x in the domain of H .

- 2a. Determine what H does to x obtaining $H(x)$.
- 2b. Determine what G does to x obtaining $G(x)$.
- 2c. Construct a process that will always transform $G(x)$ to $H(x)$.

The third kind of problem [3] may be solved by doing the following for each x in the domain of H .

- 3a. Determine what H does to x obtaining $H(x)$.
- 3b. Determine value(s) y having the property that the process of F will transform y to $H(x)$.
- 3c. Construct a process that will transform any x to such a y .

Comparing 2b with 3b (the only point of significant difference), we can see that 2b is a direct application of the process of G whereas 3b requires a reversal of the process of F . It is perhaps interesting to note that this difference in difficulty (between [2] and [3]), which is observed empirically and explained epistemologically, is completely absent from a purely mathematical analysis of the two problems. They are, from a mathematical point of view, the calculation of $H \circ G^{-1}$ and $F^{-1} \circ G$, respectively, which appear to be problems of identical difficulty. This seems to be another important example in which the psychological and mathematical natures of a problem are not the same (cf. p. 113).

Another situation in which relative difficulty can be explained by the requirement of reversing a Process occurs in the development of children's ability in arithmetic. According to Riley, Greeno and Heller (1983, p. 157), "Problems represented by sentences where the unknown is either the first ($? + a = b$) or second ($a + ? = c$) number are more difficult than problems represented by equations where the result is the unknown ($a + b = ?$)."

The first two problem types involve a reversal of the Process, which, in the third type can be applied directly. (Dubinsky 1991, p. 118)

Another example that Dubinsky (1991) presents about the generation of a new process by the mechanism of reversal is related to integration:

A calculus student may have interiorized the action of taking the derivative of a function and may be able to do this successfully with a large number of examples, using various techniques that are often taught and occasionally learned in calculus courses. If the process is interiorized, the student might be able to reverse it to solve problems in which a function is given and it is desired to find a function whose derivative is the original function. (p. 107)

The mechanism of coordination, in particular, how it is carried out mentally, is currently under investigation. It is hypothesized that coordination of two Processes, say P_A and P_B , can be thought of as the application of P_A to P_B (Fig. 3.2). For that to be possible, the learner first needs to encapsulate P_B into an Object, O_B , in order to be able to apply P_A to it. Once that happens, the coordination can continue in the following way: either O_B is assimilated and P_A can be applied to it, or P_A is accommodated so that the learner can apply it to O_B . An alternative is for P_B to be applied to P_A in a similar way. Whether coordination actually occurs in this way is the subject of future study.

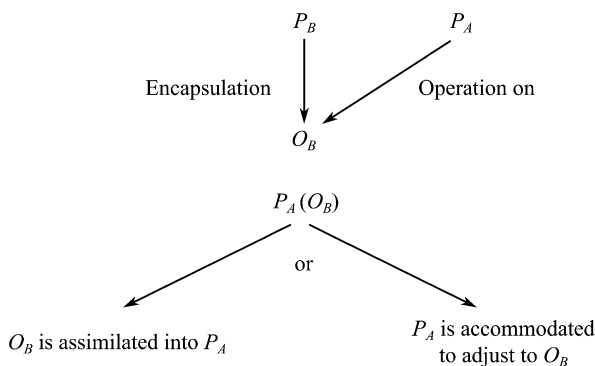


Fig. 3.2 Coordination of two Processes P_A and P_B

3.2.5 Thematization and Schemas

The interaction of the elements presented in Fig. 3.1 (Sect. 3.2) gives rise to Schemas. According to Dubinsky (1991), a Schema is characterized by its dynamism and its continuous reconstruction as determined by the mathematical activity

of the subject in specific mathematical situations. The coherence of a Schema is determined by the individual's ability to ascertain whether it can be used to deal with a particular mathematical situation. Once a Schema is constructed as a coherent collection of structures (Actions, Processes, Objects, and other Schemas) and connections established among those structures, it can be transformed into a static structure (Object) and/or used as a dynamic structure that assimilates other related Objects or Schemas.

For example, a Schema for vector space may include n -tuples and matrices as Objects and polynomials and functions as Processes. All these structures may be related by the fact that they share some properties, such as satisfying a set of axioms that define a vector space. Coherence of this Schema lies in the mathematical definition of vector space which the individual uses to determine whether or not the Schema is applicable to a given situation. The construction of a Schema as a mental Object is achieved through the mechanism of thematization. This mechanism enables an individual to apply transformations to the Schema structure.

Hence Schemas are structures that contain the descriptions, organization, and exemplifications of the mental structures that an individual has constructed regarding a mathematical concept. Studies that focus on the development of Schemas are not very numerous and more research is needed to understand better how Schemas develop and are applied. The development and application of Schemas as well as their thematization are considered in greater detail in Chap. 7.

3.3 Overview of Structures and Mechanisms

Since mathematical concepts are not constructed directly, it is necessary for an individual to construct mental structures to make sense of them (Piaget and García, 1983/1989). According to APOS Theory, individuals deal with mathematical problem situations by constructing and applying mental structures in their effort to understand mathematical concepts. This involves transforming (via Actions or Processes) previously established structures. These transformations then become new Objects via the mechanism of encapsulation. APOS Theory is based on the premise that an individual can learn any mathematical concept provided the structures necessary to understand those concepts have been built (Dubinsky 1991).

Each of the structures that make up APOS Theory is constructed via a mental mechanism: an Action is interiorized into a mental Process, a Process is encapsulated into a cognitive Object, a Process can be reversed to construct another Process, two Processes may be coordinated to form a new Process, and a Schema can be thematized into a cognitive Object. In their study of uncountable infinite Processes, Stenger et al. (2008) describe the terms structure and mechanism and the relation between the two:

A mental structure is any relatively stable (although capable of development) structure (i.e., something constructed in one's mind) that an individual uses to make sense of mathematical situations. A source for a mental structure is a description of where that structure comes from. A mental mechanism is a means by which that structure might develop in the mind(s) of an individual or a group of individuals. (p. 98)

The constructions of mathematical knowledge described in this chapter illustrate how making the most basic constructions is fundamental for an individual to construct more robust structures. Mental structures and the mechanisms by which they are constructed involve a spiral approach where new structures are built by acting on existing structures. This idea is discussed in detail in Sects. 2.1 and 2.2. Dubinsky (1997), referring to Piaget's ideas, wrote that:

Objects, once constructed, can be transformed to make higher level actions and then processes, and so on. This can continue indefinitely. Moreover, any action, process, or object can be reconstructed, as a result of experiencing new problem situations on a higher plane, interiorizing more sophisticated actions and encapsulating richer processes. The lower level construction is not lost, but remains as a part of the enriched conception. (p. 98)

Chapter 4

Genetic Decomposition

The ultimate goal of scientific research is the development of theories or models to explain and/or to predict different kinds of phenomena (Woodward 2003). As discussed in the previous chapters, research into students' learning of mathematics helps to predict what they may learn about a specific mathematical concept and the conditions by which that learning takes place. This is an important part of mathematics education as a research field and it is one of the roles of APOS Theory.

Once the constructs of the theory are defined, models that show how those constructs are related and developed. The models serve as the basis for working hypotheses that can be tested experimentally. In APOS Theory the genetic decomposition plays this role.

The role of the genetic decomposition as a hypothetical model of mental constructions needed to learn a specific mathematical concept is the subject of this chapter. Section 4.1 introduces what a genetic decomposition is, its importance, and its predictive value. The complexities involved in the design of a genetic decomposition are discussed in Sect. 4.2. In Sect. 4.3 the role of the genetic decomposition in research is presented. Section 4.4 is a discussion of whether a genetic decomposition is unique. The use of the genetic decomposition in the design of teaching activities (further discussed in Chap. 5) and as a diagnostic tool is considered in Sect. 4.5. Section 4.6 addresses the refinement of a genetic decomposition. The chapter concludes with some examples of common errors and misunderstandings about the genetic decomposition.

4.1 What Is a Genetic Decomposition?

A genetic decomposition is a hypothetical model that describes the mental structures and mechanisms that a student might need to construct in order to learn a specific mathematical concept. It typically starts as a hypothesis based on the

researchers' experiences in the learning and teaching of the concept, their knowledge of APOS Theory, their mathematical knowledge, previously published research on the concept, and the historical development of the concept. Until it is tested experimentally, a genetic decomposition is a hypothesis and is referred to as *preliminary*.

A new mathematical concept frequently arises as a transformation of an existing concept. As such a genetic decomposition consists of a description of the Actions that a student needs to perform on existing mental Objects and continues to include explanations of how these Actions are interiorized into Processes. At this point, the concept is still seen as something one does. In order to be conceived as an entity in its own right, something that can be transformed, the Process is encapsulated into a mental Object. It is entirely possible that a concept may consist of several different Actions, Processes, and Objects. A genetic decomposition may include a description of how these structures are related and organized into a larger mental structure called a Schema. Included in the description of a Schema may be an explanation of how the Schema is thematized into an Object. The genetic decomposition also explains whatever is known about students' expected performances that indicate differences in the development of students' constructions.¹

In addition to describing how a concept might be constructed mentally, a genetic decomposition might include a description of prerequisite structures an individual needs to have constructed previously, and it might explain differences in students' development that may account for variations in mathematical performance. Thus, a genetic decomposition is a model of the epistemology and cognition of a mathematical concept (Roa-Fuentes and Okaç 2010).

A preliminary genetic decomposition can guide the development of an instructional treatment (described in Chap. 5). Implementation of the instruction provides an opportunity for gathering data, usually in the form of written instruments and/or in-depth interviews. In analysis of the data, two questions are asked: (1) Did the students make the mental constructions called for by the genetic decomposition? (2) How well did the subjects learn the mathematical content? Answers to these questions may lead to revision of the genetic decomposition and/or the instruction. At this point, the genetic decomposition is no longer regarded as preliminary. Further refinements are possible, as each

¹ Some researchers use the term "cognitive path" to describe a specific ordering of concepts that students seem to follow when learning a mathematical topic. Cognitive paths are found by means of a specific statistical method using data from students (Vinner and Hershkowitz 1980). A cognitive path describes a process of learning focused on the mathematical aspects of the concept. Although it may seem that there is some similarity between "cognitive path" and "genetic decomposition," their focus and content are different.

Cognitive paths describe or suggest a linear cognitive progression based on an analysis of the mathematical aspects of the concepts involved. Instead of a linear progression, a genetic decomposition in APOS Theory describes the mental structures and the mechanisms by which those structures are constructed. Confusion with cognitive paths may explain some of the errors discussed later in this chapter.

refinement leads to further revision of the instruction, which provides an opportunity for new data analysis. Ideally, the cycle of refinement $\rightarrow$ revision $\rightarrow$ data analysis yields a genetic decomposition that reflects very closely the cognition of the concept for many individuals and that can be used in the design of instruction that positively affects student learning.

There are several issues to be clarified. When referring to a student in the context of APOS Theory, the researcher is not considering a specific student. Rather, he or she is considering a “generic student,” that is, a representative of the class of students who are learning that concept. Also, it may seem that a genetic decomposition is unique. However, as pointed out in Sect. 4.4, the genetic decomposition for a given mathematical concept may not be unique. What is important is that a genetic decomposition predicts the mental constructions deduced from the analysis of data gathered in experimental designs (Dubinsky 1991).

Because the genetic decomposition of a concept is described linearly, it might seem as though the concept develops linearly. However, this is mostly a consequence of the description, which does not reflect the possibility of different trajectories that include starts, stops, and discontinuities that occur in learning. In addition, APOS Theory does not rule out the possibility that mental structures, once developed, may not always be applied when called for. Thus, a genetic decomposition does not explain what happens in an individual’s mind, as this is probably unknowable; predict whether an individual will apply a given structure when called upon; nor offer an exclusive theoretical analysis of how mathematics is learned. APOS theory acknowledges that a student may pursue different learning paths or follow different trajectories, as a student moves from Process to Action and back to Process or from Object to Process and back to Object. Despite individual differences, a genetic decomposition describes the structures a student needs to construct in her or his learning of a concept. When verified empirically, a genetic decomposition can serve as a useful model of cognition, as evidenced by a number of empirical studies that show the efficacy of APOS theory as a tool for describing students’ conceptions and in the design of effective instruction (Weller et al. 2003).

Examples of genetic decompositions appear throughout the remainder of this chapter. Examples in this section show *preliminary* genetic decompositions. Later in the chapter refinements of some genetic decompositions, based on research data, will be shown.

4.1.1 A Genetic Decomposition for Function (Based on Ideas from Dubinsky 1991)

The construction of the function concept starts with Actions on a set. Given a set of numbers or other kinds of elements, these Actions involve taking an element from one set, explicitly applying a rule, typically an algebraic formula, and assigning to

that specific element a unique element from the second set. As these Actions are performed on different sets, say ordered pairs, points, or non-numeric objects, the individual reflects on them and perceives them as a dynamic transformation. At this point, interiorization starts, as the individual begins to see a function as a type of transformation that pairs elements of one set, called the domain, with elements of a second set, called the range. This means that the individual has constructed a mental structure that performs the same transformation as the Action, but wholly in the mind of the individual.

An individual who shows a Process conception of function can think of a function in terms of accepting inputs, manipulating them in some way, and producing outputs without the need to make explicit calculations. Evidence of a Process conception of function might include the ability to determine whether a function has an inverse, which would require a reversal of the function Process, or to describe how one would compose two functions, which would require a coordination of two function Processes.

Applications of Actions or other Processes applied to the function Process lead to its encapsulation as a cognitive Object. The mechanism of encapsulation moves the learner's focus away from the concept of function as a dynamic transformation to a static entity that itself can be examined and transformed. Indications of encapsulation might include an individual's ability to form sets of functions, or to perform arithmetic operations on functions, or to construct a function that is a limit of a sequence of functions. In the first, a function is treated as an element; in the second, as an input to binary operations; and in the third, as the transcendent Object of an infinite Process that produces a sequence of functions. In each of these cases, functions are treated as static entities to which Actions can be applied.

An individual who can determine whether the relationship between two entities defines a functional relationship, and can coordinate various Processes to determine the domain and range of a function, may be giving evidence of constructing a function Schema. An indication of the coherence of a function Schema would include an individual's ability to determine whether a particular mathematical situation defines a functional relationship.

4.1.2 A Genetic Decomposition for Induction (Dubinsky 1991, pp. 109–111)

The genetic decomposition for the induction Process assumes prior construction of the function and logic Schemas. The function Schema includes a Process for evaluating a function for a given value in its domain. The logic Schema includes the ability to construct statements in the first-order propositional calculus. For instance, the individual constructs a Process for logical necessity; that is, in certain situations, he or she will understand that if A is true, then, of necessity, B will be true. The mental development of induction from this initial starting point is illustrated in Fig. 4.1 and is described in detail in the remainder of this subsection.

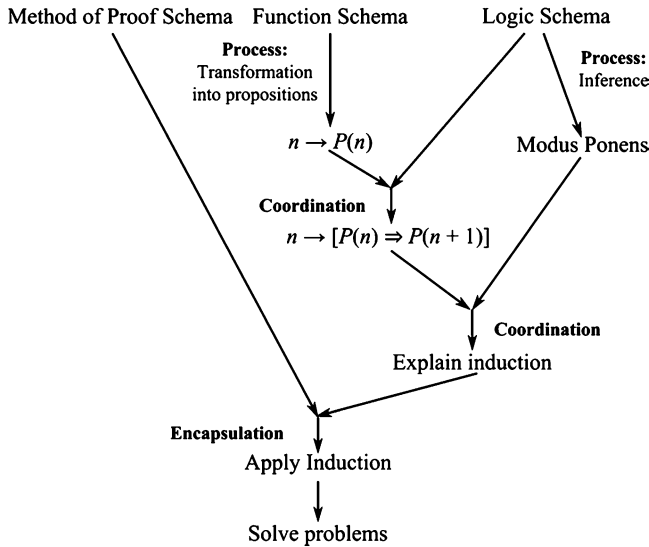


Fig. 4.1 Genetic decomposition for the concept of mathematical induction (Dubinsky and Lewin 1986)

In general, the first-order propositions in the logic Schema described above are Processes that originate from interiorizing Actions (conjunctions, disjunctions, implications, negations) on declarative statements (Objects). For example, the formation of the disjunction $P \vee Q$ can be described as an Action on the statements P , Q . The Action, which goes well beyond simply putting these symbols into a disjunction expression, involves the construction of a mental image of the two statements and the determination of the truth or falsity of the disjunction in various situations. Through interiorization, the Action is transformed into a Process for forming the disjunction of two declarative statements. If nothing further is done after this Action is interiorized, then it will be impossible for the disjunction to be combined with other statements. Specifically, to combine the disjunction $P \vee Q$ with a declarative statement R to form the statement $(P \vee Q) \wedge R$, the disjunction Process must be encapsulated to form a new Object $(P \vee Q)$ to which the statement R can be conjoined. Thus, in general, the Actions of conjunction, disjunction, implication, and negation that are applied to declarative statements must be interiorized and then encapsulated in order to construct more complex first-order propositions as Objects [note how the use of parentheses in mathematical notation corresponds here to encapsulation (Dubinsky and Lewin 1986)]. Iterating this procedure, the subject enriches her or his logic Schema to obtain a host of new Objects consisting of first-order propositions of arbitrary complexity.

The function Schema must be accommodated to enable the construction of a Process that transforms positive integers into propositions, that is, to obtain a proposition-valued function of the positive integers. Consider, for example, a statement such as “given a number of dollars it is possible to represent it with \$3 chips and \$5 chips.” To understand the meaning of such a statement, the subject must construct a Process in which the “number of dollars” (in the original statement) is replaced by that value of n . This is a proposition-valued function. In order to evaluate it, the subject must construct another Process, whereby given n a search is made to determine whether it is possible to find nonnegative integers j, k such that $n = 3j + 5k$. It is useful for the subject to discover that the value of this function is true for $n = 3, 5$, false for $n = 2, 4, 6, 7$, and true for all higher values.

Steps in the construction of proof by induction include the encapsulation of the Process of implication, which then becomes an Object that is in both the domain and the range of a function. This is followed by further assimilation of the function Schema to include implication-valued functions, as well as the interiorization of a Process of going from a proposition-valued function of the positive integers to its corresponding implication-valued function.

The logic Schema needs to include a Process called modus ponens. This Process is the interiorization of an Action applied to implications (assuming, as above, that they have been encapsulated into Objects). The Action begins with the hypothesis, determining that it is true, followed by asserting the truth of the conclusion. These constructions make it possible for a student to coordinate the function Schema, as it applies to an implication-valued function Q (obtained from a proposition-valued function P), and the logic Schema, as it applies to the Process modus ponens. This leads to construction of a function $n \rightarrow Q(n)$, where n is a positive integer and Q is the implication-valued function that transforms n to $P(n) \Rightarrow P(n+1)$. For an integer n in the domain of positive integers, one determines the value of the function Q , which, in this case, involves determination of the truth or falsity of $P(n) \Rightarrow P(n+1)$. If it has been established that Q has a constant value of true, the first step in this new Process is to evaluate P at 1 and to determine that $P(1)$ is true (or more generally to find a value n_o such that $P(n_o)$ is true). Next the function Q is evaluated at 1 to obtain $P(1) \Rightarrow P(2)$. Applying modus ponens and the fact (just established) that $P(1)$ is true yields the assertion $P(2)$. The evaluation process is again applied to Q , but this time with $n = 2$, to obtain $P(2) \Rightarrow P(3)$. Modus ponens again gives the assertion $P(3)$. The cycle is repeated ad infinitum, alternating the Processes of modus ponens and evaluation. This produces a rather complex coordination of two Processes that leads to an infinite Process. This infinite Process is encapsulated and added to the proof Schema as a new Object, proof by induction.

In the presentation of this genetic decomposition, Dubinsky noted that the student will not necessarily be aware of these Schemas, that is, the subject will be able to think in terms of plugging a value of a positive integer into a statement to ask if the result is a true statement, but will not necessarily be aware of the fact that he or she is working with a proposition-valued function or an implication-valued function. The development of the student's function Schema can only be inferred

from her or his Actions (Dubinsky 1991). In a similar vein, Dubinsky points out that “[i]n describing this construction we reiterate the point that, in the context of this theory, it is never clear (nor can it be) whether we are talking about a schema that is present or a schema that is being (re-)constructed” (p. 112).

It is interesting to note that this genetic decomposition reveals a cognitive step, which research has pointed out as providing a serious difficulty for students, that is not apparent when considering induction from a purely mathematical point of view. Specifically, if P is a proposition-valued function to be proved, $P(n)$ can be any proposition, in particular, it can be an implication. From a mathematical point of view, there is nothing new in the proposition-valued function Q defined by

$$Q(n) = P(n) \Rightarrow P(n + 1),$$

that is, once one understands P , then, as a special case, one understands Q . However, this is not the case from a cognitive point of view: implications are the most difficult propositions for students and generally the last to be encapsulated. Furthermore, there is a difference between constructing P from a given statement and constructing Q from P . From a cognitive perspective, the construction of Q from P “is a step that must be taken” (Dubinsky 1991, p. 109).

4.2 The Design of a Genetic Decomposition

One question that arises when considering genetic decompositions is their design. Specifically, how are they designed and what is needed in their design? A preliminary genetic decomposition can arise in one of several ways. The preliminary genetic decomposition for a particular concept is based on the researchers’ mathematical understanding of the concept, their experiences as teachers, prior research on students’ thinking about the concept, historical perspectives on the development of the concept, and/or an analysis of text or instructional materials related to the concept. Following are some examples.

Some preliminary genetic decompositions are designed by taking into account mathematical descriptions of a concept, together with the researchers’ experiences as learners or teachers. This is the case for the genetic decomposition for induction described in Sect. 4.1.2. Others are designed from data from previous mathematics education research, not necessarily conducted using APOS theory, into students’ difficulties in learning a particular concept. This is the case for spanning set and span that is discussed in Sect. 4.2.1. It may be recalled here that the description included in the genetic decomposition is not the same as the mathematical introduction of a concept, as is clearly illustrated by the example on mathematical induction shown in Sect. 4.1.

There are genetic decompositions based on data from observations of students who are learning a mathematical concept. Analysis of observations leads to a description of the cognition of the concept, which can be verified empirically.

Trigueros and Martínez-Planell (2010) followed this approach in their study of students' learning of two-variable functions. The authors used observations of student work to develop a preliminary genetic decomposition of two-variable functions. They then conducted interviews with students who had completed the course and used the interview data to refine their preliminary analysis.

The design of a genetic decomposition can also be based on the historical development of the concept. A study of the historical development of a concept may point to mental constructions that individuals might make. In response to historical arguments against the existence of actual infinity, Dubinsky et al. (2005a, b) used APOS Theory to explain how potential and actual infinity represent two different conceptualizations linked by the mental mechanism of encapsulation. Although their explanation did not lead directly to a genetic decomposition on mathematical infinity, it strongly influenced the design of a genetic decomposition on infinite repeating decimals (Weller et al. 2009, 2011; Dubinsky et al. 2013) and informed a study of infinite iterative Processes conducted by Brown et al. (2010). In the former case, the preliminary genetic decomposition was hypothetical; it was later tested empirically by analyzing data from students who completed specially designed instruction based on the genetic decomposition. In the latter case, the preliminary genetic decomposition was empirical; it arose from an analysis of interview data involving students who tried to solve a problem in set theory. In their efforts to make sense of the problem situation, the students constructed a variety of iterative Processes. Brown et al. (2010) used their understanding of the historical development of the concept of mathematical infinity to guide their analysis of the interviews that culminated with a genetic decomposition of infinite iterative processes.

Text materials may also inform the design of a preliminary genetic decomposition. Specifically, for a given concept the didactical approach used in the text can help researchers to determine how students might come to understand the concept. This analysis is complemented by researchers' descriptions of the mental structures students need to construct in order to succeed in learning those concepts. This will be exemplified in the discussion of linear transformations in Sect. 4.4.

Finally, genetic decompositions can be developed from data. In this case, students are interviewed and the transcription of the interview is divided into small pieces. By comparing these pieces, it is possible to find differences in students' performance on specific tasks. Differences in performance may uncover instances where certain mental constructions need to be made. Lack of success in completing a task may indicate that the student has not made the needed mental constructions while success with the task may uncover evidence that those mental constructions have been made. The totality of the results obtained by this type of analysis leads to the organization of the mental constructions that make up the genetic decomposition.

As a result of one or more of these methods, the genetic decomposition can be a simple model of the main constructions the researchers think are needed in order to learn a concept, as illustrated in Sect. 4.2.1. It can also be a model where many of the complexities involved in the construction of the concept are described. This is exemplified in Sect. 4.2.3.

In Sect. 4.1, two examples of genetic decompositions were presented; the goal was to illustrate what a genetic decomposition is and how APOS constructions are involved in describing the mental constructions involved in learning a concept. In Sect. 4.2.1 an additional example, involving the concepts of spanning set and span, is introduced in order to show how previous research informs the development of a preliminary genetic decomposition.

4.2.1 Genetic Decomposition for Spanning Set and Span (Based on Kú et al. 2011)

The design of this genetic decomposition was informed by three sources of data: results from the review of literature about the learning of basis and spanning set, a report on students' difficulties in construction of the concept of basis (Kú et al. 2011), and evidence (also from Kú et al. 2008) of differences in students' constructions of the concepts of spanning set and span.

Results of the literature review for Kú et al. (2008) indicated the importance of basis as a concept related to vector space and students' tendency to have a conceptual image of spanning set as a basis. Kú et al. (2008) found that students were not able to differentiate between the concepts of basis and spanning set, had difficulty working with vector spaces different from $\mathbb{R}^2$ and $\mathbb{R}^3$, and struggled with the concept of span. These findings suggested the need for certain prerequisite constructions. Specifically, in order to construct the concepts of spanning set and span, students need to recognize that different types of sets, such as n -tuples, polynomials, and matrices, are vector spaces and that a vector space can be generated by spanning sets of different sizes.

The analysis in Kú et al. (2008) also revealed that students who constructed an Action conception of the concept of basis were able to perform Actions to construct linear combinations using given vectors but were unable to interpret span as the set of all the linear combinations of the basis vectors. This fact was taken into account in designing the genetic decomposition for spanning set and span. Specifically, one starts with Actions involving the formation of linear combinations. These Actions are interiorized into a Process so that the individual can imagine all linear combinations of a given set of vectors. A reversal of this Process enables the individual to determine whether an arbitrary vector in a vector space can be written as a linear combination of the vectors in the given set.

Kú et al. (2008) found that many of the students in their study had difficulties with the concept of set, especially when performing Actions on sets of vectors whose elements included parameters. When solving systems of equations, these students could perform Actions involved in manipulation of the equations in order to find the solution of the system but could not differentiate between parameters as general numbers and unknowns of the equations. Students with a Process conception of basis were able to relate this concept to a vector space, showed fewer problems in interpreting solution sets of systems of equations, and

interiorized the Process for forming a basis. However, with respect to the latter, they encountered difficulties. In particular, they could not differentiate the concept of basis from that of spanning set and had trouble interpreting the span of a given basis. The reason for this difficulty lies in their inability to make sense of the solution of a system of equations $A\mathbf{x} = \mathbf{b}$ when $\mathbf{b}$ has general numbers or expressions as elements. These difficulties were taken into account in designing the genetic decomposition. Specifically, in the description of prerequisite concepts, it is stated explicitly that students need to have in their minds the concepts of solution set and variable as Objects.

4.2.2 *Prerequisite Constructions*

The mental constructions of spanning set and span assume construction of the concepts of vector space, variable, and solution set of a system of linear equations. With respect to vector spaces, students need to demonstrate the ability to work with familiar examples, such as spaces of n -tuples with real-valued entries, and recognize that other types of sets, such as sets of polynomials and sets of matrices, are vector spaces.

The solution set of a system of equations plays an important role in development of the concepts of spanning set and span. Specifically, to determine whether a subspace of a vector space is spanned by a given subset of a vector space, or to verify that a particular vector lies in the space generated by that subset, one needs to solve a system of equations. Therefore, students need to have constructed this concept as an Object.

The concept of variable plays an important role in the interpretation of solution sets as spanning sets and spans. Specifically, a subspace generated by a subset of a vector space may be represented analytically in terms of a generalized vector involving variable expressions. Therefore, students need to work with variables as mathematical Objects in order to understand variables as unknowns, general numbers, parameters, or variables in functional relationships and to move flexibly among these interpretations (Trigueros and Ursini 2003).

4.2.3 *Mental Constructions*

Given a vector space V with a specific scalar field K , students perform Actions on a given subset S of vectors from V , specifically the Action to construct linear combinations with vectors from S and scalars from K . These Actions consist of multiplying vectors by scalars and summing the result of the multiplications to obtain a new vector in V . Interiorization of these Actions yields a Process for constructing a new vector which is an element of the vector space, that is, the

Process of constructing a linear combination. The reversal of this Process allows the student to verify if a given vector can be written as a linear combination of a given set of vectors. Students who show they have constructed these processes are considered to have a Process conception of a linear combination.

By coordinating the reversal of the Process of constructing the set of all linear combinations of vectors in a subset S of V with the Process for finding the solution set of a system of equations, the learner can verify the existence of scalars in K that can be used to determine whether vectors in a subset T of V can be expressed as linear combinations of S . In short, this coordination enables the learner to determine whether a subset T of V is generated by the subset S .

When different sets are compared and considered as different possible spanning sets for a set of vectors T , the coordinated Process mentioned above is encapsulated into an Object called spanning set. A student who has an Object conception of spanning set can make comparisons to decide whether a given vector space can be generated by different spanning sets independently of the size of the set or their specific elements.

When the Process for construction of a set T generated by S is coordinated with the Process for vector spaces, the learner can verify that T is a vector space. This last Process is encapsulated into an Object that can be called generated space, spanned space, or span, of the original set of vectors S . These constructions enable students to differentiate between the concepts of span and spanning set.

This analysis does not ignore those concepts that are related to spanning sets, such as linear independence or dependence, basis, and dimension. It includes consideration of how the construction of a spanning set can help students understand concepts like those that are related to it, or if there are difficulties in the construction of this concept that act as obstacles when relating it with other linear algebra concepts (Kú et al. 2011).

The genetic decomposition does not end with the construction of Objects. It should also include a description of how links among different Actions, Processes, and Objects can develop as a Schema. The mental development of a Schema will be discussed in Chap. 7.

As research has evolved, there are many genetic decompositions published by authors around the world (see the annotated bibliography in Chap. 12). In general these genetic decompositions have been tested by research and either supported or refined and then used for new research or in the design of teaching activities. Refinements of genetic decompositions will be considered in greater detail in Sect. 4.5.

4.3 Role of the Genetic Decomposition in Research

Genetic decompositions play a central role in APOS-based research, since a theoretical model is necessary to provide researchers with hypotheses that can serve as the basis for the design of theory-based instruments to obtain and analyze

data from students. Students' constructions can be deduced from their work and their responses to interview questions compared with what is predicted in the preliminary genetic decomposition. A genetic decomposition acts as a lens, analogous to a diffraction grating² that researchers use to explain how students develop, or fail to develop, their understanding of mathematical concepts. For example, on a given task, one student may perform the task correctly, another may have difficulty, and still another may completely fail. The genetic decomposition may explain discrepancies in performance. The student who succeeds may give evidence of having successfully made one or more of the mental construction(s) called for by the genetic decomposition. The student who shows limited progress may show evidence of having begun to make the construction(s). The student who fails may not have made the construction(s) at all or may give evidence of having been unsuccessful in having made the necessary construction(s). If the differences in student performance cannot be explained by the genetic decomposition, then it may be the case that the genetic decomposition needs revision. Thus, on one hand, the genetic decomposition guides the analysis; on the other, it points out gaps in the researchers' understanding of how the concept develops in the mind of the individual. Either way, a genetic decomposition is a tool by which researchers try to make sense of how students go about learning a concept and to explain the reasons behind student difficulties. Moreover, use of a theoretical model increases the reliability of the analysis, provides a means to describe student thinking, and serves as a diagnostic and predictive tool.

When using a genetic decomposition, different researchers can analyze the same data and obtain comparable results. Working as a team they can interpret their results in terms of the model. Without a model they might have a difficulty agreeing on or negotiating their interpretations. Thus, the analysis of data becomes more reliable when it is based on a theoretical model such as a genetic decomposition.

As described in Asiala et al. (1996), a genetic decomposition needs to be tested experimentally. The goal is to test the validity of the model: Did the students make the mental constructions called for by the theoretical analysis? If the constructions described in the genetic decomposition are observed, the model is supported. If the students appear to construct the concept in a way that differs from what is described in the genetic decomposition, then the model is refined or, if the discrepancies are too great, discarded in favor of a new genetic decomposition.

The following example illustrates how the analysis of data may lead to a refinement. In an interview in a study of students' understanding of spanning sets and span (Kú et al. 2011), the authors observed that many students did not understand that the vectors that make up a spanning set are elements of the subspace generated by that set. This difficulty was attributed to a mental construction not accounted

² A diffraction grating is an instrument used to analyze the light coming from stars. It decomposes the incoming light by diffraction to obtain a pattern of colored lines. These patterns allow researchers to know the chemical composition of the star.

for in the preliminary genetic decomposition. This difficulty arose in students' responses to the following interview item taken from a linear algebra textbook:

$$\text{Let } v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ and let } H = \left\{ \begin{bmatrix} s \\ s \\ 0 \end{bmatrix} \mid s \in \mathbb{R} \right\}.$$

Therefore each vector of H is a linear combination of $\{v_1, v_2\}$ since

$$\begin{bmatrix} s \\ s \\ 0 \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Is $\{v_1, v_2\}$ a spanning set for H ?

Carlos³ was among the students who had difficulty with this item. The following excerpt exemplifies this:

Carlos: Yes. All possible linear combinations of v_1 and v_2 span H . Neither v_1 nor v_2 can generate the third element, but H doesn't have it either, so it is not necessary. . . [then he explains] It's a spanning set for H because if we take all the possible linear combinations in the reals, then clearly we can see that it can be any number. . . well, any number in H and for example H doesn't have. . . it has a zero in the third element so, no, well. . . it would be, it's not needed and we see it here. . . I mean none of the two has it so. . . If H had another s here for example [he refers to the vector $(s, s, 0)$], it wouldn't be a spanning set for H , we would need another vector, which had for example, I don't know. If it were linearly independent and if it had an element in the last position, but since these two don't have it, but H doesn't either, then H can be spanned by these two vectors.

Carlos may have grasped the idea of the span of a set being formed by "all possible linear combinations." However, it is not clear if he considers whether or not the vectors v_1 and v_2 belong to H . The interview continues with the following question, in order to provoke more reflection (Fig. 4.2):

I: Can you give another spanning set for H ?

Carlos: (writes)

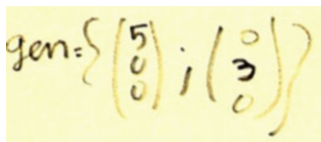


Fig. 4.2 Carlos finds another spanning set

³ All the names of interview subjects are pseudonyms. The interviewer's words are identified with "I:" throughout the text. "I" does not denote a single individual, as there were different interviewers for different studies, and sometimes multiple interviewers for the same study or even the same interview.

- I: Let's see. Why is this a spanning set?
- Carlos: Because they are two linearly independent vectors and if we take any numb. . . I mean this is in the reals, so if we take any number in H , well for example in H , I don't know, for it to be 1 and 1. We multiply this one by $1/5$ added to this one multiplied by $1/3$ and it spans H .

Then the interviewer asks Carlos to explain how he would find a spanning set for H if the question didn't provide the set $\{v_1, v_2\}$. Carlos responds as follows:

- Carlos: If I didn't have this? Well, s is in the reals, so it could be any number. Well, it would be enough to take two vectors that, I mean, with which I can generate a real number in the first one and a real number in the second one and that would be enough.

Like many of the students, Carlos did not yet understand that the elements of a spanning set necessarily belong to the span. The researchers observed that this student, and many others, had not made a construction related to the necessity of the vectors belonging to H . The researchers concluded that one of the reasons for the difficulty lay in the students' failure to perform the Action of finding the span of different spanning sets and of determining whether the elements of the spanning set are included in the span. This observation helped the researchers to guide the students to make the necessary construction, either during the interview or later in class. This finding led the researchers to make a refinement to the preliminary genetic decomposition.

The possibility of predicting the constructions students need in their learning of mathematical concepts provides researchers with a useful tool in the design of activities and teaching sequences that help students to make the constructions called for by the theoretical analysis. The way in which a genetic decomposition guides the design of instruction is considered in Chap. 5.

4.4 A Genetic Decomposition Is Not Unique

It has been reiterated several times in this book that a genetic decomposition is not unique, that is, it does not provide a single way in which all students construct a specific mathematical concept. Instead, it serves as a theoretical model which may help in understanding those constructions that appear in most students' work. Although it describes a possible trajectory for the construction of the concept, APOS Theory acknowledges that different students can follow paths different from those described in a particular genetic decomposition. Thus, the value of a genetic decomposition resides in its use as a general model which describes those constructions that are found to be needed by most of the students in the learning of a concept.

As with any general and descriptive theoretical model, several genetic decompositions can be designed by different researchers or even by the same group of researchers to describe the learning of a particular concept. If those genetic decompositions are supported by empirical studies of students'

constructions, they could all be considered reasonable descriptions of students' constructions. So far, there are only a few examples of different preliminary genetic decompositions for the same concept, and in the case of genetic decompositions supported by the results of APOS-based instruction, no diversity has been found. Of course one would expect to find some agreements among different supported models, but they may also have differences. Issues involving differences in a genetic decomposition, and how those differences may be resolved, is a subject for further research.

One exception is the work of Roa-Fuentes and Oktaç (2010), who proposed two different preliminary genetic decompositions of the linear transformation concept. The first of these aligns with how this concept is typically taught and how it appears in textbooks. The second one takes into account a suggestion reported in the literature (Dreyfus et al. 1999) that is based on instruction where nonlinear transformations are considered before the introduction of linear transformations. In Roa-Fuentes and Oktaç (2010), only Action, Process, and Object constructions were investigated; Schema did not form part of the study. As the word preliminary indicates, these theoretical analyses emerged before the collection of any data.

In Sects. 4.4.1–4.4.4, these two genetic decompositions are discussed and explained in detail. After considering the prerequisites, a general description is accompanied by the presentation of a figure that summarizes the structures that need to be constructed and the relations among those structures. Since the two theoretical analyses differ only in how the Processes of the properties of linearity are constructed, the two genetic decompositions are considered separately up to this point. After this point, the two genetic decompositions are identical.

4.4.1 Prerequisites for the Construction of the Linear Transformation Concept

Since linear transformations are defined as functions between vector spaces, the linear transformation concept depends on prior construction of a function Schema and vector space Object. In order to define a function whose domain and range are vector spaces, the function Schema needs to assimilate the vector space Object.

Since the test of whether a transformation is linear involves working with linear combinations, it is necessary for the learner to have constructed the concept of linear combination as a mental Object.

4.4.2 Genetic Decomposition 1

In the construction of the linear transformation concept according to this genetic decomposition, an individual starts by applying Actions to specific vectors from a

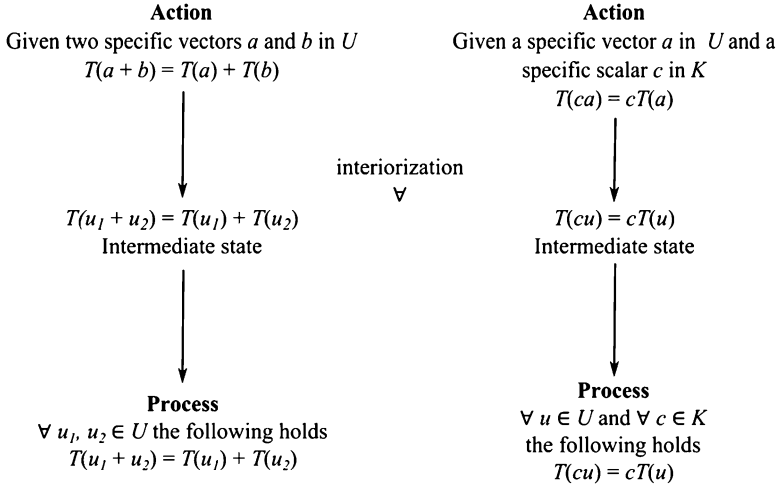


Fig. 4.3 Construction of the properties of linearity of a transformation between two vector spaces as Processes by interiorizing Actions on vectors (Roa-Fuentes and Okaç 2010, p. 105)

particular vector space. Specifically, the individual checks the addition property of linearity by applying the rule of assignment given by a particular transformation to compare the image of the sum of two vectors with the sum of their images. Limited to an Action conception, an individual cannot think beyond specific vectors or specific transformations.

As the individual reflects on these Actions and begins to think in general about this property for all the vectors in the domain space, without the need to work with specific vectors and without the need to make specific calculations, these Actions are considered to have been interiorized into a Process—the addition property of linearity of a transformation between two vector spaces.

In the preliminary genetic decomposition, as shown in Fig. 4.3, Roa-Fuentes and Okaç (2010) considered the possibility of an intermediate step between an Action conception and a Process conception. Roa-Fuentes and Okaç (2012) found that this intermediate step—checking the addition property of linearity for an arbitrary pair of vectors without consideration of all of the vectors in the domain—was not supported by empirical analysis.

As illustrated in Fig. 4.3, the Process of the scalar multiplication property of the linearity of a transformation between two vector spaces is constructed similarly.

4.4.3 Genetic Decomposition 2

This genetic decomposition begins with construction of the concept of a (general) transformation between two vector spaces. This concept has to have been constructed as an Object, since determination of the preservation of vector

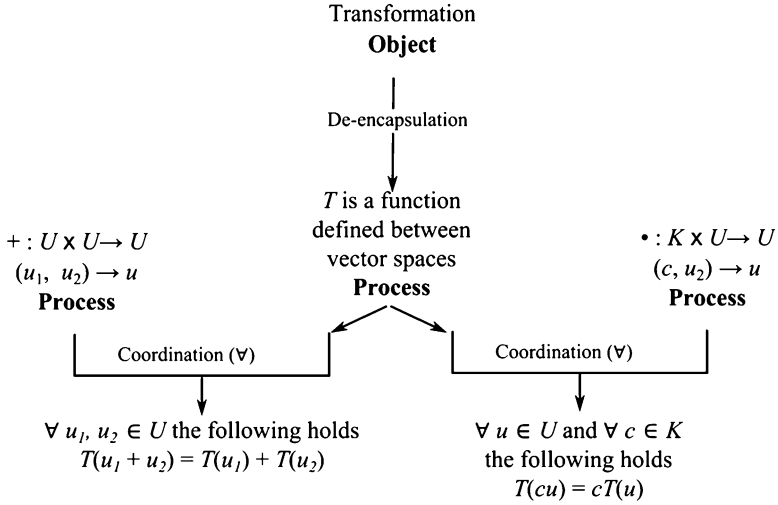


Fig. 4.4 Construction of the properties of linearity as Processes starting with transformation as an Object

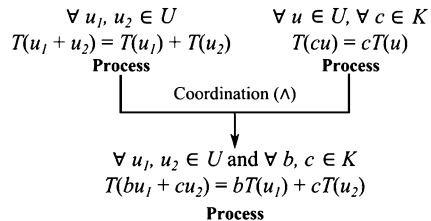
addition and of scalar multiplication under the transformation are Actions applied to the transformation.

The transformation Object is de-encapsulated so that the underlying Process can be utilized. This Process is coordinated with the Process of the binary operation “vector addition” through the universal quantifier $\forall$ to generate a new Process for determining whether the transformation satisfies the property of addition for all pairs of vectors in the domain. The Process related to the transformation allows the individual to think of the images of the domain vectors under the transformation. The Process related to vector addition allows the individual to form sums of vectors in the domain and in the range. By coordinating these two Processes, the individual can do two things: (1) form a sum of any two vectors in the domain and apply the transformation to that sum and (2) find the images of any two vectors in the domain and add them. Determination of the sum property involves comparison of the results obtained from (1) and (2) for all the vectors in the domain.

A similar coordination occurs for scalar multiplication. This is illustrated in Fig. 4.4.

The principal difference between the two genetic decompositions lies in the treatment of the transformation. In Genetic Decomposition 1, the transformation is applied to a pair of vectors (and their sum). Through interiorization, this Action is extended to all pairs of vectors in the vector space. In Genetic Decomposition 2, the binary operations of vector addition and scalar multiplication are Processes applied to the transformation Object, which is subsequently de-encapsulated so that it can be coordinated with those Processes.

Fig. 4.5 Construction of the Process of linear transformation (Roa-Fuentes and Okaç 2010, p. 106)



4.4.4 Genetic Decompositions 1 and 2: Constructing Process and Object Conceptions of Linear Transformation

Whichever genetic decomposition is followed in the construction of the two Processes in Sects. 4.4.2 and 4.4.3, once constructed, these two Processes of verifying the two linearity properties are coordinated via the logical connector “and” to give rise to a new Process. The new Process is constructed when the individual can think of the two Processes simultaneously, as in preserving linear combinations of vectors under a linear transformation, as shown in Fig. 4.5.

Encapsulation occurs with the need to apply Actions. With a conception of linear transformation as a mental Object, an individual can perform operations, such as adding or composing two linear transformations, and can ask questions about the properties linear transformations may or may not satisfy. For example: Under what conditions is a linear transformation invertible?

4.5 Refinement of a Genetic Decomposition

When researchers design a genetic decomposition, it must be tested empirically. The analysis may lead to mixed results. Some of the constructions predicted by the preliminary analysis may appear to have been made, or seem reasonable given the data, others may be lacking or different from those proposed, and others, not accounted for by the preliminary analysis, may surface as a result of the analysis. When this happens, the genetic decomposition needs to be refined to reflect what has been found empirically.

Revisions of the genetic decomposition may lead to changes in instruction, as well as provide an opportunity for further empirical analysis. The cycle of instruction $\rightarrow$ analysis $\rightarrow$ refinement can be repeated until it is determined that the refinement adequately describes students’ constructions and leads to effective instruction.

In their study of the concept of limit, Cottrill et al. (1996) devised a preliminary genetic decomposition. They based their decomposition on existing mathematics education literature, their understanding of the concept, and their instructional

experience. Students completed an instructional sequence that was informed by the decomposition. An analysis of data gathered from these students revealed the need for a refinement.

The preliminary genetic decomposition, its refinement, and the rationale for changes to the preliminary decomposition are given in Table 4.1. The preliminary decomposition is divided into six steps. The first three describe mental constructions involved in the development of an informal understanding of the concept, and the last three involve mental constructions associated with development of a formal understanding. Generally speaking, an informal understanding involves a dynamic conception, that is, the values of a function approach a limiting value as the values in the domain approach some quantity. A formal understanding is typically identified with the $\varepsilon - \delta$ definition.

An informal understanding is initially static. For a function f at a domain point $x = a$, determination of the limit starts with the Action of evaluating f at a few points, each successively closer to $x = a$ (Step 1P). As the individual reflects on these Actions, they may be interiorized into a mental Process (Step 2P); it is at this point that the static conception becomes a dynamic conception. Encapsulation occurs as the individual sees the need to apply Actions to the dynamic Process (Step 3P). The transition to more formal thinking starts with Step 4, as the Process constructed in Step 2P is reconstructed in terms of intervals. The formal definition then arises through application of a two-level quantification Schema (Step 5P) to the reconstructed Process (Step 4P).

The instructional treatment consisted of a 2-week unit that included five types of computer activities embedded in the usual topics of approximation, one- and two-sided limits, and applications of the limit. The students conducted graphical analyses, wrote and analyzed short programs related to the informal, dynamic conception, and completed tasks involving construction and analysis of $\varepsilon - \delta$ intervals.

The data analysis suggested two major revisions. The first was addition of a step that precedes the Action of evaluating a function at several points (Step 1R); specifically, the individual evaluates a single point, which may be $x = a$ itself instead of a series of selected points that are successively closer to $x = a$. The second deals with the construction of the Process conception. Rather than a single Process, the researchers uncovered evidence of a coordination of two Processes: a domain Process, in which x approaches a , and a range process, in which y approaches L . The two Processes are then coordinated through the function f . In other words, the function f is applied to the Process of x approaching a to obtain the Process of $f(x)$ approaching L . According to Cottrill et al. (1996), one source of students' difficulties with the limit concept can be attributed to this more complicated mental construction. The other principal source of difficulty, which is related to the formal concept of limit, is that students do not possess a sufficiently powerful conception of quantification (Dubinsky et al. 1988).

Table 4.1 Preliminary genetic decomposition of the concept of limit and its refinement

Preliminary genetic decomposition	Refinement	Rationale for change
	1R: The Action of evaluating the function f at a single point x that is considered to be close to, or even equal to, a	A number of students evaluated a single point, often isolating their attention on the point $x = a$
1P: The Action of evaluating the function f at a few points, each successive point closer to a	2R: The Action of evaluating the function f at a few points, each successive point closer to a	
2P: Interiorization of the Action of Step 1P to a single Process in which $f(x)$ approaches L as x approaches a	3R: Construction of a coordinated Process Schema: (a) Interiorization of the Action of Step 2R to construct a domain Process in which x approaches a (b) Construction of a range Process in which y approaches L (c) Coordination of (a) and (b) via f	
3P: Encapsulation of the Process of Step 2P so that the limit becomes an Object to which Actions can be applied	4R: Encapsulation of the Process of Step 3R(c) so that the limit becomes an Object to which Actions can be applied	The data showed numerous instances of the construction of two separate Processes. Some students only constructed one of them and this appeared to prevent them from understanding the limit concept. Students who were more successful gave evidence of having coordinated these Processes
4P: Reconstruction of the Process of Step 2P in terms of intervals and inequalities. This is done by introducing numerical estimates of the closeness approach: $0 < x - a < \delta$ and $0 < f(x) - L < \varepsilon$	5R: Reconstruction of the Process of Step 3R(c) in terms of intervals and inequalities. This is done by introducing numerical estimates of the closeness approach: $0 < x - a < \delta$ and $0 < f(x) - L < \varepsilon$	
5P: Application of a two-level quantification Schema to connect the Process described in Step 4P to the formal definition	6R: Application of a two-level quantification Schema to connect the Process described in Step 5R to the formal definition	
6P: Application of a completed $\varepsilon - \delta$ conception to specific situations	7P: Application of a completed $\varepsilon - \delta$ conception to specific situations	The reconstructed Process described in this step would be based on the coordinated Process rather than a single Process, as described in the preliminary genetic decomposition

Step 1R was not part of the preliminary decomposition; analysis of the data revealed this as a step that precedes Step 2R, which was the first step (Step 1P) in the preliminary decomposition.

The coordinated Process (Step 3R) replaced the single Process (Step 2P) in the preliminary genetic decomposition. In the data analysis, students tended to construct a separate Process for approaching $x = a$ apart from application of the function f . After constructing the domain Process, students applied the function f to the domain elements constructed through the domain Process to come up with a range Process.

The coordination of two Processes, one for approaching $x = a$ and the other for approaching L through f , shows that the dynamic conception of the limit turned out to be more complicated than first thought. Unlike some researchers, who believe that a dynamic conception hinders progress toward development of formal understanding, Cottrill et al. (1996) found that students' difficulty can be attributed, at least in part, to an insufficiently well-developed dynamic conception, which appears to need to be based on a coordinated Process Schema.

4.6 Role of the Genetic Decomposition in the Design of Teaching Activities

In addition to being a theoretical model for research, the genetic decomposition for a concept guides instruction. Since a genetic decomposition describes the constructions a student may need to make in order to learn a mathematical concept, it can be used to design activities to help students to make the proposed constructions. Although going from the genetic decomposition to the design of instructional activities is not always direct, the way in which the former informs the latter is very important since it represents a bridge between the theory and its pedagogical use (Trigueros and Oktaç 2005).

In a teaching sequence whose design is based on APOS theory, the first part of the teaching cycle, which is described in detail in Chap. 5, consists of activities for students to work on collaboratively, often in a laboratory setting. Each of the activities is designed to provide opportunities for students to repeat specific Actions and to reflect on them, to foster interiorization of Actions into Processes, to help with the coordination and reversal of Processes, and to support encapsulation of Processes into Objects. A teaching sequence can also include activities where the goal is the construction of relationships among different Actions, Processes, Objects, and previously constructed Schemas. These activities may help students construct a new Schema or, in the case of a previously constructed Schema, lead to further development or refinement of that Schema.

Research can also be conducted following instruction. The focus of the research is to determine whether students made the constructions predicted by the genetic decomposition and whether such constructions helped them to learn the mathematics in question. The data that is obtained is compared with what is

predicted by the genetic decomposition. This type of analysis may lead to refinement of the genetic decomposition. When a genetic decomposition accurately reflects the mental constructions students make in their efforts to understand a concept, comparative research can be conducted to compare the mathematical performance of students who have completed APOS-based instruction with students who have completed instruction that was not APOS based. Research of this type has been conducted and has shown the promise of instruction based on APOS Theory (Weller et al. 2003).

In order to further illustrate how a genetic decomposition informs the design of activities, examples from the vector space chapter from the textbook *Learning Linear Algebra with ISETL* (Weller et al. 2002) are presented in Sects. 4.6.1 and 4.6.2.

4.6.1 Genetic Decomposition of a Vector Space

The concept of vector space is a Schema that is constructed by coordinating the three Schemas of set, binary operation, and axiom. The set and binary operation Schemas are thematized to form Objects and coordinated through the axiom Schema.

Binary Operation. A binary operation is a function of two variables defined on a single set or on a Cartesian product of two sets. In terms of the APOS framework, there are four mental constructions involved in development of this concept:

- Action: Given a formula for a binary operation, an individual can take two specific elements of the set(s) on which the operation is defined and apply the formula.
- Process: The individual interiorizes the Actions comprising a binary operation: accepting two elements, acting on these Objects in some way, and returning a new Object.
- Object: The individual can distinguish between two binary operations, consider more than one binary operation defined on a set or on a product of sets, check whether a binary operation satisfies an axiom, and de-encapsulate a binary operation so that it can be coordinated with other Processes.
- Schema: The individual can define a binary operation on a set or on a product of two sets and/or identify whether a function defined on a set, or sets, is a binary operation.

Set. A set is a collection of Objects that satisfies a given condition. In terms of the APOS framework, there are four mental constructions involved in development of this concept:

- Action: An individual can only conceive of a set when given a specific listing of elements or when presented with a particular condition of set membership.

- Process: The Action of gathering and putting Objects together in a collection according to some condition is interiorized.
- Object: The individual can apply Actions or Processes to the Process such as determine the cardinality of a set, compare two sets (not necessarily in terms of cardinality), consider a set to be an element of another set, and define a function in which a set is one of the Objects accepted. The student can also de-encapsulate a set so that it can be coordinated with other Processes.
- Schema: The individual can apply the set Schema to a given mathematical situation. In linear algebra, this would mean being able to define sets of Objects that might later be classified as sets of vectors: tuples, polynomials, functions, and matrices. An individual may have also developed a general notion of what a set is and what it is not.

Axiom. For vector space, an axiom is a Boolean-valued function that accepts a set, or a Cartesian product of sets, and a binary operation defined on the set, or sets, and checks whether the property defined by the axiom is satisfied. Checking an axiom is a Process that involves coordination of the general notion of checking a property with the Process defined by the specific property being checked. When an axiom is applied to a set and a binary operation, the set and binary operation Schemas must be de-encapsulated and coordinated with the Process of checking the property in question. An axiom Schema includes the general notion of checking whether a set, operation pair, satisfies a property.

How the Three Schemas Work Together in the Mental Construction of a Vector Space. The axiom Schema includes the general notion that a binary operation on a set may or may not satisfy a property and the ten specific Objects obtained by encapsulating the ten Processes corresponding to the ten vector space axioms. Each axiom is de-encapsulated for individual coordinations to take place. Each coordinated Process is applied to the set and binary operation Schema. The set and binary operation Objects are de-encapsulated so that they can be coordinated with each axiom. The ten instances of this operation are then coordinated into a single Process of satisfying the axioms. The interaction of these Schemas is illustrated in Fig. 4.6.

4.6.2 Activities Designed to Facilitate Development of the Vector Space Schema

Unlike many linear algebra texts that start with systems of equations and matrices, *Learning Linear Algebra with ISETL* begins with vector spaces. This chapter is preceded by an introductory chapter on functions and structures. Although one

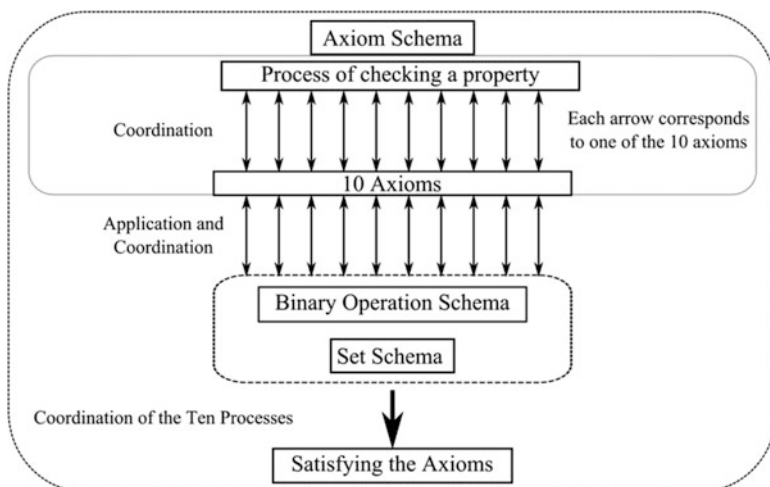


Fig. 4.6 Diagrammatic representation of the genetic decomposition of the vector space Schema

of the purposes of the introduction is to familiarize students with ISETL commands, students learn about these commands in the context of working with mathematical objects such as sets. Students study different representations for sets and convert from one representation to another (say, from a description of a set to a set former or from a set former to a list); they also construct and compare sets and distinguish sets from other objects such as tuples. The objective is development of a set Schema.

In both chapters (the introduction and the chapter on vector spaces), students interpret and write ISETL code for a variety of *funcs*. A *func* is an ISETL command for a function. A *func* accepts variable(s), whose values can be numbers, sets, tuples, and even other *funcs*. It includes a return statement that yields the output of the *func*. In two of the activities, students construct *funcs* to carry out the binary operations of addition (modulo p) and multiplication (modulo p) on a finite field (i.e., Z_p). They use these *funcs* in the construction of other *funcs* to perform addition and scalar multiplication on sets of tuples Z_p^n (n a positive integer) over the field Z_p .

Once students have constructed *funcs* for tuple addition and scalar multiplication for different pairs of scalar fields and sets of tuples, they begin to write code to test properties of the operations they have defined; specifically, to test the vector space axioms. Once they have written the code for a particular axiom, they test it on different systems $(K, V, \text{va}, \text{sm})$ ⁴ they have defined and worked with in previous activities. Interpretation of code involving the vector space axioms

⁴Here, K refers to a field, V stands for a set of tuples, va denotes addition defined on V , and sm represents the scalar multiplication operation defined on K and V .

encourages interiorization of the Process underlying each axiom. Applying a `func` for an axiom facilitates encapsulation of the set and binary operation concepts since they are accepted as inputs to the `func`. Writing a `func` for an axiom supports development of the axiom Schema since the individual needs to coordinate the Process of checking a property with the Process associated with the specific axiom being worked with.

Eventually, students are presented with all ten axioms and asked to explain how each axiom works. They then apply the axioms to 12 different systems (Z_p, Z_p^n, va, sm) (for different values of p and n). They generate a table in which they record, for each system, which axioms are satisfied. In the subsequent activity, they summarize their findings. The section culminates with construction of the `funcis_vector_space`, a Boolean-valued function that accepts a set V , a field of scalars K , an operation va defined on V , and an operation sm defined on the pair (K, V) . The `func` tests whether the system (K, V, va, sm) satisfies all ten axioms. It returns `true` if all ten axioms are satisfied and `false` otherwise. The purpose of having students write and use this `func` is to support the mental constructions called for by the preliminary analysis and to coordinate the ten Processes underlying each axiom into a single Process that establishes whether the system constitutes a vector space.

4.7 What Is Not a Genetic Decomposition

By now it may be quite clear what a genetic decomposition is. However, given its complexity, in some research projects and papers and also in students' work, it happens that what authors or students call a genetic decomposition is not really one. In what follows, some examples, stated verbatim, of such "genetic decompositions" are presented and discussed.

It is a common error to confound a genetic decomposition with a description of a teaching sequence or a mathematical description of a concept where APOS terminology is used. Example 1 on matrices, offered by a teacher who participated in a seminar on APOS Theory, illustrates this:

Example 1 of what a genetic decomposition is not:

1. Students do Actions to define a matrix.
2. Students do Actions to define size of matrices.
3. Process for adding two matrices.
4. Process to multiply matrices by scalars.
5. Processes to multiply two matrices with restrictions on this operation.
6. Processes to verify properties of operations.
7. Actions to verify if the inverse of a matrix exists.
8. Process to find the inverse of a matrix.
9. Encapsulation of the concept of matrix.

For Example 1, instead of offering a description of specific mental constructions needed to learn the concept of matrix, this teacher described a class plan that consisted of a list of mathematical topics to cover. Although APOS terminology was used, it was not related to the cognitive structures of APOS Theory, that is, what was proposed as a genetic decomposition does not specify the Objects on which Actions or Processes are performed, does not include any description of how Actions are interiorized into Processes, nor describes how Processes are encapsulated or coordinated. Moreover the steps do not even identify what the Actions, Processes, or Objects are.

Another common error is to describe a “genetic decomposition” that simply lists operations a student is to perform. This is exemplified in Example 2 on transformations of real-valued functions presented by a graduate student in a seminar:

Example 2 of what a genetic decomposition is not:

Action: Can see specific examples as representing a transformation of a known function. Can draw the graphs of translations of a real-valued function when the graph of the function is well known, such as a linear or quadratic function. Can introduce values into the rule of a transformed function to obtain its value. Can find the graph of a transformed function using points

Process: Can draw general basic transformations of a given function (translate it vertically or horizontally, stretch it). Can find transformations of a given function. Can determine the original function if given a certain transformed function. Understands the difference between horizontal and vertical translations, and of stretching functions in general, and sees if translations modify the domain and range of the original function

Object: Can operate on transformed functions to obtain new functions such as the composition of transformations. Can draw the graph of any transformed function. Can relate any transformed function to the original function. Can predict the function that results from a composition of transformations. Understands the difference between diverse transformations of functions in general and how the domain and range of a function change when the function is transformed

Although the terms Action, Process, and Object are included in the graduate student’s proposed “genetic decomposition,” they are stated in terms of students’ conceptions, that is, they are not stated in terms of the constructions a student needs to make in order to perform the activities listed. For example the description does not specify the Actions involved in graphing transformations point by point, explain how those Actions are interiorized so that students can recognize transformations graphically, or tell how to reverse the Process involved in recognizing the original function when given a transformed function.

Some of the difficulties encountered in the design of a genetic decomposition can be related to misunderstandings of the theory that are reflected in the way the constructions are described. This can be observed in Example 3 on the derivative that was presented by a graduate student in a seminar:

Example 3 of what a genetic decomposition is not

Preliminary Knowledge

Geometric Rate as an Object. The students need to have assimilated it, be aware of the meaning of the rate as a totality and be able to do Actions on it, and see it as a trigonometric rate: the tangent of a linear function.

Secant as an Action. The student must have assimilated it so that he or she is able to manipulate it physically or mentally.

Tangent to a Circle. It must be assimilated as a Process because the student must be capable of coordinating different definitions of tangents to a circle.

Linear Function. Given the slope and a point, determine the equation of a linear function and think of it as a coherent collection of Objects (slope and point), Actions (on those Objects), and Processes (manipulations to find the equation of a line and its graph), so it must be a Schema. Functions: $R \rightarrow R$. The student should have developed a Schema for functions as a collection of Objects (graphs and algebraic expressions), Actions (on the Cartesian plane and algebraic expressions), and Processes (manipulation of algebraic expressions together with Cartesian plane to draw the graph).

Limit. Process of approaching closer and closer.

The following comments can be made about the preliminary knowledge in Example 3. In the paragraph *Secant as an Action*, the author is not aware that if the student can do manipulations mentally, it means that he or she has interiorized the Actions into a Process. In addition, nothing is said about the Actions that are included in such manipulations. In *Tangent to a circle*, the author applies the notion of coordination to definitions instead of to Processes. In *Linear function* the conception of Schema is not clear in the last two points.

Continuation of Example 3

“Genetic decomposition” of derivative

1. The geometric rate as an Object, the function for which the derivative has to be obtained as a Schema, and the secant to a circle as an Action are coordinated in a new Schema to construct the secant of a function which must be considered as a new Object, that is, it has to be thematized into a Schema.
2. The Action limit is applied to the Object secant.
3. When the tangent to a point A of the function is interiorized into a simple Process to construct the coefficient that determines the slope of the tangent line to the function on the point A.
4. A Schema is created with the Object tangent to the point A of the function and the Schema linear function: a coherent collection of Objects (slope and point), Actions (on the slope and point), and Processes (manipulation to find the linear function and to draw its graph).

5. The Processes of the Schema in 4 are coordinated to the interiorized Action in 3 to create the Process, derivative of the function in the point A of the domain of the function.
6. All the Processes of the derivatives at each point of the domain of the function are coordinated to encapsulate them in the Schema derivative function of function f .
7. The Schemas for limit of a secant line when the denominator is approaching zero, equation of the derivative of a function, and that of graphical representation of the derivative function.

The problem in Example 3 is that the meaning of the constructions does not appear to be understood; for example, a “simple” Process is mentioned without reference to the Actions from which it arises; a coordination of Actions with Processes is mentioned, but it is not clear what is meant by “the Process of the Schema” in number 3; the meaning of number 5 is not clear; a different Process for each of the points of the function is considered in number 6 when this should be the interiorization of Actions; in the same item, there is no mention of the Actions that led to encapsulation of the Process; there is a misuse of the term Schema; and finally, it is very difficult to make sense of what the student wrote in number 7.

In some cases, a “genetic decomposition” consists of a description of what researchers consider an Action, Process, and Object conception of the concept, as is exemplified in Example 4, taken from a final presentation of a student in a graduate program.

Example 4 of what a genetic decomposition is not:

If a student has an **Action conception**, he or she is limited to do Actions. The Actions that the student shows are:

- A.1. Has memorized that three noncoplanar vectors in $\mathbb{R}^3$ is a basis for this space.
- A.2. Can find a basis for a subspace of $\mathbb{R}^3$ by manipulating a given equation, for example, the equation of a plane or a line.
- A.3. Can perform Actions on a given set of vectors to verify if they are or not linearly independent.
- A.4. Can verify that given sets of three vectors span or not $\mathbb{R}^3$.

The **Process conception** of a student is demonstrated by her or his possibility to show that he or she has interiorized these Processes:

- P.1. Can find a basis for any vector space.
- P.2. Can verify the linear independence of any set of vectors given.
- P.3. Can verify if the vectors of a given set span or not a given vector space.
- P.4. Demonstrate that he or she has not coordinated the previous Processes and has not encapsulated basis as an Object because he or she has difficulties to distinguish spanning sets from basis.

The **Object conception** of a student is demonstrated by

- O.1. Can perform operations on any given basis

- O.2. Can compare and distinguish different sets and decide if they are or not a basis for a given vector space
- O.3. Can consider and find basis for infinite spaces such as $\mathbb{R}^n$
- O.4. Can use the concept of basis for vector spaces different from $\mathbb{R}^n$

From a methodological point of view, Example 4 cannot be considered a genetic decomposition since the mental constructions needed to learn the concept are not described. For example, there is no description of the Actions that are interiorized into the Process of finding a basis for any vector space, nor an explanation of the result of coordination of Processes P2 and P3.

While these examples of erroneous “genetic decompositions” include APOS terminology, each falls short in some fundamental respect. What is important to learn from these examples is that a genetic decomposition is far more than a sequence of steps for instruction or a list of conceptions students may have. Rather, it is a description of the mental constructions students may need to make in their learning of a mathematical concept. In this sense, a genetic decomposition is a guide for the design of instruction that aligns with how students come to understand a mathematical concept.

Chapter 5

The Teaching of Mathematics

Using APOS Theory

This chapter is a discussion of the design and implementation of instruction using APOS Theory. For a particular mathematical concept, this typically begins with a genetic decomposition, a description of the mental constructions an individual might make in coming to understand the concept (see Chap. 4 for more details). Implementation is usually carried out using the ACE Teaching Cycle, an instructional approach that supports development of the mental constructions called for by the genetic decomposition.

The ACE cycle includes activities, which students typically work on cooperatively, sometimes with use of a *mathematical programming language* such as the Interactive Set Theoretic Language (ISETL). The phrase mathematical programming language refers to a program that satisfies three properties:

1. The syntax is close to standard mathematical notation.
2. Certain mathematical features are supported together with their usual mathematical properties.
3. Important data types, such as procedures and functions, can be operated on and called and returned by procedures and functions.

The components of the ACE Teaching Cycle and the features of ISETL are described in Sects. 5.1 and 5.2. The remainder of the chapter is devoted to examples of APOS-based instruction on groups in abstract algebra for mathematics majors (Sect. 5.3) and on infinite repeating decimals for a content course for preservice¹ elementary and middle school teachers (Sect. 5.4).

¹ Preservice refers to college or university students who are preparing to become school teachers.

5.1 The ACE Teaching Cycle

The ACE Teaching Cycle is a pedagogical strategy that consists of three components: (A) Activities; (C) Classroom Discussion; and (E) Exercises.

For Activities, which constitute the first step of the cycle, students work cooperatively in teams on tasks designed to help them to make the mental constructions suggested by the genetic decomposition. The focus of these tasks is to promote reflective abstraction rather than to obtain correct answers. This is often achieved by having students write short computer programs using a mathematical programming language.

The Classroom Discussion, the second part of the cycle, involves small group and instructor-led class discussion, as students work on paper and pencil tasks that build on the lab activities completed in the Activities phase and calculations assigned by the instructor. The class discussions and in-class work give students an opportunity to reflect on their work, particularly the activities done in the lab. As the instructor guides the discussion, he or she may provide definitions, offer explanations, and/or present an overview to tie together what the students have been thinking about and working on.

Homework exercises, the third part of the cycle, consist of fairly standard problems designed to reinforce the computer activities and the classroom discussion. The exercises help to support continued development of the mental constructions suggested by the genetic decomposition. They also guide students to apply what they have learned and to consider related mathematical ideas.

The ACE Cycle and its relationship to the genetic decomposition are illustrated in Fig. 5.1.

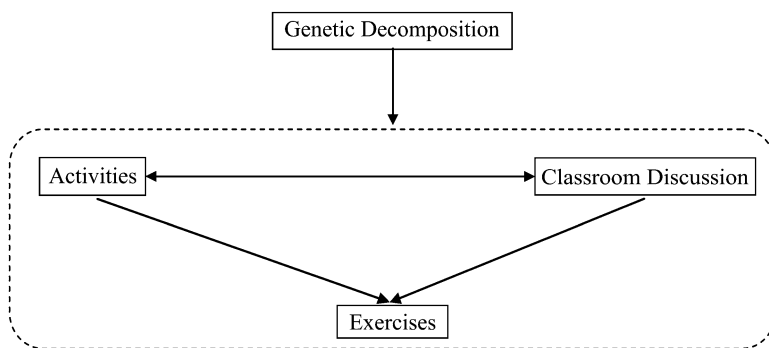


Fig. 5.1 Relation between the ACE Teaching Cycle and a genetic decomposition

The arrow from the Genetic Decomposition to the dotted box illustrates the fact that the genetic decomposition affects each component of the ACE Teaching Cycle. The bidirectional arrow between Activities and Classroom Discussion shows that, on the one hand, the activities are the principal subject of the class discussion

and, on the other, that the classroom discussion provides an opportunity for the students to reflect on the activities. The arrows to Exercises from Activities and Classroom Discussion reflect the principal purpose of the exercises—to reinforce the mental constructions the students make or have begun to make as they work through the Activities and participate in the Classroom Discussion.

The Activities phase involves completion of cooperative tasks informed by the genetic decomposition. Although computers have frequently been involved, their use is not required. It is simply the case that activities involving use of a mathematical programming language have been effective in helping students in learning a mathematical concept using the mental constructions called for by a genetic decomposition for the concept (see Weller et al. 2003). ISETL, the language typically used, is described in the next section.

5.2 ISETL: A Mathematical Programming Language

5.2.1 *A Brief Introduction to ISETL*

ISETL is a freeware mathematical programming language. What separates ISETL from other programming languages is its ability to represent mathematical concepts using mathematical notation and the ability of the language to operate on the concepts represented by that notation. The program can be obtained online from one of the following URLs:

<http://titanium.mountunion.edu/isetlj/isetlj.html>

<http://homepages.ohiodominican.edu/~cottrilj/datastore/isetl/>;

Dautermann (1992) wrote a manual that provides details regarding the use of ISETL, including its commands and features. The reader may wish to download and use ISETL to work through the examples that appear in this section.

ISETL has proven to be a powerful tool in helping students to learn mathematics. The syntax of the language is very close to standard mathematical notation, the language supports certain mathematical features, and all data types can be acted on as objects. Each of these aspects of ISETL is discussed in Sects. 5.2.2–5.2.4. Use of ISETL as a pedagogical tool is considered in Sect. 5.2.5.

5.2.2 *The Syntax Is Close to Standard Mathematical Notation*

In ISETL, syntax resembles standard mathematical notation. For example, in mathematical set former notation, the prime numbers from 2 to 100 can be represented in the following way:

$$\{x : x \in \{2, 3, \dots, 100\} \mid (\nexists y \in \{2, 3, \dots, x-1\} \ni x \bmod y = 0)\}.$$

The ISETL representation is nearly identical:

$$\{x : x \text{ in } \{2..100\} \mid (\text{not exists } y \text{ in } \{2, 3..x-1\} \mid x \bmod y = 0)\}.$$

In ISETL the word “in” stands for the “element of” symbol $\in$, the phrase “not exists” replaces the “not exists” symbol $\nexists$, and the character “l” represents the “such that” symbol $\ni$.

To return a set in list form, the user types the code for the set former at the ISETL prompt $>$, places a semicolon at the end of the code, and then presses enter. For the set of prime numbers less than 100, the screen display for returning the set as a list is

```
> !setrandom off
> {x : x in {2,3..100} | not exists y in {2,3..x-1} | x mod
    y = 0};
{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47,
 53, 59, 61, 67, 71, 73, 79, 83, 89, 97};
or
> !setrandom on
> {x : x in {2,3..100} | not exists y in {2,3..x-1} | x mod
    y = 0};
{17, 19, 29, 23, 11, 13, 7, 5, 2, 3, 41, 37, 31, 47, 43, 53,
 59, 61, 67, 71, 73, 97, 89, 83, 79};
```

The command `!setrandom off` ensures that the elements of the set will be listed in numerical order, while the command `!setrandom on` returns the listing of the elements in random order.

The syntax for sets of tuples is treated in a similar way. In standard mathematical notation, the set-builder representation for Z_2^3 , the set of all three tuples with entries in Z_2 (the set of all integers modulo 2), is

$$Z_2^3 = \{(a, b, c) : a, b, c \in \{0, 1\}\}.$$

In ISETL, the set is represented almost identically:

$$Z2_3 := \{ [a, b, c] : a, b, c \text{ in } \{0, 1\} \};$$

There is, however, a subtle, though important, difference between the two representations. In the mathematical representation, the ordered triples appear with parentheses. In ISETL, the ordered triples appear with square brackets. The reason is that ordered sequences in ISETL are expressed using bracket `[ ]` notation. When using mathematical notation, the set Z_2^3 in list form appears as

$$Z_2^3 = \{(1, 1, 1), (1, 0, 1), (1, 0, 0), (0, 1, 0), (0, 1, 1), (0, 0, 1), (0, 0, 0), (1, 1, 0)\},$$

while, in ISETL, it is represented as

```
z2_3:={ [1,1,1], [1,0,1], [1,0,0], [0,1,0], [0,0,1], [0,0,0], [1,1,0] }.
```

Numbers and operations resemble scientific or graphing calculator displays. The only difference is the prompt given by the symbol `>`. Users enter code on lines that begin with this symbol or with the symbol `>>`, which indicates input to be completed. Completion of input is typically indicated with a semicolon. For example, in Line 4 following the entry `13*(233.8)`, the program prompt `>>` appears because no semicolon was added when the expression was entered. Once a semicolon is entered, the result of the computation is returned (Line 5). Lines without prompts indicate what the computer returns as output²:

```
Line 1: > 7+18;
Line 2: 25;
Line 3: > 13*(233.8)
Line 4: >> ;
Line 5: 3039.400;
Line 6: > !rational off
Line 7: > 27/36;
Line 8: 0.750;
Line 9: > !rational on
Line 10: > 27/36;
Line 11: 3/4;
Line 12: > 5-9
Line 13: >> ;
Line 14: -4;
```

The command `!rational off` (Line 6) instructs ISETL to return rational numbers in decimal form (the number of decimal places can be set using a simple command), while the command `!rational on` (Line 9) calls for fractional representations.

ISETL supports variables, which are case sensitive and defined using the symbol `=`. For example, entry of `x:=2;` sets the variable `x` equal to 2. When the variable `x` is called (indicated in ISETL by `x;`), ISETL returns 2:

```
> x:=2;
> x;
2;
```

²The notation Line # does not actually appear on an ISETL screen. It is being used for convenience here and in other examples with longer lines of code.

In addition to variables, ISETL supports Boolean operators (and, or, not, and impl, where impl is shorthand for implies), the usual comparison operators ($=$, $\neq$ (inequality), $<$, $\leq$, $>$, $\geq$), and quantifiers (forall, exists, not exists).

5.2.3 Supporting Mathematical Features

Certain control statements such as if statements and for loops can be used to carry out mathematical procedures without the need to invoke special commands. In Line 1 of the following example, the variable x is set equal to 4. For the if command, which appears in Lines 2 through 6, the beginning statement is of the form if [condition] then and ends with the line end if;. Because the condition $x > 2$ is satisfied and precedes the condition $x > 3$ in Line 4 where the elseif statement appears, ISETL only returns the first phrase, x is larger than 2. For a value of x less than or equal to 2, ISETL would not return anything:

```
Line 1: > x:=4;
Line 2: > if x>2 then
Line 3: >> writeln "x is larger than 2";
Line 4: >> elseif x>3 then
Line 5: >> writeln "x is larger than 3";
Line 6: >> end if;
Line 7: x is larger than 2
```

Like an if statement, a for loop has both a beginning and a concluding statement. The beginning statement is of the form for [element(s) in some set] do, and the concluding statement is given by the end for command. In the sample code below, Lines 1 and 2 define variables that will be used in execution of the for loop: the set S , which represents the first three counting numbers, and the variable a , which is set equal to 0 (as an initial value). The for loop, given in Lines 3 through 5, builds a sum, represented by the variable a , consisting of all possible pairs of elements from S . The value of a is reported in Line 7 as a result of the ISETL "call" for the value of the variable, which appears in Line 6.

```
Line 1: > S:={1..3};
Line 2: > a:=0;
Line 3: > for x, y in S do
Line 4: >> a:=a+x+y;
Line 5: >> end for;
Line 6: > a;
Line 7: 36;
```

Important mathematical objects such as functions can be defined and evaluated in ISETL. In the following case, the command func (Line 1) accepts variable(s),

whose values can be numbers, sets, tuples, and even other funcs. It includes a return statement (Line 2) that yields the output of the func. The code represents a function f that accepts a number x and returns the sum of x and 3 modulo 6. Line 5 gives the code necessary to evaluate f when $x = 4$. The actual value of f at $x = 4$ is given in Line 6.

```

Line 1: > f:=func(x);
Line 2: >> return (x+3) mod 6;
Line 3: >> end;
Line 4: >
Line 5: > f(4);
Line 6: 1;

```

A proc or procedure is the same as a func except that it has no return statement and does not return a value. It is used to perform internal operations. In the example below, the proc called SetNot accepts a tuple of length 2 called pair. The variable G , the first component of the tuple (a set, defined in Line 7), is assigned the value of $\text{pair}(1)$, and the variable $\circ$, the second component of the tuple (a func, defined in Lines 8–11), is assigned the value of $\text{pair}(2)$. SetNot assigns to the variable e the identity with respect to $\circ$ and G . The value of the variable inv is a set that consists of all the tuples of the form $[g, g']$, where g is an element of G and g' is the left inverse of g with respect to the operation $\circ$. The “ $\circ$ ” notation that appears in Lines 3 and 4 in the equations $x \circ g = g$ and $g' \circ g = e$ indicates use of $\circ$ as a binary operation on G in infix form. The infix notation can be used with any func of two variables in ISETL. If $\circ$ is a func that accepts two inputs a and b , the call for $\circ$ can be expressed as $\circ(a, b)$ or as $a \circ b$. This latter notation acknowledges that a function of two variables defined on a set is a binary operation on that set:

```

Line 1: > SetNot:=proc(pair);
Line 2: >> G:=pair(1); o:=pair(2);
Line 3: >> e:=choose x in G | (forall g in G | x.o g = g);
Line 4: >> inv:={ [g, choose g' in G | g'.o g = e] : g in G };
Line 5: >> end;
Line 6: > pair:= [ ];
Line 7: > pair(1):={0..5};
Line 8: > pair(2):=func(x,y);
Line 9: >> if (x in G and y in G) then
Line 10: >> return (x*y) mod 6;
Line 11: >> end;
Line 12: >> end;
Line 13: > pair;
Line 14: [{0, 1, 2, 3, 4, 5}, !func(6)!];
Line 15: > SetNot(pair);
Line 16: > G; e; inv(5); inv(2);
Line 17: {1, 0, 2, 3, 5, 4};

```

```

Line 18: 1;
Line 19: 5;
Line 20: OM;

```

In Line 13, where `pair` appears, ISETL returns a tuple which consists of two elements, the set defined by `pair(1)` and an expression `!func(6)!`, which denotes the fact that `pair(2)` is a `func`.

Line 15 denotes the “call” of `SetNot` for `pair`. Nothing is returned because the `proc` merely performs the operations given in its lines of code. Results of the execution of `SetNot(pair)` are given in Lines 16–20. Line 16 calls for return of the set `G`, the element `e`, and the inverses of 5 and 2, mod 6. Line 17 yields the set `G` in list form, Line 18 returns the identity element of `G` with respect to the operation `o`, Line 19 returns the inverse of 5 with respect to the operation `o`, and Line 20 returns `OM` to reflect the fact that 2 has no inverse with respect to the operation `o`.

5.2.4 Operations on Data Types

ISETL is powerful in part because it can support operations on data types. From a technical point of view, this means that certain ISETL data types are first-class objects. `Funcs` represent one of the most important examples. For instance, a `func` can actually return another `func`. This can be seen in the following lines of code for the `func D`, which accepts a function f and returns a `func`, which is the difference quotient of f within 0.001 units of the value of the variable x :

```

> D:=func(f);
>>     return func(x);
>>         return (f(x+0.001)-f(x))/0.001;
>>     end;
>> end;

```

The `func D` enables ISETL to compute the difference quotient for any specified function f at any domain point x using a difference of 0.001. For instance, if the function f is defined by $f(x) = x^2 + 1$, which would appear in ISETL as

```

> f:=func(x);
>>     return x**2+1;
>> end;

```

the call of `D(f)(3)`; returns the value of the difference quotient for f at $x = 3$ using a difference of 0.001:

```

> D(f)(3);
6.001;

```

A similar example is the `func Sum`, given by the code which follows. This `func` accepts two `funcs` representing functions f and g and returns a `func` that represents their sum, which is given by `Sum`, which is defined in Lines 1–5.

Two particular functions f and g are defined in Lines 7–13. The call for Sum, given by `Sum(f,g)` (Line 15), returns the value `!func(7)!` (Line 16) to indicate that `Sum(f,g)` is itself a func. The call for `Sum(f,g)(2)` (Line 18) yields the value of $(f+g)(2)$ (Line 19):

```

Line 1: > Sum:=func(f,g);
Line 2: >>   return func(x);
Line 3: >>   return f(x)+g(x);
Line 4: >>   end;
Line 5: >> end;
Line 6: >
Line 7: >   f:=func(x);
Line 8: >>   return x**3+2;
Line 9: >> end;
Line 10: >
Line 11: >   g:=func(x);
Line 12: >>   return 3*x;
Line 13: >> end;
Line 14: >
Line 15: >   Sum(f,g);
Line 16: !func(7)!;
Line 17: >
Line 18: >   Sum(f,g)(2);
Line 19: 16;
```

5.2.5 ISETL as a Pedagogical Tool

At a functional analysis conference in 1969, Dubinsky (1995) first heard about the programming language SETL, the forerunner to ISETL, from its developer Jack Schwartz (see Chap. 2 for more details). Schwartz wanted to express complex mathematical relationships using computer programs. He believed the best way to achieve this goal was to base a programming language on fundamental mathematical concepts. This idea served as the inspiration for the development of APOS Theory and its related research framework. Specifically, Dubinsky reformulated Piaget's ideas about reflective abstraction into a cognitive theory and connected instruction based on that theory with a pedagogical approach in which students write short computer programs using a mathematical programming language. As noted in Asiala et al. (1996),

...students gain experience constructing actions corresponding to selected mathematical concepts. This experience is built upon in subsequent activities where students are asked to reconstruct familiar actions as general processes. Later activities presented exemplify those that are intended to help students encapsulate processes to objects; these activities typically involve writing programs in which the processes to be encapsulated are inputs and/or outputs to the program. (p. 16)

The idea of this pedagogical approach is that computer activities support the activation of mental mechanisms (i.e., interiorization and encapsulation) that lead to the development of mental structures (i.e., Processes and Objects) that underlie the cognitive formation of mathematical concepts. This is illustrated in Fig. 5.2.

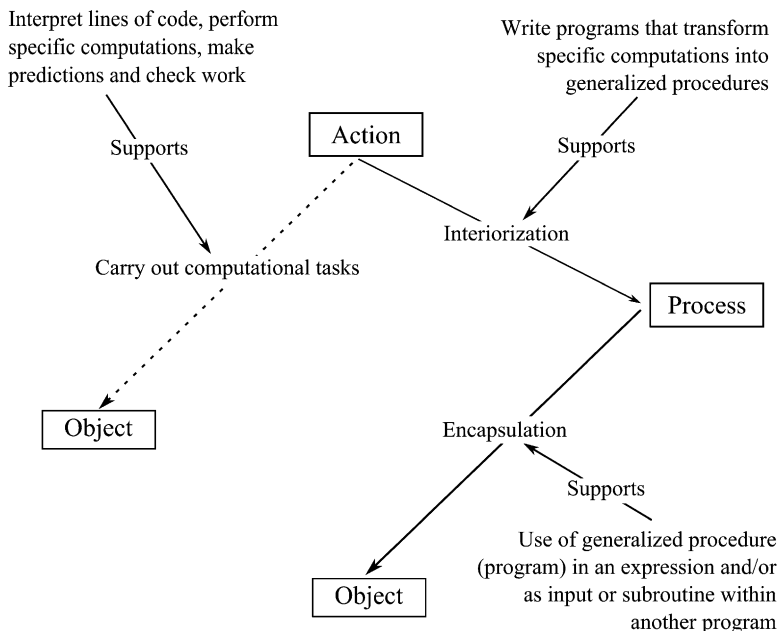


Fig. 5.2 Computer-based pedagogical approach, mental mechanisms, and mental structures

Typically, the construction of a new concept that starts as an Action is applied to an existing physical³ or mental Object (the dotted arrow denotes the fact that the Action is not part of the Object itself). From an instructional perspective, this involves computational tasks with explicit instructions and specific examples. Students construct Actions as they repeat on their computer screens what is written in the text, predict the result of running code, or modify code they have been given. This is represented by the “left arm” of Fig. 5.2.

Reflection on an Action leads to interiorization of the Action into a mental Process. In terms of instruction, interiorization is supported by replacing code that performs a specific calculation by a short program that carries out the calculation for unspecified values; that is, the computation is transformed by the learner from a specific calculation to a general procedure. This is represented by the top half of the “right arm” of Fig. 5.2.

As an Action is applied to a Process, the Process may be encapsulated into a cognitive Object (represented by the bottom half of the “right arm” of Fig. 5.2).

³The application of Actions to physical (real world) Objects is considered in detail in Chap. 9.

From an instructional perspective, encapsulation may occur when a Process is treated as the input or output of a program, used as a subroutine in a more elaborate program, or operated on within a program.

Examples of how the computer works as a pedagogical tool within the larger framework of the ACE Teaching Cycle are considered in detail in Sects. 5.3 and 5.4.

5.3 Teaching and Learning Groups

Abstract algebra is often a mathematics majors' first encounter with the study of abstract mathematics. Although many abstract algebra objects include familiar examples (for instance, the integers as an example of a commutative ring or the rational numbers as an example of a field), students have not considered these types of structures in their previous course work. As a result, students often experience significant frustration when they take an abstract algebra course.

As one response to this problem, Dubinsky and Leron (1994) developed a course based on APOS Theory that is delivered using the ACE Teaching Cycle and involves use of ISETL. In this section, examples of instruction on the group concept will be described.

The genetic decomposition of the concept of group can be understood as a Schema that consists of three Schemas: set, binary operation, and axiom. The set and binary operation Schemas are thematized to form Objects and coordinated through the axiom Schema (Brown et al. 1997). The interaction of these three Schemas is illustrated in Fig. 5.3.

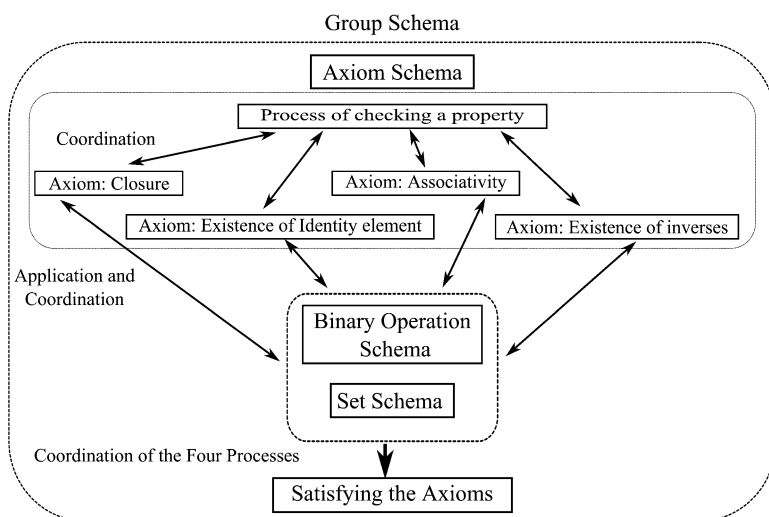


Fig. 5.3 Diagrammatic representation of the genetic decomposition of the group schema

The axiom Schema includes the general notion that a binary operation on a set may or may not satisfy a property, which is essentially the Process of checking the property. It also includes four specific Objects obtained by encapsulating the four Processes corresponding to the four group axioms. Checking an axiom consists of coordinating the general notion of checking a property with the specific Process for the axiom. Each axiom is de-encapsulated so that the individual coordinations can take place. Each coordinated Process (general property of checking an axiom and the axiom Process) is applied to the set and binary operation Schema. The set and binary operation Objects are de-encapsulated so that they can be coordinated with each axiom. The four instances of this operation are then coordinated into a single Process of satisfying the axioms (this description is taken from Brown et al. 1997, p. 192). The way in which this preliminary theoretical description is used to motivate instruction on groups is considered in Sects. 5.3.1–5.3.3.

5.3.1 Activities

5.3.1.1 Sets

Early activities in the abstract algebra course provide students with opportunities to develop the concept of set as Process. This occurs as students write set former code in ISETL or reflect on code provided in the text. For instance, in order to express a set in set former notation for the subset H of even elements of $\mathbb{Z}_{20}$ (integers modulo 20), which is given below, a student would need to carry out in her or his mind the Action of checking the condition for set membership (determination of whether an element is even) for every element of the set. This type of activity would likely lead the student to reflect on the Action of set formation, which would lead to interiorization of that Action into a mental Process.

```
> H:={g : g in Z20 | even(g)};
```

A similar type of cognitive activity would be necessary for a student to interpret a quantification statement such as

```
> forall x in Z20 | (x+0) mod 20 = x;
```

In order to predict and then to verify the Boolean value of the quantified statement, without having to check each element of $\mathbb{Z}_{20}$ explicitly, the student would need to interiorize the Action associated with checking the condition given by the universal quantifier for each member of the set.

Dubinsky and Leron (1994) designed activities to support construction of the concept of set as a mental Object. Each involved the application of Actions on sets. Two examples follow.

Create New Sets from Old Ones: The students were assigned activities in which they formed the union and intersection of sets and determined whether one set is a

subset of the other. Sample ISETL code that accompanies these types of activities is as follows:

```

Line 1: > Z20:={a mod 20 : a in [-30..50]};
Line 2: >
Line 3: > H:={g : g in Z20 : even(g)};
Line 4: >
Line 5: > K:={(5*g) mod 20 : g in Z20};
Line 6: >
Line 7: > H union K;
Line 8: {14, 12, 10, 8, 18, 16, 15, 0, 2, 4, 6, 5};
Line 9: > H inter K;
Line 10: {0, 10};
Line 11: > H subset K;
Line 12: false;

```

For the three sets given in Lines 1, 3, and 5 above, students predict the elements of each set by interpreting ISETL code. For the union, intersection, and subset operations, students do much the same: they predict results and then check their predictions, as shown in Lines 7 through 12. Despite the similarity of the outward activity, there is a difference in the inward cognitive activity. Writing or interpreting code requires one to think in terms of a Process, that is, to carry out in one's mind the Action of running through the elements of the set to check the set membership condition. On the other hand, predicting the elements of unions and intersections, and determining whether one set is a subset of the other, requires one to think of the sets as mental Objects since forming a union, taking an intersection, or determining a subset relationship are Actions performed on sets.

Write funcs that Accept Sets as Inputs: Students were asked to define binary operations on sets. In ISETL, this means writing the code that constructs a func that accepts two elements of the set and returns the result of the operation. This type of activity constitutes an Action applied to the set. In order to carry out this Action successfully, the set would need to be encapsulated into a mental Object. A sample activity with accompanying code illustrates this:

Example: Write code that constructs a func that accepts two elements of a set and returns the result of a binary operation defined on that set applied to the two elements. Then, write code to determine whether the func defined on G is commutative.

```

Line 1: > G:={1..12};
Line 2: > G;
Line 3: {1, 2, 4, 3, 12, 11, 10, 9, 6, 5, 8, 7};
Line 4: > o:=func(x,y);
Line 5: >> if (x in G and y in G) then
Line 6: >> return (x*y) mod 13;
Line 7: >> end;
Line 8: >> end;
Line 9: >

```

```

Line10: >   is_commutative:=func(S,op);
Line11: >>   return forall x,y in S : x.op y = y.op x;
Line12: >>   end;
Line13: >
Line14: >   is_commutative(G,o);
Line15: true;

```

In Line 1, the set G is defined. Line 2 is an ISETL call for G , which is returned in list form in Line 3. Lines 4 through 8 give the code for the func o , which is a binary operation defined on G . In order for a student to write the func o , he/she must write code to select two arbitrary elements from the set G and then apply the operation to those elements. The selection of arbitrary elements constitutes an Action applied to the set. As a result, the set must first be encapsulated into a mental Object so that the Action can be applied. Lines 10 through 12 are the ISETL code for the func $is_commutative$. This func accepts a set S and binary operation op defined on S . The func tests whether the binary operation op defined on S is commutative. The func is called for the set G and operation o . Since o is commutative, the func returns `true` (Line 15). Since the func $is_commutative$ is an Action applied to its inputs, determining the code for $is_commutative$ and then applying it may encourage encapsulation of the set and binary operation concepts.

5.3.1.2 Binary Operation

For a specific finite set, a specific binary operation is an Action applied to the set. Writing a func in ISETL may lead to interiorization of that Action. An example of such an activity follows:

Example: For the set S_3 of all permutations on three elements, write the code for a func $comp$ that accepts two elements from S_3 and returns the composition of those elements.

```

Line 1: >   S3:={ [a,b,c] : a,b,c in {1..3} | #{a,b,c}=3 };
Line 2: >   comp:=func(p,q);
Line 3: >>   if (p in S3 and q in S3) then
Line 4: >>   return [p(q(i)) : i in [1..3]];
Line 5: >>   end;
Line 6: >> end;

```

The set S_3 is defined in Line 1. Lines 2 through 6 give the ISETL code for the func $comp$, which accepts two elements of S_3 and returns the composition of those permutations. The act of writing this and other binary operation funcs supports the interiorization of the binary operation Action as the learner begins to think in terms of general steps: the selection of arbitrary elements together with a means of assigning another element of the set to the given pair. This type of activity involves the essence of Process—to move from consideration of specific pairs to thinking about arbitrary pairs and to think about how a binary operation operates generally.

Interiorization may be supported further by interpretation of quantified statements. For instance, when asked to reflect on a quantified statement such as

```
> forall g1, g2 in G | g1 .o g2 in G;
```

a student needs to think in terms of checking an arbitrary pair. This necessitates interiorization of the Action of testing the binary operation for a specific pair of elements in the set G .

Testing whether a binary operation satisfies certain properties can also lead to encapsulation. To determine whether a binary operation satisfies a property, a student writes a Boolean-valued `func` that accepts the set and binary operation as inputs, tests the property for every pair of elements in the set, and returns true or false. Since this constitutes an Action on both the set and the binary operation, both need to be encapsulated as Objects. Since the actual test involves a coordination of the set and binary operation Processes, the two Objects are de-encapsulated so that the property can be checked for every pair of elements. The example and sample code that follows illustrates how this might work for the set S_3 and the binary operation `comp`.

Example: Write a `func` that accepts as input a set and a binary operation defined on that set, and that tests whether the binary operation is commutative.

Sample Code:

```
Line 1: > S3:={ [a,b,c] : a,b,c in {1..3} | #{a,b,c}=3 };
Line 2: > comp:=func (p,q) ;
Line 3: >> if (p in S3 and q in S3) then
Line 4: >> return [p(q(i)) : i in [1..3]] ;
Line 5: >> end;
Line 6: >> end;
Line 9: >
Line 10: > is_commutative:=func (S,op) ;
Line 11: >> return forall x,y in S : x .op y = y .op x;
Line 12: >> end;
Line 13: >
Line 14: > is_commutative(G,o) ;
Line 15: false;
```

The `func is_commutative` (lines 10–12) is a Boolean-valued function that accepts a set S and a binary operation `op` and checks whether the operation `op` defined on S is commutative. In line 14, the `func is_commutative` is applied to the set S_3 and to the operation `comp`. The result of that test appears in Line 15.

Several kinds of binary operations appear in ISETL as predefined operations. These include `mod`, `div`, `min`, `max`, `and`, `or`. Students work with these without explicit mention that they are binary operations. Activities involving these commands are designed to help students to construct a binary operation Schema.

In alignment with standard mathematical notation, and as mentioned earlier, ISETL supports the use of infix notation. If `op` is any `func` of two variables in ISETL, the expression `a .op b` may be used instead of `op(a,b)`. For the

following example, `add_Z20(3,5)` (Lines 8–9) returns the same result as `3.add_Z20 5` (Lines 10–11):

```

Line 1: >   Z20:={0..19};
Line 2: >   add_Z20:=func(x,y);
Line 3: >>   if (x in Z20 and y in Z20) then
Line 4: >>       return (x+y) mod 20;
Line 5: >>   end;
Line 6: >> end;
Line 7: >
Line 8: >   add_Z20(3,5);
Line 9: 8;
Line 10: >   3.add_Z20 5;
Line 11: 8;

```

The use of infix notation helps students to see that any function of two variables defined on a set is a binary operation. This is another feature of ISETL that promotes development of a binary operation Schema.

5.3.1.3 Group Schema

A group Schema is constructed mentally through coordination of the axiom Schema with the set and binary operations Schemas. The axiom Schema includes two principal components:

1. Checking a property of a binary operation defined on a set
2. The four axioms of the group concept constructed as Objects

The general Process of satisfying a property is coordinated with the specific Process for the axiom, which is de-encapsulated from the axiom Object. This coordinated Process is then applied to a particular set and binary operation. This involves de-encapsulation of the set and binary operation Objects followed by coordination of the set, binary operation, and axiom Processes to establish the validity of an axiom for a given set, binary operation pair. These types of mental constructions are facilitated by activities in which students write funcs such as `is_closed`, `is_associative`, `has_identity`, and `has_inverses`. Each func accepts a set and a binary operation as inputs and returns a Boolean result. Students test these funcs on specific set and binary operation pairs. For example, to test associativity (Lines 1–3) of the function composition operation `comp` defined on `S3` (Line 5), students would enter the code `is_associative(S3, comp)` (Line 11), where `S3` denotes the set and `comp` (Line 7–9) represents the composition operation:

```

Line 1: >   is_associative:=func(G,op);
Line 2: >> return forall x,y,z in G | (x.op y) .op z = x.op
      (y.op z);
Line 3: >> end;
Line 4: >

```

```

Line 5: > S3:={ [a,b,c] : a,b,c in {1,2,3} : #{1,2,3} = 3 };
Line 6: >
Line 7: > comp:=func(p,q);
Line 8: >> return [p(q(i)) : i in [1..3]];
Line 9: >> end;
Line 10: >
Line 11: > is_associative(S3,comp);
Line 12: true;

```

The four instances of the coordination of set, binary operation, and axiom that underlie the closure, associativity, existence of identity, and existence of inverses axioms are then coordinated into a total Process of satisfying the axioms for a group. This mental construction is advanced by having students write a func called `is_group`. This func accepts a set and a binary operation and returns a Boolean value. One possibility for `is_group` is the following:

```

> is_group:=func(G,op);
>> return is_closed(G,op) and
        is_associative(G,op) and
        has_identity(G,op) and
        has_inverses(G,op);
>> end;

```

As students apply `is_group` to different set and binary operation pairs, they build a collection of examples and non-examples of groups. Consideration of different examples supports coherence of the group Schema. This includes the ability to recognize those relationships that are included in the Schema and to decide, when facing a problem situation, if the characteristics of the problem are within the scope of the Schema.

Reflection on the components and relations that make up a Schema enables the individual to perform conscious Actions on it. The ability to construct such Actions is an indicator of thematization of the group Schema. Activities and exercises that promote thematization include determination of whether a particular set and binary operation forms a group, checking various properties of a group, or considering whether two groups are isomorphic. As one example of checking properties, students construct a func `is_commutative` that accepts a group and its accompanying binary operation, checks the commutative condition for each pair of elements, and returns `true` or `false`. In a later chapter in the book, students are asked to construct homomorphisms. Central to this construction, which supports thematization of the group Schema, is construction of a Boolean-valued func called `is_hom` that accepts a representation `f` of a function between two groups (G, op) and (G', op') and determines whether the homomorphism condition is satisfied. The `is_hom` func might take the following form:

```

> is_hom:=func(f);
>> return forall x,y in G | f(x.op y) = f(x).op'
        f(y);
>> end;

```

5.3.2 Class Discussion

The class discussion phase of the ACE Cycle consists principally of two parts: full-class consideration of examples and non-examples of groups that were constructed in the Activities phase and construction of different symmetry groups that can be represented with geometric models.

In the activities, students construct the following set and binary operation pairs and then test whether those pairs satisfy certain properties and form groups:

- (Z_{12}, a_{12}) , where Z_{12} is the set of integers $\{0,1,2,3,4,5,6,7,8,9,10,11\}$ and a_{12} is addition modulo 12
- (Z_{12}, m_{12}) , where Z_{12} is the set of integers $\{0,1,2,3,4,5,6,7,8,9,10,11\}$ and m_{12} is multiplication modulo 12
- $(\text{two}Z_{12}, m_{12})$, where $\text{two}Z_{12}$ is the set of integers $\{0,2,4,6,8,10\}$ and m_{12} is multiplication modulo 12
- $(Z_{12}-\{0\}, m_{12})$, where $Z_{12}-\{0\}$ is the set of integers $\{1,2,3,4,5,6,7,8,9,10,11\}$ and m_{12} is multiplication modulo 12
- (S_3, op) , where S_3 is the set of permutations of the set $\{1,2,3\}$ and op is composition of permutations

Individual groups of students share their results, with the goal of reaching a class consensus. Included in these discussions are consideration of the properties of binary operations and groups.

The class discussion calls for students to construct the group D_4 , the set of all symmetries of the square. Students take a square, number its corners, determine the possible rotations and reflections, form an operation table, verify that the set and binary operation pair forms a group, and consider the relation between this group and the set of all permutations of the set $\{1,2,3,4\}$. This exercise, which involves work with a geometric representation, helps the students to expand their understanding of the group Schema.

5.3.3 Exercises

The exercises phase of the ACE Cycle reinforces the Activity and Classroom Discussion phases. Specifically, students continue to build and to expand their group Schemas by working with sets of matrices, permutation groups, and the rational numbers and by proving certain properties of groups, particularly conditions that guarantee that a group is Abelian, as well as showing that every group of order four is Abelian. The exercises also point to future work with orbits and subgroups. In one example related to the former, students determine the value of n for which $p^n, p \in S_6$ (the set of all permutations on six elements), yields the identity. For the latter, students determine all two, three, and four element subgroups of S_3 , the set of all permutations of $\{1,2,3\}$.

5.3.4 Results of the Instruction

Brown et al. (1997) reported on the results of students' learning of groups in courses based on the APOS instructional approach detailed in Dubinsky and Leron (1994).

With respect to the genetic decomposition, the data seemed to support the preliminary genetic decomposition. The data also revealed specific issues that arise in the development of the construction of the coordination between the axiom Schema and the set and binary operation Schemas. The issues have the potential to highlight specific aspects of learning about groups that might be helpful to instructors:

1. At the early stages of coordinating these Schemas, students have a tendency to assume that a feature that appears in one part of an environment applies throughout the entire environment (e.g., a student might express the opinion that a given subset of a group is closed because the group itself is).
2. In working with sets and conditions for set membership, students find it easier to see that an element satisfies the condition of being a member of the set than it is for them to grasp the idea that being an element of the set implies that the condition is satisfied.
3. In their construction of a group Schema, students must construct an understanding of the notion of a generic group and be able to perform calculations therein.

Brown et al. (1997) also conducted a comparative analysis of students who completed the APOS-based course with students who completed a traditional abstract algebra course. The students in the APOS group performed better on mathematical tasks related to binary operations, groups, and subgroups than did their traditionally instructed counterparts. In addition, the data also showed that at least one-third of the students who received the APOS instruction succeeded in understanding the concepts, while most of the rest made significant progress, and all showed an ability to deal with the material.

5.4 Application of the ACE Teaching Cycle in a Unit on Repeating Decimals

Rational numbers are studied extensively at the elementary and middle school levels. As a result, it would stand to reason that elementary and middle school teachers need to have a strong foundational understanding of rational number concepts, particularly their representations, which include repeating decimal expansions. However, Yopp et al. (2011) found that many preservice elementary and middle school teachers have considerable difficulty with repeating decimals. The participants in their study asserted the existence of infinitesimals ("there's a wee bit missing"), that real numbers correspond directly and solely to physical experiences, and that approximations are sufficient. Tall and Schwarzenberger (1978) also found that college students think in

terms of infinitesimals and are confused over the fact that two different decimals can correspond to the same rational number. Given teachers' difficulties, it is not surprising that K–12 students hold misconceptions. Gardiner (1985) reported that K–12 students often view infinite decimals as being like finite decimals and claim that the smallest positive number is $0.0 \dots 01$.

Repeating decimals (discussed earlier in Sect. 4.2) are one of several mathematical conceptions that stand at the nexus of the paradoxical duality between potential and actual infinity. On one hand, a repeating decimal can be thought of as an instance of potential infinity—a process of continually forming digits to express a rational number through long division. On the other hand, a repeating decimal is an instance of actual infinity—the representation of a number with fixed value.

Dubinsky et al. (2005a, b) studied the apparent tension between these seemingly contradictory notions in an APOS-based analysis of the historical development of the concept of mathematical infinity. In their analysis, they explained how potential and actual infinity represent two different conceptualizations linked by the mental mechanism of encapsulation. Potential infinity, the notion of infinity presented over time, is the conception of infinity as a Process. Because an infinite process has no final step, and hence no obvious indication of completion, the ability to think of an infinite process as mentally complete is a crucial step in moving beyond a purely potential view. As an individual reflects on a completed infinite process, he or she can conceive of it as a totality, a single operation freed from temporal constraints.⁴ At this point, the individual can apply the mechanism of encapsulation to transform the Process into a mental Object, an instance of actual infinity.

The authors used these ideas to explain individuals' difficulties with the repeating decimal $0.\bar{9}$ and 1. An individual may think of $0.\bar{9}$ as a Process, a dynamic view of continually adding 9s, or something one does, whereas he or she may conceive of the number 1 as a mental Object, a static entity that can be transformed. Given that it does not make sense, at least mentally speaking, to compare a dynamic Process with a static Object, one who sees $0.\bar{9}$ as a Process and 1 as a mental Object may see $0.\bar{9}$ and 1 as being unequal. Alternatively, an individual may view an infinite repeating decimal as an incomplete Process. In this case, a repeating decimal is a finite string of digits with indeterminate length. With such a conception, one might think of $0.\bar{9}$ as infinitesimally close to but less than 1.

This analysis was used to develop a preliminary genetic decomposition for infinite repeating decimals that informed the design of an instructional unit on repeating decimals for preservice elementary and middle school teachers. A student begins by constructing certain Actions on whole numbers. This involves reciting, either verbally or in writing, an initial sequence of digits, which may be seen as the beginning of a repeating decimal expansion. These Actions are interiorized into a

⁴ In Dubinsky et al. (2005a, b), the ability to see a Process as a Totality was considered to be a part of encapsulation. The instruction on which the study was based (Weller et al. 2009, 2011; Dubinsky et al. 2013) showed evidence of Totality as a separate stage between Process and Object. This distinction is considered later in this chapter and explored in depth in Chap. 8.

Process of forming sequences of digits of indeterminate length that is extended to form an infinite string. Specifically, the student grasps the idea that from some point on the decimal repeats forever to form an infinite string. As the student reflects on the Process and begins to see an infinite string as an entity to which mental Actions or Processes can be applied, the Process of forming an infinite string may be encapsulated into a mental Object. The Actions that may be applied to an infinite string include various arithmetic and comparison operations, determination of whether an infinite string satisfies certain relations or arithmetic equations, and the ability to see a repeating decimal as a number that equals a fraction or integer. The preliminary genetic decomposition for repeating decimals is illustrated in Fig. 5.4.

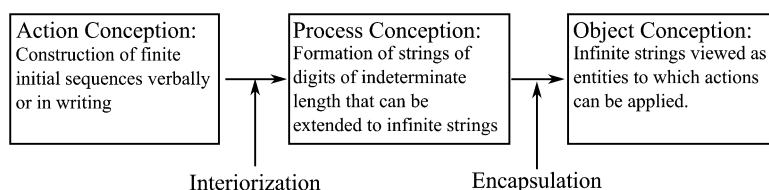


Fig. 5.4 Diagram of the genetic decomposition of infinite repeating decimals

The instructional treatment of repeating decimals consisted of three iterations of the ACE cycle. Each iteration of the cycle spanned two class days, one for computer activities and one for classroom discussions. Homework exercises were assigned at the end of each session and collected at the beginning of the next session. Two additional classroom meetings were reserved for time overruns. Analysis of this instructional unit appears in three reports, Weller et al. (2009, 2011) and Dubinsky et al. (2013).

For the activities phases of the cycle, students worked in cooperative groups in a computer lab where they used ISETL. In this particular instance, the use of ISETL differed from previous instruction based on APOS Theory. Typically, students use ISETL to write short computer programs. This type of activity supports the mental mechanisms that lead to construction of the mental structures called for by a genetic decomposition. For instance, writing and then reflecting on programs that carry out Actions supports interiorization. Writing programs that perform Actions on Processes supports encapsulation. For the instructional sequence on repeating decimals, the students performed calculations in ISETL using a preloaded decimal expansion package developed by the researchers (Weller et al. 2009, 2011; Dubinsky et al. 2013). Although the students were not asked to write computer programs, the calculations supported the mental mechanisms of interiorization and encapsulation. Students used preloaded `funcs` to look at a single place or finite range of places of a repeating decimal. This type of activity supported interiorization by helping students to reflect on the Action of writing out the terms of a decimal expansion. The students used predefined `funcs` to perform arithmetic operations and comparisons on repeating decimals and fraction-to-decimal and decimal-to-fraction conversions. These types of activities supported encapsulation

by having students perform Actions on repeating decimals. In addition to predefined funcs, the decimal expansion package stored several examples of repeating decimals for use in different activities. For many of the activities, students performed calculations by hand and then checked their results with the computer. Examples of these types of activities are given in Sects. 5.4.1–5.4.3.

The purpose of the first iteration of the ACE Cycle was twofold: to help students (1) to interiorize the Action of listing digits to a mental Process (in order to conceive of an infinite string⁵ of digits comprising a repeating decimal) and (2) to begin to see a repeating decimal as a mental Object by agreeing on a notational scheme for its representation. The second iteration of the cycle focused on encapsulation—to help students to transform infinite digit strings conceived as Processes into mental Objects to which Actions could be applied. The third iteration emphasized development of the relation between an infinite digit string and its corresponding fraction or integer. Development of this relation is an important part of an individual's rational number Schema. The construction and subsequent encapsulation of different rational number representations enables an individual to expand her or his Schema, offers the potential to develop the coherence of the Schema, and increases the likelihood for an individual to see a rational number as an entity that has value that can be compared with other numbers, that has a position on the number line, etc.

The instructional treatment did not rely on limits. Although preservice elementary and middle school teachers who specialize in the teaching of mathematics typically complete a calculus course, the instructional treatment was designed for a content course on number and operation required by all preservice elementary and middle school teachers, regardless of their area of specialization. Given that numerous studies have documented college students' difficulties with limits (e.g., Cornu 1991; Cottrill et al. 1996; Sierpinska 1987; Williams 1991, 2001), the APOS-based instructional treatment avoided calculus treatment of limits as well as instruction on conversion techniques that assume knowledge of limits and infinite series. Each of the three iterations of the ACE Cycle is described in detail in the sections that follow.

5.4.1 *First Iteration of the Cycle*

The first iteration of the ACE Cycle was designed to encourage the development of a Process conception of a repeating decimal and to begin to help students to see a repeating decimal as a mental Object by considering notational schemes to represent them.

⁵ In these discussions, decimal expansions are referred to as strings. This means finite or infinite sequences of digits that correspond to the decimal expansion of a rational number.

5.4.1.1 Activities

In the computer lab, students completed activities using eight preloaded decimal expansions⁶ whose identity⁷ was purposely left a mystery. In the activities with these strings, students made extensive use of a predefined func called `View`. The `View` command works in two ways: `View(ds,n)` yields the n th place of the decimal string `ds` and `View(ds,n,k)` returns the n th through k th entries of `ds`. The underlying idea behind the func `View` was its potential to support interiorization of the Action of listing the digits of a decimal string. This goal may be more apparent by considering a specific example. To determine the identity of the mystery string `m5`, one might apply `View` to several individual values of n :

```
> View(m5,1);  
"1";  
> View(m5,2);  
"0";  
> View(m5,3);  
"3";  
> View(m5,4);  
"5";  
> View(m5,5);  
"8";  
> View(m5,6);  
"5";  
> View(m5,7);  
"8";  
> View(m5,-1);  
"1";  
> View(m5,-2);  
"2";  
> View(m5,-3);  
"";
```

For positive values of n , `View` returns the n th place after the decimal point. For negative values of n , `View` returns positions to the left of the decimal point: $n = -1$ corresponds to the one's place, $n = -2$ to the ten's place, etc. The fact that `View` returns 5 for $n = 4$, 8 for $n = 5$, 5 for $n = 6$, and 8 for $n = 7$ suggests the possibility of a repeating decimal involving the digits 5 and 8 that starts in the ten-thousandth's

⁶ ISETL recognized decimal expansions using the notation $a.b(c)$. Here a , b , and c are nonnegative integers, where a denotes the integer part of the decimal expansion, b the decimal portion that appears before the repeating cycle, and c the repeating cycle. For repeating digits such as $0.\bar{3}$ and $0.\overline{35}$, where the cycle begins in the tenths place, the computer recognized the notation $0.3(3)$ and $0.3(53)$, respectively.

⁷ The preloaded mystery strings were denoted `m1`, `m2`, `m3`, `m4`, `m5`, `m6`, `m7`, and `m8`.

As a means of encouraging encapsulation, the final activity asked the students to figure out a notational scheme for repeating decimals, with particular focus on notation that would be used to represent decimal expansions in ISETL.

5.4.1.2 Class Discussion

Since the purpose of the first iteration of the cycle was to facilitate the development of a Process conception of decimal strings, the instructor's goal was to support reflective activity that would help students to interiorize the external Action of listing the digits of a decimal string. Beyond having the students share their descriptions of the mystery strings, this called for the class discussion to focus squarely on consideration of the functioning of `View`, with particular attention paid to what `View` returns for different values of n . The class discussion also included introduction of the notion of the meaning of decimal string. This discussion culminated with consideration of the question of "What is a repeating decimal?"

To begin to move students toward encapsulation, the instructor engaged the class in a discussion regarding notation. The instructor invited student groups to share their notational schemes leading to a class notational scheme. This discussion served as the basis for consideration of what ISETL accepts.

5.4.1.3 Exercises

Several of the exercises involved application of the command `View` to determine unknown strings. Students were presented with several lines of output and asked to provide a description expressed in the notation agreed upon during the class discussion.

Additional exercises asked for descriptions of unknown strings subject to certain conditions. Although only partial information was provided, it was sufficient to specify a string uniquely. The example below captures the essence of this type of exercise:

Exercise: Describe the decimal string `str3` if the following conditions must be satisfied:

- a. The digits 223 go before the decimal point.
- b. The first string position after the decimal point is equal to 5.
- c. The second string position after the decimal point is equal to 0.
- d. The third string position after the decimal point is equal to 4.
- e. The fourth string position after the decimal point is equal to 6.
- f. For every integer $n \geq 0$, the digit in the $2n + 5$ position after the decimal point is equal to 8.
- g. For every integer $n \geq 0$, the digit in the $2n + 6$ position after the decimal point is equal to 3.

In addition to your description, express `str3` using the notation devised in class.

The final exercise asked students to offer their own explanation of how `View` works. Although discussed in class, this exercise provided an opportunity for students to reflect on the construction of a decimal string as they described the ISETL command.

5.4.2 *Second Iteration of the Cycle*

The second iteration of the cycle was designed to help students make encapsulations—to transform infinite digit strings conceived as Processes into mental Objects to which Actions could be applied. Thus, the work in this session focused on having students perform a variety of operations on strings.

5.4.2.1 Activities

In the computer lab, the students were presented with 16 infinite strings (whose identity was revealed, unlike the first iteration) that were represented as repeating decimals and stored in the computer in the form $a.b(c)$. In the first set of activities, students performed standard arithmetic operations by hand on different pairs of decimals from the predefined list. They then used ISETL to compare the results of their written work. Students used commands from the decimal expansion package developed by the researchers to carry out their operations. Examples of the 16 predefined strings, along with operations performed on them, appear below:

```
Line 1: > ds1;
Line 2: "0.23(4)";
Line 3: > ds2;
Line 4: "2.125(0)";
Line 5: > ds3;
Line 6: "11.1(428571)";
Line 7: > ds4;
Line 8: "0.1(41)";
Line 9: > ds5;
Line 10: "0.7(867)";
Line 11: > ds6;
Line 12: "10.0(0)";
Line 13: > ds7;
Line 14: "100.0(0)";
Line 15: >
Line 16: > ds1.AddString ds2;
Line 17: "2.359(4)";
Line 18: >
Line 19: > ds1.SubString ds4;
Line 20: "0.09(30)";
Line 21: >
Line 22: > ds5.MultString ds6;
Line 23: "7.8(678)";
Line 24: >
Line 25: > ds3.DivString ds7;
Line 26: "0.11(142857)";
```

The variables `ds1`, `ds2`, `ds3`, `ds4`, `ds5`, `ds6`, and `ds7` (Lines 1–14) denote some of the decimal strings. The remaining lines of code show operations on those strings: `AddString` (Lines 16–17) represents addition, `SubString` (Lines 19–20) denotes subtraction, `MultString` (Lines 22–23) stands for multiplication, and `DivString` (Lines 25–26) signifies division. Each of the string operations are `funcs` that are part of the decimal expansion package.

The next set of activities called for comparisons. Students ordered the 16 predefined strings by hand, checked their work using comparison commands included in the decimal expansion package, and explained how one determines whether one string is larger than another. Another related activity called for students to make comparisons after applying arithmetic operations. For example, given the three decimal strings `s1=0.1(41)`, `s2=0.5(0)`, and `s3=0.2(132)`, students determined by hand whether the sum of `s1` and `s3` exceeds the sum of `s1` and `s2`. Once they performed these computations, they tested their results using ISETL:

```
> s1:="0.1(41)";
> s2:="0.5(0)";
> s3:="0.2(132)";
>
> (s1.AddString s2).LessString (s1.AddString s3);
false;
```

The activity involved carrying out arithmetic operations (via `AddString`), order of operations (through use of parentheses), and comparison (by application of `LessString`). `LessString`, as the name suggests, is a Boolean operator that tests whether the first entry (in this case, `s1.AddString s2`) is smaller than the second (`s1.AddString s3`). It was one of three comparison `funcs` defined by the researchers and included in the decimal expansion package. The other two `funcs`, `EqualString`, which tests whether two decimal strings are equal, and `GreatString`, which tests whether the first entry is larger than the second, are also Boolean operators.

The remaining activities emphasized the connection between representations, that is, that each fraction or integer has a corresponding decimal expansion. The researcher-developed decimal expansion package included a `func` called `Dec2Frac` that accepts a string and returns the fraction that corresponds to the given string. The command `Frac2Dec` does the reverse—given a fraction, it returns the corresponding decimal expansion. For the given list of 16 strings, students were asked to think of a number (fraction or integer) that would correspond to each string. They then applied `Dec2Frac` to each string in the list to test their predictions. The students were then asked to perform long division on the fractions returned by ISETL to verify by hand the correspondence revealed by `Dec2Frac`. In the code given below, the `func` `Dec2Frac` shows that the repeating decimal $0.23\overline{4}$ corresponds to the fraction $\frac{211}{900}$ and that the decimal $0.78\overline{67}$ corresponds to the fraction $\frac{262}{333}$.

```

> Dec2Frac("0.23(4)");
[211, 900];
>
> Dec2Frac("0.7(867)");
[262, 333];

```

This correspondence/conversion activity set the stage for two questions: (1) What are the characteristics of a number, that is, what makes a number? (2) In what sense do decimal strings fit your description of a number? Both of these questions encouraged the students to reflect on the operations they had performed, to begin to see repeating decimals as Objects that correspond to other Objects (fractions and integers), and to which operations can be applied.

5.4.2.2 Class Discussion

Since the purpose of the second iteration of the cycle was to facilitate encapsulation, the instructor's role was to help students see how decimal strings behave like numbers, and to consider why it is reasonable to attach integers or fractions to decimal strings. The former involved reflection on the arithmetic operations the students performed in their lab activities, specifically, a discussion regarding the characteristics of number and the way in which the activities demonstrated how decimal strings behave like numbers. The latter required a discussion of the conversion funcs, `Dec2Frac` and `Frac2Dec`, with the goal of helping the students to see the correspondence between decimal strings and integers and fractions.

The activities and supporting class discussion also opened the door for consideration of the connection between the repeating decimal $0.\bar{9}$ and 1, which was dealt with at great length in the third and final iteration of the ACE Cycle.

5.4.2.3 Exercises

In the exercises, students performed arithmetic operations using both repeating decimal and fraction/integer representations as a means of seeing the connection between the different representations. Specifically, they solved by hand simple equations of the form $s \text{ .op } t = x$, where s and t are strings (in the form $a.b(c)$) and op is an arithmetic operation. Then they performed the same arithmetic operations using the integer/fraction representations for s and t . After making these computations, they compared the representations of the result x by long division (where they transformed x from integer/fraction to decimal form) and through use of the ISETL func `Dec2Frac` (to convert the string representation of x to its fraction/integer form) to see that the operations yield equivalent results no matter the representation. Students were also given a list, in random order, of six strings and a second list, in a different random order, of six fractions. They ordered

both lists by size and then determined the correspondence between the individual strings and the fractions, first by comparing the order and then by long division. These exercises were designed to encourage encapsulation: by having students connect repeating decimal expansions with familiar representations that were more likely to be seen as Objects and by reinforcing the idea that repeating decimals, when operated on, behave like other numbers.

5.4.3 *Third Iteration of the Cycle*

The third iteration of the cycle emphasized the relation between infinite decimal strings and their corresponding fraction or integer representations, with special attention paid to the relationship between $0.\bar{9}$ and 1. This focus supported encapsulation in three ways: (1) by connecting infinite strings with fraction or integer representations that are more likely to be seen as Objects; (2) by performing Actions on infinite decimals, in which strings are converted to their fraction or integer equivalents; and (3) by seeing that infinite decimal strings yield the same results in arithmetic operations as their corresponding fraction/integer counterparts.

5.4.3.1 Activities

In the computer lab, the students were asked to perform long division on fractions and to use commands from the ISETL decimal expansion package to check their answers. In one such activity, students were asked to perform long division on the fractions $\frac{2}{7}$, $\frac{4}{9}$, $\frac{3}{11}$, $\frac{2}{13}$ and then apply the command `Frac2Dec` to verify their findings. After making these computations, they explained why each string, obtained either by hand or by application of the ISETL command, is equal to the fraction on which they performed long division. The ISETL code for making these conversions follows:

```
>          Frac2Dec(2,7);  
"0.2(857142)";  
>  
>          Frac2Dec(4,9);  
"0.4(4)";  
>  
>          Frac2Dec(3,11);  
"0.2(72)";  
>  
>          Frac2Dec(2,13);  
"0.1(538461)";
```

These lines of code verify the results of the long division carried out by hand. The reverse process, moving from decimal string to fraction, is illustrated next:

```
> Dec2Frac("0.2(857142)");
[2, 7];
>
> Dec2Frac("0.4(4)");
[4, 9];
>
> Dec2Frac("0.2(72)");
[3, 11];
>
> Dec2Frac("0.1(538461)");
[2, 13];
```

In a second series of activities, students performed a number of arithmetic and ordering operations, first on strings and then on the corresponding fraction/integer representations. They completed this work by hand and then used commands from the ISETL decimal expansion package to check their answers. In each case, they reflected on whether the fraction/integer representations are equal to the corresponding strings. For example, for sums, students were given the decimals 0.3125 and $0.\bar{1}$ and their corresponding fraction representations $\frac{5}{16}$ and $\frac{1}{9}$. They applied Dec2Frac to each decimal and Frac2Dec to each fraction to see the connection between the different representations. Then, they found the sums, first for the two decimals and then for the two fractions. After making these calculations, they applied Frac2Dec and Dec2Frac to their results to see that the operations yield equivalent results regardless of the representation being worked with. The ISETL code is shown below:

```
> Dec2Frac("0.3125(0)");
[5, 16];
>
> Dec2Frac("0.1(1)");
[1, 9];
>
> "0.3125(0)".AddString "0.1(1)";
"0.4236(1)";
>
> 5/16 + 1/9;
61/144;
>
> Dec2Frac("0.4236(1)");
[61, 144];
>
> Frac2Dec(61, 144);
"0.4236(1)";
```

These lines of codes are reflected in a commutative progression illustrated in Fig. 5.5.

Figure 5.5 shows a generalization of the steps of the procedure given in the ISETL example. Specifically, for two fractions $\frac{a}{b}$ and $\frac{c}{d}$ and their corresponding decimal expansions, the ISETL commands `Frac2Dec` and `Dec2Frac` transform one representation into another, that is, `Frac2Dec` returns the decimal expansion that corresponds to a given fraction and `Dec2Frac` returns the fraction corresponding to a given decimal. The left side of the diagram shows the addition of the two fractional representations. The right side of the diagram shows the sum of the decimal

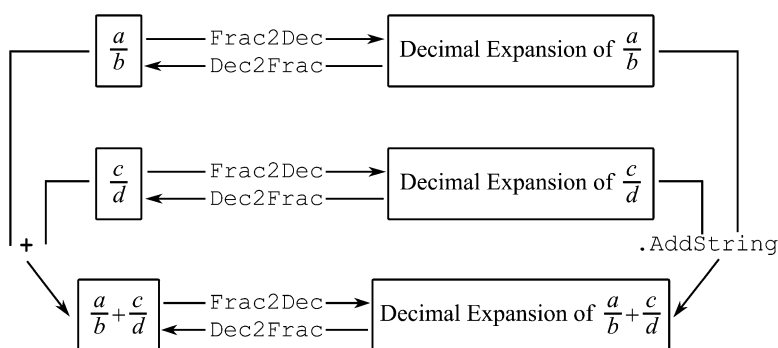


Fig. 5.5 Verification of preservation of operations for different representations

expansions using the ISETL command `.AddString`. The bottom of the diagram shows how the ISETL commands `Dec2Frac` and `Frac2Dec` can be used to show the correspondence between the sums of the two representations.

Carrying out decimal-to-fraction and fraction-to-decimal conversions, both by hand and with ISETL, and showing the equivalence of the results of arithmetic operations, no matter the representation, was seen as a means of supporting encapsulation of repeating infinite decimal representations.

The next series of activities dealt with repeating nines. In one of these activities, the students were given several lines of code in which a terminating decimal with 9s is subtracted from the decimal equivalent of a familiar fraction. The subtraction was carried out with the func `SubString`, which was part of the decimal expansion package. The purpose of the activity was to help the students to see that a fraction with a terminating decimal expansion could also be represented by a repeating decimal with an infinite sequence of 9s. The following activity exemplifies this.

Activity: Suppose the following appears on an ISETL screen:

```

> Frac2Dec(1,2) .SubString "0.4(0)";
"0.1(0)"
> Frac2Dec(1,2) .SubString "0.49(0)";
"0.01(0)"
> Frac2Dec(1,2) .SubString "0.499(0)";
"0.001(0)"

```

```
>      Frac2Dec(1,2) .SubString "0.4999(0) ";
"0.0001(0) "
```

1. Use ISETL to continue this for several steps.
2. Use your results together with the screen output that appears above, to answer the following questions:
 - a. Do you agree with the statement that each of 0.4, 0.49, 0.499, 0.4999 is close to $\frac{1}{2}$?
Do the decimal strings get closer to $\frac{1}{2}$ as you increase the number of decimal places?
 - b. If we imagine taking the entire decimal string, what is the relation between $\frac{1}{2}$ and 0.49999999...?

Building on the exercises from the second iteration of the cycle, students completed activities where they performed operations on equivalent representations. For instance, they solved for x equations of the form $ax + b = c$ where a, b , and c were first given in fraction/integer form and then in decimal form. This enabled comparisons in which students could see that equivalent representations yield equivalent solutions. The special case of $0.\bar{9}$ was included. This is illustrated in the following activity that involves use of the predefined funcs `MultString` and `DivString` that are used to carry out multiplication and division operations, respectively, on decimal strings.

Activity: Perform the following calculations using the ISETL code `.MultString`:

$0.0(769230) \times 0.9$

(type "0.0(769230)" .MultString "0.9(9)"; and press enter)

$\frac{19}{15} \times 0.9$

(type `Frac2Dec(19,15)` .MultString "0.9(9)"; and press enter)

253×0.9

(type `Frac2Dec(253,1)` .MultString "0.9(9)"; and press enter)

$0.2(342) \div 0.9$

(type "0.2(342)" .DivString "0.9(9)"; and press enter)

$713 \div 0.9$

(type `Frac2Dec(713,1)` .DivString "0.9(9)"; and press enter)

It would be very cumbersome to try to perform the operations above by hand. However, it would be very easy to do them by hand if you replaced "0.9(9)" by "1.0(0)". Explain why or why not you think this would be correct.

`MultString` and `DivString`, two funcs included in the decimal expansion package, accept two decimal strings and return the product and quotient, respectively. Sample lines of code appear below:

```
>      "0.0(769230)" .MultString "0.9(9) ";
"0.0(769230) ";
>
>      "0.2(342)" .DivString "0.9(9) ";
"0.2(342) ";
```

Activities such as these, where $0.\bar{9}$ is substituted for 1 in arithmetic operations, were developed to support students' reflection on the relationship between $0.\bar{9}$ and 1.

5.4.3.2 Class Discussion

To promote further the goal of students' understanding of the relation between an infinite decimal string and its corresponding fraction or integer, the class discussion focused on five "reasons to believe":

1. Answers in ISETL: For any pair of decimal and fraction representations that stand for the same rational number value, the funcs `Dec2Frac` and `Frac2Dec` illustrate the correspondence.
2. Performing long division on fractions: Performing long division is an action on a fraction that gives the corresponding decimal as a result.
3. Approximation involving initial segments of a decimal representation: Successively smaller arithmetic differences between a rational number representation and the initial segments of a repeating decimal suggest equality between the two.
4. Effect on operations: A fraction and its corresponding decimal string have the same effect on the results of various arithmetic operations.
5. Solutions to algebraic equations: Solutions for x of equations such as $ax = b$ are equal in value regardless of the representation.

In addition to more general considerations, the instructor led a discussion regarding the relationship between $0.\bar{9}$ and 1. Students were invited to express their belief regarding the equality and to offer reasons for that belief. This provided a means by which students on each side of the argument could try to convince their peers. The goal of this discussion was to generate a list of justification statements to mirror the more general discussion that preceded it.

5.4.3.3 Exercises

In the exercises, students compared operations on strings with those performed on their corresponding fraction/integer representations. They approximated fractions/integers using sequences of finite strings and considered what happens when one passes to the entire string. The exercises also included items where students converted a repeating decimal to a fraction or integer and vice versa. They first performed these computations by hand and then verified their results using the ISETL decimal expansion package.

Students were also asked to reflect on the special case of the relationship between $0.\bar{9}$ and 1. In one of the exercises, students had to determine whether there is a decimal string between $0.\bar{9}$ and 1 and, if not, to consider what this suggests regarding equality. In another exercise, students wrestled with the fact that one cannot obtain $0.\bar{9}$ from 1 via long division, although other Actions suggest equality. In encouraging reflection on this dilemma, students wrote a short essay in which they were asked to offer a rationale for the equality.

5.4.4 *Results of the Instruction*

Two studies (Weller et al. 2009, 2011) reported on results of a comparative analysis of the APOS-based instructional sequence with traditional instruction on repeating decimals. A third study (Dubinsky et al. 2013) analyzed students' thinking from the perspective of the genetic decomposition.

The subjects for the studies included 204 students (77 in APOS-based instruction; 127 in traditional instruction) who were enrolled at a major university in the southern United States. The instruction for the repeating decimals unit, which took place after the final drop date, was part of a required course on number and operation for preservice elementary and middle school teachers.

In the first study, Weller et al. (2009) compared the gains in procedural and conceptual understanding of the two groups. They discovered that students who received the APOS-based instruction made substantial gains when compared with students who had received the traditional instruction, particularly in their conceptual grasp of infinite repeating decimals.

The second comparative study (Weller et al. 2011), conducted several months after the instructional sequence, focused on the strength and stability of the students' beliefs. The analysis revealed that students who received the APOS-based instruction developed stronger and more stable (over time) beliefs that a repeating decimal is a number, a repeating decimal has a fraction or integer to which it corresponds, a repeating decimal equals its corresponding fraction or integer, and $0.\bar{9} = 1$.

The third study (Dubinsky et al. 2013) sought to answer two questions:

1. Does the genetic decomposition provide a relatively objective and reasonable explanation of student thinking about $0.\bar{9}$, or does the data suggest revision of the genetic decomposition?
2. How does progress in the genetic decomposition relate to belief in the equality $0.\bar{9} = 1$?

The analysis showed the need to revise the genetic decomposition, specifically, it called for the introduction of a new stage, Totality (discussed in Chap. 8), as an intermediate stage between Process and Object. The data also suggested the need for a finer-grained decomposition to understand the progression from Action to Process, from Process to Totality, and from Totality to Object. This led to the introduction of *levels* (also discussed in Chap. 8) to describe the transition from one stage to the next. The revised genetic decomposition described the development from Action toward Object for 83% of the students. Therefore, Dubinsky et al. (2013) supported a positive response to the first research question (Item 1 above) when considered in the context of the refined genetic decomposition.

With regard to the second research question (Item 2), the data revealed that the participants expressed belief or disbelief in ways that correlated with their emerging conceptions of $0.\bar{9}$. This showed, at least for the data for this study, that students

who gave evidence of Totality or Object tended to believe more readily in the equality.

Several authors (e.g., Yopp et al. 2011) argue that preservice teachers need to understand rational number concepts to avoid teaching false notions to their students. The APOS-based instructional unit on repeating decimals made progress in this regard, particularly when compared with traditional instruction on the topic.

5.5 Analysis of Instruction Using the Research Framework

Each implementation of an instructional sequence provides an opportunity to gather data. The analysis of the data has two purposes: to gauge students' mathematical performance, that is, how much mathematics the students learned as a result of the instruction, and to determine whether the students made the mental constructions called for by the preliminary genetic decomposition. This provides an opportunity for researchers to test empirically the preliminary genetic decomposition and to evaluate the effectiveness of the APOS-based instruction, particularly when compared with other instructional approaches. Issues involving the framework, that is, research involving the analysis of data, are the subject of Chap. 6.

Chapter 6

The APOS Paradigm for Research and Curriculum Development

In the Merriam-Webster online dictionary, the word *paradigm* is defined in the following way: “a philosophical and theoretical framework of a scientific school or discipline within which theories, laws, and generalizations and experiments performed in support of them are formulated; *broadly*: a philosophical or theoretical framework of any kind.” This definition reflects the contemporary meaning of the term coined by Kuhn (1962), who spoke of two characteristics of a “paradigm”: A theory powerful enough to “attract an enduring group of researchers” (p. 10) and to provide enough open ends to sustain the researchers with topics requiring further study. In light of these considerations, the overarching research stance linked to APOS Theory is referred to as a paradigm, since (1) it differs from most mathematics education research in its theoretical approach, methodology, and types of results offered; (2) it contains theoretical, methodological, and pedagogical components that are closely linked together; (3) it continues to attract researchers who find it useful to answer questions related to the learning of numerous mathematical concepts, and (4) it continues to supply open-ended questions to be resolved by the research community.

Some, but not all, studies that adopt APOS as a theoretical framework make use of all the elements of the paradigm. Depending on the particular project, the reasons may be methodological or practical, and it would be impractical to consider all variations of the way in which the APOS paradigm is used in mathematics education research. What we are describing as the methodological framework in this chapter can be considered as an “ideal” organization of an APOS-based research study.

6.1 Research and Curriculum Development Cycle

An APOS-based research and/or curriculum development project involves three components: theoretical analysis, design and implementation of instruction, and collection and analysis of data. Figure 6.1 shows how these components are related.

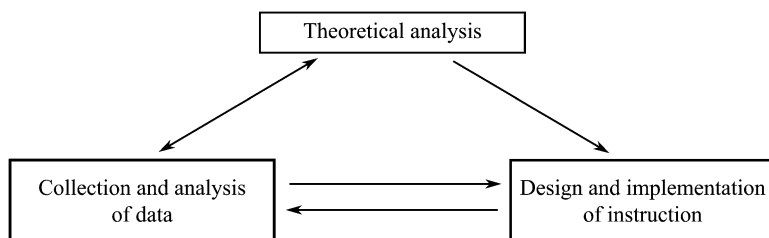


Fig. 6.1 Research cycle (adapted from Asiala et al. 1996)

According to this paradigm, research starts with a theoretical analysis of the cognition of the mathematical concept under consideration. This gives rise to a *preliminary genetic decomposition* of the concept. As discussed in Chap. 4, a genetic decomposition is a description of the mental constructions and mental mechanisms that an individual might make in constructing her or his understanding of a mathematical concept.

As indicated by the arrows in Fig. 6.1, the three components of the research cycle influence each other. The theoretical analysis drives the design and implementation of instruction through activities intended to foster the mental constructions called for by the analysis. Activities and exercises are designed to help students construct Actions, interiorize them into Processes, encapsulate Processes into Objects, and coordinate two or more Processes to construct new Processes. A variety of pedagogical strategies such as cooperative learning, small group problem solving, and even some lecturing can be highly effective in helping students learn the mathematics in question. The implementation of instruction provides an opportunity for the collection and analysis of data, which is carried out using the theoretical lens of APOS Theory. The purpose of the analysis is to answer two questions: (1) Did the students make the mental constructions called for by the theoretical analysis? (2) How well did the students learn the mathematical content? If the answer to the first question is negative, then the instruction is reconsidered and revised. If the answer to the first question is positive and the answer to the second question is negative, the theoretical analysis is reconsidered and revised. In either case, the cycle is repeated until these questions are answered positively and the instructor/researcher is satisfied that the students have learned the mathematical concepts sufficiently well. In other words, the cycle continues until the empirical evidence and theoretical analysis point towards the same mental constructions. Finally, as part of its conclusions, each study offers pedagogical suggestions for implementation and directions for future research.

The component related to theoretical analysis is explained in detail in Chaps. 3 and 4, while the component related to design and implementation of instruction is the topic of Chap. 5. The remaining component, namely, collection and analysis of data, is discussed in detail in this chapter.

6.2 Data Collection and Analysis

The data collection and analysis phase is crucial for APOS-based research, since without empirical evidence, a genetic decomposition remains merely a hypothesis. As mentioned in Sect. 6.1, the purpose of data analysis is to answer two questions: (1) Did the students appear to make the mental constructions described by the genetic decomposition? (2) How well did the students learn the concept in question?

Different kinds of instruments are used to investigate these two questions. Depending on the objectives of the particular study, these may include written questionnaires, semi-structured interviews (audio- and/or videotaped), exams, and/or computer games. The methodological design may also include classroom observations, textbook analyses, and historical/epistemological studies. Examples from the literature that illustrate the kinds of instruments that form part of the methodological design of a research study are considered in this chapter.

In all of these cases, the analysis is triangulated through collaborative research, as researchers negotiate results until they reach consensus on their interpretations and/or by implementing more than one research instrument for a study.

6.2.1 Interviews

Interviews are the most important means by which data is gathered in APOS-based research. Although an interview may be used to gauge students' attitudes and to compare mathematical performance among different approaches to instruction, the main objective is to determine whether students have made the mental constructions set forth by the genetic decomposition used in a particular study.

Interview subjects may be selected on the basis of their responses to a written questionnaire or a previously administered exam, instructor feedback, or a combination of these criteria. The idea is to access data that shows a range of mathematical performance on different mathematical tasks in order to compare the thinking of students who had difficulty with the thinking of students who succeeded. These differences enable the researchers to determine whether the mental constructions called for by the theoretical analysis account for differences in performance or whether other mental constructions not accounted for by the theoretical analysis are called for.

In designing the interview questions, different sources may be used. The responses to a previously administered written exam or questionnaire might form the basis of such questions. In this case, students are asked to clarify their responses and/or to expand on them. Another possibility might be pilot interviews whose results might uncover certain issues that can be probed more deeply through an interview protocol that is administered to a larger group of students. Observations may also play a role. In this case, difficulties that arise in lab sessions, in classroom discussions, or in homework exercises may be used as a basis for the construction of

interview questions. In all of these instances, the genetic decomposition informs the design; the goal of the analysis is to determine whether the students made the proposed mental constructions and to find supporting evidence.

In APOS-based research, as in most semi-structured interviews, the interviewer follows a prepared outline of questions. Depending on a student's responses, the interviewer might ask follow-up questions. The interviewer asks these types of questions to seek clarification or to probe a student's thinking more deeply. If these questions fail to elicit sufficient responses, the interviewer may take a more didactical route and give a hint to see, with a little bit of prompting, where the student is in terms of her or his progress in making a particular mental construction. This practice aligns with the paradigm, where the aim of an interview in particular and of APOS-based research in general is not to organize students into categories but to determine and explain how individuals construct their understanding of mathematical concepts. Such an approach, that includes follow-up questions and prompting, enables the interviewer to observe the construction process as it unfolds. This can be thought of as an application of the notion of *zone of proximal development* introduced by Vygotsky (1978).

Once the instruments have been administered, the data is organized so that the researchers can easily work with it. In the case of interviews, the audio and/or video recordings are transcribed. Everything is carefully noted, including the sounds that students make, intervals of silence, words that cannot be heard clearly, and gestures (in the case of video recordings). The transcriptions are divided into segments of related content, with descriptors used to indicate the general idea of that content. Each member of the research team analyzes the transcriptions separately. They then convene to discuss their findings and, if necessary, to negotiate differences in interpretation until they reach consensus. This is a form of triangulation that has proven to be effective in APOS-based research and is one of the main reasons why the majority of published papers that use this paradigm have multiple authors. The steps of interview analysis as a method are given in Asiala et al. (1996) as follows:

1. Script the transcript: The transcript is organized in a two-column format where the original transcript appears in the first column in segments. The second column contains a brief explanation of what happens in each segment.
2. Make the table of contents: Each item in this table summarizes one or more of the explanations appearing in the second column.
3. List the issues: "By an issue we mean some very specific mathematical point, an idea, a procedure, or a fact, for which an interviewee may or may not construct an understanding. For example, in the context of group theory one issue might be whether the student understands that a group is more than just a set, that is, it is a set together with a binary operation" (Asiala et al. 1996, p. 27). Generally after each researcher makes a list of issues, discussion and negotiation gives rise to a single set of issues to be considered.
4. Relate to the theoretical perspective: At this point, interpretation of issues is made through the lens of APOS Theory.
5. Summarize performance.

Table 6.1 Scripting of an interview transcript

I: Alright so this represents one (base ten flat). It happens to be split into a hundred equal pieces, which makes one square 1/100. If I were going to shade to represent .9 repeating, what would I shade?	45. How much of base 10 flat (=1) should be shaded?
Anita: Well. Just pretty much everything down to there.	
I: That tiny little bit.	
Anita: Yeah.	
I: Alright, so let's say we've got that.	

That tiny little bit though creates a gap again. How do you see that gap?	46. 0.9(9) means "all the nines," so no gap; 0.9(9) = 1.
Anita: You know I think I guess the gap that I am seeing in there is just adding another nine, but .9 repeating, you can't add another nine, so I guess it does equal 1, because we can't add – yeah.	
I: Ok. Repeat everything you just said.	
Anita: Ok. I guess because what I was thinking when I just said that, is that it could keep getting closer to the entire one if we just added another nine. But .9 repeating is essentially all the nines, I mean, infinite. So, I guess, you can't get in between there so I guess it is one. I keep saying there is a little gap.	
I: Yeah you do.	
Anita: But there can't be, because you can't add just another nine, because there is already all the nines on there, right?	
I: Yeah.	
Anita: Ok.	

I: But where did this come from? I mean how did this come to light like this? That's what I'm curious about.	47. Process of shading led to seeing all of the 9s at once; accepts statement D; no gap.
Anita: Well, it just says that I was supposed to shade that in, and I was thinking, well, it could go all the way down to there and you add another nine and you could just keep getting closer. But we're not adding nines, because all the nines are already on there.	

Following are illustrations of this analysis.

Table 6.1, taken from the data obtained for the study reported in Dubinsky et al. (2013), illustrates a portion of the first step, a scripted interview. An interview extract appears in the first column, and the second column provides a summary/ brief explanation of the contents of the first column.

Table 6.2, taken from the same study, illustrates the relation between Step 2 (list of issues) and Step 4 (relate to the theoretical perspective) for Anita, one of the students who participated in the study. Column 1, denoted Entry, indicates the

Table 6.2 Issues for Anita

Entry	Script Comment	Issue	Interpretation (relation to theory)
1	Meaning of the symbol 0.35454. . .	IA: Process ED: Process KW: Process R: Process	IA: Writing out a repeating decimal ED: Sees process in repeating decimal a Little weakly KW: Appears to see process ion repeating decimal R: Appears to see process in repeating decimal
2	Fraction that corresponds to 0.35454....?	IA: Correspondence; influence of authority ED: Correspondence KW: Correspondence R: Correspondence; influence of authority	IA: There is a corresponding fraction; they learned a method to find it, which the student does not remember ED: A fraction corresponds to a repeating decimal, but the student is a Little unsure KW: The student believes there is a fraction for the repeating decimal and some “formula” to find it; the student does not recall the formula R: The student believes there is a fraction for the repeating decimal and some “formula” to find it but does not recall it
3	Item 10, written instrument	IA: Approximation ED: Object KW: Object R: Object	IA: Confusion: A repeating decimal is a definite number or an approximation? ED: A repeating decimal is a definite number KW: The student says a repeating decimal is a “definite number of anything,” but her indecision suggests that such a conception may be tenuous R: The student says that a repeating decimal is a “definite number of anything,” but her indecision suggests that such a conception may be tenuous
4	$\frac{1}{4} \approx 0.25$ or $\frac{1}{4} = 0.25$	IA: Action; approximation ED: Decimals; equality KW: Equality; approximation R: Equality; approximation; division	IA: Action of long division: $\frac{1}{4} = 0.25$, not an approximation because the division “worked out evenly” ED: A finite decimal equals its fraction KW: Student says that 0.25 is an approximation of $\frac{1}{4}$, but after doing long division, she changes her mind R: The student says that 0.25 is an approximation of $\frac{1}{4}$ but after doing the long division changes her mind
5	0.333. . . an approximation of $\frac{1}{3}$	IA: Action; approximation ED: Equality KW: Approximation; process R: Approximation; process	IA: Action of long division: 0.33. . . is but an approximation of $\frac{1}{3}$ because in the conversion of 0.33. . . you do not get exactly $\frac{1}{3}$ ED: $\frac{1}{3}$ is not exactly equal to 0.33. . .; the student is a Little uncertain KW: The student sees long division of $\frac{1}{3}$ as a process in the sense that she understands that the same remainder recurs over and over. She claims that the quotient is only an approximation

(continued)

Table 6.2 (continued)

Entry	Script Comment	Issue	Interpretation (relation to theory)
			R: The student sees long division of $1/3$ as a process in the sense that she understands that the same remainder recurs over and over. She believes that the quotient is only an approximation

number of the interview segment. Column 2, labeled Script Comment, gives the comment that accompanies the interview segment. Column 3 reports the issue(s) for each interview segment. The initials “IA,” “ED,” and “KW” indicate one of the researchers, and the word or phrase that appears after each initial is the issue(s) the researcher attributes to the interview segment. “R” indicates the resolution or the result of the researchers’ joint discussion and negotiation. Column 4 gives the theoretical interpretation. Each researcher’s individual analysis appears after the initials “IA,” “ED,” and “KW,” and the joint interpretation appears after “R”.

Summarizing the data for Anita (Step 5), she reached the Process stage but not the Object stage. She made progress towards the latter and, in terms to be discussed in Chap. 8, was considered to have reached the level of an emerging Totality conception, which lies between Process and Object. Finally, regarding the equality $0.\bar{9} = 1$, she tended towards disbelief, which is consistent with her difficulty in seeing the infinite repeating decimal $0.999\dots$ as a mental Object.

In this study, and many other APOS studies, the steps of organization and analysis described and illustrated in this section are often used to test the validity of the genetic decomposition: if there is a mathematical issue that some students appear to understand, but others have difficulty with, the researcher uses the data to see if the difference can be explained in terms of the presence or absence of one or more specific mental structures and/or relations between mental structures. If this is the case, then the presence of these structures in the genetic decomposition is supported. If this does not seem to be the case, then the researchers may need to consider a revision of the genetic decomposition.

To illustrate the discussion in the above paragraph, an example from Cottrill et al. (1996) will be revisited. In Sect. 4.5, a genetic decomposition for limit and its refinement based on Cottrill et al. (1996) were presented. Table 6.3 showing both genetic decompositions is repeated below for easy reference.

In this study, the need for refinement was revealed when data indicated that some students did not appear to have been making some of the constructions proposed by the genetic decomposition. For example, the interiorization of Step 2P of the preliminary genetic decomposition was revised to include a three-step construction of a Schema, as shown in Step 3R of the refined genetic decomposition.

Below is an interview excerpt from this study revealing the need to introduce Step 3R(a) of the refined genetic decomposition.

Table 6.3 Preliminary and refined genetic decompositions of limit (based on Cottrill et al. 1996)

Preliminary genetic decomposition	Refinement
	1R: The Action of evaluating the function f at a single point x that is considered to be close to, or even equal to, a
1P: The Action of evaluating the function f at a few points, each successive point closer to a	2R: The Action of evaluating the function f at a few points, each successive point closer to a
2P: Interiorization of the Action of Step 1P to a single Process in which $f(x)$ approaches L as x approaches a	3R: Construction of a coordinated Process Schema: (a) Interiorization of the Action of Step 2R to construct a domain Process in which x approaches a. (b) Construction of a range Process in which y approaches L. (c) Coordination of (a) and (b) via f
3P: Encapsulation of the Process of Step 2P so that the limit becomes an Object to which Actions can be applied	4R: Encapsulation of the Process of Step 3R(c) so that the limit becomes an Object to which Actions can be applied
4P: Reconstruction of the Process of Step 2P in terms of intervals and inequalities. This is done by introducing numerical estimates of the closeness approach: $0 < x - a < \delta$ and $0 < f(x) - L < \epsilon$	5R: Reconstruction of the Process of Step 3R(c) in terms of intervals and inequalities. This is done by introducing numerical estimates of the closeness approach: $0 < x - a < \delta$ and $0 < f(x) - L < \epsilon$
5P: Application of a two-level quantification Schema to connect the Process described in Step 4P to the formal definition	6R: Application of a two-level quantification Schema to connect the Process described in Step 5R to the formal definition
6P: Application of a completed $\epsilon - \delta$ conception to specific situations	7P: Application of a completed $\epsilon - \delta$ conception to specific situations

Norton: Um, if you start, if, if a was, say, 3 and you started with x as 10, you go 10, 9, 7, 6, whatever, then you get 3.2, 3.1, and get smaller intervals between x and a , the function will approach a limit and the limit will be L .

I: ... How close do things have to get before you're willing to call L a limit?

Norton: Oh, tricky. Um... If, well, it should—you want get, start getting smaller intervals between x and a and if you do a sequence of points, you will see it approaching a limit L . If it doesn't approach a certain number, then the limit doesn't exist. But in this function, it says that it has the limit L , so as your x approaches a , you should, the function should assume the limit L .

In this excerpt, Norton explicitly selects values of x that approach a but does not do this for values of the function. Although he mentions “it approaching a limit L ,” which is a reference to values of the function, the focus is on the domain and on x approaching a in isolation from the values assigned to each domain.

6.2.2 Written Questions

Written questions can be administered to large groups of students during an exam or in the form of a questionnaire. They provide basic information on students'

mathematical performance. They can also be used in the design of interview questions because of their ability to reveal student difficulties that require further analysis.

Asiala et al. (1997b), in a study of students' learning of the concepts of cosets, normality, and quotient groups, illustrate how a genetic decomposition informs the design of written instrument questions. Their analysis was guided by the preliminary genetic decomposition developed by Dubinsky et al. (1994). This decomposition is simply the progression *Action to Process to Object*. The three stages can be described as follows:

An action conception of coset has to do with forming a coset in a familiar situation where formulas or recipes can be used such as the multiples of 3 in $\mathbb{Z}$ or in $\mathbb{Z}_{18}$. The action conception is not strong enough to handle formation of cosets in more complicated situations such as in S_n beyond the familiar examples of S_3 or S_4 , where cosets are not generally representable by formulas or simple recipes.

A process conception of coset will allow an individual to think of the (left) coset of a subgroup by an element by imagining the product of that element with every number of the subgroup – without having to actually form the products. In a process conception of coset, the main thing one thinks about is the formation of the coset.

With an object conception of coset, an individual can think about, name and manipulate a coset without necessarily focusing on how it is formed. Actions can be performed on cosets such as comparing the cardinality of two cosets or counting the number of cosets, both of which arise in the proof of Lagrange's theorem. . .

There are other actions which can be applied to cosets as objects, such as considering the relations among elements, subgroups and cosets expressed by the various equivalent conditions for normality or the formation of product of cosets. (Asiala et al. 1997b, pp. 247–248)

Students in two introductory abstract algebra classes participated in the study. One class followed an APOS-based instructional strategy using the ACE cycle, and the other class followed a traditional approach.

The research instruments included two written exams and interviews. These tools were used to determine whether students made the mental constructions called for by the preliminary genetic decomposition. Students worked on the first exam in their cooperative groups and completed the second one individually.

In APOS-based research, written questions are carefully designed to help gather evidence for the presence of the mental constructions predicted by the preliminary genetic decomposition and to suggest modifications of the pedagogical strategies and/or the genetic decomposition when these constructions are not present. They also allow researchers to focus their attention on the aspects of the construction of knowledge that they are studying. Some examples of these questions related to Action, Process, and Object conceptions in abstract algebra, taken from the Asiala et al. (1997b) study, follow.

Responses to Question 5 from Test 1 provided evidence of the construction of specific actions:

Find a subgroup of S_4 that is the same as S_4 . Calculate the left cosets of your subgroup. (Asiala et al. 1997b, p. 305)

In this question, once the subgroup is specified, an Action conception of coset consists of listing the elements of each coset. Here specific cosets of a particular subgroup which is familiar to the student are to be formed. The student can do this by applying a formula, which is an external cue.

Responses to Question 4(a) from Test 2 were used to determine the construction of Process conceptions:

There are many conditions that are equivalent to a subgroup H of a group G being normal. One is,

$$\text{for all } g \in G, \text{ it is the case that } gHg^{-1} \subset H$$

Give another condition for normality and show that it is equivalent to this statement. (Asiala et al. 1997b, p. 306)

The answer to this question may reveal how the student “thinks of a coset and also perhaps the way in which manipulations with cosets are performed” (p. 306). Ability to use set former notation in formation of cosets, such as in $\{gh : h \in G\}$, was considered to be an indicator of Process, since this provides evidence that the student can think about forming the cosets in her or his mind, without actually having to form them. The generality of the question, without the mention of a specific group or a specific coset provides an appropriate context to check for interiorization of Actions.

Responses to Question 6 from Test 2 were used to determine whether encapsulation had occurred (Asiala et al. 1997b, p. 307):

Let S_3 be the group of permutations of three objects.

- (a) Find a normal subgroup N of S_3 .
- (b) Identify the quotient group S_3/N .

Finding the subgroup of a given group is an Action applied to that group. Similarly forming a quotient group is an Action applied to the group. Identification of a quotient group is an Action applied to the quotient, which is itself an Action applied to a set of cosets. The ability to apply such Actions indicates an Object conception of coset.

6.2.3 Classroom Observations

Classroom observations can reveal interesting data, especially when the instruction is not based on pedagogical elements related to APOS Theory or is implemented by instructors with little or no experience with this approach.

Kú et al. (2008) observed an introductory one semester “linear algebra for engineers” course with 24 students. Although the course was designed according to elements of APOS Theory, the classroom observation brought to light the fact that there were several aspects of an APOS-based pedagogical strategy that were

missing from the course design. For example, while the computer activities were done with a computer algebra system, the emphasis was put on its use as a tool for calculating answers instead of programming with the purpose of facilitating mental constructions. None of the students showed either Process or Object conceptions of the concept of basis of a vector space. This may be attributed to the fact that the pedagogy did not encourage these types of mental constructions.

6.2.4 Textbook Analyses

Textbooks can be analyzed to account for the pedagogical strategy followed, to determine which results, rules, and theorems make use of the concept, and to investigate the notation employed that might have bearing on students' understanding. All these elements can be useful when analyzing student responses and can be used to inform the interpretation of data. Roa-Fuentes and Oktaç (2010) performed such an analysis for the linear transformation concept. They proposed two genetic decompositions for the concept of linear transformation. In their analysis of student data, they found evidence for one of the genetic decompositions but not the other. Their analysis of linear algebra textbooks uncovered the reason. As explained in Chap. 4, one of the preliminary genetic decompositions that they proposed started with a previously constructed (general) transformation Object and built on that by de-encapsulating it and coordinating the underlying Process with the related binary operation Processes. On the other hand, the textbook used in the course, as well as the instructor, defined a linear transformation as a function between vector spaces; neither of them introduced the general transformation concept before that. Therefore, it was unsurprising to find no empirical evidence for the general transformation concept. Thus, it is necessary to implement an instructional approach based on the idea proposed in this genetic decomposition in order to test whether the linear transformation concept can be constructed in the way not considered in the textbooks.

6.2.5 Historical/Epistemological Analysis

For studies of students' conceptions of mathematical infinity (e.g., Dubinsky et al. 2005a, b), historical/epistemological analyses were conducted. The study of the historical development of mathematical infinity helped in contextualizing student difficulties in terms of obstacles that mathematicians faced throughout the development of this mathematical concept as well as in explaining these difficulties in cognitive terms. For example, Weller et al. (2004) describe "how a particular theory about how people came to understand mathematics, APOS Theory, can be helpful in understanding the thinking of both novices and practitioners as they grapple with

the notion of infinity” (p. 741). In that article and in Dubinsky et al. (2005a, b), several issues and paradoxes related to the mathematical development of the concept of infinity are analyzed. These include the paradoxes of Achilles and the Tortoise and analogues of Hilbert’s Hotel, infinitesimals, the relation between the union of all finite segments $\{1, 2, \dots, n\}$ of natural numbers and the set of all natural numbers, and the possibility of using a countable mental procedure to obtain the mental construction of an uncountable set. Underlying most of these issues is the distinction between potential infinity and actual infinity. This latter issue has puzzled mathematicians and philosophers for millennia, from Aristotle to at least Poincaré in the beginning of the twentieth century. In these articles, the authors argue that the paradoxes might be resolved and the issues might be settled in the minds of the learners by an analysis that considers examples of potential infinity to be Processes and examples of actual infinity to be the mental Objects obtained by encapsulating those Processes. They go on, in several studies (Weller et al. 2009, 2011; Dubinsky et al. 2013), to use genetic decompositions based on this analysis to provide empirical data in support of their argument.

6.3 Types of APOS-Based Research Studies

Weller et al. (2003) report on student performance and attitudes in courses based on APOS Theory. They classify research studies into four types:

- Comparative studies in which the performance of students who received instruction using APOS Theory and the ACE Teaching Cycle is compared with the mathematical performance of students who completed traditional lecture/recitation courses;
- Non-comparative studies measuring the performance of students who completed courses using APOS Theory and the ACE Teaching Cycle;
- Studies of the level of cognitive development of students who completed courses based on APOS Theory and the ACE Teaching Cycle or courses using a traditional lecture/recitation model;
- Comparisons of student attitudes and the long-term impact of courses based upon APOS Theory and the ACE Teaching Cycle to that of students who completed traditional lecture/recitation courses. (p. 98)

A study may fall into more than one category depending on the type of data used. The study conducted by Asiala et al. (1997b) is an example that falls into the first three categories. Explanations and examples of all four categories follow.

6.3.1 *Comparative Studies*

The first category corresponds to studies where students who received instruction other than an APOS-related approach were interviewed for comparative purposes. This gives a general idea of how successful (or not) the APOS pedagogical strategy was in helping students to construct their understanding of a specific mathematical concept. This was the case with Asiala et al. (1997b), where the students who learned about cosets, normality, and quotient groups following the pedagogical approach based on APOS Theory performed better than the group that followed a traditional approach.

Also belonging to this category is the longitudinal study reported in Weller et al. (2009, 2011). There were several factors that led the researchers to adopt a comparative analysis. The fact that there were few studies on experimental approaches on the topic suggested the need for a comparative analysis, and the large number of student participants made such a study possible. Weller et al. (2009) found that preservice elementary and middle school teachers who completed a specially designed unit on repeating decimals based on APOS Theory and implemented using the ACE teaching cycle made considerably more progress in their development of an understanding of the equality $0.\bar{9} = 1$ and the more general relation between a rational number and its decimal expansion(s) than their control group counterparts (see Chap. 8 for details regarding the study and Chap. 5 for information about the instructional design). A second study (Weller et al. 2011), based on interviews conducted four months after the instruction, reports that the students “who received the APOS-based instruction developed stronger and more stable (over time) beliefs that a repeating decimal is a number; a repeating decimal has a fraction or integer to which it corresponds; a repeating decimal in general equals its corresponding fraction or integer; and, in particular, $0.\bar{9} = 1$ ” (p. 129).

6.3.2 *Non-comparative Studies*

The study conducted by Asiala et al. (1997b) also corresponds to the second category, since non-comparative as well as comparative data were used. The non-comparative data came from exams, as described in Sect. 6.2.2, that were administered only to those students in the experimental course. Two sets of interviews were conducted, the first one providing non-comparative data that was carried out only for the students who followed the APOS-based course.

6.3.3 *Studies of the Level of Cognitive Development*

The study conducted by Asiala et al. (1997b) is also considered as belonging to the third category because it gives a detailed account of how students construct their understanding of certain abstract algebra concepts, namely, cosets, normality, and quotient groups.

Other studies belonging to the third category are those where the goal was to observe the viability of a preliminary genetic decomposition in the absence of previous studies about the learning of a particular mathematical topic or of a specifically designed APOS-related instructional sequence. As explained in Chap. 4, Roa-Fuentes and Okaç (2012) have conducted such interviews for the purpose of looking for evidence for one or both of the preliminary genetic decompositions that they had proposed for the linear transformation concept (Roa-Fuentes and Okaç 2010). They found evidence for only one of them, commenting that this may be due to the kind of instructional treatment that the students had received.

In another study about functions of two variables, Trigueros and Martínez-Planell (2010) designed an interview in order to find information about the components of the preliminary genetic decomposition that they proposed. In their study, they focused on the analysis of students' responses to those questions related to subsets of R^3 and graphs of functions of two variables. They conducted nine interviews after the students had finished a course on multivariate calculus for undergraduate mathematics students. In their preliminary decomposition, they suggested that the construction of a Schema for R^3 and the Processes involved in drawing graphs of functions of one variable were prerequisites for the learning of the concept of functions of two variables. In the preliminary genetic decomposition, a Process to construct fundamental planes was included.

Results of the study showed that most of the interviewed students had indeed constructed a Schema for R^3 as predicted. However, their Schema included neither the construction of subsets of R^3 nor the coordination of the Schema for R^3 and that of a function of one variable. Most students in this study were not able to perform Actions on any Object in space but points. The researchers concluded that it was necessary for students to construct subsets of points in space as Objects as well as to coordinate between the Schema for R^3 and that for functions of one variable in order to be able to construct the concept of function of two variables. Comparison of results achieved by different students showed that the Action of intersecting surfaces with planes and the interiorization of this Action into a Process in which the result of the intersection can be predicted played an important role in students' learning of these functions. Since these constructions were not predicted in the preliminary genetic decomposition, they were introduced in its refinement.

The refined genetic decomposition was tested in a second study (Martínez-Planell and Trigueros 2012) where results showed that it was a good model of students' mental constructions. This genetic decomposition has been tested in instruction.

6.3.4 Comparisons of Student Attitudes and the Long-Term Impact of APOS-Based Instruction

The fourth category consists of studies which focused attention on students' attitudes. The purpose of these studies was to investigate the effect of APOS-based pedagogical strategies on students' attitudes when compared with the attitudes of students who completed learning units based on other types of instruction. Weller et al. (2003) summarized the results of a study on student attitudes in abstract algebra in which the researchers found that students who completed an APOS-based experimental course felt more positive about mathematics than those who completed a traditional course covering similar content. However, they qualified these results, noting that factors related to the instructors, the interviewers, and the grade distribution may have affected the findings. They also pointed out that no attempt was made to conduct a statistical analysis.

6.4 Scope and Limitations of APOS-Based Research

APOS is a cognitively oriented theory and as such provides a useful tool for modeling student understanding of mathematical concepts. It also has a social component that relies on cooperative learning, as the context of group work is more likely to give rise to more explicit questions, doubts, and explanations by students than what would typically transpire in individual contexts (Vidaković 1993). Moreover it provides a context that facilitates learning. That is, APOS Theory functions under the premise that working in groups makes a difference in the affective domains of the individuals.

APOS Theory has been successful in proposing models to explain the learning of numerous mathematical concepts, including those which pose serious difficulties for students such as linear independence, quotient groups, functions, and repeating decimals. The literature does not mention very many (if any) other such success stories about the learning of these topics.

The questions that can be asked in research studies where APOS Theory is used as a tool for analysis are generally of the following types:

- How might an understanding of the concept be constructed by students?
- What are the mental constructions involved in the development of a Schema and its components?

Some auxiliary questions can be used to help in answering the main questions or can aid in making suggestions for didactical approaches based on research. Examples are the following:

- What are the prerequisite concepts necessary to construct understanding of a particular mathematical concept?

- How is a particular conception characterized in the learning of a particular concept?
- How is the transition from one conception to another characterized?
- What are some pedagogical strategies that can help students in the mental construction of a particular concept?

This does not mean that the research questions have to follow a specific format if APOS Theory is used but rather emphasizes the kind of phenomena that arise in working with this paradigm. The following are examples of topics that could be researched using APOS Theory:

- The use of APOS Theory to find out how mathematicians perceive the mathematical concepts they teach (see Stenger et al. 2008).
- Strength and stability of constructions of mathematical concepts learned using an APOS-based learning sequence (see Weller et al. 2011).
- How to teach APOS Theory to pre or in-service teachers, so as to help them to use APOS Theory methodology in their practice.
- Is APOS Theory applicable/adaptable to other humanistic or scientific domains?

The research trend using APOS Theory indicates that the construction of mathematical concepts will continue to be studied using the lens of APOS Theory and more research questions will be added to the repertoire as researchers discover the need for it.

Chapter 7

Schemas, Their Development and Interaction

APOS Theory has been successful in describing and predicting the types of mental structures students need to construct in order to learn abstract concepts. As new research is carried out and complex research projects are undertaken, it has become necessary to widen the scope of the theory. This has been achieved by expanding the researchers' understanding of various theoretical constructs. Although there has been less research using these constructs, they already form part of the theory or are being tested in current research. One of these constructs is Schema; another is the mechanism of thematization and another, to be discussed in Chap. 8, is a possible new stage, Totality, between Process and Object.

Schema is the central subject of this chapter. A general description appears in Sect. 7.1. Several examples are considered in Sect. 7.2. How Schemas develop is the subject of Sects. 7.3 and 7.4. Sections 7.5 and 7.6 detail how a Schema changes, either through the introduction of new information or through the interaction of two Schemas. Finally, Sect. 7.7 deals with the issue of thematization, the mechanism involved in the construction of an Object from a Schema.

7.1 Schemas in Piaget's Work and in APOS Theory

Piaget and Inhelder (1966/1969) relate formal schemas to "... the concepts which the subject potentially can organize from the beginning of the formal level when faced with certain kinds of data, but which are not manifest outside these conditions" (p. 398). That is, schemas are related to important ways of reasoning and refer to structures brought to bear on certain learning situations. Piaget considered schemas as a way to work with classes of situations in order to make sense of them and to achieve various goals.

For Piaget, schemas are instruments that structure knowledge:

A schema is only constructed when it is functioning, and it only functions through experience: then, that which is essential is, not the schema as structure in itself but the structuring activity that gives rise to schemas. (Piaget 1975/1985)

Any particular schema in itself does not have a logical component, but Schemas are coordinated with each other, and this fact results in the general coordination of actions. These coordinations form a logic of actions that is the beginning of logico-mathematical structures. Piaget said that a schema could include subschemas or subsystems. The subschemas are included in the total Schema in the same way that a logico-mathematical structure of classification into subclasses is included within the whole class. At a later stage, this relationship of class inclusion gives rise to certain concepts (Piaget 1975/1985).

Piaget's ideas are reflected in APOS Theory. Asiala et al. (1996) describe an individual's Schema for a mathematical topic as all of her or his knowledge connected (explicitly or implicitly) to that topic. Specifically, an individual's Schema for a certain mathematical concept is the individual's collection of Actions, Processes, Objects, and other Schemas which are linked by some general principles or relations to form a framework in the individual's mind that may be brought to bear upon a problem situation involving that concept (see Sect. 3.2.5). This framework must be coherent in the sense that it gives, explicitly or implicitly, a means of determining which phenomena are in the scope of the Schema and which are not (Dubinsky and McDonald 2001). A Schema can be thematized to become a cognitive Object to which Actions and Processes can be applied. By consciously de-thematizing a Schema, it is possible to obtain the original Actions, Processes, Objects, and other Schemas from which the Schema was constructed (Clark et al. 1997).

A particular Schema may not necessarily be accessed in all situations, because mathematical learning is highly nonlinear. However, the structure of a Schema and its development may explain why students have difficulty with different aspects of a topic and may even have different difficulties with the same situation in different encounters.

Schema has been part of APOS Theory since its beginning (see Chap. 2 for further details). In earlier papers, it was only referred to as one of the possible constructions in the theory, but no research was conducted in which Schema had a central role. As APOS-based research progressed, it was found that the Schema structure was necessary in order to describe certain learning situations such as those that will be described in Sect. 7.4.

7.2 Examples of Schemas

Some examples may be helpful to understand Schemas in APOS Theory. In each example given below, the constructions included in the Schema are described. It is important to note that the structure of a Schema can differ among different individuals because each individual constructs different kinds of relations among

the components of a Schema. Some of these differences are discussed in this section and the sections that follow.

One example of a Schema is the function Schema. It can be composed of different types of functions such as real-valued functions, multivariable functions, vector-valued functions, and/or proposition-valued functions. These different types of functions may have been constructed as Processes or Objects, together with the operations that can be applied to them. For some students, different types of functions may be related by the common idea of a function Process: an operation applied to a set of inputs to obtain a set of outputs. Functions differ in the types of inputs involved, the nature of the operations applied to those inputs, and the results of the operations. Although individuals' Schemas may include the same types of functions, their components or the types relations constructed among them may differ.

Another example of a Schema is the vector space Schema. It is composed of vectors and operations defined on them, together with linear combinations, spanning sets, bases, and dimension, each considered as a Process or an Object (Parraguez and Okaç 2010). For some individuals, these concepts are related to the concept of vector space only because they might be defined for vector spaces. However, other individuals who have constructed a coherent Schema are aware of the nature of their relation to the vector space concept, for example, whether every combination of vectors in a vector space is linear.

A third example of a Schema is the Cartesian plane Schema. It is a structure that includes points as Objects and relations among points, such as curves, functions, and regions that are constructed as Processes that result from interiorization of the Actions of representing their points. For some individuals, these relations may consist only of specific Processes applied to points to obtain new curves, functions, or regions. For others, these relations may include the distinction between different sets of points through their definition. A different Schema of the Cartesian plane, constructed for the concept of equivalence classes of fractions, is described in Sect. 9.5.

A Schema can be considered as being composed of different components. It may include a single concept that can be applied to different situations, as in the case of function, or it may include different but interrelated concepts, as in the case of the vector space Schema. In either case, a Schema is a tool for understanding how knowledge is structured and its development through the learning process.

As an individual's mathematics learning progresses, different concepts need to be related and used in problem-solving activities. Sometimes new Actions, Processes, or Objects can be assimilated to a previously constructed Schema by establishing new relations among the components of the Schema. In other situations, a Schema may be related to one or more different Schemas that lead to the construction of a new, more extensive Schema. For example, the Schema for the Cartesian plane can be related to other higher dimensional spaces that results in the construction of a new Schema that includes n -dimensional spaces and also non-Cartesian spaces.

When a person confronts a mathematical problem situation, he or she evokes a Schema and makes use of some of its components and some relations among them to deal with the situation. When facing the same situation, different persons may use the same components but construct different relations among them. The study of the relations together with the type of constructions brought to bear in dealing with a particular problem-solving situation reveals the structure of an individual's Schema and gives information about its development. In APOS studies of the development of individual Schemas, researchers have used "the triad," a progression of three stages proposed by Piaget and García (1983/1989). The mechanism of accommodation accounts for the progression from one stage of the triad to the next. Each subsequent stage of the triad involves the development of relations and transformations an individual can make between particular constructs within the schema as well as the development of the coherence of the Schema in terms of its possible application to specific problem situations.

In APOS-based research, the triad progression of stages has been used to describe the development of students' Schemas associated with specific mathematical topics and to better understand how Schemas are thematized to become cognitive Objects. Schema development has proven to be an effective way to understand this facet of cognitive construction and has led to a deep understanding of the construction of Schemas (Trigueros 2005).

7.3 Development of a Schema in the Mind of an Individual

From the beginning, the use of Schemas in APOS Theory required the introduction of the notion of Schema development; this development is described in stages. Piaget and García call the stages involved in the development of any schema Intra-, Inter-, and Trans-. The hyphen symbol "-" is followed by the name of the Schema being discussed to indicate its application to a particular Schema. Since Schemas can be defined by structures that differ in their complexity, this terminology helps to distinguish the Schema described by the three stages. For example, a Schema for functions includes different types of functions, such as in the example described in Sect. 7.2, or it might include all the Objects and relations in differential and integral calculus. Both examples are function Schemas whose development can be described by stages, for example, the first one could be named Intra-function, Inter-function, and Trans-function, and the second Intra-calculus, Inter-calculus, and Trans-calculus in order to distinguish what is comprised by the Schema. It would be possible to use function to name the stages of the second schema, together with acknowledgment of its complexity level by making its components explicit.

The first stage, Intra-, is marked by a focus on individual components of a Schema. This consists mainly of correspondences among the system components. The individual discovers a set of properties that are common among the Objects that are included in the Schema, with all connections being local and particular. For example, in the historical development of geometric structures, the stages of

Schema development can be described in terms of the relationships between figures and space. In the Intra-figural stage, the study of geometric figures in Euclidean Geometry focuses on the representation of figures and the study of their properties. It is considered that figures are part of space, but space is regarded as a global entity where figures can be studied. In this stage, it is possible, for example, to establish a correspondence between a segment and a number to define a unit of measurement. This kind of correspondence is an internal relation between the elements of the figure (Piaget and García 1983/1989).

As knowledge develops, comprehension of local transformations starts to play a more fundamental role and access to necessary connections and the reasoning behind them begins to be developed. At this point, the Schema is said to be at the Inter-stage. In the case of geometrical structures, algebraic representations in Analytic and Projective Geometry lead to the introduction of a system of transformations; these transformations relate the figures under different perspectives. This is characteristic of the Inter-figural stage (Piaget and García 1983/1989).

Later on, it is necessary to determine the links and reasons behind the local transformations that constitute the Schema. In particular, an individual begins to see the Schema as a whole, and a structure that can account for its composition as a whole is constructed by means of synthesis. The structure is now coherent, and the individual can determine whether it is applicable or not to a given situation. This is called the Trans-stage. In the case of the development of the geometrical structures, this stage is constructed when groups of transformations are introduced (Piaget and García 1983/1989).

According to Piaget and García (1983/1989):

Passing from one stage to the other is not characterized by a period of “increments” in knowledge with respect to the previous stages, instead, a total reinterpretation of the conceptual fundamentals is involved... access to the next stage needs the reconstruction of what had been constructed in previous stages. (p. 109)

Piaget and García assert that all Schemas develop through a progression or series of stages they called the triad. They hypothesized that these stages can be found when analyzing any developing Schema. An important issue that needs to be clarified is that for Piaget the nature of the triad stages was functional, not structural. The focus on Schema in Piaget’s genetic epistemology is not the structure of the Schema but the way Schemas function in cognitive development. Central mechanisms of Piaget’s Theory, such as assimilation, accommodation, and equilibration, play a fundamental role: the incorporation of new elements to the Schema by assimilation leads to its modification through accommodation. It is through these mechanisms that the Schema reaches a new equilibrium. Equilibrium is dynamic, so that through it the Schemas are constantly changing, although they maintain their identity.

The triad was first used in APOS Theory by Clark et al. (1997) in a study of students’ understanding of the chain rule. It was also used in other studies such as sequences of numbers (McDonald et al. 2000), the chain rule and its relation to composition of functions (Cottrill 1999), and the relation between the graph of a

function and properties of its first and second derivatives (Baker et al. 2000; Cooley et al. 2007). In all of these studies, the triad helped the researchers to understand the development of Schemas and support their explanations of students' thinking that arose in the analysis of the data.

In APOS Theory, in line with Piaget and García, the development of the Schema consists of three stages, Intra-, Inter-, and Trans-, and the triad progression of stages is involved in the transition from one stage to the next through the development of relations and transformations that an individual makes among the particular constructs within the Schema: the Intra-stage of Schema development is characterized by a focus on individual Actions, Processes, and Objects in isolation from other cognitive items of a similar nature; the Inter-stage is characterized by the construction of relationships and transformations among the cognitive structures that make up the Schema where an individual may begin to group items together and even call them by the same name; at the Trans-stage, the individual constructs an implicit or explicit underlying structure through which the relationships developed in the Inter-stage are understood and by which the Schema achieves coherence that is indicated by the individual's ability to determine what is in the scope of the Schema and what is not (Dubinsky and McDonald 2001).

7.4 Examples of Development of a Schema

In this section, examples of Schema development at each of the triad stages are discussed.

7.4.1 *The Intra-Stage*

In APOS Theory, the Intra-stage of Schema development is characterized by a focus on individual Actions, Processes, and Objects in isolation from other cognitive items. At the Intra-stage, the student concentrates on a repeatable action or operation and may recognize some relationships or transformations among Actions on different components of the Schema. Some examples may help to better understand this stage:

Functions In the case of the function concept, an individual at the Intra-stage tends to focus on a single type of function and the various activities that he or she could perform with it (Dubinsky and McDonald 2001).

Derivative At the Intra-stage of the derivative Schema, the student can interpret the derivative as the slope of a tangent line at specific points and can perform Actions or Processes to find the derivative of a function on specific intervals. The individual can determine if the derivative is positive or negative and use the sign of the derivative to decide whether the function increases or decreases on those intervals.

The student can also solve some rate-of-change problems. However, the focus is on individual Actions or Processes, so the description of the behavior of a curve and the solution of rate-of-change problems are only related by the need to find the derivative of a function. The individual is unable to consider the derivative as a means to describe local variation of the function (Baker et al. 2000).

Chain Rule At the intra-stage of the chain rule Schema, the student has a collection of rules to calculate some individual cases where the chain rule is used implicitly, such as the power rule or the exponential rule, but does not see any relationship between those cases. The student considers each case as different rules that can be applied to specific situations (Clark et al. 1997; Cottrill 1999).

The following is an example of a student at the Intra-chain rule stage.¹ In this case, Tim could see the chain rule for implicit differentiation but could not recognize how it might be used in other problems where it applies (Cottrill 1999, p. 39):

- I: Could you write down the chain rule in whatever words or symbols you remember?
- Tim: The chain. . . ?
- I: The chain rule, for taking derivatives.
- Tim: You mean $y = x^2 - 7x + 5$ and you take the chain rule for this one?
- I: No, the chain rule is just a rule we have for taking certain kinds of derivatives. Do you remember using the chain rule?
- Tim: Um, y, it was something, um. . . Can I do an example out of the book? The chain rule for this one? [writes out $f(x) = y^n$, $y^{n'} = ny^{(n-1)}y'$. . .]
- I: Let me show you an example of a problem where you would use the chain rule, for instance #18 here. The original problem was $\sin(5x^4)$. [Student is shown written work from the questionnaire, which has the correct solution to the problem].
- Tim: You mean give the rules of this one, how I worked out this one?
- I: Yes, how did you find the derivative there?
- Tim: Oh, [writes and crosses out $f(x) = u(x)f'(x)$ [mumbles]. . . I don't know, you know, if you give me an example of how to do the chain rule, I know how to do products.
- I: That is what I am saying, this solution you have right here. . .
- Tim: Mm-hmm
- I: you used the chain rule
- Tim: yeah
- I: to get that solution, which is correct. OK? Does this remind you of the chain rule, then?
- Tim: Uh-huh, so what you want me to . . .
- I: We are starting with $\sin(5x^4)$ and look at what you wrote down for your answer, and try and remember how you came up with that idea.
- Tim: Oh! So first I take derivative of outside, derivative of sine is cosine
- I: Right
- Tim: so then I take derivative of inside, so inside is $5x^4$, so I write down $20x^3$.
- I: OK
- Tim: That is so easy, you know, I don't know how to get the something that you asked me to do.
- I: You don't. . . So the question was, what does the chain rule say?
- Tim: If there is a function of x , something like that, and take the derivative of that, right, first you take the derivative of outside first, then take derivative of inside.

¹ Problems for the excerpts of students' responses shown in this section appear in the Appendix at the end of this chapter.

Although Tim can remember and use correctly the power rule and, with some help, use the formula for the chain rule, he describes his work in terms of Actions related to specific problems. He describes each Action in isolation as if each rule were different.

7.4.2 *The Inter-stage*

As mentioned in Sect. 7.3, the Inter-stage is characterized by the construction of relationships and transformations among the Processes and Objects that make up the Schema. At this stage, an individual may begin to group items together and even call them by the same name. The Inter-stage is described for the examples considered in Sect. 7.4.1:

Function As an individual considers possible analogies among operations on different types of functions, as Processes, he or she may construct a relation among them and recognize types of functions as instances of the same sort of activity, for example, as a means of constructing new functions from known ones.

Derivative At the Inter-stage of the derivative Schema, the student can relate the Process of the derivative as the slope of the tangent line and the Process of the derivative as the rate of change at a given point so that he or she can consider the derivative as a means to describe local variation of the function (Baker et al. 2000).

Chain Rule In the case of the chain rule Schema, the Inter-stage is characterized by recognition that different instances of the chain rule, such as the power rule or the derivative of a composition of functions, represent something more general. That is, the individual becomes aware that special cases are related and that those rules are instantiations of a more general rule (Clark et al. 1997; Cottrill 1999).

In the next excerpt, Peg demonstrates an understanding at the Inter-chain rule stage. She groups different differentiation problems, which include problems involving compositions of functions, according to the chain rule as an initial criterion. Although she distinguishes between exponential and trigonometric functions, she keeps the chain rule as her main criterion. However, she is unable to include in her criterion exponential or trigonometric functions applied to the identity function because expressions for these functions do not include parentheses. She fails to recognize implicit functions as instances where she needs to use the chain rule, and she also fails to recognize the case where composition of functions is used in an integral. It seems that her criterion for grouping is based on the use of parentheses (Cottrill 1999, p. 46).

- I: So is that five different groups?
 Peg: [pause] Yes. Five different groups.
 I: OK, what was the discriminating features?
 Peg: OK, One is like the most straight forward, where it is just strictly using the uh, what's the name of that rule? The power rule.
 I: Uh-huh.

- Peg: It's just straight out the power rule. Uh, two and three are pretty much exactly the same thing except for two, the expression needs to be rewritten to use that rule. 6 and 7 is also just the power rule except for you have uh, uh, you have a power rule and then it's a chain rule. Uh, 4, 5, 8, & 9 all have trigonometric uh terms in them to where then you also have to know their trig functions and then 10 is the e function which also has it's separate rule.
- I: OK. When you went through that. . .
- Peg: And that's pretty much from my point of view which was from the easiest to the most difficult. Although, I guess that the e function is not really that difficult; it's just knowing that it, it's just not as, it's not difficult it's just different because it just doesn't seem to follow the format as all of the others.
- I: Uh-huh. Um, was there any other way that you considered grouping these?
- Peg: [pause] Uh. . . in the time frame, I just went with the first thought that popped into my mind. I'm sure that if I was to sit here longer, I could think of other ways to do it.
- I: OK. What might be other criteria?
- Peg: [pause] Uh, the other ones would have been, may be, the ones that just had single terms as opposed to having two terms that have to be differentiated. Or anything that that has a chain, or pulling out anything that has the chain rule and knowing that also have to add more terms to the final expression.
- I: So the last couple of questions that I have for you, write down the chain rule using whatever words or symbols that you like. It's as best as you can remember it. You have here like examples of it. Number 6 and 7 which is 16 and 17 on our papers here. You used the chain rule in your answer. So if you want to use those to help spark some ideas. . .
- Peg: [pause] [mumbles to self] [pause] OK.
- I: Read that for the tape.
- Peg: OK. Uh, when you have the derivative of $f(g(x))$ it's $f'(g(x))$ times $g'(x)$. It's basically you take a derivative of the outer term and I'm using outer term because it's just the way that I look at it, composed with the inner term and then multiply it by the derivative of the inner term.
- I: OK. Great. Um, actually the last five statements here on the list, 6 through 10
- Peg: Uh-huh.
- I: were intended to be, to use the chain rule and you named 6, 7, & 8 as using the chain rule. Do you see the chain rule being used in 9 and 10?
- Peg: Yes, I do, but I had pulled them out separately because they had the trigonometric functions also.
- I: OK, no that's fine. Does your rule that you have written there at the bottom, does it apply or does it take care of all six of those or uh, all five of those cases?
- Peg: [pause] Uh. . . in the way that I would look at it, the way that I look at the problem, it does. I look at the outer term and the inner term which would, the outer term that I'm looking at is being either how you solve strictly for the power rule or for the trigonometric functions and then going inside and actually you know. . .
- I: OK. [pause] You did a very interesting thing when you solved 19.
- Peg: I don't think that's correct.
- I: I don't think that it's very far wrong actually. But what you did is rather than use the chain rule, you expanded the expression. The expression was cosine cubed of t . And so you wrote down three products of cosine t which is cosine cubed.
- Peg: Uh-huh.
- I: And then, you took a derivative from there.
- Peg: Even with that, I didn't finish it did I?
- I: No, you needed to do the other product.
- Peg: Yeah, but, yeah, I made it to that.
- I: But, my question is, if you can recall, this was a long time ago, um, was there something that didn't say chain rule to you in that problem?

- Peg: That would be the one that I'm still the least clear on still today because it's not as straight as the chain rule because the uh, the variable t doesn't have anything associated with it like all of the other ones, that there's something inside the parenthesis to differentiate. Where this one... so would it be $3 \cos^2 t$ uh times minus $\sin$? Is that the correct answer?
- I: Uh-huh.
- Peg: Oh.
- I: That's exactly it.
- Peg: But I, it's just not as obvious of the chain rule as all of the others. ... (Cottrill 1999, p. 46)

In this example, Peg recognizes the chain rule in different problems. She thinks of the chain rule as a Process she can apply to problems where she is able to distinguish the inner and the outer functions in a composition. However, when she is not able to make this distinction, she does not recognize their relation to the general rule.

7.4.3 *The Trans-stage*

As a student reflects upon coordinations and relations developed in the Inter-stage, new structures arise. Through syntheses of those relations, the student becomes aware of the transformations involved in the Schema and constructs an underlying structure. This leads to development of the Schema at the Trans-stage. A critical aspect of the Trans-stage is development of coherence. Coherence is demonstrated by an individual's ability to recognize the relationships that are included in the Schema and, when facing a problem situation, to determine whether the problem situation fits within the scope of the Schema. In some cases, the constructions involved in the mathematical definitions of a concept show coherence of the Schema; this means the individual is able to reflect on the explicit structure of the Schema and select from it the content that is suitable in solution of the problem. The examples that follow illustrate this.

Functions At the Trans-function stage, an individual can construct various systems of transformations of functions such as rings of functions and infinite dimensional vector spaces of functions. The coherence of the function Schema consists of the recognition that any function has a domain set, a range set and a process that transforms objects in the domain set to objects in the range set (Dubinsky and McDonald 2001).

Derivative At the Trans-derivative stage, the student synthesizes problems involving variation. For example, a student can relate the derivative as the slope of a tangent line at a given point, with the rate of change of a function at a given point. The student can also construct transformations among different representations of the derivative. An individual demonstrates coherence by determining conditions for differentiability in terms of the constructions involved in its definition (Baker et al. 2000).

Chain Rule At the Trans-chain rule stage, the student can relate function composition to differentiation and recognize that various instantiations of the chain rule follow from the same general rule through function composition. The components of the Schema progress from being described as a list of “inner-outer” algorithms to a single rule, $(f \circ g)'(x) = f'(g(x))g'(x)$, that can be applied to different situations. The ability to grasp this general principle indicates coherence (Clark et al. 1997; Cottrill 1999).

In the following excerpt (Cottrill 1999, p. 49), Jack gives evidence of having constructed the chain rule Schema at a Trans-stage. This is exemplified by his ability to group all the differentiation problems in order of difficulty using the chain rule as a criterion and to distinguish among different instances of the chain rule such as the power rule, the exponential rule, and rules for trigonometric and implicit functions. He also describes the chain rule in terms of functions and their compositions instead of making reference to external features such as parentheses. In this excerpt, Jack uses the Leibnitz rule in an integral problem. Although he needs some help, he is able to work on the original problem and to generalize the chain rule to arrive at the solution.

Jack: 10, I would probably put, I don't know, I would be tempted to put 10 in a group by itself just because with it just being e it's basically the chain rule, and the chain rule and the power rule together. That's it. That one is done. Um, 7 and, let's see 7 requires it, 7 requires product rule and chain rule, and the all-powerful power rule. . . 8 just requires product rule. I know, no I probably would just go ahead and group 4, 5, 8, & 9 all together because they have the trig function and because I mean, the trig functions are the only derivatives that can throw you off real easy if you don't know them, because chain rule, if you understand chain rule and you understand the use of things like exponentials and logarithms then you aren't gonna get messed up bad on chain rule. You aren't gonna get messed up bad on the power rule if you know simple mathematics. You're not gonna get messed up too bad on product rule as long as you remember to keep everything straight. But, the trig functions, you know, you got, I can't even think of them at the moment. I haven't used them in a while, but it's like you know one it's the other and it's just negative and one it's just the other, period. And it's like if you forget that sine, the derivative of sine just by whether or not it's got a sign change in it, you just messed yourself up big time and you're gonna get a wrong answer because they cycle; and if you start off on, off with the wrong derivative of it then you have messed up the cycle already. Then, no matter, if you know all the others you are gonna be messed up anyway. So. . .

Jack: Um, huh, [pause] let's see how would I write that down? OK. Let's see. . . [pause—writing] Um, that's just the way that it runs through in my head. [had written $f(g(x))' = f'(g(x)) \cdot (g'(x)) \leq (g(x)) \leq x'$] Without any words.

I: OK.

Jack: That's just the way that I think of it.

I: OK. Read that for the tape.

Jack: Um, you take the derivative, essentially, whenever you use the chain rule you are essentially looking at a function that has got another function within it um, I don't know, it's sort of like doing a, handling a composite, um and in order to take the derivative of that composite you have to first take the derivative of the outside function and not even do anything with what's the inside of it, the function that's on the inside of it, you take the outside, it's derivative first and leave the inside function alone and then multiply that by the derivative of the inside function, and then multiply it by the derivative

of the variable, or however many times you have to break it down. Because you can have a huge function that's got a lot of stuff inside of it and you'd have to do the chain rule several times to get the x variable.

I: OK.

Jack: So, I mean you could have, that's just like a simple composite f of g , but you could have, if you have like $h(f(g))$ then you'd have to do the derivative h with f of g inside of it and then the derivative of f with g inside it and then the derivative of g and then derivative of x .

I: OK.

...

I: Can you work on this problem? [Jack was given a Leibniz rule problem with a monomial integrand.]

Jack: This, yes. I can integrate this functions [wrote the integral]

I: Can you label the function $H(x)$?

Jack: What do you mean?... this?

I: Yes

Jack: So this is $H'(x)$ (writes $H'(x) = 4\sin^2(x) \cos(x)$). So I can integrate. . . and this is $H(x)$. And I think this is the solution.

Jack applies the chain rule in various problems, including implicit functions and derivatives of integrals. In the excerpt, he provides evidence of having constructed all the elements of the chain rule definition.

Table 7.1 summarizes each of the triad stages for each of the examples discussed above.

At each stage of the triad, the student reorganizes knowledge acquired during the preceding stage. The change from one stage to the next includes not only an increase in the elements of the Schema but the construction of new forms of relations or transformations among the elements of the Schema. As is evident from the chain rule, students at the Intra-chain rule stage are able to find derivatives of composed functions by following specific rules. The change from the Intra-stage to the Inter-stage involves a shift of thinking that includes not only adding new instances of the chain rule but also constructing a transformation which enables the subject to see commonality among those rules, that is, to see those rules as specific cases of a more general phenomenon. The change from the Inter-stage to the Trans-stage involves the recognition of a single rule called the chain rule that applies to any differentiable composition of functions.

The triad provides researchers with a tool with which to analyze students' thinking and to see how it develops, taking into account the richness of problem situations by focusing attention on relationships among different mental constructions. It utilizes complexities involved in problem solving, how new relationships among ideas emerge, which relationships play an important role in newly formed structures, and development of coherence of the Schema. All of these important aspects of a Schema are shown through students' work in different related problem-solving situations.

Table 7.1 Comparison of triad stages for different Schemas

Concept	Intra-	Inter-	Trans-
Function	Tendency to focus on a single function and the various activities that can be performed with it.	Possibility to find analogies among arithmetic operations as Processes on different types of functions or differences involved in composing them as Processes. Construction of a relation among all of these individual types of functions as instances of the same sort of activity, as a means of constructing new functions from known ones.	Construction of various systems of transformations of functions such as rings of functions and infinite dimensional vector spaces of functions. The coherence of the Schema is demonstrated by the recognition that any function has a domain set, a range set, and a process that transforms objects in the domain set to objects in the range set.
Derivative	Interpretation of the derivative as the slope of the tangent line at specific points and to perform Actions or Processes to find the derivative of a function on specific intervals to determine if the function increases or decreases on those intervals. Ability to solve some types of rate-of-change problems. The focus is on individual Actions or Processes, so the description of the behavior of a curve and the solution of rate-of-change problems are only related by the need to find the derivative of a function.	Ability to relate the Process of the derivative as the slope of the tangent line and the Process of the derivative as the rate of change at a given point. Construction of a relation between them in order to consider the derivative as a means to describe local variation of the function.	Construction of a synthesis where all the problems in which variation is involved, such as slopes of tangent lines to a function at a given point or rates of change of a function at a given point, are related to the derivative. Ability to construct transformations among different representations of the derivative. Coherence is demonstrated by the ability to determine conditions for differentiability in terms of the constructions involved in the definition of derivative.
Chain rule	The student has a collection of rules to calculate some individual cases where the chain rule is used implicitly, such as the power rule or the general formula, but does not see the rules as being related.	Recognition that different instances of the chain rule such as the power rule or the derivative of a composition of functions represent something more general. Awareness that special instances of the chain rule are related by a general rule where the “outer” part of the composed function is derived and then multiplied by the derivative of the “inner” part of the composition.	Ability to relate function composition to differentiation and to recognize that various instantiations of the chain rule follow from the same general rule through function composition. Coherence is demonstrated by the ability to describe the elements in the schema by a single rule, $(f \circ g)'(x) = f'(g(x))g'(x)$, that can be applied to different situations.

7.5 Assimilation of New Constructions into a Schema

Construction of knowledge is a dynamic process. As individuals face new situations, previous knowledge can be reconstructed and new knowledge can be constructed. The notion of a Schema helps researchers understand the dynamism associated with these changes by means of different mechanisms: as new Actions, Processes, and Objects related to a mathematical concept or topic are constructed, new relations with previously constructed concepts are also established. New Actions, Processes, Objects, or Schemas can become part of a previously constructed Schema, or assimilated by a Schema that has thereby been reconstructed.

For example, an individual's function Schema may include the definition of a single variable function in terms of domain, range, and an idea of how domain elements are assigned to range elements. When studying multivariable functions, the student may assimilate these new functions as Processes into her or his previously constructed function Schema. At the same time, accommodation and re-equilibration are likely to occur as the notion of domain is extended by coordinating intervals of real numbers with regions in R^n . The study of linear transformations may lead to further reconstruction of an individual's function Schema as the conception of both domain and range is expanded to include vector spaces as domain and range sets. In this act of accommodation and re-equilibration, the student learns to differentiate among different types of functions and to integrate new kinds of functions into her or his Schema structure. As a result, her or his knowledge will grow.

In the case of the derivative Schema, work with partial derivatives may not only foster the construction of a new Schema to deal with derivatives of multivariable functions but may also lead to the development of relations between different types of functions and their derivatives. This assimilation results in an expanded Schema that can be brought to bear on a wider range of problem-solving situations that involve both single and multivariable functions.

7.6 Interaction of Schemas

In the process of learning, as knowledge develops, an individual may construct coexisting Schemas that are constantly changing and at varying stages of development. Each Schema is itself made up of Actions, Processes, Objects, and other Schemas and the relationships among those structures. When facing a problem-solving situation, a person may need to coordinate different Schemas. One goal of research is to identify the different Schemas that need to be developed and how they are coordinated or how they interact. Therefore, in understanding the development of a Schema, research must not only determine how the Schema is constructed but how it may be coordinated with other related Schemas. An example of this relationship is considered in Sects. [7.6.1](#) and [7.6.2](#).

7.6.1 Two Studies of Students' Calculus Graphing Schema

Baker et al. (2000) and Cooley et al. (2007) described students' attempts to solve a non-routine calculus graphing problem in terms of the interaction of two Schemas. In the first study, Baker et al. (2000) investigated how students coordinated information regarding the first and second derivatives, continuity, and limits to sketch the graph of a function. The following problem was given to the students during an interview:

- (a) Sketch the graph of a function that satisfies the following conditions:

h is continuous;

$$h(0) = 2, \quad h'(-2) = h'(3) = 0, \quad \text{and} \quad \lim_{x \rightarrow 0} h'(x) = \infty;$$

$$h'(x) > 0 \text{ when } -4 < -2 \text{ and when } -2 < 3;$$

$$h'(x) < 0 \text{ when } x < -4 \text{ and when } x > 3;$$

$$h''(x) < 0 \text{ when } x < -4, \text{ when } -4 < x < -2, \text{ and when } 0 < x < 5;$$

$$h''(x) > 0 \text{ when } -2 < x < 0 \text{ and when } x > 5;$$

$$\lim_{x \rightarrow -\infty} h(x) = \infty \text{ and } \lim_{x \rightarrow \infty} h(x) = -2.$$

- (b) Do there exist other graphs besides the one you drew that satisfy the same conditions? Justify your response.
- (c) If we remove the continuity condition, and the other conditions remain, does the graph change? In what way? Do other possible graphs exist? If other graphs exist, could you sketch one example?

In their attempts to deal with the problem-solving situation, students had several difficulties: they tended to work on each of the given intervals of the domain in isolation, they failed to connect Processes related to different properties, and they could not coordinate Processes for the properties across different intervals. Taken together, these difficulties kept the students from drawing a correct graph of the function. In their analysis of the data, the authors uncovered the interaction of two Schemas as the source of the difficulties. They called these Schemas the interval Schema and the property Schema. The authors developed genetic decompositions for the Schemas and their interaction, which resulted in a new Schema they referred to as the Calculus Graphing Schema.

In the second study, Cooley et al. (2007) used the same genetic decompositions to determine whether successful calculus students' made the same mental constructions while working with a series of problems that included the former

one with new problems added. This offered the opportunity to examine the way students used their knowledge when they encountered problems posed in different representational contexts and to analyze their ability to access and use the different parts of their Calculus Graphing Schema. The authors were interested in knowing how the students would apply and/or reconstruct and coordinate the interval and property Schemas when the original problem was modified. The students were presented with tasks that increased in difficulty and that differed from the types of tasks generally asked in calculus courses. The purpose of assigning these tasks was to see how students would deal with the mental structures they had constructed when encountering new problem-solving situations.

7.6.2 The Development of the Calculus Graphing Schema

The researchers found that differences in students' difficulties and performance could be attributed to the students' abilities to coordinate the property and interval Schemas. The genetic decomposition for each of the Schemas follows:

7.6.2.1 Development of the Property Schema

Intra-Property Stage: Focus on Actions or Processes corresponding to one property of the function in isolation from other properties. Recognition that there are other properties, but the Processes involved in them are not coordinated into a single Process in terms of the graph.

Inter-Property Stage: Construction of relationships among some of the Processes related to the properties of the function and transformation of these Processes into those corresponding to the graph of the function.

Trans-Property Stage: Awareness of the transformations involved in the coordination of all the Processes related to the analytic conditions with the graphical properties of the function in an interval. Coherence of the Schema is demonstrated by recognition of the aspects of the graph that may be included and the coordination of all the properties that lead to a correct graph of the function on the given interval.

7.6.2.2 Development of the Interval Schema

Intra-Interval Stage: Focus on properties of the function as Actions or Processes on isolated intervals. The coordination of the Process or Processes related to properties of the function over contiguous intervals has not been constructed.

Inter-Interval Stage: Grouping of contiguous intervals as a union of sets that is part of the domain of the function.

Trans-Interval Stage: Relation of all the intervals through intersections and unions to form the entire domain of the function. Coherence of the Schema is demonstrated by the ability to describe which behaviors of the graph are allowed by the overlap and connection of the intervals and which are not.

When these two triads are combined in a double triad, it is possible to analyze the interaction of both Schemas and to describe a single Schema that can be called the Calculus Graphing Schema. Its development can be described as follows:

At the Intra-property, Intra-interval stage, the focus is on one or a few isolated Actions on the given properties of the function on isolated intervals. The Actions result from the relation of a single property to isolated intervals of the graph.

At the Intra-property, Inter-interval stage, the focus is on one or a few isolated Actions on the given properties of the function on contiguous intervals seen as a union of sets that are part of the domain of the function.

At the Intra-property, Trans-interval stage, the focus is on one or a few isolated Actions on the given properties of the function across the domain of the function.

At the Inter-property, Intra-interval stage, some Processes related to the properties of functions have been constructed, but the focus is on one or a few isolated Actions of the given properties of the function on isolated intervals. The Actions result from the relation of a single property to isolated intervals of the graph.

At the Inter-property, Inter-interval stage, some Processes related to the properties of functions have been constructed on contiguous intervals of the domain of the function that are seen as being related.

At the Inter-property, Trans-interval stage, some Processes related to the properties of functions have been constructed across the domain of the function.

At the Trans-property, Intra-interval stage, there is awareness of the transformations involved in the coordination of all the Processes related to the analytic conditions of the function with the graphical properties of the function in an interval, but the focus is on one or a few isolated Actions of the given properties of the function on isolated intervals. The Actions result from the relation of a single property to isolated intervals of the graph.

At the Trans-property, Inter-interval stage, there is awareness of the transformations involved in the coordination of all the Processes related of the function with the analytic conditions of the function with the graphical properties of the function in an interval and relations have been constructed between contiguous intervals of the domain of the function.

At the Trans-property, Trans-interval stage, there is awareness of the transformations involved in the coordination of all the Processes related to the analytic conditions to the graphical properties of the function in an interval across the entire domain of the function.

The first study (Baker et al. 2000) revealed a wide variety of differences among students, and the genetic decomposition proved to be a useful tool to describe in detail students' constructions. In the second study (Cooley et al. 2007), the usefulness of the genetic decomposition was again proved, and more students showed evidence of effectively describing relationships and succinctly explaining their

reasoning, showing that they had constructed a graphing Schema at a Trans-property, Trans-interval stage. In both studies, examples of students at each of the stages of the Calculus Graphing Schema, except for the Trans-property, Intra-interval stage, were found. The following excerpts illustrate some of the students' responses.

One of the students, Carol, graphed the function mainly by using information from the first derivative. Near the graph, she constructed a table with the meaning of the signs of the second derivative but did not use it. She described her graph as follows:

Carol: Okay...it's increasing from negative 4 to 3...it doesn't say what it is past 3. Let's see...it's increasing because it is concave up. So, that's a point of inflection, and then past this point it's all down. And there is a local min at negative 2 because it's a limit.

When trying to coordinate two conditions on a single interval, Carol said, "I don't understand how this [the graph to the right of $x = 3$] can be decreasing when this [the graph to the right of $x = 5$] is concave up."

Even while discussing concavity and inflection points, she did not relate these properties to the graph and could not coordinate them. Her understanding of the first-derivative also allowed her to integrate the inflection point at $x = -2$ and the vertical tangent into the graph...

I: Okay, now limit as x goes to 0 is infinity. Does that satisfy ... ?

Carol: Well, that means the slope would go to infinity...

She then sketched a vertical line segment along the y -axis, demonstrating that she was using the calculus graphing Schema at the Intra-property, Inter-interval stage. (Baker et al. 2000, p. 579)

John gave evidence of operating at the Inter-property, Inter-interval stage of the Calculus Graphing Schema since he showed difficulties in coordinating the properties on some of the intervals as well as difficulties in coordinating the properties across the intervals:

John: Okay ... From negative 4 to negative 2 the slope's going to be positive, too, so this [the graph] will go like that [in a positive slope direction]. And from negative 2 to negative 3... see, that's where I was getting mixed up because if as it [the limit of $h'(x)$ approaching infinity as x approaches 0] goes to 0, the slope's going to infinity.

I: Right.

John: But what happens on this [the right] side of 0?

I: Right.

John: So, unless we just forget about that, you know it's [the graph] just gotta keep increasing ... and so forth, up to 3. And then it just kind of turns at 3... up to 5, because that's when it'll switch again, because of concavity here [at $x = 5$].

I: Okay.

John: Because, wait, it can't. All right, um, Okay. I've got a question for you, all right?

I: Hold on ...

John: All right? And then according to this [$h''(x)$ changes from negative to positive at $x = 5$], it's [the graph] going to switch the other way, all right? But the limit as it approaches infinity is negative 2.

I: Right.

John: And then at 5 it switches and then it just kinda approaches negative 2 like that. Something like that [constructs a concave-up graph dipping below the horizontal asymptote at $y = -2$].

- I: So now, does it dip down below negative 2 then?
- John: I think it has to, because if it's concave up when x is greater than 5, it's gotta keep rising, at least a little bit.
- I: Does it?
- John: Because if it was, well not to approach, to approach negative 2, it would have to go like this [sketches a horizontal line just above $y = -2$] and this would be a straight line.
- I: Could it be concave up and still decreasing?
- John: Probably. I'm sure it probably could.
- I: Okay.
- John: I just can't think of how it would be.

John connected the conditions at $x = -4$ with the graph, but he could not think of the graph as having a cusp. He drew a smooth graph that otherwise had the required properties. (Baker et al. 2000, p. 281)

Another student, Stacey, used mainly the first derivative, although she wrote notes about the concavity of the graph and used those notes as she sketched the graph. Although she worked with the union of the intervals across the domain, she needed considerable help to coordinate the limit condition and encountered difficulty in considering the conditions at $x = -4$ (Baker et al. 2000, p. 583):

- Stacy: So, how do we get [the graph] from coming in decreasing to going increasing [at $x = -4$] without a horizontal tangent? And it is continuous, so it can't do one of these things ... that point thing [cusp].
- I: Well, would that not be continuous if you had a point like that?
- Stacy: Okay. So it's ... I don't know what I was thinking. [The graph is] smooth, that's what I was thinking.

Although she was able to coordinate most of the properties, she was not able to coordinate the information at $x = 0$ and $x = -4$. She was also unable to transfer information to the graph despite coordinating properties verbally. As a result, she was deemed to be at the Inter-property, Trans-interval stage of Schema development. This is illustrated in Fig. 7.1.

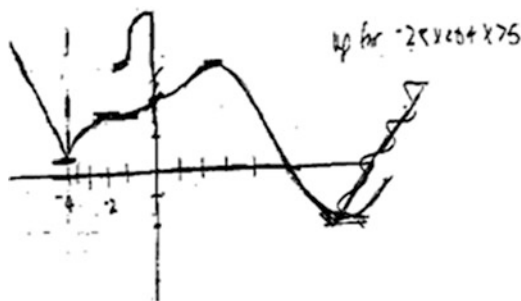


Fig. 7.1 Stacy's sketch of the graph of the function (Baker et al. 2000, p. 605)

This example shows how two Schemas interact in the construction of knowledge and demonstrates how the genetic decomposition of a Schema can be a very flexible tool to support the study of the specificities involved in the learning of concepts. This tool can also help in the design of teaching sequences that may help students overcome difficulties and develop coherent knowledge. The scope of a subject's

mathematical knowledge can be related to the development and interaction of different Schemas and her or his ability to construct new relations among different mathematical structures. The construction of Objects from Schemas is another important feature of mathematical understanding and is described in the next section.

7.7 Thematization of a Schema

Piaget spoke about thematization in several of his books. For example, he discussed thematization when talking about reflective abstraction and how “actions and operations become thematized objects of thought” (Piaget 1975/1985, p. 49). In his work with García (1983–1989), he introduced the notion of thematization of a schema:

Abstract mathematical notions have in many cases first been used in an instrumental way, without giving rise to any reflection concerning their general significance or even any conscious awareness of the fact that they were being used. Such consciousness comes about only after a process that may be more or less long, at the end of which the particular notion used becomes an object of reflection, which then constitutes itself as a fundamental concept. This change from usage or implicit application to consequent use, and conceptualization constitutes what has come to be known under the term thematization. (p. 105)

According to Piaget and García, the development of a schema is a slow process in which the individual becomes aware of its components and their relations. For some time, the individual can use a schema to solve some problems without the need to reflect on its components and the relations among them. Eventually, the individual is able to reflect on the meaning of the components and relations that make up the schema and is able to perform conscious actions on it. When this happens, Piaget and García consider the schema to have been thematized. In this sense, thematization is the mechanism by which a schema is consciously used in the solution of problems (e.g., in Piaget and García, 1983/1989, pp. 65, 113).

In APOS Theory, thematization is associated with Schema development and its meaning in a somewhat different way than the description given by Piaget and García. In early APOS papers, thematization was described as one of the six kinds of reflective abstraction (Asiala et al. 1996; see Chap. 3). In Clark et al. (1997), it was described in the following way: “We consider a schema to have been thematized if the individual can think of it as a total entity and perform actions on it” (p. 353). Later thematization was seen as the mechanism responsible for transforming a Schema into an Object (Czarnocha et al. 1999; Asiala et al. 1997a). This same idea is expressed in Sect. 2.2. Finally, in trying to clarify the APOS meaning of thematization, some specific research projects have asserted that thematization occurs when Actions can be performed on a Schema, such as operating with it, comparing it with another Schema or, as it has been described before (Clark et al. 1997), when it can be decomposed to recover its components (de-thematized), and/or to make the necessary Actions and Processes to reconstruct it when conditions of the problem situation are changed. In this latter case,

reconstruction involves comparison of Schemas with the same components but different relations among them (Cooley et al. 2007). While not all these descriptions are the same, it can generally be said that thematization is the mechanism by which a Schema is transformed into an Object so that it is possible to perform Actions on it or to apply Processes to it.

There is only one APOS Theory study focusing on thematization of a Schema (Cooley et al. 2007). In that study, college students who succeeded in their study of calculus were interviewed to determine whether they had thematized their Calculus Graphing Schema. The researchers examined how students constructed relations among properties of functions such as first and second derivatives, limits, and continuity and how they related these properties to the graphs of functions. Nine problems were used in the interview. The analysis of the data focused on students' coordination of the different properties and intervals to describe possible mental groupings within their Schemas and to determine their ability to access parts of the Schema when called upon. In the last problem, shown below, students had to reconsider the calculus graphing problem described in Sect. 7.6.2 when various conditions were changed:

Problem 9:

- (a) Sketch the graph of a function that satisfies the following conditions:

h is continuous;

$$h(0) = 2, \quad h'(-2) = h'(3) = 0, \quad \text{and} \quad \lim_{x \rightarrow 0} h'(x) = \infty;$$

$$h'(x) > 0 \text{ when } -4 < x < -2 \text{ and when } -2 < x < 0$$

and when $0 < x < 3$;

$$h'(x) < 0 \text{ when } x < -4 \text{ and when } x > 3;$$

$$h''(x) < 0 \text{ when } x < -4, \text{ when } -4 < x < -2, \text{ and when } 0 < x < 5;$$

$$h''(x) > 0 \text{ when } -2 < x < 0 \text{ and when } x > 5;$$

$$\lim_{x \rightarrow -\infty} h(x) = \infty \text{ and } \lim_{x \rightarrow \infty} h(x) = -2$$

- (b) Do there exist other graphs besides the one you drew that satisfy the same conditions? Justify your response.
- (c) If we remove the continuity condition, and the other conditions remain, does the graph change? In what way? Do other possible graphs exist? If other graphs exist, could you sketch one example?

- (d) If we remove all of the first derivative conditions, and the other conditions remain, does the graph change? In what way? Do other possible graphs exist? If other graphs exist, could you sketch one example?
- (e) If we remove all of the second derivative conditions, and the other conditions remain, does the graph change? In what way? Do other possible graphs exist? If other graphs exist, could you sketch an example?

The solution of this problem involves the coordination of several properties that change in overlapping intervals. The required coordinations, shown in schematic form, appear in Fig. 7.2.

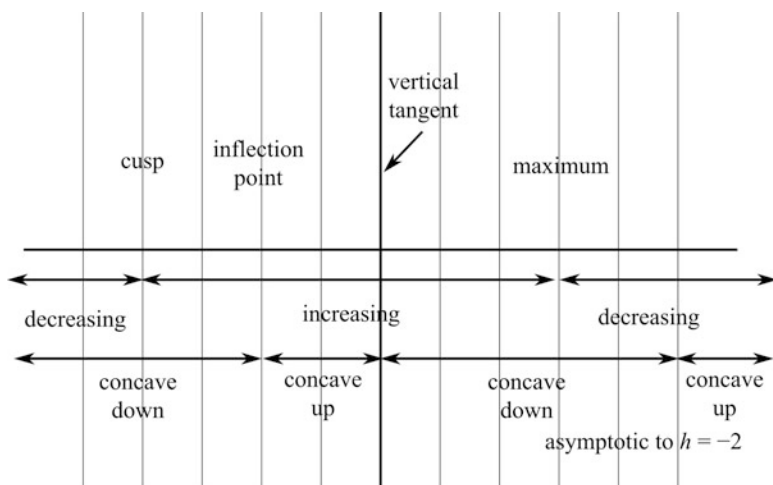


Fig. 7.2 Demonstration of the coordinations needed in the solution of the problem

The figure delineates the interaction and overlap of the various properties across the intervals of the domain needed to sketch the graph.

For students who were successful in solving the problem, the researchers determined whether they accessed the necessary parts of the Schema in a flexible way, and adapted to the demands of the specific problem situation. Once the researchers determined that some of the students could be classified as operating at what they called the Trans-property, Trans-interval stage of the Calculus Graphing Schema, they considered whether thematization of the Schema had occurred. They proposed that those students who had thematized the Schema should be sufficiently conscious of the structure of the Schema that they could reflect and act upon it while solving the given problem. To determine if this was indeed the case, the researchers asked the students to reconsider the solution to all the parts included in Problem 9. In their analysis, the researchers focused their attention on students' abilities to determine which properties of the graph would change and which would remain invariant.

In general, thematization of a Schema is indicated by an individual's awareness of the global behavior of problems related to the Schema, flexible use of it in

different situations, and the ability to perform conscious Actions on it. In the study considered in this section (Cooley et al. 2007), the researchers considered that thematization had occurred among those students who could demonstrate awareness of the global behavior of a function over its domain in terms of all the properties across all of its intervals, that is, among students who showed that they had constructed a coherent graphing Schema, as described in Sect. 7.6. They also considered that thematization had occurred if a student could decompose it into its components and analyze the relations among them to discern which of them were relevant to the solution of the problem and to reconstruct the Schema to be used as a totality² for the required purpose.

As the properties of the function differed from the problem posed during the interview, it was agreed that a student who had thematized the Schema should give evidence of having coordinated all given properties across all intervals of the function. According to the authors, the ability to make this coordination in lieu of changes made to the properties of the function demonstrated conservation in their understanding. For the students who demonstrated such evidence, the authors asserted that the Calculus Graphing Schema had become a fundamental part of the students' understanding and could be viewed as an Object, that is, the Schema had been thematized.

In this study, the focus was on finding students who had thematized the Schema and not necessarily whether they viewed it as a totality. More research is needed to understand the differences between these two concepts. In the next set of excerpts, Susan shows a Trans-interval, Trans-property level of Schema development but fails to achieve thematization. In the following passage, Susan gives evidence of coordinating all the given conditions across the intervals and of synthesizing the different transformations required to graph the function. However, when asked about the function's behavior if the continuity property is removed, she has difficulty considering which properties of the graph would remain invariant and which would change. Although she is conscious of her Calculus Graphing Schema, her inconsistency in performing Actions on the Schema suggested that thematization has not yet occurred.

We can observe this difficulty when Susan worked on the following problem (Cooley et al. 2007, p. 10).

Problem 7c.

Sketch the graph of a continuous function that complies with the following conditions:

its domain is -1 to 1 ,

²The term totality was used as part of the encapsulation of a Process into an Object before it was proposed as a new stage in Dubinsky et al. (2013). The former is the meaning of its use here. In Chap. 8, this term is used differently as a possible new stage in APOS Theory between Process and Object.

it is increasing on $(-1, 0)$, decreasing on $(0, 1)$,

concave down on $\left(-1, -\frac{1}{2}\right)$ and on $(0, 1)$, and concave up on $\left(-\frac{1}{2}, 0\right)$

What happens at the point $x = -\frac{1}{2}$ of the function?

What happens at $x = 0$?

Is this function unique? How could you change the graph if you were allowed to remove the continuity condition? Justify your response.

If other graphs exist, could you sketch an example?

I: And the continuity condition? If you remove it, what happens?

Susan: I remove it and... well... in order for it to be discontinuous I can take off the definition on this interval, the one at the middle. So it has a jump and it's not continuous and it's defined on the other two intervals of the domain.

I: Well, I think that the question is not clear. The idea is that everything remains the same, I mean the domain where it is defined, the function, and also its properties, where it increases or decreases, and so on. The only thing that can change is that it does not have to be necessarily continuous like the one you drew here.

Susan: Oh! Now I understand, then the other thing I said is also wrong. That is, the function is like this one, that I drew here and it's one function, not three. Of course, it's defined by parts and what happens, let's see, well it's still not unique, but it is so because we don't know the values it takes at each point. We only know that it grows, or decreases, and so on. Then the graph could be up here or down here and there could be a lot of graphs, a lot of functions that satisfy those conditions. And if you remove the continuity condition, well, I don't know. Let's see, I think that at $-\frac{1}{2}$ and at 1, at those points there are those sharp points and we don't know what happens, we only know what happens in the open intervals. Yes, I think there can be little holes there, aren't there? The rest... but... its domain is from -1 to 1 . That means it is defined and if there is a hole it would not be defined. Oh well, it would be a little weird, but it could have a little hole in those points and those points can be defined up here or down here so that there is not a jump. I think so. I think that's right. In reality I don't know. I don't understand very well. This is difficult, isn't it? Yes, well, I don't know, I better continue and do number 8, Okay?

I: Yes, if you prefer to, but why don't you think a little bit more on this one?

Susan: The problem is I don't know how. Because what I do know is that at these points there is a derivative there, but at these other points there is not and...

I: The fact that the derivative is defined there, does that tell something to you?

Susan: Something? Yes, well about continuity, there is a theorem, but I can't remember it. Well, I think I better do the next one.

I: Alright.

On Problem 9, Susan constructs an accurate graph (Fig. 7.3) and describes coordination of properties across intervals. However, she again struggles to integrate the properties across the intervals when some conditions were removed. Taking into account all her responses, the researchers concluded that even though she demonstrated a Calculus Graphing Schema at the Trans-interval, Trans-property level, she was unable to perform the Actions needed to break the Schema into its components and to distinguish between the aspects of the graph that remained invariant. As a result, the researchers decided that her Schema had not been thematized.

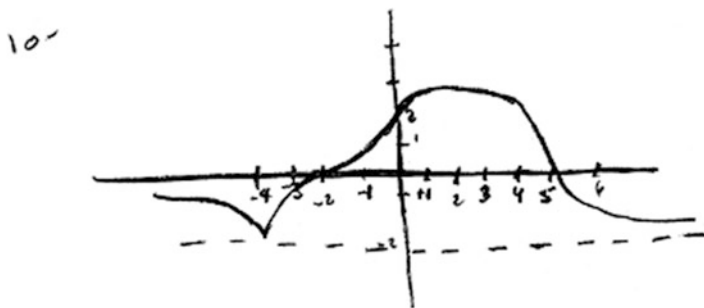


Fig. 7.3 Susan's work for Problem 9 (Cooley et al. 2007, p. 11)

Only one of the 28 interviewed students, Clara, gave evidence of thematization. She was able to describe which properties of the function remained unchanged when changes to the problem were introduced. She explained in detail the effects on the graph resulting from each change, that is, she showed that the Schema was an Object to her, as can be seen in the following excerpts from her interview (Cooley et al. 2007, p. 13):

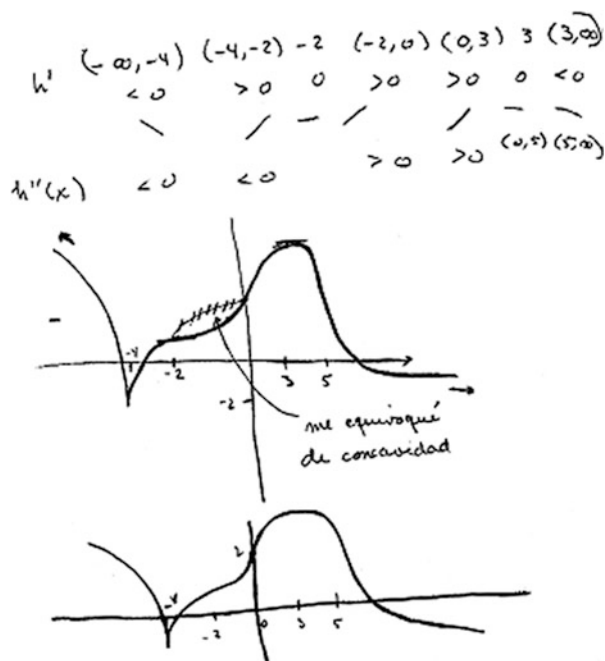
On Problem 7 described above, she responds to the removal of continuity saying:

- I: ... Then what happens if you can remove the continuity condition?
- Clara: If the continuity condition is removed at, for example, at $x = 0$, the function will still be a function that does not have derivative at 0, like this for example [sketches a possible graph around the point], or it may be like this other one, too [draws another possible graph around the point], because there would not be a derivative at $x = \frac{1}{2}$, or $-\frac{1}{2}$. Then we would not be able to say that it is an inflection point because the second derivative would not exist either.
- I: Is there a condition that says that the derivative does not exist at $-\frac{1}{2}$?
- Clara: Yes, oh no, there is no condition, but if you remove continuity you can break the curve at those points where there is no derivative and then. . . oh I see, I am wrong there, because at that inflection point the first derivative has to be something, for example 0, and then it has to be continuous. It is only at 0 where there can be a change for the function if the continuity is removed.

Clara explained which properties remained fixed and the intervals to which those properties applied. She was able to remove any condition and flexibly reassemble the information into a new graph and explain why the resulting graph is consistent with the given information. This is exemplified by her work on Problem 9:

- Clara: No it's not unique (referring to the graph of the function), but all the graphs have to have the same basic shape as this one. . .
- I: And what happens if we remove the continuity?
- Clara: Then it can change. The interval is open at -2 . We know the derivative is zero and therefore the derivative exists. So it means that the function must be continuous there [at $x = -2$], and the same happens at 3 and at $x = 0$ because of the limit condition. Because the limit of the derivative has to be infinite, we can have a discontinuous function there [at $x = 0$]. It can have an asymptote there [$x = 0$] but still h of 0 has to be 2 and the function has to grow on both sides of 0. So the conditions are satisfied.

Fig. 7.4 Work illustrating Clara's thematized schema (Cooley et al. 2007, p. 15)



I: Is x equal to 0 the only point where you can break the graph if the continuity condition is removed?

Clara: Let me see, no I skipped -4 . I don't have any condition for the derivative there [at $x = -4$] and so the graph can also be broken there. There can also be an asymptote there. (Cooley et al. 2007, p. 14).

Clara's ability to break the problem into its parts and to introduce changes dictated by the new conditions is illustrated in Fig. 7.4. She provided reasons for her decisions and showed that she was able to differentiate those parts of the Schema needed in each situation and to integrate them back into a new graph (Cooley et al. 2007, p. 16):

I: Now if we have a continuous function and we remove the conditions on the first derivative, what happens?

Clara: If we remove the conditions on the first derivative. Let me see, we still have $h(0)$ is 2 and the conditions on the second derivative are the same, the conditions on h double prime mean that at $x = -4$ the derivative is not defined, but without the conditions on h prime the function can be decreasing in the interval from -4 to -2 , and also we don't have to have the maximum at 3 and we can have a graph like this one.

I: Okay, now the last part, imagine now that the conditions on the second derivative are the ones that are removed. What would happen to the graph of the function?

Clara: Now if we don't have the conditions for the second derivative we still have the value for the function at 0 and the limit of the derivative at 0 and then the function decreases from $-\infty$ to -4 and decreases from, no, increases from -2 to 0 and from 0 to 3 and decreases from 3 to infinity, but approaches $y = -2$ at the right end of the graph. There is an asymptote there [at $y = -2$], so we could have a graph that looks like this. There can

be changes in the concavity between -2 and 0 and between 0 and 3 . Only the first derivative there will not be zero and also there has to be an inflection point because of the asymptote there at the right end of the graph but if we don't have those conditions, the change in concavity can be at any place to the right of 3 , even at 3 . Well, not at 3 exactly because there the derivative is zero and it decreases after 3 , but to the right it is possible that there is an inflection point. Well, yes, $x = 5$ does not have to be an inflection point and $x = -4$ is not necessarily a cusp. And also, there can be other inflection points and still satisfy the other conditions, this gives you more freedom to change the graph. For example, at $x = 5$ we don't have any restriction now.

Clara demonstrated that the relations among the concepts in her Schema were stable and that the different parts of her Schema could be accessed and reevaluated appropriately. She was able to act on the Schema as an Object. In her explanations, she demonstrated conscious control of the result of those Actions. The researchers considered this to be clear evidence of thematization.

The fact that it was possible to find a student who thematized the Calculus Graphing Schema provides evidence that thematization of a Schema is possible. It also shows that it is possible to find evidence of students' conscious and flexible use of mathematical knowledge, although more research is needed to see how prevalent this level of understanding is.

The only student in this study who showed evidence of thematization of the Graphing Calculus Schema had already taken three calculus courses and an analysis course. This fact may demonstrate that the thematization of a Schema takes time and that many opportunities for reflection as well as good instructional strategies are needed to accomplish it. More research is needed, however, to back this conclusion.

Appendix: Problems for the Interview in the Chain Rule Study (Cottrill 1999)

Compute the derivative of each of the following functions. Show all your work.

$$11. f(x) = 11x^5 - 6x^3 + 8$$

$$13. h(x) = (x^2 - 3)$$

$$15. y = x^2 \sin(x)$$

$$17. G(x) = 2(5x^2 + 1)^4 - 4x(5x^2 + 1)^4$$

$$19. y = \cos^3(t)$$

$$12. g(x) = 3/x^2$$

$$14. y = 3e^x - 4 \tan(x)$$

$$16. F(x) = (1 - 4x^3)^2$$

$$18. H(x) = \sin(5x^4)$$

$$20. y = e^{-t^2}$$

Additional question for interview:

$$\text{Compute } F'(x) \text{ if } F(x) = \int_0^{\sin x} e^{t^2} dt$$

Chapter 8

Totality as a Possible New Stage and Levels in APOS Theory

The focus of this chapter is a discussion of the emergence of a possible new stage or structure and the use of levels in APOS Theory. The potential new stage, Totality, would lie between Process and Object. At this point, the status of Totality and the use of levels described in this chapter are no more than tentative because evidence for a separate stage and/or the need for levels arose out of just two studies: fractions (Arnon 1998) and an extended study of the infinite repeating decimal $0.\bar{9}$ and its relation to 1 (Weller et al. 2009, 2011; Dubinsky et al. 2013). It remains for future research to determine if Totality exists as a separate stage, if levels are really needed in these contexts, and to explore what the mental mechanism(s) for constructing them might be. Research is also needed to determine the role of Totality and levels for other contexts, both those involving infinite processes and those involving finite processes. It seems clear that explicit pedagogical strategies are needed to help most students construct each of the stages in APOS Theory and that levels which describe the progressions from one stage to another may point to such strategies. Moreover, observation of levels may serve to help evaluate students' progress in making those constructions.

Evidence for levels and Totality in the studies of the relation between $0.\bar{9}$ and 1 was mentioned in Chap. 5 and is discussed in this chapter; evidence for levels from the study of fractions is considered in Chap. 9.

This chapter contains three themes: a general discussion of the progression between stages in APOS Theory and related pedagogical strategies; a description of the terms *stages* and *levels* as they appear in the work of Piaget and in APOS-based research; and a summary of the research regarding $0.\bar{9}$ that suggested the need for levels between stages and for Totality as a new stage between Process and Object. Because the idea of Totality and related notions have appeared in previous research, both within and outside of APOS Theory, it is necessary to point out the differences between those usages and the proposed meaning of Totality. Included in this discussion is a review of the tentative nature of Totality as a stage and comments on the need for future research to determine its status.

8.1 Progression Between Stages

One major issue in APOS-based research is to understand the cognitive progression from one of the stages, Action, Process or Object, to the next “higher” stage. Very often, in learning a particular concept, a subject achieves the Action stage but shows difficulty in reaching Process, or reaches the Process stage but cannot progress to Object. How can instruction help students overcome these apparent obstacles? To answer such a question, it is first necessary to understand why the difficulty occurs. That is: How do the mechanisms that lead from one stage to the next (interiorization—from Action to Process and encapsulation—from Process to Object) function?

There are some “first-tier” answers to these questions. In the case of progression from Action to Process, an individual may fail to develop a Process conception because he or she has not yet successfully interiorized the Action. One powerful pedagogical strategy to help students make this mental construction, discussed in Chap. 5, is to have them represent the action as a computer procedure which accepts an appropriate input, performs the action on it and returns the result. Research has shown that writing these types of computer programs helps students to move from Action to Process (Weller et al. 2003). For children at the age of concrete operations (in the sense of Piaget, 1975), who may not yet be capable of writing such computer programs, the teaching sequence may need to start with an Action that can be imagined. The role of imagination in the interiorization of an Action is considered in Chap. 9.

In the case of progression from Process to Object, an individual may fail to progress to an Object conception of a Process because he or she has not successfully constructed and applied a transformation to the Process. Again, a strategy, also discussed in Chap. 5, exists with computer programming: once a Process has been represented as a computer procedure, the student can write computer code to transform the computer procedure in various ways (provided the programming language is sufficiently powerful). As indicated in Chap. 5, research has shown that this type of activity facilitates encapsulation of the Process into an Object (Weller et al. 2003).

But what about the substantial number of students who are not helped by these instructional treatments (Weller et al. 2003)? As indicated above, the search for an answer to this question must begin with an investigation of a previous question: Why do difficulties in moving from Action to Process, and from Process to Object, exist? The investigation into this latter question begins with an attempt to better understand the kinds of thinking that may be taking place as an individual tries, successfully or unsuccessfully, to progress from one stage to another. But before discussing investigations of progressions between stages, determination of what a stage is, what a level is, and the differences between them must be considered.

8.2 Stages and Levels

In his work, Piaget considered not only stages, but levels between stages. In the APOS-based studies of pre-service teachers’ understanding of the relation between 0.9 and 1 (Weller et al. 2009, 2011; Dubinsky et al. 2013), and of children’s

development of the concept of fraction (Arnon 1998), the data suggested the existence of levels between stages. In this context, a stage refers to one of the mental constructions of Action, Process, or Object, and a level denotes a developmental juncture between two of these stages. In considering both levels and stages, a full investigation of an individual's development of her or his understanding of a mathematical concept would include what it means to progress between stages, between levels, from a level to a stage, and from a stage to a level. Sections 8.2.1 and 8.2.2 contain a more detailed discussion of levels and stages as they appear in Piaget's work and in the studies of 0.9 and its relation to 1.

8.2.1 *Piaget's Work on Stages and Levels*

The following comments are based on the work of Piaget (1974/1976, 1975) and Dubinsky et al. (2013).

A stage cannot be skipped. If it is, the subject's understanding of the concept will lack coherence. Thus, stages are sequential, with each stage necessary for development of successive stages.

A level may or may not be reflected in the data of a specific subject. This is because the subject may be able to move to the next level or stage rapidly so that the level is skipped, done very quickly, or is not observable in the already acquired higher level or stage.

Stages are invariant over topics and are part of the general theory. Levels will be different for different concepts (Dubinsky et al. 2013). In many works, Piaget gave examples in which the development of different concepts gave rise to different levels.

The role of the level is to analyze, and provide mechanisms for, building the next level in a stage or the stage itself; this should be reflected in the definition of the level. According to Piaget, stages, together with their levels, are sequential, each contributing to the development of its successor. In particular, every level contributes to the development of the following stage.

8.2.2 *Levels in APOS Theory*

The use of levels in APOS-based research in the study of fractions (Arnon 1998) and infinite processes (Weller et al. 2009, 2011; Dubinsky et al. 2013) is quite consistent with what Piaget wrote, as described in the above summary. In her study of fractions, Arnon investigated the learning of several fraction concepts. For these concepts she found, within the progression from Action to Process, what she called "more subtle distinctions". These constitute what she referred to as a "continuum", and can be considered to be levels within the progression from the

Action stage to the Process stage. (See Chap. 9 for more details on these studies). On the other hand, in the studies related to 0.9, different levels were found between all of the different stages.

The levels defined in APOS-based research, as discussed here and in Chap. 9, are content specific and arose from the interview data. Hence, there is no expectation that the levels reported in Dubinsky et al. (2013) and Arnon (1998) will necessarily be found in studies of other topics, even those involving infinite mathematical processes or fractions. This aligns with Piaget's ideas above and with his other work. Stages, as cognitive developments of knowledge of specific mathematical concepts, are defined in terms of major structures, which are general and do not depend on specific content. Levels, on the other hand, as indicated above, depend on the specific topic and the data collected from several subjects. For example, in the experiment called The Hanoi Towers (Piaget, 1974/1976), there are no sub-divisions of stages into levels, and all the data is presented in just the three main stages I, II, and III. In the experiment Walking on All Fours (Piaget, 1974/1976) stages I and II are each divided into two levels, denoted IA, IB, and IIA, IIB. In the experiment on Seriation (Piaget, 1974/1976), the data of stage I has two layers of levels, so that within stage I there are four levels, IA(i), IA(ii), IA(iii), IB. In both Piaget's work and in APOS-based research, the definitions of the levels within stages are based on interview evidence.

8.3 A New Stage in the Infinity Studies

In addition to levels between stages, the widespread difficulties of students in progressing from Process to Object conceptions led to consideration of another possible change in APOS Theory. Obstacles in this progression appeared in the studies of the relationship between 0.9 and 1 (Weller et al. 2009, 2011; Dubinsky et al. 2013) which report on the difficulty of this progression in specific mathematical contexts and tend to confirm the results of previous studies (e.g., Sfard 1991; Breidenbach et al. 1992). Indeed, Sfard even suggests that this progression from Process to Object "seems inherently so difficult that at certain levels, it may remain practically out of reach for certain students" (Sfard 1991, p. 1).

The results of infinity studies related to historical developments of the concept of infinity (Weller et al. 2004; Dubinsky et al. 2005a, b) suggest that the difficulty of the progression from Process to Object may be particularly strong for infinite processes. Brown et al. (2010) acknowledged this difficulty and called the Object in this case a *transcendent object*. It has the property of being very different from any of the objects in the sequence making up the infinite process. A large percentage of the subjects in Dubinsky et al. (2013) reached the Process stage but not the Object stage. Among those who did not make this progression, there were differences in their interview responses. The data suggests that one way to interpret these differences is to posit the existence of a new stage, Totality, between Process

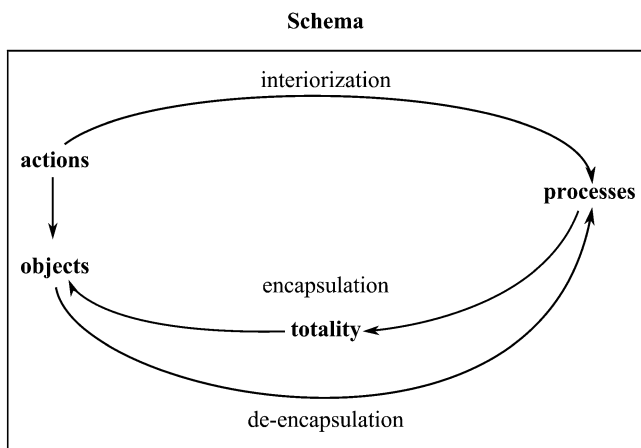


Fig. 8.1 APOS Theory with Totality

and Object, and then study the two progressions from Process to Totality and from Totality to Object. Figure 8.1 is a variation of the diagram in Fig. 2.1 that would incorporate this new stage.

8.3.1 *The Introduction of a New Stage*

The literature reports no more than limited success in helping students overcome their difficulties in progressing from a Process to an Object conception of $0.\bar{9}$. Although Zazkis and Leikin (2010) and Weller et al. (2009, 2011) are exceptions, the progress reported in these three studies fell considerably short of a complete solution to the problem. In particular, in the APOS-based studies by Weller et al. (2009, 2011) of pre-service elementary and middle school teachers, some students who completed APOS-based instructional treatments made somewhat more progress in development of an Object conception of $0.\bar{9}$ and belief that $0.\bar{9} = 1$ than did students who completed traditional instruction, but many did not. Carly is one example of the latter group. She gave substantial evidence of seeing $0.\bar{9}$ as a Process. In her interview, she repeatedly spoke about the idea that $0.\bar{9}$ “keeps going”. The following excerpt provides one such instance:

I: If I give you a decimal point and I give you one hundred 9s after that decimal point, is that .9 repeating?

Carly: To a certain point, but it ends. This one $[0.\bar{9}]$ is infinity. This one always keeps going.

Expressions that relate to $0.\bar{9}$ going on forever were considered to be indications of a Process conception. On the other hand, although given ample opportunity to

speak about $0.\bar{9}$ as an Object, many subjects like Carly spoke exclusively in Process-oriented language, as exemplified in the following passage:

Carly: Just because it's $0.\bar{9}$ still not one. One is a whole number. One is one and this is approximately.

Here, Carly is not only denying the equality of $0.\bar{9}$ and 1, but may also be rejecting the idea that $0.\bar{9}$ is even a number. Also missing with subjects such as Carly was any evidence of thinking of, or of constructing, transformations to act on $0.\bar{9}$.

The following excerpt from Tanisha is a similar example. She repeatedly stated (here and elsewhere) that $0.\bar{9}$ just keeps going on forever, but is unable to operate on it in order to solve the equation $0.\bar{9} + X = 1$ for X :

- I: Yeah. Uh-huh. Now, so if you've got this equation—.9 repeating plus X equal 1, what do you think goes in for X ?
- Tanisha: Awesome. I want to say .1 repeating. Just because that will obviously make it 10, but I don't—I don't know what X could be. That's the part that I'm missing. What's between the 9 and what's the—you know, the .9 and the 1, what's in between there.
- I: And what makes it so that you can't determine that?
- Tanisha: Because the 9 keeps going and I can—I mean—if you told me that if you really wanted it to go on till 10,000, I could find what that X would be, but since it keeps going on forever then the 9 doesn't stop, I guess, working. And so X —not that it no, not that X would always change, but it would always keep going with the 9.

These two examples contrast with subjects such as Estelle. Like Carly and Tanisha, Estelle sees $0.\bar{9}$ as a Process (as seen in the excerpt Estelle: Process), but differs from Carly and Tanisha in her ability to conceive of $0.\bar{9}$ as an Object and in her expression of a belief that $0.\bar{9} = 1$ (as seen in the excerpt Estelle: Object):

8.3.1.1 Estelle: Process

- Estelle: Well, if you keep adding 9's and the sequence like, never ends, but—I—that's a hard question.
- I: That's hard. Yeah, that's hard.
- ...
- Estelle: Because you're always gonna add one more 9.
- ...
- Estelle: No. I think it's more that since sequences go on forever, that you're forever gonna be adding one more 9.
- ...
- Estelle: Yeah. And that's gonna go on forever so you're always gonna keep borrowing and borrowing and borrowing.

8.3.1.2 Estelle: Object

- Estelle: Okay. So we have .9 repeating plus X is equal to 1. So to solve for X you subtract .9 repeating from both sides and X is going to equal 0, because you have like—yeah. Yes.
- ...
- Because with .9 repeating like, I've always been taught that like, the bar over the 9 means it goes on forever, and so if you have an infinite—if you have the bar over it,

it's gonna be infinite so you're gonna have all those 9's at once, I guess. And that's equal to 1.

I: And how do you know it's gonna equal 1?

Estelle: Because .9 repeating equals 1 because it goes on forever.
The infinite amount of 9's that's a number. Yeah.

The ability to apply operations to the Process in question, or, in the case of $0.\bar{9}$, referring to it as a “number” or a “thing” were considered to be indications of an Object conception; inability to perform such operations were considered to be evidence to the contrary.

Estelle's comment in the above excerpt that “it's gonna be infinite so you're gonna *have all those 9's at once*” seems to be a key to success in, and a better understanding of, the progression from Process to Object conceptions. Subjects who indicated an inability to see all of the 9's at once failed to reach an Object conception. Some subjects who indicated that they had this ability went on, like Estelle, to achieve an Object conception, but some, although sharing the ability to see all the 9s at once, did not. This suggested the possibility of a new stage between Process and Object. This new stage, Totality, refers to the ability, as expressed by Estelle, to see or to imagine all of the 9s present at once. Estelle appears to have achieved the stage of Totality and also to have gone on to achieve the Object stage for $0.\bar{9}$.

The following passage presents yet a different example. Here, Natasha gives evidence of Totality when she indicates that she may be conceiving of all of the 9's at once:

Natasha: If you went on forever, at the end of forever then it [$0.\bar{9}$] would be the whole thing

But, when asked to determine the solution to the equation $0.\bar{9} + X = 1$ she has difficulty:

I: Ok. Your thing is, you can't put anything else in there [for X]?

Natasha: Yeah. Even though I don't think there is a number such as that [$0.\bar{0}1$ ¹] because that means this would have to end. But the 9's never end so why would the zeros have to end? And just imagine that 1 out there, at the end of the 0s. Or, you can imagine this, and the one being like the end and the zeros going on, like just pushing the 1 back, because of all the 0s forever.

Although Natasha achieved the Totality stage, the second of her two excerpts suggests an inability to see $0.\bar{9}$ as an Object.

Subjects like Natasha, who moved beyond Process toward Totality but did not reach Object, provide support for the possibility of Totality as a definite stage between Process and Object.

¹ $0.\bar{0}1$ refers to repeating 0s, with 1 at the end.

8.4 Levels Between Stages in $0.\bar{9}$

The data in Dubinsky et al. (2013) also uncovered the existence of levels between stages, that is, incremental points of progression from Action to Process, from Process to Totality, and from Totality to Object. In Dubinsky et al. (2013), there is a very detailed description, called a *Framework for Analysis* (FFA), that gives operational definitions that provide an objective and reasonable set of criteria for determining all levels and stages in the context of an infinite repeating decimal. For most subjects, the criteria sufficed to determine the level, but in a few cases it was necessary to return to the interviews and make interpretations. Interpretations of the excerpts that follow provide the rationale for the operational definitions outlined in the FFA that were used to determine the levels between stages.

8.4.1 Action to Process

There were some subjects who gave evidence of an Action conception by writing out a finite number of 9s, but gave no evidence of a Process conception, or any stage beyond Process. These subjects had progressed to the stage of Action but not beyond. On the other hand there were subjects who gave substantial evidence of having achieved an Action conception and having progressed to the Process stage. However, there were also subjects for which the determination was less clear. For example, Maria made multiple statements in which she expressed the idea that the 9s in $0.\bar{9}$ continue forever, a strong indication of Process. But she also stated that she believed that a finite number of 9s would suffice:

I: So, do you think .999 is equal to .9 repeating?

Maria: Hmhm . . . Well, maybe not .999, but definitely after maybe four or five nines.

Excerpts such as this suggested that Maria had not fully achieved a Process conception of $0.\bar{9}$ but was in transition from Action to Process. This suggested the existence of a level between Action and Process called *Emerging Process* (EP).

Table 8.1, based on the FFA (Dubinsky et al. 2013), summarizes the operational definitions of the progressions between stages and/or levels in the transition from Action to Process. In this table, “Segment” refers to a short, coherent episode in the transcript that carries with it a very brief (most often a single sentence or phrase) *descriptor* to summarize its content. The arrows indicate transitions between levels and/or stages.

8.4.2 Process to Totality

As seen in the examples given above, among the subjects who reached the Process stage, several gave evidence of seeing, or beginning to see, the repeating decimal $0.\bar{9}$ as a totality. Others gave no such evidence. To study more closely the

Table 8.1 Progression from Action to Process

Progression	Criteria
A: Action EP: Emerging Process P: Process	
A	# Segments under Evidence of Action > 0 ; and # segments under Evidence of Process = 0
A $\rightarrow$ EP	# Segments under Evidence of Process > 0 and one or more interview segments indicating difficulty in making the transition from A to P (e.g., the subject thought in terms of action when a process was called for)
A $\rightarrow$ EP $\rightarrow$ P	# Segments under Evidence of Process > 0 and no interview segments indicating difficulty in making the transition from A to P

progression between these two stages, three levels were introduced: *Start towards Totality (ST)*, *Progress towards Totality (PT)* and *Emerging Totality (ET)*. Following are some examples.

Carlos made many statements about $0.\bar{9}$ going on forever. For example,

Carlos: Because, again .9 is . . . we're talking about .9 repeating, so you're gonna just keep on repeating and repeating that space. So it just keeps on going and going and going, so it's always changing.

So he was considered to have achieved the Process stage. For the next stage—Totality—it was a different story. One of the indicators for Totality involved analysis of the results of the following thought experiment: divide a square into 10 equal parts, shade 9 parts, divide the remaining part similarly and repeat this process indefinitely. Subjects were asked to imagine how much of the square would be shaded. Shading of the total square indicated Totality for the process (but not Object because there is no application of an action or process to the process); a part left unshaded indicated otherwise. When discussing this question, Carlos once expressed the opinion that “all of it” would be shaded, but twice asserted that a little bit was always left. The inconsistency of his responses, together with the higher relative frequency of his difficulty in seeing the terms of $0.\bar{9}$ “all at once,” suggested no more than a Start towards Totality (ST).

Rosa, on the other hand, seemed to express equally often positive and negative opinions about all of the 9s being present all at once. For example, at one point she says:

Rosa: I can't imagine all the 9's that it would take to get to forever.

But when asked about the thought experiment, she asserts that the entire square would be shaded. The relative equality of the number of positive statements (the ability to see $0.\bar{9}$ “all at once”) and negative statements (inability or difficulty in seeing $0.\bar{9}$ “all at once”) suggested Progress towards Totality (PT).

Susan made statements such as:

Susan: You can't really imagine never ending nines.

But on twice as many occasions she used phrases such as “when you actually had .9 repeating and it's never ending” and she asserted that the entire square would be shaded in the thought experiment. Although her responses revealed difficulties, the relative dominance of positive versus negative comments indicated an Emerging Totality (ET) level.

Finally, subjects who gave indications of seeing the 9s present all at once and no evidence to the contrary were deemed to have reached the Totality stage.

Table 8.2, also based on the FFA (Dubinsky et al. 2013), summarizes the operational definitions of the progressions between levels and/or stages in the transition from Process to Totality. In this table, “Segment” again refers to a short portion in the interview transcript and the arrows indicate transitions between stages and/or levels.

Table 8.2 Progression from Process to Totality

Progression	Criteria
ST: Start Toward Totality	For any level in the progression from Process to Totality
PT: Progress Toward Totality	it is required that the # segments under Totality Sees > 0
ET: Emerging Totality	(Here, “Totality Sees” means the subject sees the Process as a Totality and “Totality Does not See” means the subject does not see the Process as a Totality)
TOT: Totality	
ST	# Segments under Totality Does not See $>$ # segments under Totality Sees
ST $\rightarrow$ PT	# Segments under Totality Does not See = # segments under Totality Sees
ST $\rightarrow$ PT $\rightarrow$ ET	# Segments under Totality Does not See $<$ # segments under Totality Sees
ST $\rightarrow$ PT $\rightarrow$ ET $\rightarrow$ TOT	# Segments under Totality Does not See = 0

8.4.3 Totality to Object

In APOS Theory, moving from a Process conception to an Object conception is indicated by the individual's ability to think about and/or perform actions or processes on the Process. Among the subjects who achieved the Process stage, several achieved some of the levels in the progression from the Process to the Totality stage, and then went on to show progress toward the Object stage. However, that progress was tempered by evidence of difficulties. Similar to the examples above, which showed differing levels of progress from Process to Totality, the interview evidence suggested the need to introduce intermediate levels in

the progress to the Object stage. The following levels were indicated: *Start towards Object (SO)*, *Progress towards Object (PO)*, and *Emerging Object (EO)*. This decision was confirmed when it was seen that there were subjects whose progression went no further than one of these levels. Following are some examples.

Awaethu was able to see that $X = 0$ is the solution to the equation $0.9 + X = 1$. She had difficulties, however, with manipulating $0.\bar{9}$ directly to show that it was equal to 1 because she saw $0.\bar{9}$ as a “repeating thing”. Therefore she was designated as having made a start towards Object (SO).

Roberto, on the other hand, was evenly balanced in giving evidence of having achieved an Object conception and of not having reached that stage, so he was designated as Progress towards Object (PO).

Although Rita gave indications of an Object conception, for example by expressing her belief that $0.\bar{9}$ is equal to 1, and no indications to the contrary, the weakness of her positive comments caused her to be one of the few subjects for whom determination of level required interpretations of her comments beyond mere application of the FFA. She made statements about the 9’s in $0.\bar{9}$ and the 0s in $0.\bar{0}$ stopping or not stopping. She also said that because $0.\bar{9}$ “doesn’t stop, you’d get 1”. The last statement suggests Object; the previous one reveals confusion. For example, Rita said:

Rita: But in the case of 1 minus .9 repeating, the 0 would just continue on forever until the 9s stopped, which then would make those two numbers not equal to each other so you’d have a 1 at the very end.

Rita’s tendency to see the difference between $0.\bar{9}$ and 1 as $0.\bar{0}$ and her inability to say 0 indicated some difficulty with encapsulation. However, according to the FFA criteria, her overall progress, marked by the fact that she made exclusively positive statements, including an ability to compute with $0.\bar{9}$, suggested Object. The apparent difficulty with encapsulation necessitated the need for interpretation, which led to a designation of Emerging Object (EO).

Rita: You can give me an equation and I will believe the equation. You can give me the numbers to prove that they are equal, but the way I see it, the simplest way, is 1 minus .9. As long as the 9 doesn’t stop, the difference is going to be point zero repeating.

Finally, Clyde performed correctly at least 10 arithmetic operations on $0.\bar{9}$ and located it in an appropriate position on the number line. In the following exchange, he symbolizes $0.\bar{9}$ as an object X in an equation.

Clyde: Mentally, I think of it more as X , like in an equation.

I: Uh-huh.

...

Clyde: Because dealing with .9 repeating, just trying to even visualize it, it’s a little bit hard, so I think of it as X —an X sign. That way there’s a place holder in my head that I can think about it that’s more concrete than dealing with the .9’s repeating.

Because he gave exclusive evidence of having reached object, with no evidence to the contrary, he was designated as having an Object conception of $0.\bar{9}$.

Table 8.3 Progression from Totality to Object

Progression	Criteria
SO: Start Toward Object PO: Progress Toward Object EO: Emerging Object O : Object	For any level in the progression from Totality to Object it is required that #segments under Object Sees > 0 (Here, “Object Sees” means the subject does perceive the process as an object, “Object Tries to See” means that the subject tries to see the process as an object and “Object Does not See” means the subject does not perceive the process as an Object)
SO	#Segments under Object Does not See + #segments under Object tries to See $\geq$ #segments under Object Sees
SO $\rightarrow$ PO	$0 < \text{\#segments under Object Does not See} + \text{\#segments under Object tries to See} < \text{\#segments under Object Sees}$
SO $\rightarrow$ PO $\rightarrow$ EO	#Segments under Object Does not See = 0 and #segments under Object Tries to See = 0 and Object Sees includes exactly 1 context
Review descriptors and/or transcript to determine EO or O	#Segments under Object Does not See = 0 and #segments under Object Tries to See = 0 and Object Sees includes 2 different contexts
SO $\rightarrow$ PO $\rightarrow$ EO $\rightarrow$ O	#Segments under Object Does not See = 0 and #segments under Object Tries to See = 0 and Object Sees includes 3 or more different contexts

Table 8.3, based on the FFA (Dubinsky et al. 2013), summarizes the operational definitions of the progressions between stages and/or levels in the transition from Totality to Object.

8.5 Previous Uses of the Idea of Totality

Research much earlier than Dubinsky et al. (2013) discussed ideas similar to the introduction of a stage between Process and Object. Dubinsky referred to the notion of a function as a Totality, so that it can be a point in a function space, as opposed to a process (Dubinsky 1984), and of an individual’s inability to see a process as a “total entity” and reason about it (Dubinsky 1987); Ayers et al. (1988) referred to the encapsulation of a process into a single, total entity and thinking of it as a mental object; and Cornu and Dubinsky (1989) distinguished between performing an action in a step-by-step manner and seeing it as a totality. All but the last of these examples appear to include a notion of Totality as part of an

Object conception and not as a separate stage. In the last example, there is the following comment:

We refer to an action when the subject is able to perform it in a step-by-step manner but does not see it as a totality and cannot think about it, or explain it to another person. (Cornu and Dubinsky 1989, p. 74)

While Cornu and Dubinsky state that the notion of totality is not part of an Action conception, they do not propose totality as a new stage, separate from Object, although they are somewhat vague about where totality fits relative to the subsequent stages of Process and Object. In any case, they did not discuss the issue beyond this single comment, nor did they make any investigation of it.

A series of papers (Weller et al. 2004; Dubinsky et al. 2005a, b) concerned with the history of the concept of infinity in mathematics and the distinction between actual and potential infinity, made several references to totality in statements such as:

If one becomes aware of the process as a totality, realizes that transformations can act on that totality, and can actually construct such transformations, (explicitly or in one's imagination) then we say the individual has encapsulated the process into a cognitive object (Dubinsky et al. 2005b, p. 256).

Brown et al. (2010) discussed totality as an important preliminary to encapsulation, although they did not suggest it as a separate stage. There is also a mention of totality in Dubinsky et al. (2008), again without considering it as a new stage.

Sfard (1992) also introduced a new stage between Process and Object. She called it a *condensed operational conception*. By this she means the subject is able to see a process as an input/output machine without paying attention to the details by which an input is transformed into an output. This notion seems to be quite different from the Totality stage discussed here. In any case, perspectives related to APOS Theory, such as Sfard's, may be discussed in a follow-up book.

The discussion of Totality in Dubinsky et al. (2013) introduced two main differences with some previous studies that mentioned totality. First, as was indicated above, previous studies generally considered totality as part of an Object conception. Second, in Dubinsky et al. (2013), Totality is introduced as a possible new stage supported by data-driven descriptions of the progression from Process to Totality and from Totality to Object.

8.6 The Tentative Nature of Totality as a Stage

As of this writing, there is not sufficient evidence to decide whether Totality is really any of the following: a stage, a level between Process and Object, a part of a Process conception or a part of an Object conception. It seems clear that the Start, Progress, and Emerging levels are only categorizations of the specifics in a subject's thinking at the time of the observation and so would be very different for different concepts. Therefore they are best designated as levels. Based on the research in Dubinsky et al. (2013), however, there is a strong likelihood that, at least

in the case of infinite repeating decimals, Totality functions as a separate stage as opposed to being a level or part of another stage. This is because a Totality conception appears to represent a change in how the individual thinks about the mathematical concept. It may be that this development occurs in a similar manner in the development of thought about other mathematical concepts. This seems very likely for infinite processes and even possibly for finite processes that involve a very large number of steps. In any case, the extent to which Totality appears as a stage throughout the realm of mathematical concepts and the mechanism or mechanisms by which a Totality conception is constructed (about which little or nothing is known at present) remain matters for continuing research.

Chapter 9

Use of APOS Theory to Teach Mathematics at Elementary School

Throughout the first half of the 1990s, the mathematics team of the Center for Educational Technology, Tel-Aviv, Israel (CET), set out to revise the team's existing materials for teaching mathematics in Israeli elementary schools (Grades 1–6, ages 6–12). One important aspect of the revision was to introduce the ideas of Piaget and APOS Theory into the teaching sequences. An area of particular interest was the teaching of fractions in grades 4 and 5.

The data and conclusions described in this chapter emerge from two studies on fractions: one on students' understandings of part-whole relationships (Arnon 1998) and the other on students' work with equivalence classes of fractions (Arnon et al. 1999, 2001). They are described in this chapter in some detail in order to acquaint the reader with the application of these ideas at the elementary school level and to compare the learning of elementary school students with that of postsecondary students.

Chapter 2 of this book describes how Dubinsky adapted Piaget's ideas on learning to postsecondary mathematics. According to Piaget, postsecondary students are expected to be at the *stage of formal operations*. This means that the objects on which they perform actions are abstract rather than concrete. In elementary schools, most students are at the *stage of concrete operations*. This means that the objects on which actions are applied need to be concrete, that is, they can be perceived by one's senses (Piaget, 1975, 1974/1976). Thus, from the perspective of APOS Theory, the principal difference between the elementary and postsecondary mathematics classroom lies in the nature of the objects to which actions are applied.

The data presented in both studies and are discussed in this chapter were collected in the early 1990s. APOS Theory has developed since then, including the latest ideas about possible levels between stages and a possible new stage between Process and Object (Totality, see Chap. 8). This chapter presents an interpretation of the data collected in those studies in light of current APOS Theory, which is somewhat different from the way it was originally presented in the 1990s. In particular, this chapter includes use of the term *level* according to the meaning

ascribed to it in Chap. 8: incremental points of progression in the development of a concept between one APOS stage and its subsequent stage.

The chapter is organized as follows: The application of APOS Theory in the elementary school versus its application at the postsecondary level will be discussed in Sect. 9.1. Research about the learning of the part-whole interpretation of fractions in grade 4 is discussed in Sects. 9.2–9.5. Section 9.2 consists of a comparison of the performance of elementary school children who completed APOS-based instruction on fractions with those who completed a unit on fractions using a standard pedagogy that follows Nesher's paradigm (Nesher 1989), which is also Piagetian. Section 9.3 consists of a discussion of possible levels between the Action and Process stages that Arnon (1998) identified in her investigation of students' thinking about some fraction concepts and the part-whole interpretation. The role of imagination in the interiorization of an Action on concrete objects into a Process conception is the subject of Sect. 9.4. Section 9.5 focuses on the concept of equivalence classes of fractions in a fifth grade classroom. A summary of what is known today about the use of APOS Theory in elementary schools is presented in Sect. 9.6.

9.1 Applying APOS Theory in Elementary School Versus Applying It in Postsecondary School

Piaget's theory of cognitive development is based on the assumption that an individual constructs knowledge by reflecting on her or his own experiences. At the stage of concrete operations, these experiences need to be concrete (Piaget, 1975, 1974/1976). At the stage of formal operations, they can be abstract. The terms concrete and abstract often have different meanings. Throughout this chapter they will be used according to the meanings described in Arnon et al. (2001). Here, the term concrete involves the use of real or imagined physical objects. For example, many children, in dealing with fractions, feel comfortable when assisted by circle cutouts (circle sectors) made of real cardboard paper, drawings, or mental images (see Fig. 9.1). Other concrete representations of fractions are also described

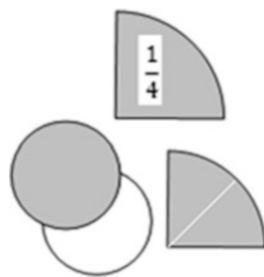


Fig. 9.1 Circle cutouts

in the literature. One can understand this interpretation of concreteness to mean that a concrete experience needs to involve the learner's senses.

In Fig. 9.1 one can see circle cutouts and whole circles. Some of the cutouts carry the fraction they represent, some do not. Mathematicians also often use concrete aids. For example, they often use Cartesian graphs when dealing with real-valued functions and use drawings of two- and three-dimensional bodies when trying to prove theorems in geometry (Arnon et al. 2001, p. 171).

The term abstract refers to the use of a mathematical concept without any physical world representation (Arnon et al. 2001, p. 171). For example, people often manipulate real-valued functions without any reference to their graphs or any physical content that could be used to represent them; instead, they use mathematical language and syntax, as well as their knowledge of the mathematical structures involved and the rules for combining their components.

The differences in APOS Theory between an abstract context (postsecondary education) and a concrete context (elementary education) are illustrated in Figs. 9.2 and 9.3.

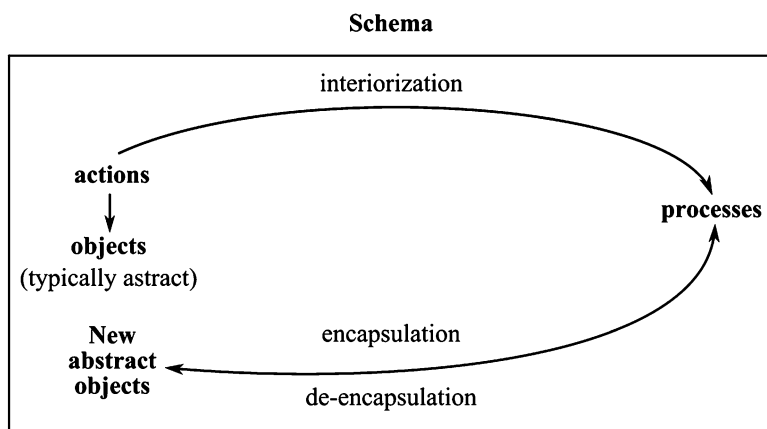


Fig. 9.2 APOS for postsecondary students

Figure 9.2 shows the postsecondary context where the initial Objects, as well as the Objects that emerge from the encapsulation of Processes, are typically abstract.

According to Piaget, children at the stage of concrete operations (approximately from the age of 2 years up to 11 or 13 years) develop mental (abstract) concepts as a result of reflection upon actions they perform in the real (concrete) world with their own hands or in their imaginations (Piaget 1975, 1976). This is illustrated in Fig. 9.3.

Figure 9.3 is a modification of Fig. 9.2 to represent the implementation of APOS Theory for children at the stage of concrete operations. The figure illustrates how Actions applied to physical Objects give rise to abstract mathematical Objects in a child's mind. At the stage of concrete operations, the Objects upon which the

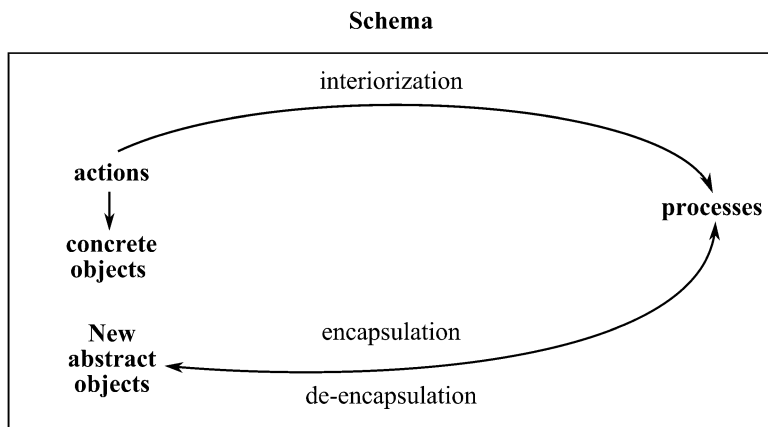


Fig. 9.3 APOS for elementary school students

student acts need to be concrete. The Objects that emerge from the encapsulation of the interiorized Actions are abstract, just as for postsecondary learners. Among the participants of the study of Arnon (1998), two students demonstrated such encapsulation, as described in Sect. 9.3.3.

9.2 Comparing a Standard Instructional Sequence to an Instructional Sequence Based on APOS Ideas

The grade 4 curriculum on fractions includes instruction on the part–whole interpretation of fractions. For a fraction $\frac{k}{n}$, $k, n \in \mathbb{Z}$, $n \neq 0$, an object that represents a whole is divided into n equal parts, with k of those n parts selected.¹ A circle and its sectors (referred to as circle cutouts) were used as representations.

In a traditional teaching sequence, fourth grade students learned about the part–whole interpretation by operating on circles as “wholes” and ready-made circle cutouts representing a variety of unit fractions up to $\frac{1}{20}$, with n cutouts for each fraction $\frac{1}{n}$. The appropriate fraction symbol $\frac{1}{n}$ was printed on each of these ready-made cardboard circle cutouts (Arnon 1998, p. 87). The following is an example of an action operated on these manipulatives: students were asked to use the separated cutouts to find out how many cuts of $\frac{1}{9}$ were needed to fill an empty circle or a cutout representing the fraction $\frac{1}{3}$. In this chapter, an instructional

¹ About the part–whole interpretation of fractions, and other interpretations, see in Arnon (1998), pp. 65–74.

sequence that used these types of materials is referred to as STN (for *standard*). Never in this instructional sequence did students actually divide a circle into equal parts. As a result, it was suspected that these activities tended to lead the students to reflect on the characteristics of the cutouts rather than the part-whole interpretation they were meant to represent. Unofficial reports from schools suggested that some of the fourth grade students who used these materials and activities developed good part-whole conceptions; however, many did not.

The mathematics team set out to investigate how to improve this situation. The ideas of Piaget and APOS Theory pointed to a direction.

Piaget distinguishes among three types of experience acquired through contact with the external world:

- A. “Simple exercise”, which “does not necessarily imply that knowledge will be extracted. . .”
- B. “Physical experience”, where the child manipulates physical objects, and by means of “a simple process of abstraction” abstracts the properties of the objects.
- C. “Logico-mathematical experience”, in which the child, manipulating objects, constructs properties of the action itself and of the transformation(s) he or she applies to the objects. In this type of experience knowledge is constructed by means of reflective abstraction, as described in Chap. 2. (see also, Piaget 1975, p. 193–194)

In Sect. 2.2 we have seen examples of how Piaget’s theory of reflective abstraction formed the antecedents to APOS Theory—the mental structures as well as mental mechanisms such as interiorization (Dubinsky 1991). Yet the activities of the STN sequence are more like Piaget’s Type B experience. As a result, instead of a reflective abstraction and interiorization of part-whole Actions, the students used “simple processes of abstraction,” or Type B experience, and tended to identify properties of the circle cutouts.

A new teaching sequence was developed, the goal of which was to increase the likelihood of Type C experiences and to enhance the construction of a part-whole interpretation of fractions. To that end the development team introduced a different set of materials, which started with a cardboard page illustrated in Fig. 9.4. This cardboard tool was called *partitioning rings*.

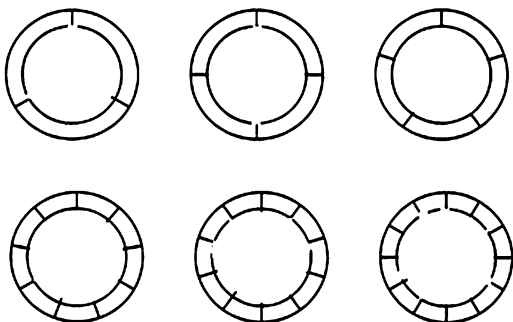


Fig. 9.4 The partitioning rings (Arnon 1998, p. 210)

In addition to the partitioning rings, the manipulative set included drawings of preprepared empty circles of the same size, with center points. The inner circles of the partitioning rings were perforated to be easily pushed out. To construct a representation of a fraction, say $\frac{k}{n}$, the student had to choose a ring that was divided into n equal parts, place it on an empty circle that showed the center point, and use the segments indicated on the outer ring to divide the circle into n equal parts and then shade k of those parts. This was the action the students were expected to interiorize. This action consisted of the following steps: For a fraction $\frac{k}{n}$, one chooses a circle (“the whole”), divides it into n equal parts (according to the denominator), and finally shades k of these parts (according to the numerator). Later in this instructional sequence, ready-made circle cutouts were introduced in class, representing a variety of fractions, including non-unit fractions $\left(\frac{k}{n} \text{ with } k > 1\right)$. These cutouts were purposely left unnamed since an important activity was to have the students name them (Arnon 1998 p. 87–89). This instructional sequence, as well as the class where it was used, is referred to in this chapter as EXP (for *experimental*).

Since the use of the ready-made cutouts encourages Type B rather than Type C experience, APOS Theory would suggest that the introduction of ready-made cutouts should be delayed until the action of drawing circle representations of fractions is interiorized. The main evidence that such interiorization has taken place is the student’s ability to describe verbally how to produce such a representation. The following excerpt from the interview with Offir provides an example.

With none of the concrete materials present, Offir was asked to compare $\frac{1}{100}$ and $\frac{1}{13}$, neither of which was included in the set of concrete materials. In his response, he explains why he thinks that $\frac{1}{100}$ is smaller than $\frac{1}{13}$:

... Now, when you divide the whole into thirteen, the parts which are narrow, are of some specific size, and when... eh... , when you divide the whole into one over a hundred, so the density is that... there is more density, and the part of the hundred, the circle-cutout of the hundred becomes more, eh, smaller [showing with a gesture of two fists the act of narrowing, of getting closer]. (Arnon 1998, p. 105.)

Another indication of interiorization is the student’s ability to produce a schematic drawing of a fraction without using the partitioning rings. This is illustrated in Figs. 9.5 and 9.6, where one can see two hand drawings performed without the tools; Yulia makes the first drawing and Gali the second:

Fig. 9.5 Approximate drawings by Yulia (Arnon 1998, p. 220)

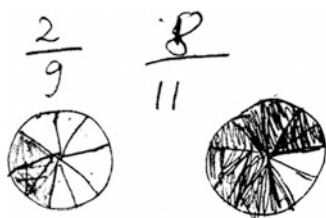
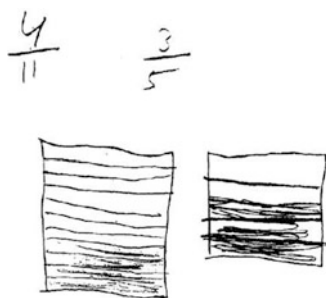


Fig. 9.6 Approximate drawings by Gali (Arnon 1998, p. 115)



Despite the clumsiness of the productions, the correct ideas are present. Drawings of this kind demonstrate a student's ability to reconstruct physical representations when asked to perform specific actions on fractions.

Both learning sequences, EXP and STN, were based on Piaget's idea that the construction of a new mathematical concept begins with actions applied to physical objects (Piaget, 1975; 1974/1976). The difference between the approaches lies in the nature of the actions used in each instructional sequence. Arnon (1998) set out to investigate the question of whether the EXP instructional sequence led to improved learning. The data showed that it did.

The following four fraction concepts were investigated in Arnon's (1998) data analysis: comparison of unit fractions, construction of non-unit fractions, comparison of non-unit fractions, and multiplication of a unit fraction by an integer. As for the arithmetic operations, the two sequences had an important common trait: their way of treating the arithmetic operations of addition, subtraction, comparison, and multiplication. These operations were not meant for the learning of the algorithms, but only to serve as catalysts to encourage encapsulation, so that the students' conceptions of fractions would develop from Process to Object. In both sequences students only practiced cases of these operations that could easily be solved with the aid of the manipulatives. For example, addition problems were presented only if they involved cutouts that existed in the set of manipulatives and where one

denominator was a multiple of the other. For a problem such as $\frac{2}{5} + \frac{1}{10}$, the students would arrange the proper cutouts next to each other, then cover the $\frac{2}{5}$ with $\frac{1}{10}$ cutouts, and conclude with the solution $\frac{3}{10}$. No computations were involved.²

Two classes, one STN and one EXP, participated in the study. Instruction for each class took place during the last trimester of grade 4. The students were interviewed when they reached grade 5, after summer vacation. The interviews were individual, audio recorded, and transcribed. The physical objects that were used in Grade 4 were not present in the interviews. They were only referred to verbally, and students often reconstructed them in the drawings they made during their interviews. The interview protocol consisted of the following types of arithmetic problems: comparison of two unit fractions, comparison of two non-unit fractions, and multiplication of a unit fraction by an integer. Information about the concepts *constructing a unit fraction* and *constructing a non-unit fraction* was derived from the discussions of the binary operations.

The data provided information about the following issues: comparative mathematical performance between the two groups, STN and EXP (in this section); the existence of possible developmental levels between Action and Process for the four investigated concepts (Sect. 9.3); preliminary genetic decompositions for the same four concepts with a detailed description for the concept *construction of a non-unit fraction* (Sect. 9.3); and the prevalence of operating on concrete objects in the children's imagination (Sect. 9.4).

In analyzing students' progression from Action to Process, responses appeared to fall into one of three categories:

- i. Not further than an Action conception: No evidence of interiorization of any Action on concrete objects
- ii. The transition from Action to Process: Evidence of interiorization of either a partially or completely incorrect Action on concrete objects
- iii. At least Process conception: Evidence of interiorization of correct Actions

Because the manipulatives were not used in the interviews, the researcher could only detect levels of interiorization, and not knowledge of the Action itself. But she could trace *No Evidence of Action* by the lack of a description or drawing that referred to either fractions or concrete representations of such or by lack of any response. Such evidence was categorized as No Evidence of Action (category i).³ The author could also trace verbalizations and/or drawings that indicated the

² According to the curriculum, in grades 5 and 6, where their fraction conceptions hopefully have developed into Objects, they will learn the proper algorithms for arithmetic operations of fractions.

³ In cases where the student provided a *No evidence of action* response, it might be the case that provided the manipulatives, the student might perform a correct Action.

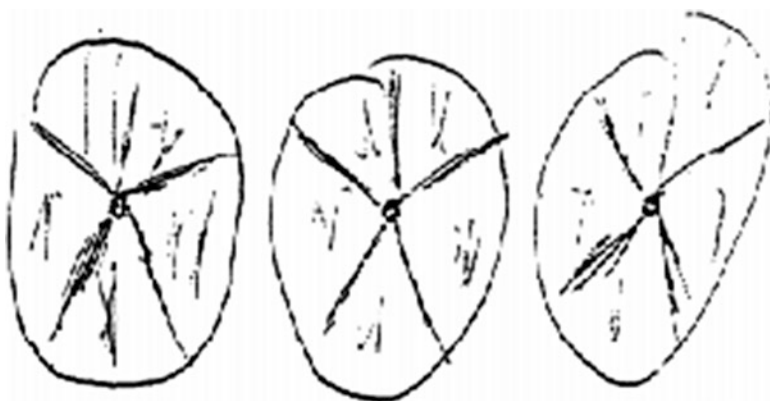


Fig. 9.7 Avi's incorrect action for construction of the fraction $\frac{3}{5}$

interiorization of a partially or completely incorrect action, which indicated that the student's conception was in transition from Action to Process. Such responses were categorized as category ii. Responses that revealed the interiorization of a complete and correct Action were categorized as category iii, indicating a Process conception.

Figure 9.7 illustrates the interiorization of a completely incorrect action (category ii) for the construction of a non-unit fraction by Avi:

In the interview Avi produced this drawing to represent $\frac{3}{5}$ (Arnon 1998, p. 111) and said the following:

- I: How many fifths do we see in your drawing?
 ...
 Avi: Ah, so in every circle there are five, and if we join them all we have fifteen.
 I: So three fifths is fifteen fifths?
 Avi: This [the drawing]? five over, eh, three fif. ...,three over.
 ...
 I: Explain again, why did you do here three circles?
 Avi: Because each one has, eh, because it is written in the numerator three.

Avi's action was incorrect in two ways: first, he misinterpreted the numerator by thinking that it could be represented by three circles, and then, he divided each of the three circles into five equal parts and shaded all 15 parts. None of these action steps were correct. Avi's response was judged as category ii.

The following example illustrates the interiorization of an action that was only partially incorrect (category ii). Dafni constructed a unit fraction. Then she explained what a student should do in order to construct the unit fraction $\frac{1}{10}$:

Dafni: In all the circle there ought to be ten. And after that he **erases** one [Author's emphasis].

Dafni's suggestion to divide the circle into ten equal parts, shade one, and leave nine unshaded was correct. Her call to erase the shaded portion was not, because it indicated her belief that the unshaded parts represent the fraction. Dafni further exhibited her incorrect conception of non-unit fractions in her attempt to compare $\frac{2}{5}$ with $\frac{4}{6}$:

I: Do you want to explain to another child what he should do in order to check [the comparison]?

Dafni: He needs to draw the, eh, nu, the, eh, circle of six, six parts, and color there four.

I: Yes.

Dafni: After that he knows what is left of it, and after that he does the same thing with two over five.

I: Yes? And, how does one know which is greater?

Dafni: According to... According to the size that came out.

I: According to the size of what he colored or according to the size of that which was left?

Dafni: Of that which was left.

Similar to her representation of $\frac{1}{10}$, Dafni again divided the circle correctly and shaded the right number of portions. However, she discarded the shaded portion and misinterpreted the fraction as "that which was left" unshaded. This again constituted a partially incorrect action (category ii). This was the case with Dafni for each of the concepts investigated: *comparison of unit fractions, constructing a non-unit fraction, comparing non-unit fractions, and multiplication*.

For each of the four concepts investigated in the study, the three categories of interiorization accounted for all of the students' responses. When comparing the achievement of the STN and EXP students for each category and for each concept, the EXP class always did better. Table 9.1 illustrates the comparative results (STN versus EXP) for the *comparison of non-unit fractions*. Notice that for this concept, a correct action consisted of the following steps: *take the two appropriate cutouts and put them one upon the other so that they are either equal or one is completely covered by the other. If the cutouts are equal then the fractions are equal; if one cutout covers completely and overflows the other, then the fraction it represents is the larger fraction*. This action is difficult to accomplish without the cutouts; nevertheless about a third of the STN students and two thirds of the EXP students succeeded in solving it.

Table 9.1 Comparing the percentage of students of each class for the concept comparing non-unit fractions

	Class STN <i>N</i> = 28 (%)	Class EXP <i>N</i> = 32 (%)
Comparing non-unit fractions, degree of interiorization		
Not further than Action conception	32	12.5
Transition from Action to Process	39	22
At least Process conception	29	66

A comparison of the results in Table 9.1 shows that the EXP class fully interiorized the action of comparing non-unit fractions at more than twice the rate of students from the STN class. Comparative results were similar for all four concepts that were investigated in the study. A cumulative comparison is shown in Table 9.2 (Arnon 1998, p. 174):

Table 9.2 Overall interiorization

Overall interiorization	STN class <i>N</i> = 28 (%)	EXP class <i>N</i> = 32 (%)
No proper actions interiorized	25	17
Exactly one proper action interiorized	21	10
Exactly two proper actions interiorized	14	3
Exactly three proper actions interiorized	18	27
Exactly four proper actions interiorized	21	43

Table 9.2 shows the percentages of students in each class who correctly interiorized exactly none, one, two, three, or four of the actions. For example, for the four concepts studied, 43 % of the EXP students versus only 21 % of the STN students were judged to have interiorized all four correct actions, namely, achieving at least Process conceptions for all of the investigated concepts; 70 % of the EXP versus 39 % of the STN students interiorized actions for more than two of the investigated concepts. As the table shows, the EXP students made correct interiorizations more readily than the STN students. This reinforces the claim that the action of constructing fraction representations with the help of the partitioning rings (the one used in the EXP class) was more easily interiorized than activities with the ready-made (and named) circle cutouts (as used in the STN class). Two of the EXP students even gave evidence of achieving Object conceptions of the concept *constructing non-unit fractions*, while none of the STN students provided such evidence (see Sect. 9.3.3).

9.3 Levels and Genetic Decompositions for the Transition from Action to Process of Some Fraction Concepts

In this section the term level is used in the sense elaborated in Chap. 8. Level refers to developmental phases that occur in the transition from one APOS stage to another.

9.3.1 Levels in the Developments of Some Fraction Concepts

In the study of elementary school students' learning of fractions, Arnon (1998) concentrated on the transition from Action to Process. For Piaget this transition was from an unconscious application of the action to full consciousness of it. In his

book *The Grasp of Consciousness*, Piaget (1974/1976) discusses the gradual development of consciousness. According to Piaget, that passage “must require constructions, and cannot be reduced to a simple process of illumination” (Piaget (1974/1976), p. 322). For many of the situations he investigated, Piaget determined the existence of incremental points of progression from the unconscious to the conscious.

Phenomena similar to what Piaget described are discussed in Chap. 8. Such incremental points are named in APOS Theory as levels. In the development of the concept of repeating decimals, the researchers identified levels of development between every two consecutive stages of APOS, including a proposed new stage, Totality (Dubinsky et al. 2013). In her study of different fraction concepts, Arnon (1998) identified levels in the transition from Action to Process.

Similar to what Piaget found and what is reported in Chap. 8, Arnon (1998) determined that levels, unlike stages, are not general but are content specific. Levels for the concept *constructing a non-unit fraction* are described in Table 9.3.

Table 9.3 Levels in the interiorization of *constructing a non-unit fraction*

Constructing a non-unit fraction	
The transition from Action to Process	• Drawing k (numerator) circles, each divided into n (denominator) equal parts—a completely incorrect action • The rest (shading = erasing)—only part of the action incorrect

For the transition from Action to Process of this concept, the researcher showed the existence of two levels, which are given in the right-hand column of Table 9.3. Levels for the other investigated fraction concepts were also found. For example, for the concept of *comparing two non-unit fractions*, six different levels were identified on the transition from Action to Process. These levels appear in Table 9.4 (Arnon 1998):

Table 9.4 Levels in the interiorization of *comparing two non-unit fractions*

Comparing two non-unit fractions	
The transition from Action to Process	• Larger numerator $\Rightarrow$ larger fraction (counting circles) • Larger denominator $\Rightarrow$ larger fraction (counting equal parts) • Smaller integers (numerator and denominator), mean larger parts, and hence a larger fraction • Smaller denominator means larger parts, and hence a larger fraction • Larger non-shaded area $\Rightarrow$ larger fraction • Larger number of non-shaded parts $\Rightarrow$ smaller fraction

Notice that identifying a student’s conception as being at some level within the transition from Action to Process does not imply that the student necessarily went through all the previous levels. He or she might have skipped some of them. More research is needed to establish the existence of levels in this case.

9.3.2 Genetic Decompositions for the Concepts That Were Investigated

The data in the tables for the different concepts could be interpreted as levels in the transition of each concept from Action to Process. A preliminary genetic decomposition arising from each of the tables for each of the concepts that were investigated can be useful in continuing research, as described in Asiala et al. (1996) and in Chaps. 4 and 6.

Figure 9.8 presents one such genetic decomposition (for the transition from Action to Process) derived from Table 9.3 for *constructing a non-unit fraction*.

The findings described above contain also some data about the transition from Process to Object, yet not enough to design a full genetic decomposition. Nevertheless, some interesting findings about an Object conception will be presented in Sect. 9.3.3.

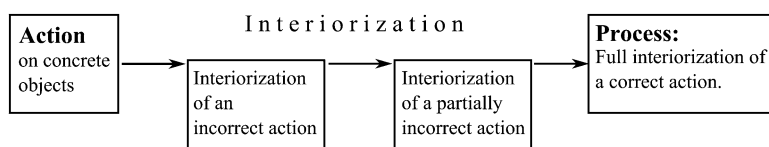


Fig. 9.8 A genetic decomposition from Action to Process for constructing a non-unit fraction

9.3.3 Additional Achievements: Abstract Objects

As indicated above, both the EXP and STN instructional sequences included simple arithmetic operations and comparisons of fractions that were performed with physical manipulatives. These actions were intended to support the encapsulation of the conception of fraction as Process into the conception of fraction as mental Object. The students were not taught any algorithms. Although the study set out to investigate the development from Action to Process, two students gave evidence of having reached the Object stage when they performed formal actions of comparison on fractions as abstract objects. For example:

Judi, an EXP student, used $\frac{1}{2}$ as an abstract benchmark when comparing $\frac{2}{5}$ and $\frac{4}{6}$:

- I: What about two fifths and four sixths, which is larger?
 Judy: It seems to me that four sixths.
 I: Why?
 Judy: Because two fifths is smaller than a half, and four sixths is already more, because three sixths is a half, and two fifths has not yet reached a half. (Arnon 1998, p. 129)

By way of comparison, Yulia also used $\frac{1}{2}$ as a benchmark for comparing $\frac{4}{10}$ and $\frac{3}{4}$:

- Yulia: Because four tenths is approximately such a thing [gesture of an arc smaller than 180°] and this will be approximately like this... more [gesture of an arc larger than 180°]. (Arnon 1998, p. 155)

Unlike Judi, Yulia referred to $\frac{1}{2}$ with gestures of imaginary arcs, smaller and larger than 180° . Comparing these two excerpts highlights Judi's response as evidence of having constructed an Object conception of fractions.

Dan, also an EXP student, used fraction equivalence and transitivity to explain why $\frac{4}{6}$ was larger than $\frac{2}{5}$:

Dan: $\frac{2}{5}$ is $\frac{4}{10}$, and $\frac{4}{6}$ is larger than $\frac{4}{10}$. (Arnon 1998, p. 129)

Dan and Judi, both EXP students, appeared to operate on the non-unit fractions as abstract objects. It seems that they had encapsulated the Process into an Object. They solved the comparison problem properly and made no reference to concrete objects. Although they started their learning with actions on concrete objects, they constructed abstract objects, as predicted by APOS Theory. No student in the STN class gave evidence of this type of development.

9.4 Manipulating Concrete Objects in the Imagination

For Piaget, the interiorization of actions on concrete objects is demonstrated by the emergence of the learner's consciousness of the actions. This is described in detail in *The Grasp of Consciousness* (Piaget, 1974/1976). In APOS Theory, interiorization is described as the emergence of an individual's ability to gain internal control over the Action. For learners at the stage of concrete operations, this, according to APOS Theory, might be evidenced by the learner's ability to carry out the Action in her or his imagination. Also, the levels that emerged from the analysis described above were defined in terms of Actions operated in the imagination on imaginary concrete Objects. In order to understand the scope of the newly defined levels, it was interesting to investigate the prevalence of manipulating concrete Objects in the imagination.

In the interviews (Arnon 1998), students completed tasks they originally learned to perform with manipulatives. However, the interviews took place without manipulatives. One of the purposes of the study was to collect data about the methods the students used for solving such problems in the absence of their manipulatives.

Some of the students produced drawings that approximated their use of manipulatives. The ability to produce such drawings suggested that the student had performed the Action in her or his imagination. Other subjects used terminology and gestures that indicated evidence that they carried out these Actions in their imaginations. Arnon developed six criteria to indicate instances of a learner using her or his imagination to carry out an action on imaginary concrete objects:

9.4.1 *Criterion 1: The Student Declared Explicitly That the Answer He or She Had Provided Was a Result of Actions Which He or She Had Performed on Imaginary Concrete Objects*

Following are some examples.

EXP student Gil:

...if I draw two fifths, then I'll have...eh,...eh,...a third?...Because four sixths, a sixth and a sixth, is two sixths, and two sixths equals a third. So it is as if two thirds. (Arnon 1998, p. 128)

Gil's words, "*if I draw two fifths*," constitute a declaration she imagined, or could imagine, drawing a representation of $\frac{2}{5}$. Yet no drawing was present in the interview.

EXP student Roni said that $\frac{4}{6}$ was larger than $\frac{2}{5}$. In his explanation he declared that he had used an imaginary drawing:

I: How do you know?

Roni: Ah,...I did the circle,[no drawing present] and it came out more,...mm,..in the comparison it came out bigger.

I: You did the circle in your head?

Roni: Yes. (Arnon 1998, p. 151)

In the next excerpt, EXP student Lina tries to explain why she thinks $\frac{2}{5}$ is larger than $\frac{4}{6}$.

Lina: A fifth is a bigger part. Two fifths and the,...a sixth.

I: How are you trying to find out? I can see that you are thinking.

Lina: ...

I: What are you trying to do in your head in order to know?

Lina: To see what each form looks like, to arrange it.

In Lina's last response, she gives evidence of constructing a representation ("what each form looks like") and of carrying out an Action in her imagination ("to arrange it").

Her use of imagination continued to be apparent in the following excerpt:

I: Maybe you would like to draw it?

Lina: No.

I: No? You don't have to? Then what do you draw in your head? What do you draw in your head? Describe to us what do you try to, to arrange in your head?

Lina: The circle.

I: Yes...

Lina: How the circle,...together...

I: What does the circle that... what does your drawing look like?

Lina: You divide into five parts

- I: Yes, and
 Lina: And... one takes two fifths, yes.
 I: What do you mean? What does one do? You took a circle and divided into five parts. What do we do now?
 Lina: Eh... the second circle, divide into six.
 I: Still with the first circle. What do you do with the first circle in order to see two fifths.
 Lina: ...
 I: Is it enough to divide it into five parts?
 Lina: No.
 I: But?
 Lina: ...
 I: What do you do?
 Lina: To color?
 I: To color? What do you color?
 Lina: Eh, two, two fif, eh, two fifths.
 I: O.K. Now, what do you do in the second circle?
 Lina: You divide into six parts, six parts, and color four sixths.

Since no drawing was present throughout the entire conversation, the researcher judged this to fall under Criterion 1 (Arnon 1998, p. 151).

9.4.2 Criterion 2: Activating Imaginary Circle Cutouts That Did Not Exist in the Original Set of Manipulatives

Offir's comparison of $\frac{1}{100}$ and $\frac{1}{13}$, which was discussed in Sect. 9.2, is an example of a Criterion 2 response. This idea coincides with Piaget and Inhelder's (1966/1971) notion of *authentic anticipatory image*, which occurs when the subject has to imagine an object unknown to her or him in advance and anticipate actions applied to that object. In the following excerpt, Sharon, an STN student, appears to offer authentic anticipatory imagery:

- I: ...How, if you had these hard-paper objects, how would you use them to check [the answer that $\frac{1}{7}$ was larger than $\frac{1}{11}$?]
 Sharon: I'd see, let's say, one, eleven, one over eleven, eh, we see that it is so small, even in the classroom, even with that thing.
 I: Did you have one over eleven in the classroom?
 Sharon: Eh, no, there was not one over eleven
 I: Yes, so how do you know about one over eleven?
 Sharon: ...
 I: Was there one over eleven in the classroom?
 Sharon: No.
 I: So how do you know about one over eleven if it was not in the classroom?

Sharon gives evidence of authentic anticipatory image in her response to the interviewer's question:

- Sharon: Ah, there was also a one over ten. One over ten is close to eleven. So we checked with the ten, and we also saw with six, with that thing, which was bigger.

She provides additional evidence somewhat later:

- I: How did you check?
 Sharon: We put the one over eleven, and we put the one over seven like that, above it, like that and you check. If you have a space left of the one over seven, you know that the one over seven is bigger. (Arnon 1998, pp. 153–154)

9.4.3 *Criterion 3: The Use of Drawings*

The students' drawings were necessarily inaccurate approximations of the real manipulatives. In the case of comparing fractions, inaccurate drawings by themselves were of little use, because of the way comparison was carried out in class: put one circle cut upon the other to see which is physically larger (see Sharon's description of such comparison in the last part of her excerpt). Arnon describes conditions for drawings to be counted as evidence of performing an action in one's imagination. Rikki, an EXP student, made useful drawings that constituted evidence of Criterion 3. In the following example, Rikki tries to solve a multiplication problem by using the drawing she provided. In her effort to solve the problem $\frac{1}{5} \times 3$, she makes a drawing to represent $\frac{1}{5}$ (Fig. 9.9):



Fig. 9.9 Rikki's drawing of $\frac{1}{5}$

Then she proceeds verbally, describing a mental operation:

Now it is, eh,...as if multiplied by three, so it is three, eh,...it is two more like this, as if, as if it equals three fifths? (Arnon 1998, pp. 155–156)

9.4.4 *Criterion 4: Verbal Indications That Involve the Use of Language That Refers to the Concrete Manipulatives*

Arnon (1998) provides a long list of expressions that were accepted as indicators of an individual manipulating concrete objects in her or his imagination (p. 160).

A special case of verbal indicators was use of the terms “part” and “the whole.” These expressions do not necessarily indicate concrete connotations. Use of these

terms was considered an indication of an Action carried out in one's imagination only if somewhere in the interview, there was evidence of the student using the terms "part" and "the whole" to symbolize concrete entities. An example follows.

Effi, an STN student, when explaining his comparison of unit fractions, used the terms *parts* and *the whole* with no indication of concrete meaning. Yet, later in the interview, when comparing two non-unit fractions, he said:

Effi: the two and the four are, we learned that the four one should color out of the six parts, and out of the five one should color two parts. (Arnon 1998, pp. 156–157)

The use of the expression "color two parts," although it came in a different part of the interview, ensured that for this student, "parts" were concrete objects.

9.4.5 Criterion 5: Gestural Indications

Following are some examples of gestural indicators. In the following excerpt, Yulia uses hand gestures to represent fractions larger and smaller than a half:

I: Which is larger $\left[\frac{4}{10} \text{ or } \frac{3}{4}\right]$?

Yulia: Three quarters.

I: Why?

Yulia: ...

I: How do you know?

Yulia: Because four tenths is approximately such a thing [gesture of an arc smaller than 180°] and this will be approximately like this, ... more [gesture of an arc larger than 180°]. (Arnon 1998, p. 155)

Maya, an EXP student, uses gestures in her construction of one seventh $\left(\frac{1}{7}\right)$:

Maya: One does a whole [with her finger 'draws' a circle in the air].

...

Maya: One divides it into seven parts [with her finger 'draws' radii of the imaginary circle in the air]. (Arnon 1998, p. 158)

Criteria 1–5 correspond to three of the criteria distinguished by Piaget and Inhelder to identify Actions in the imagination: verbal expressions, drawings, and body gestures (Piaget and Inhelder 1966/1971). In the present study, an additional criterion was used, as described below.

9.4.6 Criterion 6: Prompting

Often in the interview the interviewer encouraged the student to refer to concrete Objects used in class. Arnon called this type of interference *prompting*. Evidence of operating imaginary Actions upon imaginary Objects was counted according to the

different chronological relations between the evidence and the prompt (Arnon 1998). For more about the role of prompts in APOS-based research, see Chap. 6.

Altogether, 44 students, or 70 % of the interviewees (of both EXP and STN classes), provided evidence of manipulating concrete Objects in their imaginations when solving formally presented problems, an indication that they had interiorized these Actions. Also, all the students who interiorized correct Actions for all four concepts of the experiment provided such evidence (Arnon 1998).

These results support the role of imagination in the transition from Action on concrete Object to the development of a Process conception. When this Process is itself encapsulated, the resulting Object is an abstract Object.

Also, the high percentage of students who operated on concrete Objects in their imagination gives hope that the levels defined above have some generality.

One can find evidence in the literature of similar behavior by adults. For example, Hatano et al. (1977) studied the performance of expert abacus users and found that users of intermediate skill employed imitative finger movements when solving problems without an abacus, while advanced users testified to having used finger movements earlier, but no longer needed to do so. Similar behavior was exhibited by Arnon's "experts": all (100 %) of the students who interiorized correct Actions for all four concepts of the experiment provided evidence of performing concrete Actions in their imaginations.

9.5 Equivalence Classes of Fractions in Grade 5

This section discusses an experiment designed to investigate the use of APOS Theory in the teaching of formal concepts of advanced mathematics at the elementary school level, in this case the concept *Fractions as Equivalence Classes*⁴ in grade 5 (Arnon et al. 1999, 2001). A software program was developed specifically to serve as a concrete (graphical) representation for the teaching of this concept. This representation consists of points and lines in a discrete Cartesian coordinate system, which will be presented in detail in Sect. 9.5.2.

This study is an APOS-based study in the following sense: the learning of the mathematical concept began with an Action on concrete Objects (drawings); the learning sequence consisted of small group activities, class discussions, and additional exercises; 20 (out of 30) students participated in individual interviews after the instruction; the interview data was analyzed using APOS Theory; the software and the learning sequence were constructed so that students first learned to construct representations of fractions, classes, and binary operations in a step-by-step manner, and only after interiorizing these constructing Actions did they start working with ready-made representations (similar to the learning sequence of the

⁴ See the appendix at the end of this chapter for the definition of fractions as equivalence classes and $Q(R)$, the quotient field of a commutative ring.

EXP class in the study of fractions); and the algorithms for the binary operations were not taught in class. The operations were used to enhance encapsulation of Process conceptions to Object conceptions. In the search for evidence of Process conceptions of Equivalence Classes, Arnon et al. (1999, 2001) found situations where students used imaginary lines and points when solving arithmetic problems such as comparisons, again similar to the findings of the fraction study described in Sects. 9.2–9.4.

9.5.1 *Equivalence Classes of Fractions in the Literature*

Several studies have dealt with middle grade students, preservice teachers, and university mathematics students' difficulties in learning about equivalence classes (e.g., Asghary and Tall 2005; Chin and Tall 2001; Hamdan 2006; Mills 2004). Moreira and David (2008) claim that the study of fractions as equivalence classes is important for prospective teachers' understanding of the real numbers, even though it is not part of the school syllabus. Although Chin and Tall (2001) report on the representations of equivalence classes by means of points and lines on a discrete grid, they did not use it in their teaching because they estimated that it was too complex, even for postsecondary students. Arnon et al. (1999, 2001) report on a teaching experiment with 5th graders, who learned about equivalence classes of fractions using this representation.

9.5.2 *The Experiment*

Software was designed to provide students with a concrete environment in which they could work with visual representations of fraction-related concepts. The environment consists of a Cartesian coordinate system, where a single fraction $\frac{a}{b}$ is represented by a discrete point on the system, with the denominator b represented on the horizontal axis and the numerator a represented on the vertical axis. The origin, as well as the entire vertical axis, is inaccessible (because the denominator is 0).

An example screen appears in Fig. 9.10. Formal mathematical expressions are on the left (here a green isolated fraction $\frac{2}{3}$ and the two equivalence classes: for $\frac{1}{5}$, given in yellow, and $\frac{1}{3}$, given in purple), and each numeric object is represented, with its corresponding color, by a graphical representation on the right.⁵ A fraction appears as a discrete point on the grid, and its equivalence class is represented by a line that passes through the origin and the point, passing through all and only the points of the equivalence class. During the first part of the learning sequence, the

⁵The colors are displayed in the electronic version of the book.

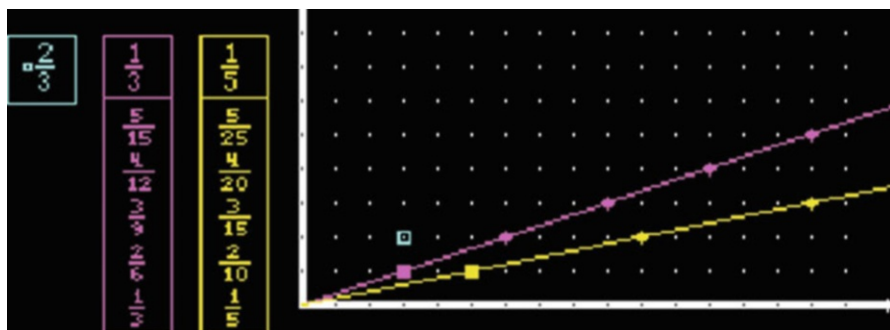


Fig. 9.10 The concrete environment (Arnon et al. 1999, p. 35)

numeric part of the screen is inaccessible to the student. The students work in the graphics window (the right-hand side of the window). Here they can construct points that represent fractions and lines that represent equivalence classes, and watch the corresponding arithmetic expression appearing automatically on the left-hand side of the screen following their constructions. The graphics window also allows them to compare fractions and perform arithmetic operations. The software and activities were designed to help the students see that arithmetic and comparative operations are independent of the equivalence class representatives selected. Similar to the instruction sequence of the study of *fractions with circle cutouts* (Arnon 1998), arithmetic operations were not taught using algorithms. Rather, computer activities involving comparison, addition, and subtraction were meant to encourage encapsulation of the Process of forming equivalence classes.

Although the software can also include fractions with negative denominators or numerators, the 5th grade students involved in this study worked only with fractions with positive denominators and nonnegative numerators.⁶

Thirty 5th graders (ages 11–12) participated in the experiment. When the instruction sequence was over, 20 of these students participated in audio-recorded individual interviews. The interview problems were presented to them in either formal language or drawings. Only after solving a problem without the computer was the interviewee allowed to check her or his result with the software. The study describes in detail students' responses and behavior in these interviews and analyzes them according to APOS Theory.

The data reveal that most of the interviewees developed an Action or Process conception or were in transition from Process toward Object conception of equivalence class.

In the instructional sequence, students solved traditional fraction problems (addition, subtraction, and comparison) using equivalence classes and the software tools instead of numerical algorithms. This is illustrated in examples of student work given in Figs. 9.11 and 9.12.

⁶The representation of fractions described above was also dealt with by Kalman (1985), Kaput and Hollowell (1985), Kieren (1976), and Lemerise and Côté (1991).

Fig. 9.11 Dora, a low-achieving student, solved an addition problem with different members of the relevant equivalence classes

$$\frac{1}{4} + \frac{2}{5} = \frac{9}{20}$$

$$\frac{1}{4} + \frac{1}{5} = \frac{3}{20}$$

$$\frac{1}{4} + \frac{1}{5} = \frac{18}{40}$$

$$\frac{1}{4} + \frac{1}{5} = \frac{27}{60}$$

① $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

② $\frac{3}{4} + \frac{1}{4} = \frac{4}{4}$

③ $\frac{4}{8} + \frac{2}{8} = \frac{6}{8}$

④ $\frac{6}{12} + \frac{3}{12} = \frac{9}{12}$

⑤ $\frac{4}{16} + \frac{8}{16} = \frac{12}{16}$

⑥ $\frac{5}{20} + \frac{10}{20} = \frac{15}{20}$

⑦ $\frac{12}{24} + \frac{6}{24} = \frac{18}{24}$

⑧ $\frac{16}{32} + \frac{8}{32} = \frac{24}{32}$

⑨ $\frac{18}{36} + \frac{9}{36} = \frac{27}{36}$

⑩ $\frac{20}{40} + \frac{10}{40} = \frac{30}{40}$

⑪ $\frac{22}{44} + \frac{11}{44} = \frac{33}{44}$

⑫ $\frac{24}{48} + \frac{12}{48} = \frac{36}{48}$

⑬ $\frac{26}{52} + \frac{13}{52} = \frac{39}{52}$

⑭ $\frac{28}{56} + \frac{14}{56} = \frac{42}{56}$

⑮ $\frac{30}{60} + \frac{5}{60} = \frac{35}{60}$

⑯ $\frac{32}{64} + \frac{16}{64} = \frac{48}{64}$

⑰ $\frac{34}{68} + \frac{17}{68} = \frac{51}{68}$

⑱ $\frac{36}{72} + \frac{18}{72} = \frac{54}{72}$

⑲ $\frac{38}{76} + \frac{19}{76} = \frac{57}{76}$

⑳ $\frac{40}{80} + \frac{20}{80} = \frac{60}{80}$

㉑ $\frac{42}{84} + \frac{21}{84} = \frac{63}{84}$

㉒ $\frac{44}{88} + \frac{22}{88} = \frac{66}{88}$

㉓ $\frac{46}{92} + \frac{23}{92} = \frac{69}{92}$

㉔ $\frac{48}{96} + \frac{24}{96} = \frac{72}{96}$

Fig. 9.12 Limor, an above average achieving student, solved an addition problem with different members of the relevant equivalence classes

This experiment established the plausibility of teaching the concept *equivalence classes of fractions* in elementary schools and perhaps other advanced mathematical concepts. As for this concept, more research is needed to establish a preliminary genetic decomposition that will serve as a beginning of a series of research studies that will produce an adequate genetic decomposition according to the APOS framework, described in Asiala et al. (1996).

9.6 What Is Known About the Use of APOS Theory in Elementary School

Following is a summary of what was learned from the studies reported in this chapter about the use of APOS Theory in elementary school.

Children did better in developing a Process conception of fractions when starting with an Action of producing a fraction representation that corresponds to the *part-whole* interpretations of fractions than children who started with Actions on ready-made concrete representatives. For example, in the fractions study discussed in Sect. 9.2, only 27 % of the STN students interiorized proper Actions for all four concepts studied versus 43 % of the EXP students. The study (Arnon 1998) showed that the Action of producing concrete representations of a mathematical concept is more effective than Actions on ready-made representations.

In the case of fractions, there might be several levels (substages)⁷ in the transition from Action to Process. For example, the data showed the existence of two such levels—*interiorization of a completely incorrect action* and *interiorization of a partially incorrect action*—between the Action and Process for the concept *non-unit fraction*. As is typical to levels, they are specific to each concept.

A criterion for the interiorization of actions performed on concrete Objects was established: when shifting from Action to Process or within the Process stage, the learner performs the Action in her or his imagination when solving problems presented formally. The data showed that 70 % of all the interviewees of Arnon (1998) (both EXP and STN students) provided evidence of that.

In Sect. 9.3 criteria were suggested for determining when an Action takes place in the imagination: by body gestures, use of language, or use of approximate drawings [all of which were found in both studies, that of elementary fraction concepts and that of fractions as equivalence classes (Arnon 1998; Arnon et al. 1999, 2001)]. These criteria may be useful in future research.

Advanced mathematical concepts, such as equivalence classes of fractions, can be adapted to the elementary school level by means of appropriate concrete Objects and adequate Actions. APOS Theory was used to construct a teaching sequence that enabled the students to develop meaningful conceptions of the topics in question.

⁷ About the difference between level and stage, see Chap. 8.

Appendix: Fractions as Equivalence Classes: Definition

Let R be a commutative ring without zero divisors.

- (a) We define a relation on $R \times (R \setminus \{0\})$ by $(a, b) \sim (c, d) \Leftrightarrow ad = bc$.

This is an equivalence relation. The equivalence class of (a, b) is denoted by $\frac{a}{b}$.

- (b) The set $Q(R) := \left\{ \frac{a}{b} \mid a \in R, b \in R \setminus \{0\} \right\}$ of equivalence classes, endowed with the operations $\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}$ and $\frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}$, is a field, called the **quotient field** of R (Spindler 1994, V. II, p. 40).

Chapter 10

Frequently Asked Questions

This chapter consists of answers to questions about APOS Theory that either have appeared in print or have arisen in personal communications with the authors. The format for this chapter is similar to that of an interview: there is a question or statement followed by a response from the authors. Where appropriate, the response will include a reference to one or more of the chapters in this book.

10.1 Questions About Structures, Mechanisms, and the Relationship between APOS Theory and the Work of J. Piaget

Q: Is a Process a generalization of the Action to which it corresponds?

A: A process is not a generalization; it is a reconstruction of a transformation of Objects onto a higher plane—from the plane of external, physical transformations (Action) to the plane of mental transformations (Process). In this sense reconstruction of a transformation refers to the learner's ability to internalize an external Action through the mechanism of interiorization so that the transformation is wholly under the learner's control.

Q: What is the difference between a mental structure and a mental mechanism?

A: A mental structure is any relatively stable (although capable of development) transformation that an individual uses to make sense of a mathematical situation. A mental mechanism is the means by which a mental structure is constructed in the mind of an individual.

In APOS Theory, the mental structures are Actions, Processes, Objects, and Schema. These structures are constructed through mental mechanisms such as interiorization, coordination, encapsulation, and thematization. Specifically, an Action is interiorized into a mental Process, two mental Processes can be coordinated to form a new Process, a Process is encapsulated to form a mental Object, and a Schema can be thematized into a mental Object.

See Chap. 3 for more on mental structures and the mechanisms by which they are constructed.

Q: In the mental development of a mathematical concept, must the learner run through every stage of APOS, and if so, doesn't that necessarily mean that each stage must be constructed (or passed through) linearly?

A: Since a Process involves the reconstruction of an Action (via interiorization) and since an Object arises (via encapsulation) out of the desire (or need) to apply an Action or Process to a Process, full development of a mathematical concept necessitates the construction of each stage.

This seems to suggest that the development of mental constructions always proceeds linearly. However, this may not always be the case, particularly when the learner is asked to apply a concept to an unfamiliar situation. Although the learner first tries to assimilate the new situation, that is, to use existing structures to make sense of the situation, this may not be possible. When this is the case, the existing structures need to be reconstructed, that is, the learner needs to accommodate the existing structures in order to assimilate the new learning situation. The reconstruction typically involves a nonlinear progression through the Action—Process—Object sequence. For example, a learner may have previously constructed the concept of binary operation as a mental Object. When presented with a new situation, say a function defined on a set with which the learner is unfamiliar, he or she would need to de-encapsulate the binary operation Object back to its underlying Process and reconstruct her or his Process conception in order to assimilate the new context.

Q: Can a learner encapsulate the “wrong” Process?

A: In general, any mental Process can be encapsulated. For instance, in the mental construction of infinite repeating decimals, it is possible for a learner who has constructed a repeating decimal as an infinite Process to encapsulate a finite Process. This occurs when the learner does not yet conceive of an infinite repeating decimal Process as a Totality. In an effort to apply an Action (or Actions) to a repeating decimal (or repeating decimals), the learner may encapsulate a finite Process. For the repeating decimal $0.\bar{9}$, a learner who does not see the infinite decimal in Totality might see the decimal as infinitesimally close to but not equal to 1 (see Chaps. 5 and 8 for more on the mental construction of infinite repeating decimals).

Q: Often it is said that Actions or Processes can be applied to mental Objects. Can you give an example of a Process applied to an Object? How does this differ from an Action applied to an Object?

A: A Process is an Action which has been interiorized and is under the learner's control. As such, its steps do not need to be carried out explicitly. Thus, any interiorized Action that a learner applies to an Object is necessarily a Process applied to the Object. For example, given propositions A and B , where both are conceived as Objects, an individual might apply an implication, that is, if A is

true, then B is true. This thought does not require knowing whether A or B or both are true, so it is not being done explicitly following any algorithm. By definition, this is a Process.

Another example might involve operations on cosets. If the learner is given a set of cosets and asked in general how to define an operation on a coset, the operation would be conceived as a Process if no explicit expression for carrying out the operation is given.

Q: It is sometimes difficult to apply APOS Theory in nonfunction-related contexts. How can one distinguish between a Process and a function?

A: In a sense, since both Actions and Processes are transformations of Objects just about everything to which APOS Theory is applied can be considered to be a function. But each concept has its own features that must be considered explicitly, so thinking of everything only as a function may not be useful. Also, while mathematicians do not always distinguish between a Process and a function (out of convenience), one must be aware of the distinction (i.e., a Process is only one part of a function) and maintain it where appropriate.

It is also important to note that APOS Theory has been applied successfully to a variety of concepts that do not explicitly involve the concept of function (see Chaps. 4, 5, 7–9, as well as Weller et al. 2003).

Q: Where can one learn more about the relation between APOS Theory and Piaget?

A: APOS is an extension of Piaget's theory of reflective abstraction applied to advanced mathematical thinking. If one accepts the idea that mathematics is the study of mental objects and how they are transformed, APOS provides a language and methodology that can be used to describe how individuals construct and transform such objects, the mechanisms by which they are constructed, and the role of instruction in helping students to make those constructions. The relationship between Piaget's theory of reflective abstraction and APOS Theory is discussed in detail in Chap. 2.

Q: Does APOS Theory take into account Actions applied to physical objects?

A: This is discussed in Chap. 9.

10.2 Questions Related to Genetic Decomposition

Q: Is it possible that different genetic decompositions lead to different understandings of the same concept?

A: A genetic decomposition is a description of the mental structures an individual may need to construct in coming to understand a mathematical concept. As discussed in Chap. 4, a genetic decomposition for a concept may not be unique, that is, there may be multiple paths by which individuals construct their understandings. The issue is whether a particular path can be verified empirically. If student data show evidence of mental constructions that align with a

particular genetic decomposition, then that description is validated. As pointed out in Chap. 4, APOS-based research has not found very many examples of different genetic decompositions for a single concept except in the case of preliminary genetic decompositions.

Q: Reference to a genetic decomposition means that the mathematical object of interest can be decomposed, but it is impossible to decompose many interesting mathematical objects. How can it be possible that a cognitive process can be decomposed in stages?

A: In APOS Theory, it is cognitive concepts and not Processes that are decomposed. A genetic decomposition is a model that describes the mental structures that an individual may need to construct in order to learn a mathematical concept. This means that a genetic decomposition is not about decomposing an Object or a Process. Rather, according to the theory, the construction of cognitive Objects begins with Actions applied to known physical or mental Objects. As an individual reflects on these Actions, a new type of construction develops, as the Action is transformed into a mental Process. Reflection on the Process and the need to perform Actions on the Process result in the encapsulation of the Process into a cognitive Object. Thus, Actions, Processes, and Objects are not parts into which a mathematical object is decomposed.

In the case of Schema, construction of relations among different Actions, Processes, Objects, and Schemas makes the construction of a new Schema possible, that is, a Schema is developed in terms of relations among its components. While it is true that one can think of a Schema as composed of these elements and can think of decomposing the Schema into its components, the relations among the components are as important as the components themselves.

10.3 Questions About Instruction and Performance

Q: Can APOS Theory be applied everywhere? Does it work for every topic and concept?

A: APOS-based instruction has been designed and implemented for a wide variety of concepts in the undergraduate curriculum. APOS Theory has been used in the design of instruction and study of student thinking in the areas of mathematical induction, quantification, calculus, functions, linear algebra, abstract algebra, mathematical infinity, and repeating decimals. It has also been applied at the K–12 level with fractions and algebraic thinking. So far, the theory has proven to be effective for all of the concepts to which it has been applied (Weller et al. 2003).

Since mathematics involves the study of mental Objects and since the study of mental objects involves transforming them, it is conceivable that APOS Theory

can be applied to the study of the cognition and instruction of any mathematical concept.

Whether APOS can be applied to concepts outside of mathematics is an open question. At this point, there are some efforts to see whether APOS can be applied to questions of cognition and instruction for concepts in computer science.

Q: For performance on tasks involving a given concept, does limitation to an Action conception necessarily suggest difficulty whereas an Object conception implies success?

A: Action, Process, and Object are mental structures; they do not represent levels of mathematical performance. However, it is possible to conflate these ideas. A learner who is limited to an Action conception would have difficulty with tasks that call for a Process or an Object conception. For instance, a learner who is limited to an Action conception of the function concept would have difficulty composing two functions that are not given by explicit formulas. Similarly, a student who is limited to a Process conception of the function concept would have difficulty determining the supremum of a set of functions. So, limitation to an Action conception means that the learner is confined to tasks requiring no more than Actions whereas a Process conception enables the learner to work on a wider variety of tasks.

Q: How do traditional and reform-oriented approaches to instruction differ from an APOS-based approach to instruction?

A: A traditional approach to instruction, usually with a focus on lecture as the principal instructional strategy, emphasizes the dissemination of information. The instructor's role is to organize ideas related to a concept and then to present those ideas as clearly as possible.

A reform-oriented approach to instruction, with a focus on non-lecture instructional strategies, emphasizes learning through engagement. The instructor's role is to design individual and collaborative activities that help students learn about a concept through experience.

An APOS-based approach to instruction emphasizes the construction of mental structures that may be needed in the learning of a concept. The role of the instructor is to identify the mental structures that might be needed in learning the concept and to design activities that help students make the proposed mental constructions.

Elements of traditional and reform-oriented approaches may be used in the design of APOS-based instructional activities. This typically involves use of the ACE Teaching Cycle, which is discussed in detail in Chap. 5.

Q: Different APOS papers say, "In our work, we have used cooperative learning and implemented mathematical concepts on the computer...." Since APOS-based pedagogy is not necessarily tied to cooperative learning and programming, can other pedagogies be applied in the implementation of APOS Theory?

A: Other pedagogies have been tried, for example, for elementary school students (see Chap. 9), but it seems that for students at the collegiate level, the best results are obtained when cooperative learning, along with writing and running programs to represent Processes and Objects, is used.

10.4 Questions Related to Topics Discussed in Mathematics Education: Representations, Epistemology, Metacognition, Metaphors, Context

Q: What is the role of APOS Theory in metacognition and reflection?

A: The underlying concept in APOS Theory is Piaget's notion of reflective abstraction (Dubinsky 1991). According to Piaget, reflective abstraction has two aspects. One is the reflection (and possible reconstruction) of a concept onto a higher plane of thought. The other is reflection on an individual's thinking about a concept. Reflection and relations to metacognition are contained in the second aspect.

Q: Does APOS Theory take into account representations of mathematical concepts? If so, how would this occur?

A: Most of the mathematics education literature on representations concerns issues involving the transition from one representation to another. APOS Theory considers representations and transitions among them in a somewhat unique manner (as illustrated in the accompanying figure for the concept of function). Figure 10.1 shows an apex with several downward pointing arrows. The apex represents the genetic decomposition. The arrows from the apex point down to different representations. The key idea is that the student constructs the concept via the genetic decomposition. In dealing with a problem situation, which may call for a particular representation of the concept, the learner thinks of the concept in terms of that representation. An arrow from the apex to one of the representations of the concept accounts for this. If a student needs to change representations, that is, to transfer from representation A to representation B, he or she moves from representation A (in her or his thinking) to the apex and then from the apex to representation B. In the example of the function process illustrated below, the student uses the

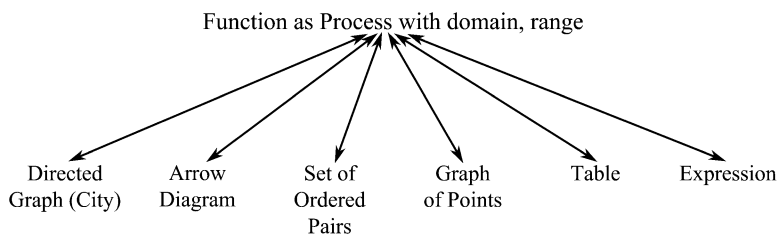


Fig. 10.1 Transitions among function representations

given representation to figure out the Process of the function that is represented. Then, using her or his Process conception, the individual moves down (most likely, in an unconscious way) along the line corresponding to the desired new representation to express the process in terms of the new representation. Instruction based on this idea has been used in a high school class with promising results (Dubinsky and Wilson 2013).

According to APOS Theory, the reason students have so much trouble making the transition from one representation to the next is that they (are taught to) go directly from one representation to another without passing through the cognitive meaning of the concept (given by the genetic decomposition). Considerably more research needs to be conducted to determine whether the point of view suggested by APOS Theory is useful.

Q: Does APOS Theory take into account the epistemology of mathematical concepts?

A: The answer is yes. In fact, a genetic decomposition is an epistemological analysis, in line with the genetic epistemology of Piaget. In it the nature of a mathematical concept appears through mental structures and mechanisms that might give rise to its construction.

Q: In APOS Theory, what does it mean to construct an understanding of a mathematical concept?

A: APOS Theory considers that mathematical concepts are the building blocks of mathematics. From the point of view of APOS Theory, constructing an understanding of a mathematical concept means that students are capable of dealing with certain types of familiar and unfamiliar problem situations involving the concept. According to the types of situations a person can deal with, an analysis using APOS Theory describes that understanding in terms of mental structures and their relationships. However, construction of a particular mental structure does not necessarily mean that a student will deal successfully with a problem-solving situation where the structure is needed. Rather, construction of a particular structure would suggest that the student has the capability to deal with problem-solving situations in which construction of the structure is called for. A person who demonstrates a deep understanding of a concept is capable of dealing with unfamiliar and even new situations using the concept or concepts in question. See Chaps. 5, 6, 8, and 9 for a discussion related to this issue.

Q: APOS studies do not appear to pay particular attention to the contextual interaction of subjects with mathematical concepts. Why?

A: APOS Theory deals with the construction of mental structures that may be needed in the development of mathematical knowledge. The theory implicitly assumes that individuals need to construct certain mental structures before they can deal with mathematical contexts involving the concept. The importance of context in the learning of mathematical concepts is worthy of further inquiry, although there seems to be little evidence that studying mathematics in context improves learning and even some indication that it might be counterproductive in some cases.

Q: Does APOS Theory take into account the construction of metaphors or Grundvorstellungen (simple ideas)?

A: Simple ideas, or Grundvorstellungen, are taken into account in APOS Theory in the sense that the construction of new cognitive Objects starts from previously constructed Objects that may be based on simpler ideas. Metaphors are not taken into account in APOS Theory, which is an alternative to metaphors as a means of describing the construction of new knowledge.

10.5 A Question About Intuition

Q: What role does intuition play in APOS Theory?

A: According to Piaget, intuitions, like all thought, are constructed (Beth and Piaget 1965/1974). Generally speaking, the usefulness of an intuition depends on its effect on a student's ability to make the specific mental constructions called for by a theoretical analysis. In APOS Theory, instruction for a concept is based on its genetic decomposition, a description of the mental constructions a student needs to make in order to learn the concept. According to the theory, a student can develop understanding of any concept for which he or she has made the necessary mental constructions. Within this context, there are two types of intuitions—those that support development of the proposed mental constructions or those that impede its development.

For example, when comparing the cardinality of infinite sets, students sometimes try to extend to infinite sets the part-whole notion developed in comparing finite sets. In working with finite sets, students see that the cardinality of B exceeds the cardinality of A if A is a proper subset of B . In seeing that this relationship holds for every pair of finite sets A and B (for which A is a proper subset of B), a student constructs an intuitive scheme, that is, the student naturally and unconsciously thinks of and applies this relationship when comparing the cardinality of finite sets. When presented with two infinite sets A and B , with A a proper subset of B , the learner may apply the constructed intuition, believing that a part-whole comparison, which has worked successfully for finite sets, applies to infinite sets. A familiar case involves comparison of the even counting numbers and the natural numbers. Many students believe the former has a smaller cardinality than the latter because the even numbers are a proper subset of the natural numbers. In a situation such as this, the role of APOS Theory is not to dismiss the usefulness of the intuition but to enhance accommodation of the existing structure of the intuition, so it treats infinite sets differently. Large finite sets and infinite sets have a common characteristic: neither can be physically enumerated. The two sets differ, though, in how one *imagines* their enumeration—a large finite set has a last element while an infinite set does not. For a large finite set, the last element signifies completion; the two sets can be counted and the results of each count compared. For an infinite set, since there is no last element, completion is not indicated by a last element but

by *completion of the Process*, that is, by the ability to conceive of the Process of enumeration in the past tense. As a result, the usual notion of counting does not apply. The difference in the meaning of completion between finite and infinite sets explains why part-whole, which works for finite sets, fails for comparison of infinite sets and why a different means of comparison is called for.

In both cases, comparison is an Action applied to sets. For this Action to be applied, the Process must be encapsulated. For finite sets, this is triggered by enumeration of the last element. For infinite sets, things are more complicated. Specifically, the completed Process of enumeration must be viewed in its totality, that is, as a single operation freed from temporal constraints. This transition from Process to Object is what makes the comparison of infinite sets so difficult and is what helps to explain the persistence of part-whole thinking, even when it does not apply.

On the other hand, intuitions can prove to be useful, even when crossing domains such as the transition from finite to infinite. For a finite sequence of iterated actions in construction of a set (e.g., a nested sequence of sets), one would conclude that an element that arises at step n and for all successive steps beyond step n would be contained in the final set constructed by the iteration (here called the “principle of accumulation”). This principle guides the iterative construction of any finite set (e.g., the first k counting numbers) and also applies to construction of the set of natural numbers N : once a natural number appears, as one iterates, it appears in the set constructed at each subsequent step and in the resulting set, which is N itself. This construction is rooted in one of the most elemental mathematical activities—counting—which is the basis for any process of incremental accumulation. Thus, it is reasonable to say that individuals likely construct the principle of accumulation as an intuitive notion.

Radu and Weber (2011) provide some confirmation for this in their study of students’ thinking about completed infinite iterative processes. On one task, students are asked to determine the state at infinity, or resultant state, for the Vector Problem, which follows:

Let $v = (1, 0, 0, \dots) \in N^N$. This vector will be modified in the following ways:

Step 1: $v = (0, 1, 2, 0, 0)$

Step 2: $v = (0, 0, 1, 2, 3, 0, 0)$

Step 3: $v = (0, 0, 0, 1, 2, 3, 4, 0, 0)$

If this process is continued ad infinitum, what form does v take after all of the steps have been completed? (p. 167)

Without prior instruction, the students solved the problem correctly, explained why their solutions made sense, and used their work to solve other related infinity tasks. Radu and Weber attributed the students’ success, in part, to their construction of correct intuitions, in particular, to the principle of accumulation.

The students did not perform as well on a similar problem, referred to here as the Tennis Ball Problem:

Suppose that an infinite set of numbered tennis balls and a large table are available. Place balls numbered 1 and 2 on the table and remove number 1. Next, place balls 3 and 4 on

the table and remove number 2. Then place balls 5 and 6 on the table and remove number 3. And so on, *ad infinitum*. What happens after all of the steps have been carried out? (Radu & Weber 2011, p. 172)

The difference between the Vector Problem and the Tennis Ball Problem lay in the context of each problem. According to Radu and Weber, the Vector Problem focuses attention exclusively on the elements and positions of the natural numbers (in the representation of each vector) whereas the Tennis Ball Problem includes cardinality (the number of balls that remain on the table at each step increases by one). The issue of cardinality is what makes the Tennis Ball Problem paradoxical: it seems that one cannot determine how many balls are on the table at the end of the activity, because, on the one hand, the number increases by one at each step, which implies an infinite quantity, but, on the other hand, given any tennis ball, one can say exactly when that ball is removed so that none is left. The latter explanation is correct: since ball n is removed from the table at step n and remains removed for every subsequent step, it follows that every ball is removed.

APOS Theory can be used to explain why the principle of accumulation supports student thinking in the Vector Problem and why the paradox seems to blunt its effect in the Tennis Ball Problem. According to Brown et al. (2010), an iterative process is based on understanding iteration through N : the process, which begins at 1 and at each successive step adds 1, results in the sequence 1, 2, 3, ..., which leads to the construction of sets: $\{1\}$, $\{1,2\}$, $\{1,2,3\}$, ... Embedded in this construction is the idea that once a natural number appears, it appears for every subsequent step. Since the construction is cumulative, encapsulation produces a final object that includes every natural number. Thus, the principle of accumulation supports the mental construction of the set. The same idea stands behind the construction of the infinite zero vector in the Vector Problem—the number 0 is added at step n in position n and appears in every subsequent vector constructed at every subsequent step. The accumulation principle then leads one to conclude that the state at infinity consists of the infinite zero vector.

On the other hand, the issue of cardinality confounds one's ability to solve the Tennis Ball Problem. Since the number of balls on the table increases by one at each step, one is tempted to conclude that there are infinitely many balls that remain on the table when the procedure has been fully carried out. This view is plausible if the iterative process of placing and removing balls is viewed as incomplete. Determination of the result of the procedure is an Action applied to the iterative Process. If that Process is not seen as complete, the individual encapsulates what is, for all practical purposes, a finite Process. As a result, he or she concludes that balls remain on the table. To move beyond that, the individual needs to see the infinite process as a completed totality. This enables her or him to see that the relevant correspondence, the removal of ball n at step n , is a single operation applied to each ball. Although the Vector Problem and the Tennis Ball Problem are similar mathematically (they both involve a movement of natural numbers that leads to a cumulative result), the Tennis Ball Problem

represents that movement in a way that makes it more difficult to solve. This is consistent with what Tirosh and Tsamir (1996) found in their study of infinite comparison tasks—the representation of the task strongly influences students’ reasoning about the task. The APOS analysis provides a theoretical explanation for why the Tennis Ball Problem is difficult to solve and shows why the intuitive principle of accumulation supports the mental construction of the natural numbers N , which makes the Vector Problem more readily solvable.

Finally, an individual’s Schema for different infinity concepts necessarily includes different intuitions regarding the concept. The role of what in APOS Theory is called *coherence of the Schema* provides the mechanism by which an individual decides which intuition to use in given problem situations. Whether a particular intuition is useful depends on whether it supports the other mental structures that constitute the Schema as well as the relationship among those structures.

10.6 Questions About How Specific Concepts Can Be Approached with APOS Theory

Q: In performing Actions on Objects, some researchers wonder why many learners do not connect activities with concrete manipulatives with formal operations. This question is often raised in relation to fractions (see Freudenthal 1973; Herman et al. 2004; and Chap. 9). Can APOS Theory be used to explain why this connection may or may not take place?

A: APOS Theory claims that such a connection does not develop automatically but as a result of an appropriate choice of manipulatives and learning sequences (see Chap. 5 for a detailed discussion of APOS-based instruction) that enhance the development of a specific concept through the stages Action–Process–Object. Without evidence of these two—appropriate manipulatives and adequate learning sequences—the connection between manipulatives and formal activities cannot be assumed.

Q: For a given set S , a binary operation o defined on S , and an operation problem $a \circ b = c$ for $a, b, c \in S$, is the element c , the result of the operation, the object that results from encapsulation of the process of applying o to $a, b \in S$?

A: The Object obtained from the encapsulation of a binary operation is not the same as the Object that results from the application of that binary operation to two specific elements of the set. For example,

[I]n the addition of two numbers, say 2 and 6, the number 8 is obtained from the process of adding 2 and 6 but is not the object that results from encapsulation of the binary operation process. Instead, the encapsulation allows the addition to be considered as an object that can be acted on; for example, it could be compared to other processes like $6 + 2$ and $8 - 6$. The number 8, like other natural numbers, is an object that is constructed by encapsulating processes *other* than the basic arithmetic operations (Piaget, 1952 [sic][1941/1965]). (Dubinsky et al. 2005b, p. 260)

If the number 8 had not been previously constructed, that is, by encapsulating the Processes that would lead to its construction, then the individual would probably be unsuccessful in performing arithmetic operations to obtain 8 as a result.

Q: In studying proportions, fractions $\frac{a}{b}$ and non-numeric ratios such as “ a is to b ” arise. A possible APOS interpretation is that such a relation is a transformation from a to b which could be an Action or a Process. The question is how does an observer decide which it is?

A: One response is that it is not easy to tell what the subject is able to do in working with the situation. As a result, one must conduct an interview with the subject to see how he or she is thinking about the situation.

Another, perhaps better, response is that if the subject is unable to reverse the relation or coordinate it with other relations, then it is likely that the subject has no more than an Action conception.

Q: Many students are confused about the difference between a set A being an element of a set B and the set A being a subset of the set B . How is this explained by APOS Theory?

A: One needs to establish the notion of a set as an Action and later, as a Process, of placing objects into a container. This can be done initially using physical containers or bags of physical objects. Then A is an element of B if the entire container (not just its contents) is contained in B . The meaning of A being a subset of B is that every time an individual selects an element of A and tries to determine whether it is contained in B , the individual finds that it is already contained in B . Activities involving objects and containers can help learners to make this distinction. Whether these types of activities would result in construction of the proposed mental constructions is a subject for future research.

Q: How can APOS Theory be used to answer the previous question for the case in which A is the empty set?

A: The response is the same as for the previous question. The empty set is represented by an empty container.

Q: Many subjects have difficulty accepting the fact that the empty set is a subset of every set. How is this dealt with using APOS Theory?

A: One way that helps with such issues is to always (at least until fundamental ideas about sets are well established) have a universal set explicitly involved when talking about sets. Then the empty set is the complement of the universal set.

Another approach is to make use of the notion of a set as an Action and later as a Process of placing objects into a container. Then, as discussed in the previous two questions, the meaning of A as a subset of B is that every time an individual selects an element of A and tries to put it in B , he or she finds that it is already contained in B . If A is the empty set, this condition is always satisfied, but vacuously, which may make it harder to understand. Activities in which one

tests for a subset by emptying the contents of bag A into a bag B to see that nothing in B has changed can help. The condition is always satisfied if A is empty.

- Q:** APOS studies of the learning of cosets do not appear to take into account geometric representations of cosets, that is, as objects that are points in a geometric space. Would the consideration of this type of representation require the need for a different genetic decomposition of the coset concept?
- A:** There do not seem to be any studies that consider cosets as points in a geometric space. Whether construction of this representation would be described by a different genetic decomposition is a topic for further research.

Chapter 11

Conclusions

The discussion concludes with three themes that have been developed throughout the preceding chapters. The notions of the developmental/evaluative dichotomy, the development of APOS Theory through mechanisms similar to those used in the theory, and the future of APOS Theory are discussed. The chapter provides “APOS Theory at a glance” and ends with some final thoughts.

11.1 Developmental vs. Evaluative Nature

The research question of how a person may learn a particular concept (or topic or subject area) is readily seen as too complex to be able to address in a single study. The variables at work are too wide ranging and sometimes cannot be directly observed and measured. In response, researchers must choose a lens, or a filter, through which to explore how learning takes place. Theories of learning and associated frameworks serve as guides for the researcher to explore more finely focused aspects of the research question. Some frameworks analyze the learning environment, pedagogical strategies, or curricular materials to be employed. Others explore student–student interactions or student–instructor interactions in problem situations. A third type of framework analyzes the types of mental constructions that may be made by an individual as learning takes place.

APOS Theory primarily falls into this third type, cognitive studies. It proposes certain mental structures that may be necessary for the construction of a concept along with the mechanisms for building those structures. In this aspect, APOS Theory serves as an evaluative framework as individuals are observed in problem situations in which the researcher attempts to describe their level of understanding as well as the mental structures at work in their learning of the concept. Chapter 2 described in detail the connections to Piaget’s work and gave the background for using APOS Theory to analyze how an individual might learn a concept. Chapters 3 and 7 elaborated on the structures and mechanisms used by APOS Theory to

evaluate a particular individual's state of learning. Chapter 4 explained that a genetic decomposition describes what a general learner might need to construct the concept under study. The genetic decomposition becomes the working hypothesis that is used to evaluate the degree to which learning has taken place.

APOS Theory also may be considered to be a developmental framework for the design and implementation of instructional materials and settings. These pedagogical strategies are typically constructed based on a genetic decomposition (preliminary or revised). They use problem situations often based on computer activities that help the students consider new mathematics or nuances in previously seen ideas. The ACE Teaching Cycle used by APOS Theory was discussed in Chap. 5. In Chap. 6, it was seen how the research methodology is used to assess the effectiveness of APOS-based pedagogy as well as to evaluate the theoretical description used in its development.

11.2 Macro-Level Consistency

Reflective abstraction is the mechanism that drives construction of new mental structures in APOS Theory. As discussed in Chap. 3, generalization allows a learner to construct a new Process from existing Processes. Both assimilation and accommodation are examples of generalization. With assimilation, new situations are dealt with by using existing structures in a new way. Accommodation involves the reconstruction of an existing structure in order to deal with an unfamiliar situation.

The evolution of APOS Theory as discussed in Chap. 2 and observed in Chaps. 8 and 9 seems to have progressed by means of similar mechanisms. Notable progress was found in the study of limits (Cottrill et al. 1996), the chain rule project (Clark et al. 1997), and the work with graphing via derivatives (Baker et al. 2000; Cooley et al. 2007). The obstacle that the researchers found in the limit study and in the chain rule study was an inability to identify the Processes that might be encapsulated into an Object conception. The solution in the limit study was to recognize that two Processes are coordinated. This was an assimilation of our existent understanding of APOS Theory. A reconstruction was necessary in the case of the chain rule where the limitations of the theory caused an in-depth exploration of what a Schema is and how it might be described. Thus, APOS Theory was accommodated in order to assimilate the triad of stages found in the work of Piaget and Garcia (1983/1989) as discussed in Chap. 7.

The projects that studied how students come to learn to apply derivative information to construct the graph of a function found the triad descriptions lacking in the attempt to evaluate the observations. The first study analyzed the interaction between Schemas, extending the theory's use of the triad stages (Baker et al. 2000). In the second paper (Cooley et al. 2007), the researchers then found it necessary to unpack the triad descriptors and reconstruct them along with the mechanism of thematization, which is described in Chap. 7. This sort of accommodation to APOS

Theory was also seen in the introduction of Totality in the study of infinite repeating decimals (Dubinsky et al. 2013) described in Chap. 8.

Thus, the fundamental impetus for modifying APOS Theory is also reflective abstraction. As a hypothesis (genetic decomposition) is tested, the data may (1) support the proposed constructions, (2) offer an obstacle that presses the theoretical description and requires an assimilation by reworking the description, or (3) require a reconsideration of the theory underlying the description that results possibly in an accommodation of the theory to account for the data.

11.3 A View of the Future of APOS

As seen in Chaps. 8 and 9, APOS Theory continues to evolve as a potential new structure (Totality) is investigated and is being applied to other areas of mathematics (as it moves into elementary mathematics). Analyses of students' understanding of infinity suggest the need for a mental structure apart from Process and Object. These studies, described in Chap. 8, also suggest the need for levels between the stages of Action and Process, between Process and Totality, and between Totality and Object. Studies will need to be designed and implemented to clarify whether this new stage exists in topics other than that of infinite repeating decimals.

In Chap. 9, the work of Arnon and her colleagues on students learning elementary mathematics was described. The investigations deal with concrete objects in order to begin the development of mental Objects. The research into students' understanding of fraction also explores the notions of levels between stages. Their work involves students at the stage of concrete operations prior to moving to formal operations. APOS Theory might be used to explore learning of other elementary mathematical topics.

There remain many topics in undergraduate mathematics that have not been investigated via the lens of APOS Theory. Other topics have been studied and the results published before our current understanding of thematization of Schemas. One example is the paper on limits (Cottrill et al. 1996), which has served as a useful example of our research paradigm (discussed in Chap. 6). It would be beneficial to reconsider the topic of the limit of a function at a point from the point of view of constructing a Limit Schema, which may include other types of limits as well.

11.4 APOS Theory at a Glance

APOS Theory is based on the following principle:

An individual's mathematical knowledge is her or his tendency to respond to perceived mathematical problem situations by reflecting on problems and their solutions in a social context and by constructing or reconstructing mathematical actions, processes and objects and organizing these in schemas to use in dealing with the situations. (Asiala et al. 1996, p. 7)

This statement was made by Dubinsky early in the work on APOS Theory and normalized for the article mentioned.

11.4.1 Structures and Mechanisms

Three basic types of knowledge—Actions, Processes, and Objects—are involved in mathematical concept construction and are organized into structures called Schemas. An Action is any repeatable physical or mental manipulation of Objects to obtain other Objects. It is a transformation that is a reaction to stimuli that the individual perceives as external.

As an individual reflects on an Action, it is interiorized and becomes a Process. With a Process conception, the learner perceives the Action as part of her or him and has control over it. As the individual realizes that an Action can be brought to operate on a Process, the Process is encapsulated to become an Object. The Object can be de-encapsulated back to the Process as needed. Processes may also be constructed by reversal and coordination mechanisms.

A Schema is a coherent¹ collection of Actions, Processes, Objects, and other Schemas that is invoked to deal with a new mathematical problem situation. A Schema can be thematized to become another kind of cognitive Object to which Actions and Processes can be applied. By consciously unpacking a Schema, it is possible to obtain the original Processes, Objects, and other Schemas from which the Schema was constructed (Chap. 7).

The triad, introduced by Piaget and Garcia (1983/1989), distinguishes three stages in the development of a Schema: Intra-, Inter-, and Trans-. In APOS Theory, the Intra-stage is characterized by a focus on a single Object in isolation from any other Actions, Processes, or Objects. The Inter-stage is characterized by recognizing relationships between different Actions, Processes, Objects, and/or Schemas. It is useful to call a collection at the Inter-stage of development a pre-schema. Finally, the Trans-stage is characterized by the construction of an overall structure underlying the relationships discovered in the Inter-stage of development.

It is worth noting that it is only when a schema reaches the Trans-stage of development that it can properly be referred to as a Schema in APOS Theory. The reason is that at the Trans-stage, the underlying structure is constructed through reflecting on the relationships among the various Actions, Processes, Objects, and/or Schemas from the earlier stages. This structure provides the necessary coherence in order to identify the collection as a Schema, that is, as a coherent whole. This coherence consists in deciding what is in the scope of the Schema and what is not.

¹ The definition of coherent is found two paragraphs below.

11.4.2 Research Methodology

A genetic decomposition is a hypothetical model that describes the mental structures and mechanisms that a student might need to construct in order to learn a specific mathematical concept. It typically starts as a hypothesis, called a preliminary genetic decomposition, based on the researchers' experiences in the learning and teaching of the concept, their knowledge of APOS Theory, their mathematical knowledge, previously published research on the concept, and the historical development of the concept (see Sect. 4.1).

A genetic decomposition might be used in two ways: (1) to develop pedagogical materials and settings or (2) as the hypothesis for collecting data to test the model. In the first case, instruction takes place followed by data collection. The data are then explored to determine how learning has taken place—evaluating the instructional approach—as well as testing the genetic decomposition. In the second case, the genetic decomposition acts as the research hypothesis to be tested by empirical data (see Chap. 6).

Data are collected via instruments that may include written questionnaires, semi-structured interviews (audio- and/or videotaped), exams, and/or computer games. All written work from questionnaires and interview tasks are collected. Triangulation of observations is enhanced by collaborative research as each researcher negotiates her or his analysis with the others. Chapter 6 describes a process for scripting interview transcripts and analyzing trends that emerge.

11.4.3 Pedagogical Approach

The design and implementation of instruction is one of the three components of the research paradigm of APOS Theory. This component involves the ACE Teaching Cycle and cooperative learning techniques. These provide the “perceived mathematical problem situations” and “social context” mentioned in the statement of the principle in the beginning of this section.

The ACE cycle begins with activities (typically involving computer programming experiences) that ask the students to consider problems that lie just beyond their shared experience. The activities may be extensions of previous experiences or situations that lead the students through an algorithm. The classroom discussion is based on the shared experience of the activities. The discussion allows students to analyze the experience together and reflect on the important aspects of the activity set. The third component of the cycle consists of homework exercises that continue the reflections on the activities and discussion as well as extend the students' experiences with the topic at hand.

The students work cooperatively on the activities in order that mathematical notions may be learned through discussion. As the students work in groups, the activities may involve a more complex situation than might be reasonable for an

individual to handle. Alternatively, the activity may be ambiguously stated so that a negotiation of its meaning leads to the necessary reflection on the mathematics being studied.

11.4.4 An Integrated Theory

APOS Theory provides descriptions of the mental structures and mechanisms that may be necessary to describe a student's apparent learning of a concept. With such tools, it is possible to build a genetic decomposition of a concept for a generic student. Using the genetic decomposition to design instruction based on the ACE cycle enhances the value of the model. One might employ cooperative learning techniques and activities in the classroom, but without a guiding model, these may not lead to the type of learning that is desired. The research component provides the empirical evidence to test the validity of the model and the efficacy of the instruction. The analysis is based on the genetic decomposition and at the same time informs it. As the model is refined, the activities, discussion, and exercises are modified and improved. Thus, all three components of the research and curriculum development cycle—*theoretical analysis, design and implementation of instruction, and collection and analysis of data* (Sect. 6.1)—are essential to APOS Theory.

11.5 Last Word

The intent of this book is to better explain issues that sometimes are not clear for readers or students who try to understand or use APOS Theory. In particular, Chaps. 2, 4, and 7 collect and synthesize ideas that had been spread among many prior publications. Chapter 2 gives as complete a history as possible of APOS Theory, written, as is the entire book, by those who have lived it. It is, of necessity, incomplete because APOS Theory still lives and its story continues with little sign of an ending.

Chapter 4 stresses that a genetic decomposition is a predictive model that can be refuted or supported by experimental data. Data can also be used in the refinement of a genetic decomposition; this possibility gives empirical support for the model. The design of a genetic decomposition is one of the most difficult aspects of applying APOS Theory to research on students' understanding of mathematical concepts and to teaching.

Chapter 7 demonstrates how the investigations into the notion of Schema show consistently that even though students can use specific concepts to solve even difficult mathematical problems, their understanding consists (possibly for a long time) of separate domains and that the development of relations and transformations between them is fundamental to achieving a deep understanding of mathematics. These studies also show how Schema development mechanisms are useful tools to

understand students' needs to develop richer Schemas to be able to develop the kinds of tasks that may be needed to help them construct these richer Schemas that can flexibly interact with other Schemas and can be thematized.

As shown throughout this book, APOS Theory, as a developmental and an evaluative approach, can be useful to answer research questions about the learning of a variety of mathematical concepts and to design and implement pedagogical approaches so that learning occurs. As an active theory in continuous development itself, the feedback it receives from research is reflected through new elements, and this in turn gives rise to new research studies. The mathematics education community, through the numerous publications listed in the annotated bibliography of this book and many other works that are in development, has contributed to this growth and, it is hoped, will continue to do so.

Chapter 12

Annotated Bibliography

This chapter contains over 120 publications about APOS Theory. These publications span a period of over 25 years and include research studies conducted all over the world. The list of studies is not exhaustive but is representative. Some publications that appear in this chapter might present perspectives on APOS Theory that differ somewhat from those presented in this book. The reader is invited to discern such differences, if found. The annotations that appear in this chapter either were written by authors of the study or have been adapted from various sources, for example, Dubinsky and McDonald (2001).

12.1 A Through B

One annotation appears for the following three studies:

Arnon, I. (1998). *In the mind's eye: How children develop mathematical concepts—Extending Piaget's theory*. Unpublished doctoral dissertation, School of Education, Haifa University.

Arnon, I., Nesher, P., & Nirenburg, R. (1999). What can be learnt about fractions only with computers. In O. Zaslavsky (Ed.), *Proceedings of the 23rd Conference of the International Group for the Psychology of Mathematics Education* (Vol. 2, pp. 33–40). Haifa, Israel.

Arnon, I., Nesher, P., & Nirenburg, R. (2001). Where do fractions encounter their equivalents? Can this encounter take place in elementary school? *International Journal of Computers for Mathematical Learning*, 6, 167–214.

The studies by Arnon and her colleagues deal with the development of mathematical concepts by elementary school children. The authors describe the difference between the use of APOS at the postsecondary level and its use in elementary school. Using a framework that combines APOS Theory with Nesher's theory of Learning Systems, they investigate the introduction of mathematical concepts as *Actions on concrete Objects* versus their introduction as *ready-made concrete representations*. Yerushalmy's ideas (1991) of multiple representations were added to the above perspectives to develop

software adapted to the learning of the concept of equivalence classes of fractions. In these studies, developmental paths for certain fraction concepts are established. It was found that students who received instruction in which fractions were introduced as Actions on concrete Objects progressed better along these paths than students who received instruction in which fractions were introduced as ready-made concrete Objects. The findings also established the following characteristic of students' development of Actions on concrete Objects into abstract Objects: after abandoning concrete materials, and before achieving abstract levels, children perform the Actions in their imaginations. This corresponds to interiorization (the passage from Action to Process) in APOS Theory.

- Artigue, M. (1998). Enseñanza y aprendizaje del análisis elemental: ¿qué se puede aprender de las investigaciones didácticas y los cambios curriculares? *Revista Latinoamericana de Investigación en Matemática Educativa*, 1(1), 40–55.

In the first part of this paper, the author discusses a number of student difficulties using various theories of learning including APOS Theory. She explains that students are generally unwilling to accept the equality $0.999\ldots = 1$ because they see the former as a Process and the latter as an Object. To accept the equality, both $0.999\ldots$ and 1 must be conceived as Objects. However, as Artigue points out, it is very difficult for students to make the necessary encapsulation. In the second part of the paper, the author discusses the measures that took place in France during the twentieth century to help students overcome these difficulties.

- Asiala, M., Brown, A., DeVries, D., Dubinsky, E., Mathews, D., & Thomas, K. (1996). A framework for research and curriculum development in undergraduate mathematics education. In *Research in Collegiate mathematics education II*. CBMS issues in mathematics education (Vol. 6, pp. 1–32). Providence, RI: American Mathematical Society.

In this paper, the authors give a complete description of APOS Theory as it stood at the time; a description of the ACE Teaching Cycle, the main pedagogical strategy for APOS-based instruction; and the methodology used in APOS-based research. The combination of the three has become known as “the APOS research framework.”

- Asiala, M., Brown, A., Kleiman, J., & Mathews, D. (1998). The development of students' understanding of permutations and symmetries. *International Journal of Mathematical Learning*, 3, 13–43.

The authors examine how abstract algebra students might develop their understandings of permutations of a finite set and symmetries of a regular polygon. They give an initial theoretical analysis of these topics, expressed in terms of APOS Theory, describe an instructional approach designed to encourage development of the mental constructions postulated by the theoretical analysis, and discuss the results of individual interviews and performance on written examinations. The results indicate that the pedagogical approach was reasonably effective in helping students develop strong conceptions of permutations and symmetries. The authors also used the data to propose a revised epistemological analysis of permutations and symmetries and to offer pedagogical suggestions.

- Asiala, M., Cottrill, J., Dubinsky, E., & Schwingendorf, K. (1997a). The development of students' graphical understanding of the derivative. *Journal of Mathematical Behavior*, 16, 399–431.

In this study, the authors explore calculus students' graphical understanding of a function and its derivative using APOS Theory. They present an initial theoretical analysis of the cognitive constructions for development of the concept, outline an instructional treatment designed to foster formation of the proposed mental constructions, discuss the results of

interviews conducted after the implementation of the instructional treatment, and describe a revised epistemological analysis based on analysis of the data. Comparative data suggest that students who received instruction based on the theoretical analysis were more successful in developing a graphical understanding of a function and its derivative than students who received traditional instruction.

Asiala, M., Dubinsky, E., Mathews, D., Morics, S., & Oktaç, A. (1997b). Development of students' understanding of cosets, normality and quotient groups. *Journal of Mathematical Behavior*, 16, 241–309.

Using an initial epistemological analysis from Dubinsky et al. (1994), the authors determine the extent to which APOS Theory explains students' mental constructions of the concepts of cosets, normality, and quotient groups. They evaluate the effectiveness of instructional treatments developed to foster students' mental constructions and compare the performance of students receiving this instructional treatment with those completing a traditional course.

Asiala, M., & Dubinsky, E. (1999). *Evaluation of research based on innovative pedagogy used in several mathematics courses*. Unpublished report, available from the authors.

During three academic years from Fall 1997 through Spring 2000, APOS Theory was used to teach a number of mathematics courses at Georgia State University. This study attempts to assess the effectiveness of that approach in terms of improvement in students' learning and students' attitudes toward mathematics. The manuscript contains an overview of the literature that describes similar attempts by others at different universities. The results of this study show an improvement, sometimes over time, of students' attitudes and their learning.

Ayers, T., Davis, G., Dubinsky, E., & Lewin, P. (1988). Computer experiences in the teaching of composition of functions. *Journal for Research in Mathematics Education*, 19, 246–259.

Students from two sections of a college mathematics lab ($n = 13$) who were given computer experiences to encourage reflective abstraction scored higher on a test of their understanding of functions and composition of functions than students from another section ($n = 17$) who were taught using traditional methods. The comparison was based on questions intended to indicate whether reflective abstraction had taken place.

Badillo, E., Azcárate, C., & Font, V. (2011). Análisis de los niveles de comprensión de los objetos $f'(a)$ y $f'(x)$ en profesores de matemáticas. *Enseñanza de las Ciencias*, 29(1), 191–206.

This paper describes the level of understanding of the relation between $f'(a)$ (the derivative of a function at a specified point) and $f'(x)$ (the derivative at an unspecified point) among five mathematics teachers who were teaching 16–18-year-olds in different schools in Colombia. The analysis is based on APOS Theory with the addition of certain semiotic aspects. The five teachers responded to an indirect questionnaire about their understanding of $f'(a)$ and $f'(x)$ and were subsequently interviewed in relation to a series of vignettes. Results illustrate how the comprehension of these two macro-objects, $f'(a)$ and $f'(x)$, can be related to the structure of both graphic and algebraic schemes and the associated semiotic conflicts.

Baker, B., Cooley, L., & Trigueros, M. (2000). A calculus graphing schema. *Journal for Research in Mathematics Education*, 31, 557–578.

The authors used APOS Theory to analyze students' understanding of a complex calculus graphing problem that involved sketching the graph of a function on specific intervals of the domain when given certain analytical properties. The data analysis uncovered the three-

tiered development of Schema referred to as the *triad of Schema development* and two Schemas that were interacting in the solution of the problem. One Schema involved intervals and the second involved analytical properties. The authors also showed that the interaction of these two different Schemas played an important role in the explanation of many of students' known difficulties.

- Baker, B., Trigueros, M., & Hemenway, C. (2001). On transformations of functions. In *Proceedings of the Twenty-Third Annual Meeting, North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol. 1, pp. 91–98).

This study focuses on the analysis of student understanding of transformations. APOS Theory was used as a theoretical framework to come up with a genetic decomposition for the concept of transformation. The genetic decomposition was used to analyze class work and interviews with 24 college students who had taken a precalculus course based on transformations of functions. The course included writing and the use of graphing calculators. This paper analyzes students' difficulties related to the concept of transformation and the efficacy of writing and calculators as teaching tools. Results showed that students tend to develop a strong dependency on calculators to visualize functions, yet the use of calculators together with writing assignments seemed to help with development of the transformation concept. Results also suggest that this concept proved to be difficult for students.

- Barbosa Alvarenga, K. (2003). La enseñanza de inecuaciones desde el punto de vista de la teoría APOE. *Revista Latinoamericana de Investigación en Matemática Educativa*, 6(3), 199–219.

This study, which is based on APOS Theory, discusses mental constructions that undergraduate students might make when trying to understand the inequality concept. This involves many notions that must be coordinated: order of real numbers, factorization, functions, function roots, 1–1 correspondence of real numbers with the number line, equations, graphs, and graphical analysis of functions, implication, and equivalence. Based on the construction of a Schema for inequality, the authors elaborate on a methodology that will improve the teaching and learning of inequalities.

- Baxter, N., Dubinsky, E., & Levin, G. (1988). *Learning Discrete Mathematics with ISETL*. New York: Springer.

This is the first textbook based entirely on the use of computer programming together with APOS Theory. It was written before the ACE pedagogical structure was developed. The subject matter includes topics for a college-level course in discrete mathematics: propositional and predicate calculus, sets and tuples, functions, combinatorics, matrices, determinants, mathematical induction, and relations and graphs. For each concept, the authors developed a genetic decomposition. The genetic decompositions guided the design of laboratory activities involving use of the mathematical programming language ISETL that students used to write short computer programs. The purpose of the programming activities is to encourage reflective abstractions, for example, interiorization, by having students write programs that perform Actions on appropriate input, and encapsulation, by having students use a program as input and/or output in another program.

- Bayazit, I. (2010). The influence of teaching on student learning: The notion of piecewise function. *International Electronic Journal of Mathematics Education*, 5(3), 146–164.

This paper examines the influence of classroom teaching on student understanding of piecewise defined functions. The participants consisted of two experienced mathematics teachers and their 9th grade students. Using a theoretical framework that emerged from an APOS analysis, the author illustrated that the teachers differed remarkably in their approaches to the essence of piecewise functions and that this, in turn, substantially affected their students' understanding of this notion. The author found that Action-oriented

teaching, which is distinguished by the communication of rules, procedures, and factual knowledge, confines students' understanding to an Action conception of piecewise functions, whereas Process-oriented teaching, which places a priority on conceptual development and involves consideration of multiple representations, encourages development of a Process conception of the function concept.

Bayazit, I., & Gray, E. (2008). Qualitative differences in the teaching and learning of the constant function. *Mediterranean Journal for Research in Mathematics Education*, 7, 147–163.

This paper examines two experienced Turkish teachers' teaching of the constant function and their students' resulting understandings. Using a theoretical framework based on APOS Theory, the authors illustrated that the teachers differed markedly in their approaches to the essence of the concept. Though their personal subject matter knowledge and understanding of the potential difficulties and misconceptions associated with the acquisition of aspects of the function concept were similar, and although they assigned similar tasks, their classroom presentations focused on qualitatively different aspects of the concept. This had a considerable influence on their students' construction of knowledge.

Bodí, S., Valls, J., & Llinares, S. (2005). El análisis del desarrollo del esquema de divisibilidad en N. La construcción de un instrumento. *Números*, 60, 3–24.

The aim of this study is to use APOS Theory to validate an instrument built to evaluate the development of the comprehension of divisibility. The paper includes an analysis of the activities and the problems from different textbooks, as well as a review of previous research on the comprehension of divisibility, that was used to prepare a questionnaire that includes the mathematical content of the secondary school curriculum. A subsequent psychometric analysis that was validated by clinical interviews was performed about the index of difficulty of the questionnaire. The analysis enabled the discrimination of different ways secondary students understand the notions of divisibility.

Breidenbach, D., Dubinsky, E., Hawks, J., & Nichols, D. (1992). Development of the process conception of function. *Educational Studies in Mathematics*, 23, 247–285.

The authors show that APOS Theory, and how it applies to the concept of function, point to an instructional treatment, using computers, that results in substantial improvements in students' understanding of the concept of function. The data analysis shows that students appear to develop a Process conception of function that they use to perform certain mathematical tasks.

Brown, A., DeVries, D., Dubinsky, E., & Thomas, K. (1997). Learning binary operations, groups, and subgroups. *Journal of Mathematical Behavior*, 16, 187–239.

APOS Theory was used to study students' learning of binary operations, groups, and subgroups. The authors propose preliminary genetic decompositions of these topics, describe an instructional treatment designed to foster development of the proposed mental constructions, discuss the results of interviews and performance on examinations, suggest revisions of the genetic decompositions as a result of their analysis of the data, and offer pedagogical suggestions. The results suggest that the pedagogical approach, based on application of the instruction detailed in Dubinsky and Leron (1994), was reasonably effective in helping students to develop strong conceptions of binary operations, groups, and subgroups.

Brown, A., McDonald, M., & Weller, K. (2010). Step by step: Infinite iterative processes and actual infinity. In *Research in Collegiate mathematics education VII*. CBMS issues in mathematics education (Vol. 16, pp. 115–141). Providence, RI: American Mathematical Society.

Students in two introduction to abstract mathematics courses were interviewed while trying to determine whether the set $\bigcup_{k=1}^{\infty} P(\{1, 2, \dots, k\})$ equals the set $P(\mathbb{N})$, where $\mathbb{N}$ denotes the set of natural numbers and P denotes the power set operator. An APOS analysis of the data

describes the role of interiorization, coordination, and encapsulation in the development of infinite iterative Processes and their states at infinity. The theoretical analysis is illustrated through a series of case studies and is compared to what is predicted by the Basic Metaphor of Infinity of Lakoff and Nunez (2000).

Brown, A., Thomas, K., & Tolias, G. (2002). Conceptions of divisibility: Success and understanding. In S. Campbell & R. Zazkis (Eds.), *Learning and teaching number theory: Research in cognition and instruction* (pp. 41–82). Westport: Ablex Publishing.

The authors report on an examination of prospective elementary teachers' understanding of the concept of multiples, with a particular focus on the least common multiple. Students' understanding is examined using APOS Theory combined with a stage model adapted from Piaget's work in *Success and Understanding* (Piaget 1978).

12.2 C Through De

Carlson, M. (1998). A cross-sectional investigation of the development of the function concept. *Research in Collegiate mathematics education III*. CBMS issues in mathematics education (Vol. 7, pp. 114–162). Providence, RI: American Mathematical Society.

In this study, the author investigates students' development of the function conception. An exam measuring students' understandings of major aspects of the function concept was developed and administered to students who had just received A's in college algebra, second-semester honors calculus, or first-year graduate mathematics courses. Follow-up interviews were conducted with five students from each of these groups. APOS Theory was one of several theoretical frameworks used to classify students' conceptual views of function. The author reaches a number of conclusions, including agreement with Breidenbach et al. (1992), that students' understanding of functions was improved as a result of engaging students in certain types of construction activities.

Çetin, İ. (2009). *Students' understanding of limit concept: An APOS perspective*. Doctoral Thesis, Middle East Technical University, Turkey.

The main purpose of this study is to investigate first-year calculus students' understanding of the formal limit concept and the change in their understanding after following an instruction designed by the researcher and based on APOS Theory. The case study method was utilized to explore the research questions. Twenty-five mathematics majors from Middle East Technical University in Turkey who were taking first-year calculus participated in the study. The students had five weeks of instruction in the fall semester of 2007–2008. Each week they met for 2 hours in a computer laboratory to study in groups and then they attended 4 hours of class. In the computer lab, they worked on programming activities in order to reflect on the limit concept before they received formal lecture in class. A questionnaire on limits including open-ended questions was administered as a pretest and posttest to determine changes in students' understanding of this concept. At the end of the instruction, a semi-structured interview protocol developed by the researcher was administered to all of the students to explore their understanding in depth. The students' responses in the questionnaire were analyzed both qualitatively and quantitatively. The interview results were analyzed using the APOS framework. The results of the study showed that students thinking reflected what was predicted by the preliminary genetic decomposition. The instruction was found to play a positive role in facilitating students' understanding of the limit concept.

Clark, J. M., Cordero, F., Cottrill, J., Czarnocha, B., DeVries, D. J., St. John, D., Tolias, G., & Vidaković, D. (1997). Constructing a schema: The case of the chain rule. *Journal of Mathematical Behavior*, 16, 345–364.

Based on a preliminary genetic decomposition of how the chain rule concept may be developed, the authors used APOS Theory together with Piaget and García's ideas on the development of Schema to develop a triad of Schema development that provided a structure to interpret students' understanding of the chain rule and to classify their responses. The results of the data analysis allowed for a revised epistemological analysis of the chain rule.

Clark, J., Hemenway, C., St. John, D., Tolias, G., & Vakil, R. (1999). Student attitudes toward abstract algebra. *Primus*, 9, 76–96.

The authors conducted a comparative study of student attitudes in an abstract algebra course. One group of students completed an APOS-based instructional sequence that included computer programming activities and cooperative learning. The other group of students received traditional instruction. Students from both groups shared their impressions about the course and of abstract algebra in general in individual interviews conducted at the conclusion of the course. The students' responses favored the APOS approach in many ways, even though the content of the APOS course was at least as rigorous and demanding as the traditional course.

Cooley, L., Trigueros, M., & Baker, B. (2007). Schema thematization: A theoretical framework and an example. *Journal for Research in Mathematics Education*, 38, 370–392.

Although the idea of a thematization of a Schema emerged very early in the development of APOS Theory (Dubinsky 1986b), it was not studied extensively until this and a related study (Baker et al. 2000). In these studies, the authors studied the development of a “Calculus Graphing Schema” via the *triad of Schema development*. The present investigation builds on this previous work and focuses on the thematization of a Schema. Successful calculus students were interviewed. They appeared to be operating at different stages of development of the “Calculus Graphing Schema”. Only one student showed to have thematized this Schema.

Cordero, F. (1998). El entendimiento de algunas categorías del conocimiento del cálculo y análisis: el caso de comportamiento tendencial de las funciones. *Revista Latinoamericana de Investigación en Matemática Educativa*, 1, 56–74.

In the school-teaching context, the author encountered an argument given by students on the subject of graphs of functions. He calls this argument the “tendencial behavior of functions” because of its nature. The author shows some constructions of this argument that were made by the student participants and analyzes the data using a version of APOS Theory.

Cordero, F., & Miranda, E. (2002). El entendimiento de la transformada de Laplace: una epistemología como base de una descomposición genética. *Revista Latinoamericana de Investigación en Matemática Educativa*, 5(2), 133–168.

In this paper, two issues are considered: the didactical mathematical discourse related to the Laplace transform and a theoretical questioning of the notion of genetic decomposition that could possibly be reformulated with an epistemological basis. The research points to the absence of a reference frame related to the meaning of this concept and the origin of the conditions that would allow its construction. This fact questioned any formulation of the genetic decomposition, since it would imply a learning model for students and a genetic decomposition formulated in terms of mental constructions to be aware only of the

definition of the Laplace transform. Then an epistemology of Laplace transform is formulated and its role as a basis for a genetic decomposition is discussed with the intention to enlarge its conceptual frame.

- Cottrill, J. F. (1999). *Students' understanding of the concept of chain rule in first year calculus and the relation to their understanding of composition of functions*. Unpublished doctoral dissertation, Purdue University, West Lafayette.

This is a follow-up study to Clark et al. (1997). The author finds that the triad mechanism describes the observations of student behaviors and can be used to develop instruction to help students make certain mental constructions. It presents more detailed descriptions of the Intra-, Inter-, and Trans-levels of the development of the chain rule Schema than those given in Clark et al. (1997).

- Cottrill, J., Dubinsky, E., Nichols, D., Schwingendorf, K., Thomas, K., & Vidaković, D. (1996). Understanding the limit concept: Beginning with a coordinated process schema. *Journal of Mathematical Behavior*, 15, 167–192.

This is the first publication in which the acronym APOS was used as a name for the theory. The authors reinterpret some points in the literature about the concept of limit and suggest a new variation of the dichotomy, considered by various authors, between dynamic or Process conceptions of limits and static or formal conceptions. They also propose explanations of why these conceptions are so difficult for students to construct. They present a genetic decomposition for the limit concept and then describe the evolution of the genetic decomposition using examples from their analysis of interviews with 25 students from a calculus course.

- Czarnocha, B., Dubinsky, E., Prabhu, V., & Vidaković, D. (1999). One theoretical perspective in undergraduate mathematics education research. In O. Zaslavsky (Ed.), *Proceedings of the 23rd Conference of the International Group for the Psychology of Mathematics Education* (Vol. 1, pp. 95–110). Haifa, Israel.

The core of this study is the presentation of the belief that research in undergraduate mathematics education should, on the one hand, be closely connected to curriculum development and teaching practice and, on the other hand, be used as a source for empirical data for one or more theories of learning. The study continues to describe how this belief plays a major role in APOS-based research for both the postsecondary and K-12 levels. The study ends with a consideration of alternative perspectives to APOS Theory.

- Davis, G., & Tall, D. (2002). What is a scheme? In D. Tall & M. Thomas (Eds.), *Intelligence, learning and understanding in mathematics: A tribute to Richard Skemp* (pp. 141–160). Flaxton, QLD: Post Pressed.

This chapter is dedicated to, and fundamentally influenced by, Richard Skemp's pioneering work on schemes. The authors discuss examples of scheme formation; schemes and symbols; schemes as mental Objects; perceptual, social, and conceptual categorization; and the connection to APOS Theory.

- DeVries, D., & Arnon, I. (2004). Solution—What does it mean? Helping linear algebra students develop the concept while improving research tools. In M. Hoines & A. Fuglestad (Eds.), *Proceedings of the 28th Conference for the International Group for the Psychology of Mathematics Education* (Vol. 2, pp. 55–62). Bergen, Norway.

Twelve linear algebra students were interviewed after completing the course about the concept of a solution of a system of equations. The interviews were analyzed using an

APOS genetic decomposition of the topic. The analysis of the interviews revealed several misconceptions of solution (some of which might be related to misconceptions reported in the literature on the equality sign). The analysis also revealed shortcomings of the questionnaire that was used in the interviews: it did not create a distinction between total lack of knowledge and partial knowledge. Research tools were improved (genetic decomposition, suggestions for teaching materials, and the questionnaire) and prepared for the next cycle of research.

12.3 Dubinsky (as Lead Author)

Dubinsky, E. (1984). The cognitive effect of computer experiences on learning abstract mathematical concepts. *Korkeakoulujen Atk-Uutiset*, 2, 41–47.

This is the first publication concerning Dubinsky's ideas about incorporating Piaget's ideas of reflective abstractions into postsecondary mathematics. In this talk, the author discusses the distinction between thinking about a function as a Process and as an Object and using the experience of computer programming to help students understand that distinction. He also describes the application of Actions to mental Objects, gives an example of a Pascal program to represent a proof by mathematical induction, and expresses the view that if students write, debug, and use such a program, their development of an understanding of induction is enhanced. For the first time, Dubinsky spoke about Actions, Processes, and Objects; how an external Action is transformed to an internal Process; and how an Action is applied to a Process or to a mental Object.

Dubinsky, E. (1985, March). *Computer experiences as an aid in learning mathematics concepts*. Working paper for the Conference on the Influence of Computers and Informatics on Mathematics and its Teaching, Strasbourg.

The term encapsulation, as the mental mechanism for transforming a Process to a mental Object, is introduced for the first time.

Dubinsky, E. (1986a). On teaching mathematical induction I. *Journal of Mathematical Behavior*, 5, 305–317.

This study presents a prototype version of what was at the time (1986) a novel approach for teaching mathematical induction. An instructional treatment using computer activities was introduced in a small class of 8 college students. The instructional treatment, based on an early version of what would develop into APOS Theory, was designed to help students to make certain mental constructions through reflective abstraction. Computer activities were already used to enhance reflective abstractions. The method seemed to be reasonably effective and several areas of possible improvement were indicated.

Dubinsky, E. (1986b, September 25–27). Reflective abstraction and computer experiences: A new approach to teaching theoretical mathematics. In *Proceedings of the Eighth Annual PME-NA Meeting*, East Lansing, MI.

This manuscript introduced interiorization as the mechanism for converting an Action to a mental Process. The idea of transforming a Schema to an Object, which could be acted on by another Schema, was also discussed. In this article, the transformation of a Schema to an Object was treated as an instance of encapsulation, as opposed to the term thematization, which would arise later.

- Dubinsky, E. (1989). On teaching mathematical induction II. *Journal of Mathematical Behavior*, 8, 285–304.

In this paper, a continuation of Dubinsky and Lewin (1986) and Dubinsky (1986a), the author details two classroom experiments in which a theoretically based instructional approach (not yet called APOS) using computer experiences with SETL and ISETL was implemented. The data showed that students seemed to develop a more positive attitude and were totally successful in solving straightforward induction problems. When presented with more difficult, unfamiliar problems, they tended to set up most problems correctly, knew how to use induction, and intended to do so but continued to exhibit difficulty in completing the proofs.

- Dubinsky, E. (1991a). The constructive aspects of reflective abstraction in advanced mathematics. In L. P. Steffe (Ed.), *Epistemological foundations of mathematical experience*. New York: Springer.

The author presents a brief discussion of APOS (not yet named so) as a developing theory of mathematical knowledge and its acquisition. He also describes specific methods of construction observed with students. He presents an analysis of studies of induction, quantification, and function according to this point of view.

- Dubinsky, E. (1991b). Reflective abstraction in advanced mathematical thinking. In D. Tall (Ed.), *Advanced mathematical thinking* (pp. 95–123). Dordrecht, The Netherlands: Kluwer.

The author makes the case that the concept of reflective abstraction can be a powerful tool in the study of advanced mathematical thinking, can provide a theoretical basis that supports and contributes to an understanding of what this thinking is, and suggests how students can be helped to develop the ability to engage in this type of thinking.

- Dubinsky, E. (1992). A learning theory approach to calculus. In Z. Karian (Ed.), *Symbolic computation in undergraduate mathematics education*. MAA Notes 24 (pp. 48–55). Washington, DC: Mathematical Association of America.

The author outlines APOS Theory (not yet named so) and discusses some of the choices about teaching that seem to follow from the theory. In particular, he discusses how computers can be used in teaching and learning.

- Dubinsky, E. (1994). A theory and practice of learning college mathematics. In A. Schoenfeld (Ed.), *Mathematical thinking and problem solving* (pp. 221–243). Hillsdale: Erlbaum.

In mathematics education, there are many dichotomies. The author is interested in *binary syntheses* for each of these dichotomies. Specifically he refers to the following dichotomies: theory and practice, research and development (where development means curriculum development), and beliefs and choices. In addition to discussing these syntheses, he describes the beginning of a theory and its application and goes on to describe some examples of instructional treatments derived from the theory (where the term APOS, again, has not yet emerged). Results of research that accompanied these treatments suggest that it might be possible to design instructional treatments based on a theory of learning that involves use of computers.

- Dubinsky, E. (1995). ISETL: A programming language for learning mathematics. *Communications in Pure and Applied Mathematics*, 48, 1–25.

The author gives a brief history of the development of a pedagogical strategy for helping students learn mathematical concepts at the postsecondary level. The method uses ISETL to implement instruction designed on the basis of APOS Theory (not yet named so). ISETL is

described in some detail and examples of its application are given for its use as a pedagogical tool in abstract algebra, calculus, and mathematical induction.

Dubinsky, E. (1996a). *Applying a Piagetian perspective to post-secondary mathematics education*. Second International Workshop on Mathematics Education for Engineers (pp. 25–29). Havana.

The author begins with a brief introduction on Piaget's ideas about education. He shows how these ideas form the theoretical foundation for curriculum development activities and inform the overall structure of a particular pedagogical strategy. The author describes how this constructivist approach uses analysis of data from students to determine the mental structures that might be needed for the development of mathematical knowledge and gives examples of how computer activities can be used to help students construct the needed structures. The manuscript does not report on existing research but refers to relevant publications.

Dubinsky, E. (1996b). Aplicación de la perspectiva piagetiana a la educación matemática universitaria. *Educación Matemática*, 8(3), 24–41.

This report describes Dubinsky's work on curricular development at the college level. It is based upon Piaget's ideas about the way teaching can help a child to learn. The article begins with a brief introduction to Piaget's ideas and shows how they form the foundation of Dubinsky's activities in curricular development. The author shows how a theoretical perspective can be used to explain students' answers to an interview question about the order of the elements in a group. The author also includes examples of computer tasks that appear in the activities.

Dubinsky, E. (1997a). On learning quantification. *Journal of Computers in Mathematics and Science Teaching*, 16(2/3), 335–362.

In this study, the author examines students' learning of universal and existential quantification in a specially designed course based on the theoretical analysis of quantification found in Dubinsky, Elterman, and Gong (1988). The instruction was designed to assist students in making mental constructions using the computer program ISETL. Students' responses to written questions suggest that the pedagogical approach helped students to develop their understanding of quantification, even when working on difficult problems.

Dubinsky, E. (1997b). Some thoughts on a first course in linear algebra on the college level. In D. Carlson, C. Johnson, D. Lay, D. Porter, A. Watkins, & W. Watkins (Eds.), *Resources for teaching linear algebra*. MAA Notes 42 (pp. 85–106). Washington, DC: Mathematical Association of America.

This chapter is a reaction to the recommendations of two programs for teaching linear algebra and a proposal for an alternative instructional approach based on APOS Theory. The proposal includes specific descriptions of how ISETL activities can be incorporated in the proposed teaching sequence. The study contains detailed descriptions of the three approaches.

Dubinsky, E. (2000a). Mathematical literacy and abstraction in the 21st century. *School Science and Mathematics*, 100(6), 289–297.

In this paper, the author explains the growing need for abstraction as an important component of literacy for life in the twenty-first century. He explains how abstraction occurs and why its development needs to be taught. He reinforces his perspective through example, specifically of instruction on the mathematical concept of *function*, and in

consideration of economical/political issues such as the *changing rate of change of the national debt*. The author proposes APOS as one possible instructional tool for encouraging abstraction in postsecondary education and reflects on the necessity to develop similar tools at the K-12 level.

- Dubinsky, E. (2000b). Meaning and formalism in mathematics. *International Journal of Computers for Mathematical Learning*, 5, 211–240.

This essay is an exploration of possible psychological sources of mathematical ideas, specifically the relation between meaning and formalism. Two possible relations between the two are suggested. Although the use of formalism to construct meaning is very difficult for students, the author suggests the possibility that this is the only route to learn large portions of mathematics in upper high school and tertiary levels. The essay concludes with an outline of APOS as a pedagogical strategy for helping students to deal with formalism.

- Dubinsky, E., & Harel, G. (1992). The nature of the process conception of function. In G. Harel, E. Dubinsky (Eds.), *The concept of function: Aspects of epistemology and pedagogy*. MAA Notes 25 (pp. 85–106). Washington, DC: Mathematical Association of America.

The authors interviewed 13 students who received APOS-based instruction on the concept of function that included programming activities involving use of ISETL. After examining the students' thinking from a number of different perspectives, the authors found that students have difficulty constructing a Process conception of function because of the complexity of the construction.

- Dubinsky, E., & Leron, U. (1994). *Learning abstract algebra with ISETL*. New York: Springer.

This is a textbook for a course in Abstract Algebra. It is intended to support APOS Theory as a constructivist (in the epistemological, not mathematical sense) approach to teaching, although the term APOS was still not in use at the time. In the introduction (Comments for the Instructor, pp. xvii–xix), the authors present their constructivist approach: “students construct, for themselves, mathematical concepts”; “the ideas in the textbook are not presented in a completed, polished form.” They discuss the use of ISETL and the ACE Teaching Cycle and consider issues related to the covering of the course material.

- Dubinsky, E., & Lewin, P. (1986). Reflective abstraction and mathematics education: The genetic decomposition of induction and compactness. *Journal of Mathematical Behavior*, 5, 55–92.

The authors formulate a precursor to APOS Theory by interpreting Piaget's epistemology, especially equilibration and reflective abstraction. They apply Piaget's epistemology to describe genetic decompositions of mathematical induction and compactness.

- Dubinsky, E., & McDonald, M. (2001). APOS: A constructivist theory of learning in undergrad mathematics education. In D. Holton (Ed.), *The teaching and learning of mathematics at the university level: An ICMI study* (pp. 273–280). Dordrecht, The Netherlands: Kluwer.

The authors present six ways in which a theory in mathematics education can contribute to research and suggest how those ways can be used as criteria for evaluating a theory. Then they describe how members of RUMEC (Research in Undergraduate Mathematics Education Community) and others use APOS Theory to design instruction, develop curricula, and conduct research as well as explain how the theory is used as a language to communicate ideas about learning. The chapter includes an annotated bibliography that presents further details about the theory and its use in research in undergraduate mathematics education.

Dubinsky, E., & Schwingendorf, K. (1990). Calculus, concepts, and computers—Innovations in learning calculus. In T. Tucker (Ed.), *Priming the calculus pump: Innovations and resources*. MAA Notes 17 (pp. 175–198). Washington, DC: Mathematical Association of America.

This report describes a three-semester innovative calculus course developed at Purdue University with support from the US National Science Foundation. The materials were used at Purdue and other universities in the USA and other countries during the decade of the 1990s. The course is still being used in some universities at the time of this writing.

The design of the course is based on APOS Theory and involves students writing and running programs in a mathematical programming language and making calculations on the computer using a symbolic computing system. The pedagogical strategy consists of students working in cooperative groups in a computer lab where they are expected to construct mental structures proposed by theoretical analyses of the mathematics they are trying to learn, small group problem solving in a classroom where students are confronted with problem situations designed to get them to use the mental structures developed in the computer lab to construct their understanding of mathematical concepts, and homework exercises intended to reinforce their understandings and provide practice with standard calculus problems.

The report describes APOS Theory and how it is used in design of the course. Detailed examples are given of the treatment of graphs and functions, the fundamental theorem of calculus, infinite sequences, and infinite series. This article includes some of the programs the students are asked to write, comparative data on students' learning of calculus, and comments from students and administrators.

Dubinsky, E., & Schwingendorf, K. (1991a). Constructing calculus concepts: Cooperation in a computer laboratory. In C. Leinbach, J. R. Hundhausen, A. M. Ostebee, L. J. Senechal, & D. B. Small (Eds.), *The laboratory approach to teaching calculus*. MAA Notes 20 (pp. 47–70). Washington, DC: Mathematical Association of America.

Dubinsky, E., & Schwingendorf, K. (1991b). Calculus, concepts, and computers: Some laboratory projects for differential calculus. In C. Leinbach, J. R. Hundhausen, A. M. Ostebee, L. J. Senechal, & D. B. Small (Eds.), *The laboratory approach to teaching calculus*. MAA Notes 20 (pp. 197–212). Washington, DC: Mathematical Association of America.

This is a pair of papers that extend Dubinsky and Schwingendorf (1990) with a full discussion of the theoretical background for the laboratory approach and a description of the cooperative learning environment in the first paper. The second paper provides 14 pages of sample lab and homework assignments.

Dubinsky, E., Schwingendorf, K. E., & Mathews, D. M. (1995a). *Calculus, concepts & computers* (2nd ed.). New York: McGraw-Hill.

This is a first-year course in calculus employing the ACE Teaching Cycle using ISETL activities. This is the textbook used in the course described above in Dubinsky and Schwingendorf (1990, 1991a, b). The text covers limits, differential and integral calculus, sequences and series, and polar and parametric curves in a two-semester sequence.

Dubinsky, E., & Schwingendorf, K. E. (1995b). *Calculus, concepts, and computers: Multivariable and vector calculus* (Revised Preliminary Version). New York: McGraw-Hill.

This is the textbook used in a third semester follow-up to the two-semester course described above in Dubinsky et al. (1995a). The follow-up course treats the calculus of functions of several real variables and uses all of the theoretical background and pedagogical strategies employed in the first two semesters.

- Dubinsky, E., & Wilson, R. (2013). High school students' understanding of the function concept. *Journal of Mathematical Behavior*, 32, 83–101.

This paper is a study of part of the Algebra Project's program for high school students from the lowest quartile of academic achievement and social and economic status. The study focuses on learning the concept of function. APOS Theory is used here as a strictly analytical evaluative tool. Fifteen high school students from the project's target population participated in the research. Immediately after instruction, a written instrument was administered. Several weeks later, in-depth interviews were conducted and then analyzed using APOS-based tools. The results indicate that with appropriate pedagogy, students from the project's target population are able to learn a substantial amount of nontrivial mathematics at the high school level.

- Dubinsky, E., & Yiparaki, O. (1996, July 25–26). *Predicate calculus and the mathematical thinking of students, international symposium on teaching logic and reasoning in an illogical world* (Report). Centre of Discrete Mathematics and Theoretical Computer Science, Rutgers University.

This report is based on two related projects. The first was an attempt to apply ideas taken from Piaget to analyze how students might come to understand predicate calculus and how to apply this analysis in the design and implementation of instruction (Dubinsky et al. 1988; Dubinsky 1997a). The second project returns to earlier work in an attempt to apply what today is called APOS Theory (Asiala et al. 1996). This report begins with a statement about what the authors think about the role of predicate calculus in understanding mathematics and concludes with the claim that this role does not appear to work for many students. The authors found that students who participated in an APOS-based implementation did develop some understanding of quantification and the ability to work with it. This view is supported by the overall performance of the students in the experiment.

- Dubinsky, E., Dautermann, J., Leron, U., & Zazkis R. (1994). On learning fundamental concepts of group theory. *Educational Studies in Mathematics*, 27, 267–305.

This is a systematic investigation using APOS Theory of students' construction of the concepts of group, subgroup, coset, normality, and quotient group. The authors make general observations about learning these topics and discuss the complex nature of "understanding" and the role of errors and misconceptions.

- Dubinsky, E., Elterman, F., & Gong, C. (1988). The student's construction of quantification. *For the Learning of Mathematics—An International Journal of Mathematics Education*, 8, 44–51.

As part of a course in discrete mathematics, the authors designed a unit on quantification using computer activities with SETL (the forerunner of ISETL). Using data collected from students who completed the unit, the authors propose a genetic decomposition for the concept of quantification.

- Dubinsky, E., Weller, K., & Armon, I. (2013). Preservice teachers' understanding of the relation between a fraction or integer and its decimal expansion: The Case of $0.999\ldots$ and 1. *Canadian Journal of Science, Mathematics, and Technology Education*, 13(3).

This is the third of a sequence of three studies based on data collected from the same experiment (Weller et al. 2009, 2011). The authors devise a genetic decomposition of students' progress in their development of an understanding of the decimal $0.\overline{9}$ and its relation to 1. The genetic decomposition appears to be valid for a high percentage of the study participants and suggests the possibility of a new stage in APOS that would be the first substantial change in the theory since its inception (Dubinsky and Lewin 1986). The analysis includes a relatively objective and highly efficient methodology that might be

useful in other research and in assessment of student learning. For further analysis of this study, see Weller et al. (2009, 2011) in this bibliography.

- Dubinsky, E., Weller, K., McDonald, M. A., & Brown, A. (2005a). Some historical issues and paradoxes regarding the concept of infinity: An APOS analysis: Part 1. *Educational Studies in Mathematics*, 58, 335–359.

This paper applies APOS Theory to suggest a new explanation of how people might think about the concept of infinity. The authors propose cognitive explanations and, in some cases, resolutions of various dichotomies, paradoxes, and mathematical problems involving the concept of infinity. These explanations are expressed in terms of the mental mechanisms of interiorization and encapsulation. The purpose for providing a cognitive perspective is that issues involving the infinite have been and continue to be a source of interest, of controversy, and of student difficulty.

- Dubinsky, E., Weller, K., McDonald, M. A., & Brown, A. (2005b). Some historical issues and paradoxes regarding the concept of infinity: An APOS analysis: Part 2. *Educational Studies in Mathematics*, 60, 253–266.

This is the second part of a study on how APOS Theory may be used to provide cognitive explanations of how students and mathematicians might think about the concept of infinity. The authors discuss infinite Processes, describe how the mental mechanisms of interiorization and encapsulation can be used to conceive of an infinite Process as a completed totality, explain the relationship between infinite Processes and the Objects that may result from them, and apply their analyses to certain mathematical issues related to infinity.

- Dubinsky, E., Weller, K., Stenger, C., & Vidaković, D. (2008). Infinite iterative processes: The tennis ball problem. *European Journal of Pure and Applied Mathematics*, 1(1), 99–121.

In this paper, the authors use APOS Theory to describe the mental constructions needed to understand and to solve the Tennis Ball Problem, a paradoxical problem that involves the coordination of three infinite Processes. Of the 15 interview subjects, only one solved the problem correctly. His responses indicated that he had made the mental constructions called for by the theoretical analysis, while the other students gave evidence of not having made those constructions. On the basis of the data analysis, the authors offer various pedagogical suggestions and avenues for future research.

12.4 E Through M

- Ely, R. (2011). Envisioning the infinite by projecting finite properties. *Journal of Mathematical Behavior*, 30, 1–18.

Twenty-four postsecondary students were interviewed as they worked on the Tennis Ball Problem. The author presents a framework for making sense of the participants' responses. The author does not assume that this framework is a universal or comprehensive framework to describe the learning of infinite Processes, but suggests that other existing frameworks, such as the BMI¹ and APOS, are limited in accounting for the participants' responses.

¹ For BMI, see: Lakoff, G., & Nunez, R. (2000). *Where mathematics comes from*. New York: Basic Books.

Fenton, W., & Dubinsky, E. (1996). *Introduction to discrete mathematics with ISETL*. New York: Springer.

Intended for first- or second-year undergraduates, this introduction to discrete mathematics covers the usual topics of such a course but applies constructivist principles that promote—indeed, require—active participation by the student. Working with the programming language ISETL, whose syntax is close to that of standard mathematical language, the student constructs the concepts in her or his mind as a result of constructing them on the computer in the syntax of ISETL. This dramatically different approach allows students to attempt to discover concepts in a “Socratic” dialog with the computer. The discussion avoids the formal “definition-theorem” approach and promotes active involvement by the reader by its questioning style. An instructor using this text can expect a lively class whose students develop a deep conceptual understanding rather than simply manipulative skills.

Font, V., Malaspina, U., Giménez, J., & Wilhelmi, M. R. (2011). Mathematical objects through the lens of three different theoretical perspectives. In E. Svoboda (Ed.), *Proceedings of the Seventh Congress of the European Society for Research in Mathematics Education*. Rzeszow.

In this paper, a link between the onto-semiotic approach (OSA) to mathematics cognition and instruction, APOS Theory, and the cognitive science of mathematics (CSM) is established as it regards use of the concept “mathematical object.” It is argued that the notion of object used in the OSA does not contradict that employed by APOS Theory or the CSM, since the latter two theories highlight partial aspects of the complex process through which, according to the OSA, mathematical objects emerge out of mathematical practices.

Gavilán, J. M., García, M. M., & Llinares, S. (2007a). La modelación de la descomposición genética de una noción matemática. Explicando la práctica del profesor desde el punto de vista del aprendizaje potencial en los estudiantes. *Educación Matemática*, 19(2), 5–39.

The construct “modeling of the genetic decomposition of a notion” is introduced in order to explain mathematics teachers’ practices from the point of view of the construction of mathematical knowledge that seems to develop in students. This construct is used to analyze two teachers’ teaching when introducing the notion of derivative to high school students (16–18 years). The results of the analysis allowed characterization of the principles on which the teacher drew in her or his practice. Reflections about this construct are made.

Gavilán, J. M., García, M. M., & Llinares, S. (2007b). Una perspectiva para el análisis de la práctica del profesor de matemáticas. Implicaciones metodológicas. *Enseñanza de las Ciencias*, 25(2), 157–170.

In this study, the authors emphasize that analysis of mathematics teachers’ practice involves making explicit a model of students’ learning (construction of mathematical knowledge) and generating analytic tools that allow the explanation of teachers’ practice in a way coherent with the chosen learning model. In this article, the following notions are introduced: the analytic tool “modeling of mechanisms for the construction of knowledge” in order to carry out this analysis and the “vignette” notion as a way to make it explicit in the analysis of teachers’ practice. The description and interpretation of this practice is based on two elements: the notion of practice, which provides the sociocultural focus, and APOS Theory, which offers a theoretical model of knowledge construction.

Hähkiöniemi, M. (2005). Is there a limit in the derivative?—Exploring students’ understanding of the limit of the difference quotient. *Proceedings of CERME 4*, 1758–1767.

Task-based interviews with five postsecondary students were arranged to investigate students’ understanding of the limit of the difference quotient (LDQ). The students’ procedural knowledge was analyzed using APOS Theory, and their conceptual knowledge

was analyzed by examining the kind of representations they had of the limiting process and how these were connected to LDQ. It was found that students had two kinds of connections: change from one representation to other or the explanation of one representation with the other. Among the students, all combinations of good or poor procedural and conceptual knowledge of LDQ were found.

Hamdan, M. (2006). Equivalent structures on sets: Equivalence classes, partitions and fiber structures of functions. *Educational Studies in Mathematics*, 62, 127–147.

This study reports on how students can be led to make meaningful connections between structures on a set such as a partition, the set of equivalence classes determined by an equivalence relation, and the fiber structure of a function on that set (i.e., the set of pre-images of all sets $\{b\}$ for b in the range of the function). The author presents an initial genetic decomposition, in the sense of APOS Theory, for these concepts, and suggests and applies instructional procedures that reflect the proposed genetic decomposition. The author suggests the need for a revised genetic decomposition based on informal interviews with students at different stages in their learning of these concepts.

Harel, G., & Dubinsky, E. (1991). The development of the concept of function by preservice secondary teachers: From action conception to process conception. In F. Furinghetti (Ed.), *Proceedings of the Fifteenth Conference of the International Group for the Psychology of Mathematics Education* (Vol.2, pp. 133–140). Assisi, Italy.

A group of 22 students participated in a course in discrete mathematics using an instructional treatment based on the constructivist theory that was later named APOS. This instructional treatment was meant to help the students improve their conceptions of function. Their starting points ranged from very primitive conceptions to Action conceptions. As a result of the instructional treatment, all of the students progressed toward a Process conception of function. The authors list four factors that play a role in the progression to a full Process conception.

Hernández Rebollar, L. A., & Trigueros, M. (2012). Acerca de la comprensión del concepto de supremo. *Revista educación Matemática*, 24(3).

The main goal of this work was to study how university students construct the supremum concept. The authors used APOS Theory as a theoretical framework and presented a genetic decomposition of the supremum concept. To validate the genetic decomposition, the authors designed a questionnaire for mathematics and physics students at a public university and analyzed the data using the theory. Results revealed that most of the students involved in the study did not construct an Action conception of this concept. The analysis has been useful in explaining the difficulties students face when they try to demonstrate that a number is the supremum of a given set.

Hollebrands, K. F. (2003). High school students' understandings of geometric transformations in the context of a technological environment. *Journal of Mathematical Behavior*, 22, 55–72.

This study investigated the nature of students' understandings of geometric transformations—translations, reflections, rotations, and dilations. Instruction involved use of The Geometer's Sketchpad. The author implemented a seven-week instructional unit on geometric transformations within an honors geometry class. Students' conceptions of transformations as functions were analyzed using APOS Theory. The analysis suggests that students' understandings of key concepts such as domain, variables, and parameters, as well as relationships and properties of transformations, were crucial in the support of deeper understandings of transformations as functions.

- Kabael, T. (2011). Generalizing single variable functions to two-variable functions, function machine and APOS. *Educational Sciences: Theory and Practice*, 11(1), 484–499.

The study examines how students generalize the concept of function from the single-variable case to the two-variable case. The author uses APOS to analyze data collected from 23 students in an Analysis II course in an elementary mathematics education program. As a result of the data analysis, the author concludes that construction of the two-variable function concept depends on understanding the one-variable function concept and development of a Schema for three-dimensional space.

- Kú, D., Okaç, A., & Trigueros, M. (2011). Spanning set and span—An analysis of the mental constructions of undergraduate students. In S. Brown, S. Larsen, K. Marrongelle, & M. Oehrtman (Eds.), *Proceedings of the 14th annual conference on research in undergraduate mathematics education* (pp. 176–186). Washington, DC: Special Interest Group of the Mathematical Association of America (SIGMAA) for Research in Undergraduate Mathematics Education.

The authors present a genetic decomposition for the construction of the concepts of spanning set and span in Linear Algebra. They used the genetic decomposition to analyze data from interviews with 11 students who completed an introductory linear algebra course. The authors concluded that it is easier in general for students to decide whether a given set spans a given vector space than to construct a spanning set for a given vector space. Some modifications to the preliminary genetic decomposition are suggested.

- Kú, D., Trigueros, M., & Okaç, A. (2008). Comprensión del concepto de base de un espacio vectorial desde el punto de vista de la teoría APOE. *Educación Matemática*, 20 (2), 65–89.

The authors use APOS Theory to develop a genetic decomposition of the concept of basis. They test the genetic decomposition empirically by interviewing six undergraduate students who completed a linear algebra course. The results showed that it was easier for the students to determine if a given set is a basis of a vector space than to find a basis for a given vector space. The authors attribute the difference to students' inability to coordinate the Processes for linear independence and spanning set.

- Llinares, S., Boigues, F., & Estruch, V. (2010). Desarrollo de un esquema de la integral definida en estudiantes de ingenierías relacionadas con las ciencias de la naturaleza. Un análisis a través de la lógica Fuzzi. *Revista Latinoamericana de Investigación en Matemática Educativa*, 13, 255–282.

This research describes the triad development of a Schema for the concept of definite integral. Data for the study was gathered from earth science engineering students who were using fuzzy metrics. The results demonstrate students' difficulty in linking a succession of Riemann sums to the limit, which forms the basis for the meaning of the definite integral.

- Mathews, D., & Clark, J. (1997, March). *Successful students' conceptions of mean, standard deviation, and the Central Limit Theorem*. Paper presented at the Midwest Conference on Teaching Statistics, Oshkosh, WI.

The authors present an APOS-based analysis of audiotaped clinical interviews with college freshmen immediately after they completed an elementary statistics course and obtained a grade of "A." The authors found that APOS is a useful way of describing students' understanding of mean, standard deviation, and the Central Limit Theorem. In addition, they conclude that traditional instruction in statistics does not help students make the mental constructions appropriate for development of these concepts. In particular, traditional instruction seems to inhibit students from moving from a Process to an Object conception of standard deviation.

Mamolo, A. (2009). Accommodating infinity: A leap of imagination. In *Proceedings of the 31st annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol. 5, pp. 65–72). Atlanta, GA: Georgia State University.

This paper presents first results of a study which seeks to identify the necessary and sufficient features of accommodating the idea of actual infinity. Data was collected from university mathematics majors' and graduates' engagement with the Ping-Pong Ball Conundrum. APOS Theory was used in the analysis of the data. The paper focuses on the following feature: the leap of imagination required to conceive of actual infinity and its associated challenges.

Mamolo, A., & Zazkis, R. (2008). Paradoxes as a window to infinity. *Research in Mathematics Education*, 10(2), 167–182.

This study examines approaches to infinity of two groups of university students with different mathematical background: undergraduate students in Liberal Arts Programs and graduate students in a Mathematics Education Master's Program. Data are drawn from students' engagement with Hilbert's Grand Hotel paradox and the Ping-Pong Ball Conundrum. Two frameworks were used for the interpretation of students' responses as well as their emergent ideas of infinity: reducing abstraction (Hazzan 1999) and APOS. While graduate students found the resolution of Hilbert's Grand Hotel paradox unproblematic, responses of students in both groups to the Ping-Pong Ball Conundrum were surprisingly similar. Consistent with prior research, the work of participants in this study revealed that they perceive infinity as an ongoing Process, rather than a completed one, and fail to notice conflicting ideas. The contribution of this work is in describing specific challenging features of these paradoxes that might influence students' understanding of infinity, as well as the persuasive factors in students' reasoning, that have not been unveiled by other means.

Martínez-Planell, R., & Trigueros, M. (2012). Students' understanding of the general notion of a function of two variables. *Educational Studies in Mathematics*, 81, 365–384.

In this study, the authors continue their research on the different components of students' understanding of two-variable functions. In particular, they consider students' understanding of the concepts of domain, the possibility of an arbitrary nature of function, the uniqueness of function image, and range. (Trigueros and Martínez-Planell, 2010) The thinking of 13 college students was analyzed using APOS Theory and a semiotic representation theory. The authors concluded that many of the students' notions of function could be considered pre-Bourbaki.

McDonald, M., Mathews, D., & Strobel, K. (2000). Understanding sequences: A tale of two objects. In *Research in Collegiate mathematics education IV*. CBMS issues in mathematics education (Vol. 8, pp. 77–102). Providence, RI: American Mathematical Society.

The authors used APOS Theory to examine how students construct the concept of sequence. The authors show that students tend to construct two distinct cognitive Objects and refer to both as a sequence. One construction, which the authors call SEQLIST, is what one might understand as a listing representation of a sequence. The other, which they call SEQFUNC, is what one might interpret as a functional representation of a sequence. In this paper, the authors detail students' constructions of SEQLIST and SEQFUNC and characterize the connections between them using the detailed descriptions of the Intra-, Inter-, and Trans-levels of the development of the chain rule Schema triad introduced by Clark et al. (1997).

Meel, D. (2003). Models and theories of mathematical understanding: Comparing Pirie and Kieren's model of the growth of mathematical understanding and APOS Theory. In *Research in Collegiate mathematics education V. CBMS issues in mathematics education* (Vol. 12, pp. 132–187). Providence, RI: American Mathematical Society.

This paper focuses on two theoretical frameworks for understanding student thinking in mathematics: Pirie and Kieren's model of the growth of mathematical understanding (known as The Onion Model) and Dubinsky's APOS Theory. The author explains how these two perspectives satisfy criteria for classification as a theory and discusses a variety of interconnections between these two theories.

Mena, A. (2011). *Estudio epistemológico del teorema del isomorfismo de grupos*. Doctorate thesis, Cicata-IPN, Mexico.

The author presents a pedagogical approach for the teaching of the isomorphism theorem for groups that is based on a version for sets followed by a version that incorporates the group structure. The author proposes a genetic decomposition for construction of the theorem that involves development of a Schema structure.

Moreira, R. N., & Wodewotzki, M. L. L. (2004). A perspective on the conceptions of college freshmen regarding absolute value of real numbers. *Boletim de Educação Matemática*, 17(22), 63–81.

The authors start by discussing how students try to understand the concept of absolute value. Based on an initial cognitive model, the authors attempt to interpret interview data using APOS Theory. The results of the analysis seem to suggest that starting college students' ability to make abstractions enables them to develop an adequate understanding of the absolute value concept. The analysis also pointed out that graphical representations and cooperative learning were relevant factors in the students' learning of the concept.

12.5 Works of Piaget

Although these items do not mention APOS Theory, much of the foundation for the creation and development of APOS Theory comes from works of Piaget such as these.

Beth, E. W., & Piaget, J. (1974). *Mathematical epistemology and psychology* (W. Mays, Trans.). Dordrecht, The Netherlands: D. Reidel. (Original work published 1966).

The book is in two parts. The first part was written by Beth and the second by Piaget. The second part is a major source for the foundation of APOS Theory. Piaget argues for a constructivist epistemology, which he calls genetic epistemology, and explains why it is superior to other theories such as Platonism, empiricism, apriorism, logical reductionism, and nominalism (linguistic). He explains how the development of an individual's mathematical thought applies to advanced mathematical thinking as well as to the thinking of children. In particular, he relates some of his ideas to the three "mother structures" of Bourbaki. The key ingredient of genetic epistemology is reflective abstraction, which is discussed throughout the book and involves actions, operations (processes), and objects.

Piaget, J. (1975). Piaget's theory (G. Cellier & J. Langer, Trans.). In P.B. Neubauer (Ed.), *The process of child development* (pp. 164–212). New York: Jason Aronson.

Here Piaget describes his genetic epistemology which was developed over more than half a century and continued until his death in 1980. The survey begins with a discussion of the biological origins of the cognitive functions involving several biological and cognitive mechanisms such as adaptation, assimilation, accommodation, and equilibration. These bio-cognitive mechanisms are the sources of the cognitive structures that, according to Piaget, produce all knowledge and intelligence. The paper reiterates Piaget's belief in the continuity of the development of thought from infancy to adult scientific thinking. Piaget describes in some detail his theory of stages, which relates to thinking at the early ages and then goes on to consider the logico-mathematical aspects of mental structures and their construction at all ages.

Piaget, J. (1976). *The grasp of consciousness* (S. Wedgwood, Trans.). Cambridge, MA: Harvard University Press. (Original work published 1974).

Piaget calls this book the completion of his long study of causality. The main issue he discusses here is the relation between a child's ability to complete a task and her or his understanding of how that task is completed. It turns out that there is a considerable time delay, observable in many experiments reported in the book, between the former and the latter. Piaget's explanation is that the subject takes time to make the mental constructions by which he or she develops an understanding of the success in performing a task. He analyzes the mental constructions his subjects appear to be making as he gradually moves from students who succeeded in the action without consciousness of it to students with growing levels of cognizance.

The format of the book is a description and analysis of 15 different experiments followed by a conclusion.

Piaget, J. (1978). *Success and understanding* (A. J. Pomerans, Trans.). Cambridge, MA: Harvard University Press. (Original work published 1974).

Much of Piaget's contributions to learning consist of theoretical descriptions of how knowledge and intelligence develop in the mind of an individual. But his work is far from purely theoretical. Piaget also wrote several books that report on his empirical studies. *Success and Understanding* is one of these. It consists of thirteen chapters, the first 12 of which report on 12 different experiments and a final chapter in which he summarizes the first 12 and presents his general conclusions. Each of the first 12 chapters displays Piaget's remarkable ability to construct interesting tasks that use "apparatuses" cleverly crafted out of locally purchased materials that embodied the particular concepts he wished to study. The subjects, who were children of different ages, engaged in the activities, and Piaget reported on the successes they did or did not have. Then he interviewed each subject to understand the subject's thinking. An interesting theme that runs through the entire book is that young children very often succeed with a task long before they understand why they succeeded. Piaget's analyses of the interview transcripts, many of which appear in the book, give rise to the points in his theoretical descriptions. One can learn several different things from reading this book in addition to getting a better understanding of Piaget's epistemology and its source in empirical data. For example, the tasks themselves are powerful tools for analyzing the thinking of children, and the interview excerpts teach us much about how to conduct in-depth interviews with children at various ages. Finally, when his often subtle and opaque theoretical points are couched in the concrete activities of children, they sometimes become a little easier to understand.

Piaget, J., & García, R. (1989). *Psychogenesis and the history of science* (H. Feider, Trans.). New York: Columbia University Press. (Original work published 1983).

The main purpose of this book is to compare the historical development of scientific thought to its cognitive development in the mind of an individual. The concern of the authors is not with the content of concepts but the common mechanisms by which they are constructed mentally. According to their analysis, both scientific and cognitive developments proceed by the mental construction of a sequence of stages. In some cases, the developmental sequences in history and in cognition are parallel, but in other cases, they are different, even in at least one case, directly opposite. On the other hand, the authors argue, and support their arguments by reference to empirical evidence, that the basic mechanisms for mental constructions in history and cognition are the same.

These mechanisms are reflective abstraction; an interaction between subject and object in which experience arises out of interpretation and construction; differentiation and integration; a search for “reasons,” which means relating phenomena to a mental structure or coordinated schema; and a sequence of stages each of which is made possible by the preceding ones and each of which in turn prepares those that follow. The transitional mechanisms for this sequence of stages exhibit two characteristics that are common between the history of science and psychological development.

The first common transitional mechanism is that each stage is integrated in the succeeding structure. The second is a new mechanism introduced for the first time in this book. It is a dialectical triad that leads from intra- (object analysis) to inter- (analyzing relations or transformations) to trans- (building of structures) levels of analysis.

Piaget and Garcia apply these very general considerations to several topics: the development of mechanics in physics from Aristotelian to Newtonian thinking, geometry, and algebra.

12.6 P Through T

Parraguez, M., & Oktaç, A. (2010). Construction of the vector space concept from the viewpoint of APOS theory. *Linear Algebra and its Applications*, 432, 2112–2124.

APOS Theory is used to propose a genetic decomposition of the vector space concept. Empirical results are based on an analysis of interview and questionnaire data with 10 undergraduate mathematics students. The analysis focuses on the coordination between the two operations that form the vector space structure and the relation of the vector space Schema to other concepts such as linear independence and binary operations.

Pegg, J., & Tall, D. (2005). The fundamental cycle of concept construction underlying various theoretical frameworks. *International Reviews on Mathematical Education (Zentralblatt für Didaktik der Mathematik)*, 37, 468–475.

In this paper, the authors consider the development of mathematical concepts over time. Specific attention is given to the shifting of the learner’s attention from step-by-step procedures that are performed in time to symbolism that can be manipulated as mental entities on paper and in the mind. The analysis uses different theoretical perspectives, including the SOLO model, APOS Theory, and various other theories of concept construction. The analysis reveals a fundamental cycle underlying conceptual development from Actions in time to concepts that can be manipulated as mental entities. This cycle appears widely in different ways of thinking that occur throughout mathematical learning.

Possani, E., Trigueros, M., Preciado, J. G., & Lozano, D. (2010). Use of models in the teaching of linear algebra. *Linear Algebra and its Applications*, 432, 2125–2140.

The authors present the results of an approach to teaching linear algebra using models. Their interest lies in analyzing the use of two theories of mathematics education, namely, Models and Modeling and APOS Theory. These two theories are used in the design of a teaching sequence that starts with presenting “real-life” decision-making problems to students. The possibilities of this methodology are illustrated through the analysis and description of classroom experience involving a problem related to traffic flow that elicits the use of a system of linear equations and different parameterizations of this system to answer questions on traffic control. Cycles of students’ work on the problem and the advantages of this approach in terms of students’ learning are described. The possibilities for extending it to other problems and linear algebra concepts are also discussed.

Ramirez, A. (2009). *A cognitive approach to solving systems of linear equations*. Ph.D. Dissertation, Illinois State University.

In this study, the author investigated the ways in which students come to understand systems of linear equations. Data were collected from observations of a teaching sequence with a small linear algebra class and from written tasks presented during interviews with four students from the same class. In her analysis of the data, the author used APOS Theory to conclude that systems of linear equations are a part of one’s Object conception of equivalent systems. This finding represented a modification of the author’s preliminary genetic decomposition.

Reynolds, B. E., & Fenton, W. E. (2006). *College geometry: Using the geometer’s sketchpad*. Hoboken, NJ: Wiley.

In this book, APOS Theory is used as a grounded learning theory for college-level courses on Euclidean and non-Euclidean geometries. The textbook relies on the use of Geometer’s Sketchpad that provides a dynamic interactive environment for students to explore the properties of geometric figures and their relationships.

Reynolds, B. E., Przybylski, J., Kiaie, C. C., Schwingendorf, K. E., & Dubinsky, E. (1996). *Precalculus, concepts & computers*. New York: McGraw-Hill.

This is a course in precalculus employing the ACE Teaching Cycle using ISETL activities. This is part of the calculus series with Dubinsky et al. (1995) and Dubinsky and Schwingendorf (1995).

Roa Fuentes, S. (2012). *El infinito: un análisis cognitivo de niños y jóvenes talento en matemáticas*. Doctorate thesis, Cinvestav-IPN, Mexico.

In this research study, an approach to mathematical talent is presented from a cognitive point of view, based on three pillars: APOS Theory, the construction of mathematical infinity as an iterative Process, and academic programs in Colombia and Mexico that focus on maximizing mathematical talent.

A genetic decomposition of infinity is presented, where the mental structures and mechanisms that an individual might develop in order to construct mathematical infinity in different contexts are described. Particular analyses for the Tennis Ball Paradox, the Hilbert’s Hotel, and the construction of the Koch curve are proposed. In all of these situations, the specificity of the iterative Processes and the role of the context in the construction of the infinity concept are analyzed.

- Roa-Fuentes, S., & Oktaç, A. (2010). Construcción de una descomposición genética: Análisis teórico del concepto transformación lineal. *Revista Latinoamericana de Investigación en Matemática Educativa*, 13(1), 89–112.

In this article, two preliminary genetic decompositions for the linear transformation concept are presented in detail, one that aligns with the treatment of this concept in most textbooks and another that starts with the construction of the concept of (general) transformation between vector spaces.

- Roa-Fuentes, S., & Oktaç, A. (2012). Validación de una descomposición genética de transformación lineal: Un análisis refinado por la aplicación del ciclo de investigación de APOE. *Revista Latinoamericana de Investigación en Matemática Educativa*, 15(2), 199–232.

In this paper, the third component of APOS Theory, namely, the “collection and analysis of data” phase is developed for the linear transformation concept. The authors design a diagnostic test and an interview with college students that are based on the theoretical analysis suggested by Roa-Fuentes and Oktaç (2010). Analysis of data shows that the properties of addition of vectors and multiplication of a vector by a scalar must be coordinated in order for the student to construct the concept of linear transformation as a Process. A refined genetic decomposition and didactic suggestions in relation to the construction of properties and the preservation of linear combinations follow.

- Salgado, H., & Trigueros, M. (2009). Conteo: una propuesta didáctica y su análisis. *Educación Matemática*, 21, 91–117

This paper uses APOS Theory to study the learning of the concepts related to combinations and permutations. The authors present a preliminary genetic decomposition for the construction of these concepts and consider a didactical approach to teach them at the university level. After teaching a course, the authors refined the genetic decomposition and the didactical sequence and tested them in the next semester. The authors analyzed the production of the students during the two semesters and the results of the exam corresponding to that topic. Results show how students’ mental constructions develop while they work with the activities and conclude that students of the second experience showed a better understanding of the concepts related to combinations and permutations.

- Schwingendorf, K. E., McCabe, G. P., & Kuhn, J. (2000). A longitudinal study of the C4L calculus reform program: Comparisons of C4L and traditional students. In *Research in Collegiate mathematics education IV*. CBMS issues in mathematics education (Vol. 8, pp. 63–76). Providence, RI: American Mathematical Society.

The authors present results of a statistical comparison between 205 students who took the course Calculus, Concepts, Computers, and Cooperative Learning (a reform course designed using APOS Theory) and 4431 students who took a traditional calculus course at Purdue University. When compared with the traditionally taught students, the students who received the reform course earned higher grades in further calculus courses, were as well prepared for math courses beyond calculus, as well as all other academic courses, took more calculus courses, and completed about the same number of non-calculus mathematics courses.

- Stenger, C., Weller, K., Arnon, I., Dubinsky, E., & Vidaković, D. (2008). A search for a constructivist approach for understanding the uncountable set $P(N)$. *Revista Latinoamericana de Investigación en Matemáticas Educativas*, 11(1), 93–126.

This study considers the question of whether individuals build mental structures for the set $P(N)$ that give meaning to the phrase “all subsets of N .” The contributions concerning this question are twofold. First, constructivist perspectives were identified and described, such that have been or could be used to describe individuals’ thinking about infinite sets,

specifically the set of natural numbers N . APOS was one of the perspectives considered. Second, to determine whether individuals' thinking about the set $P(N)$ can be interpreted in terms of one or more of these perspectives, eight mathematicians were interviewed. Their ideas about N and $P(N)$ were analyzed in terms of the chosen perspectives. The authors found that APOS Theory seems to explain more readily the mental structures for N . For $P(N)$, the same analysis cast doubt on whether individual understanding of the set $P(N)$ extends beyond the formal definition.

Tabaghi, S. G., Mamolo, A., & Sinclair, N. (2009). The effect of DGS on students' conception of slope. In *Proceedings of the 31st annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol. 5, pp. 226–234). Atlanta, GA: Georgia State University.

This report is the first installment of a broader study that investigated university students' conceptualizations of static and dynamic geometric entities. In this part, a refined look at the conceptualizations of two groups of students is offered—one group which was taught using Dynamic Geometric Software and the other in a “traditional” fashion. Both APOS Theory and the notion of reification were used to interpret learners' understanding of the slope of lines. Data revealed that students using DGS developed a strong proceptual understanding of slope, which enabled them to solve problems in which slope could be seen as a conceptual Object. This report sets the stage for a look forward to how DGS may influence learners' Process-Object conceptualization of other geometric representations of algebraic equations.

Thomas, K. S. C. (1995). *The fundamental theorem of calculus: An investigation into students' constructions*. Unpublished doctoral dissertation, Purdue University, West Lafayette.

This study was designed to investigate the question, “How can the fundamental theorem of calculus be learned, and how do computer activities and the pedagogy of a particular kind of nontraditional calculus course affect this learning?” The nontraditional calculus course was based on APOS Theory and used the ACE Teaching Cycle. The study found that the participants' function Schemas contained a misconception. The students believed that the name of the independent variable was a significant characteristic of a function and that it was a characteristic which was subject to being changed when a Process such as differentiation was applied to the function.

Tossavainen, T. (2009). Who can solve $2x=1$?—An analysis of cognitive load related to learning linear equation solving. *The Montana Mathematics Enthusiast*, 6(3), 435–448.

Using $2x = 1$ as an example, the cognitive load related to learning how to solve linear equations is discussed. Intrinsic cognitive loads needed in arithmetical, geometrical, and real analytical approaches to linear equation solving are considered using the framework of the Cognitive Load Theory. This is done from the point of view of the conceptual and procedural knowledge of mathematics and APOS Theory. A design of a setting for teaching linear equation solving is offered.

Trigueros, M. (2004). Understanding the meaning and representation of straight line solutions of systems of differential equations. In D.E. McDougall & J.A. Ross (Eds.), *Proceedings of the Twenty-sixth Annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol. 1, pp. 127–134). Toronto.

The main purpose of this study is the analysis of student responses to questions related to their understanding of the meaning and representation of straight-line solutions of systems of differential equations. Students' responses to questions involving the linearity theorem in the context of systems of linear differential equations and the geometric representation of straight line solutions to these systems were analyzed using APOS Theory with particular

focus on the development of Schema structures. Students' responses provided evidence of difficulties in relating concepts that come from different areas of mathematics even when the students could apply certain solution methods. Some instructional activities that seem to be successful are suggested.

- Trigueros, M. (2005). La noción del esquema en la investigación en matemática educativa a nivel superior. *Educación Matemática*, 17 (1), 5–31.

Piaget's work is the epistemological source of some of the theories that are used in the field of mathematics education research. In this paper, the fundamental ideas of one of these theories, APOS Theory, are presented. It is shown how this theory is evolving dynamically and continuously through the investigation of university students' understanding of advanced mathematical concepts and whether students are able to integrate several concepts in the solution of specific problem situations.

- Trigueros, M., & Campero, J. (2010). Propuesta didáctica en optimización dinámica. Investigación en el aula. *Educación Matemática*, 22(3), 87–117.

The purpose of this paper is to present the results of a research study on a didactical proposal to teach dynamical optimization, in particular, calculus of variations. The proposal design was based on APOS Theory and was tested at a private Mexican university. Results obtained from the analysis of students' responses to a questionnaire and an interview show that students construct Process conceptions, and in some cases, Object conceptions, of the related concepts. However, some obstacles were difficult for the students to overcome.

- Trigueros Gaisman, M., & Escandón, C. (2008). Los conceptos relevantes en el aprendizaje de la graficación. Un análisis a través de la estadística implicative. *Revista Mexicana de Investigación Educativa*, 13, 59–85.

Various studies show that students experience difficulties in understanding specific concepts of differential calculus. Some studies point to the obstacles students have in integrating different concepts into solving specific problems, including the writing of functions. The current study uses an instrument based on a genetic decomposition that was used in previous APOS studies (Cooley et al. 2007). Responses from 40 students were analyzed using implicative and cohesive statistics as an analytical tool. The results show that it is important for students to understand the second derivative and the intervals into which the domain is subdivided. The use of the particular statistical tool was found to be both pertinent and highly useful since the results that were obtained were similar to results obtained in previous studies using qualitative analysis.

- Trigueros, M., & Lage, A. (2006). An analysis of students' ideas about transformations of functions. In S. Alatorre, J. L. Cortina, M. Sáiz, & A. Méndez (Eds.), *Proceedings of the 28th annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (pp. 23–30). Mérida, México: Universidad Pedagógica Nacional.

This study contributes to researchers' and instructors' understanding of students' difficulties with transformations of functions. Students were interviewed while solving problems involving such transformations. The results, which were analyzed using APOS Theory, show that few students can work confidently with these problems involving transformations of functions. The analysis showed limited evidence of students who had interiorized the Actions involved in transformations of functions into Processes or who had encapsulated those Processes into Objects.

Trigueros, M., & Martínez-Planell, R. (2010). Geometrical representations in the learning of two-variable functions. *Educational Studies in Mathematics*, 73, 3–19.

This study is part of a project concerned with the analysis of how students work with two-variable functions, a topic of fundamental importance in mathematics and its applications. The authors investigate the relationship between students' notion of subsets of Cartesian three-dimensional space and the understanding of graphs of two-variable functions. APOS Theory and Duval's theory of semiotic representations are used in the analysis. Nine students who had taken a multivariable calculus course were interviewed. Results show that students' understanding can be related to the structure of their Schema for $\mathbb{R}^3$ and to their flexibility in the use of different representations.

Trigueros, M., & Okaç, A. (2005). La théorie APOS et l'enseignement de l'Algèbre Linéaire. *Annales de Didactique et de Sciences Cognitives. Revue internationale de didactique des mathématiques* (Vol. 10, pp. 157–176). IREM de Strasbourg, Université Louis Pasteur.

The authors use APOS Theory to describe the students' mental constructions of linear algebra concepts. Special emphasis is placed on the concept of vector spaces since it is one of the fundamental concepts of linear algebra and constitutes the beginning of an introductory course.

Trigueros, M., Okaç, A., & Manzanero, L. (2007). Understanding of systems of equations in linear algebra. In D. Pitta-Pantazi & G. Philippou (Eds.), *Proceedings of the 5th Congress of the European Society for Research in Mathematics Education* (pp. 2359–2368). Larnaca, Cyprus: University of Cyprus.

In this study, six students who were taking a course based on APOS Theory were interviewed at the beginning of the course and at the end of the course in order to study the viability of a proposed genetic decomposition of the concept of linear systems of equations. The study also focused on students' difficulties, their reasoning patterns, and the evolution of their development of Schema (as defined in APOS Theory). Results show that the students' progress depended strongly on development of their Schema for variable. The data also showed that a course based on APOS Theory helps students in the development of their systems of equations Schema.

12.7 V Through Z

Vidaković, D. (1996). Learning the concept of inverse function. *Journal of Computers in Mathematics and Science Teaching*, 15, 295–318.

This report is a part of a study that was conducted with five individual students and five groups of students who were assigned to work together in the first course of the experimental calculus classes at Purdue University during the fall of 1992. The goal of the study was to “discover” how the concept of inverse function can be learned, and hence taught, as well as to investigate the differences between group and individual mental constructions of that particular concept. The research followed the APOS research paradigm. It used the data to obtain a genetic decomposition of the concept. On the basis of the genetic decomposition, an instructional treatment was proposed. The instructional treatment consisted of computer activities designed to encourage students' development of a Schema for inverse functions. This instructional treatment has not yet been implemented.

Vidaković, D. (1997). Learning the concept of inverse function in a group versus individual environment. In E. Dubinsky, D. Mathews, & B. E. Reynolds (Eds.), *Cooperative learning for undergraduate mathematics*. MAA Notes 44 (pp. 175–196). Washington, DC: Mathematical Association of America.

The study was conducted with five groups of students working together on learning activities and five individuals working alone on the same tasks. The mathematical issue was the concept of function, in particular inverse and composition. The author was interested in knowledge about the mental structures that underlie the cognitive development of these concepts and differences between group and individual learning. An APOS analysis of the data resulted in the development of a genetic decomposition for the concept of inverse function and a related instructional treatment. The author also discovered that students' mental constructions were similar, regardless of whether they worked in an individual or collaborative setting, but that the quality and quantity of their learning was enhanced by working in collaboration with others.

Vidaković, D., & Martin, W. O. (2004). Small-group searches for mathematical proofs and individual reconstructions of mathematical concepts. *Journal of Mathematical Behavior*, 23, 465–492.

The authors investigate and report on cooperative learning situations, specifically, how individual ideas develop in a social context. Students with experience doing proofs in group situations were videotaped working collaboratively on three mathematical statements. Later, the students viewed segments of the group video and reflected on the activity of their group. The authors observed changes in understanding that may have resulted from parallel and successive interiorization and externalization of ideas by individuals in a social context.

Vizcaíno, O. (2004). *Evaluación del aprendizaje del cálculo desde una perspectiva constructivista*. Doctorate Thesis, CICATA, Instituto, Politécnico Nacional, Mexico.

This thesis is based on the idea that evaluation of the process of teaching and learning is very important but also very complex and difficult to do. Moreover, simplistic methods can even be counterproductive. Traditional methods of evaluation, which assign a number to a student's ability, can provide a distorted picture of the learning that may or may not have taken place or is about to take place. As a result, such methods of evaluation may not provide a reliable guide to improving pedagogy and its results in terms of learning. An alternative to traditional methods of evaluation is the method of interviewing students individually. Unfortunately, this requires more time and energy than is available to most teachers.

In this thesis, the author tests a third method based on APOS Theory and the ACE Teaching Cycle. In this approach, evaluation is done through a series of instruments that can be designed, administered, and scored within the normal progress of the class. To test this method, the author used APOS and the ACE Teaching Cycle to teach a course in calculus. The final grades (on a scale of 0 to 100) of the students were determined using the evaluation methods proposed in APOS/ACE. Then an interview was conducted with each student and the final grade was determined (using the same scale) again. Thus, the comparison was between an evaluation method that was practical to use and a method that was considered very accurate but highly impractical to use. The correlation between the two sets of grades was very high (about 0.87), suggesting that the practical method could be used to obtain high quality evaluations, at least in the case when the teaching was based on APOS/ACE.

Weller, K., Arnon, I., & Dubinsky, E. (2009). Preservice teachers' understanding of the relation between a fraction or integer and its decimal expansion. *Canadian Journal of Science, Mathematics, and Technology Education*, 9, 5–28.

This article reports on the mathematical performance of preservice elementary and middle school teachers who completed a specially designed unit on repeating decimals, including the relation between $0.\bar{9}$ and 1. The teaching sequence was based on APOS Theory and implemented the use of the ACE Teaching Cycle. The quantitative results suggest that the students who received the experimental instruction made considerable progress in their development of an understanding of the equality between $0.\bar{9}$ and 1 as well as between any rational number and its decimal expansion(s). Students in a control group, who received traditional treatment on these topics, made substantially less progress. For further analysis of this study, see Weller et al. (2011) and Dubinsky et al. (2013) in this bibliography.

Weller, K., Arnon, I., & Dubinsky, E. (2011). Preservice teachers' understanding of the relation between a fraction or integer and its decimal expansion: Strength and stability of belief. *Canadian Journal of Science, Mathematics, and Technology Education*, 11, 129–159.

In an earlier study of preservice elementary and middle school teachers' beliefs about repeating decimals, the same authors reported on a comparison of the mathematical performance of 77 preservice teachers who completed an APOS-based instructional unit with 127 preservice teachers who completed traditional instruction. The study was based on interviews conducted 4 months after the instruction with 47 of these students. The interviews revealed that the students who received the APOS-based instruction developed stronger and more stable beliefs (over time) regarding their beliefs about repeating decimals and the connection between repeating decimals and other rational number representations. In their analysis, the authors develop a number of indices and categories that may prove useful in other comparative studies involving interview and questionnaire data with a large number of students. For further analysis of this study, see Weller et al. (2009) and Dubinsky et al. (2013) in this bibliography.

Weller, K., Brown, A., Dubinsky, E., McDonald, M., & Stenger, C. (2004). Intimations of infinity. *Notices of the AMS*, 51, 741–750.

The purpose of this article is to show how APOS, being a theory about how people come to understand mathematics, can be helpful in understanding the thinking of both novices and practitioners as they grapple with the notion of infinity.

Weller, K., Clark, J. M., Dubinsky, E., Loch, S., McDonald, M. A., & Merkovsky, R. (2003). Student performance and attitudes in courses based on APOS theory and the ACE teaching cycle. In *Research in Collegiate mathematics education V*. CBMS issues in mathematics education (Vol. 12, pp. 97–131). Providence, RI: American Mathematical Society.

The authors examine the effectiveness of instruction based on APOS Theory and the ACE Teaching Cycle using data from 14 previous studies in the areas of calculus, abstract algebra, concept of function, quantification, induction, and the affective domain. The results suggest that instruction based on APOS Theory may be an effective tool in helping students to learn mathematical concepts.

Weller, K., Montgomery, A., Clark, J., Cottrill, J., Trigueros, M., Arnon, I., & Dubinsky, E. (2002). Learning Linear Algebra with ISETL. Available from <http://homepages.ohiodominican.edu/~cottrillj/datastore/linear-alg/LLAWI-P3.pdf>. Accessed 9 Jan 2013.

This is a first course in linear algebra employing the ACE Teaching Cycle using ISETL activities. A matrix environment is implemented in ISETL to allow work in matrices over finite fields and matrix algebra. The text covers vector spaces, solutions to systems of

equations, basis of vector space, and linear transformations. Extended topics include matrices of transformations, change of basis, diagonalization, and eigenvectors/eigenvalues. The text has 23 sections that allow for customization in a one-semester course or the possibility of a two-semester sequence.

- Zazkis, R., & Campbell, S. (1996). Divisibility and multiplicative structure of natural numbers: Preservice teachers' understanding. *Journal for Research in Mathematics Education*, 27, 540–563.

This study contributes to a growing body of research on teachers' content knowledge in mathematics. The domain under investigation was elementary number theory. The main focus concerned the concept of divisibility and its relation to division, multiplication, prime and composite numbers, factorization, divisibility rules, and prime decomposition. The APO (Action-Process-Object) framework was used for analyzing and interpreting data acquired in clinical interviews with preservice teachers. Participants' responses to questions and tasks indicated pervasive dispositions toward procedural attachments, even when some degree of conceptual understanding was evident. The results of this study provide a preliminary overview of cognitive structures in elementary number theory.

- Zazkis, R., & Gunn, C. (1997). Sets, subsets and the empty set: Students' constructions and mathematical conventions. *Journal of Computers in Mathematics and Science Teaching*, 16, 133–169.

This study investigates preservice elementary school teachers' understandings of basic concepts of set theory. The students' understandings are analyzed after instruction using ISETL. Analysis of the data, which is based on APOS (not yet so called), reveals students' difficulties with the idea of a set as an element of a set and the idea of the empty set.

- Zazkis, R., & Khoury, H. (1994). To the right of the “decimal” point: Preservice teachers' concepts of place value and multidigit structures. In *Research in Collegiate mathematics education I*. CBMS issues in mathematics education (Vol. 4, pp. 195–224). Providence, RI: American Mathematical Society.

The focus of this work is on preservice elementary teachers' understanding of concepts related to place value through the lens of the APOS framework. Special emphasis is put on the de-encapsulation mechanism through a conversion task used in an interview. A genetic decomposition for the construction of non-decimal number is presented.

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